

MAP-MAKING GUIDE

Everything human-made: where settlements go, how big they are, and where to put capitals, castles, roads, borders, farms, mines, churches and ruins.



Based on real history, c. 1000–1500, with notes to 1650 · continent, kingdom and local maps

HOW TO USE THIS GUIDE

1. **Start with [Chapter 0: Step by Step](#).** It tells you what to draw, in order, from a blank continent down to a single valley. Each step links to the chapter that explains it.
2. **Keep [Chapter 13: Quick Reference](#) open while you draw.** It has every key number and the master legend of map symbols.
3. **Read [Chapters 1–11](#) when you want the reasons,** the real-world examples and the fine detail.
4. **See it done in [Chapter 12: Worked Example](#),** which builds a whole kingdom with the guide's numbers and shows the finished map.
5. **Check your map against [Chapter 11: Common Mistakes](#)** when you are done.

THE FOUR KINDS OF NOTE BOX

MAP TIP

What to draw, where, and which symbol to use.

RULE OF THUMB

A short rule you can remember while drawing.

LATER ERA (1500s+)

What changes after the Middle Ages.

FANTASY TWIST

How magic, monsters or other races change the rule.

Distances are given in kilometres and miles. Populations are ranges, because historians' estimates differ. Blue text is a link: click it to jump to that chapter or section.

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CHAPTER 0

STEP BY STEP: FROM BLANK MAP TO LIVING WORLD

This chapter is the master method of the guide. It puts the other chapters in order: what to decide first, what to show at each scale, what to place next and how to test each result. It assumes your terrain is already drawn: coasts, mountains, rivers, forests, marshes and climate. Every historical number is copied from the chapter section linked in that step or table.

IN THIS CHAPTER

Step 0: Choose the date, realm type and scale

What to draw at each scale

The order of work

Part A: Continent and world scale

Part B: Kingdom and region scale

Part C: Local area scale

Finishing: fantasy and the final check

Printable checklist

Quick summary

Sources and further reading

Step 0: Choose the date, realm type and scale

RULE OF THUMB

Make four decisions before you draw a single town: the date, the kind of realm, the scale and the amount of magic. Each one changes the numbers you use.

Why: The same kingdom looks very different in 1100 and in 1600. A strong kingdom with one capital looks different from a loose empire of princes. And the scale decides what you can draw at all: on a kingdom map a whole village is smaller than a pen dot.

Choose the date

Pick one year and keep to it. These four dates give four quite different maps. Every feature has a first date, so check [Mixing eras without reason](#) before you add anything new.

Date	People and towns	Castles and war	Church and learning	Rulers and links
c. 1100	Growing, but thin. England had about 1.7 million people in 1086, about a third of its 1300 peak. Fewer and smaller towns.	Mostly mottes (man-made earth mounds) and ringworks of earth and timber; a few great stone keeps (the White Tower, London, c. 1075–1100).	No friaries yet (they arrive in the 1210s–1220s). Universities are only beginning (Bologna, late 11th century). The Cistercian order is new (founded 1098).	Most western kings travel: draw 5–15 royal residences, not one capital. Watermills, but no windmills (first certain record 1185).
c. 1300	The crowded peak: England about 4.75 million. Most towns and markets have been founded (1,746 English markets recorded by 1300, many tiny).	Stone castles, concentric castles and town walls.	Friaries in every big town; about 20 universities in Europe.	Fixed capitals in the strongest kingdoms (Paris, Westminster and London). Windmills common in flat, dry country.
c. 1450	A third to a half fewer people after the Black Death (1347–51) and later plagues; England's low point was about 1.9 million c. 1450. Deserted villages and more sheep pasture; many small markets are failing.	Many 12th-century castles half-ruined; tower houses in raided borders; by the mid-1400s cannon decide sieges.	Universities spreading (60–80 by 1500).	The first canal pound locks (Low Countries, northern Italy).
c. 1600	Back near the 1300 level (England about 4.1 million). Capitals and ocean ports 2–4 times bigger than in 1500 (London about 200,000).	Star forts at frontiers, capitals and main ports; interior castles become ruins or palaces.	In Protestant lands most monasteries are closed (England 1536–1541); draw abbey ruins, and show each faith as a soft colour tint, not a new border line.	Fixed capitals everywhere (Madrid from 1561); post roads with stations every 20–40 km (12–25 mi); ocean trade.

All figures come from the guide's chapters: [Change over time](#), [When and why capitals became fixed](#) and [What Changes After 1500](#).

Choose the realm type

Most real realms sit somewhere between the two columns below. France c. 1200, for example, had a royal core around Paris and six great fiefs that were nearly independent. Decide which end your realm is closer to.

	Centralised kingdom	Fragmented realm
Real examples	England; France (more loosely) by about 1300	The Holy Roman Empire (Germany); the city-states of northern Italy
Capital	One fixed capital that holds 1–2% of the realm's people and is 3–7 times the size of the second city	No real capital, or a court in a middling town. Several rival cities of similar size. The largest city holds about 0.5% of the people or less (Cologne in Germany c. 1300: about 0.4–0.5%)
Inner borders	A regular network of counties, mostly 50–70 km (30–45 mi) across, each with a king's officer (a sheriff in England)	A patchwork of princes, bishops and free cities, with enclaves (the Empire had about 400 imperial estates in 1521)
Castles	Fewer, mostly royal or great baronial: about 4 active castles per 1,000 km ² (390 sq mi) in England and Wales	Many small lords' castles and toll castles: in hilly German-type lands about 25 castle sites per 1,000 km ² , 8–12 of them active (estimate)
Mints	3–4, controlled by the king	One in every princely seat, many bishops' cities and the big free towns

Read: [Capital is not always the biggest city](#); [Kinds of states and how they look on a map](#); [How many mints?](#)

Choose the scale and the amount of magic

Decide	Options	What it changes
Scale	Continent, kingdom or local area	What you draw and how big it is on paper: see What to draw at each scale , then Part A, B or C below.
Magic	None, a little or a lot	Nothing at first. Use the real rules, then adjust in Step 17 .

If you draw only one kingdom, do Part A for that kingdom, then Parts B and C.

Check: Can you write one line under the title, such as “The Kingdom of X, Year 1312: a strong kingdom, 1:1,000,000, little magic”?

MAP TIP

Write the date in the title box (“The Kingdom of X, Year 1312”) and draw a scale bar first. If you are unsure which date to choose, choose c. 1300: most of this guide’s numbers are for that date.

What to draw at each scale

RULE OF THUMB

Divide the scale number by 1,000 to get metres per millimetre. At 1:1,000,000, 1 mm on paper is 1,000 m (1 km) on the ground. If a thing is smaller than about 1 mm at your scale, draw a symbol; if it is bigger, you can draw its outline.

Why: A map cannot show everything. As the scale gets smaller, mapmakers leave features out and replace the rest with symbols (this is called *generalisation*). The chapters of this guide each give advice for their own features. This section puts it in one table, so each map shows the right things in the right numbers.

Scale arithmetic: how big things are on paper

A scale of 1:50,000 means that 1 cm on the map stands for 50,000 cm (500 m) on the ground. A **large-scale** map has a small number after the colon and shows more detail; a **small-scale** map has a big number and shows less.

Map	Typical scale	1 mm on paper =	1 cm =	One A3 sheet (297 × 420 mm, 11.7 × 16.5 in) covers	How big things are
Continent	1:10,000,000	10 km	100 km (about 1 inch = 160 mi)	3,000 × 4,200 km (1,850 × 2,600 mi): about the whole Mediterranean, which is about 4,000 km (2,500 mi) from west to east	An England-sized kingdom (130,000 km ² , 50,000 sq mi) is only about 36 × 36 mm. A great city 2–4 km across is 0.2–0.4 mm: a dot. A county is 5–7 mm.
Kingdom	1:1,000,000	1 km	10 km (about 1 inch = 16 mi)	300 × 420 km (185 × 260 mi), about 125,000 km ² : one England-sized realm only if it were square	A great city (2–4 km) is 2–4 mm; a city (1–2 km) 1–2 mm; a town (0.5–1 km) 0.5–1 mm. Use symbols. A county is 50–70 mm across. Market towns are 9–16 mm apart; villages only 1.5–4 mm apart.
Local	1:50,000–1:100,000	50–100 m	0.5–1 km (1:100,000 is about 1 inch = 1.6 mi)	15 × 21 km (9 × 13 mi) to 30 × 42 km (18 × 26 mi)	A village (250–450 m) is 2.5–9 mm; a town (0.5–1 km) 5–20 mm; a great abbey precinct (350–600 m) 3.5–12 mm. Draw outlines. A moated manor (40–80 m) is under 2 mm, so use a symbol.

Real realms are rarely square. England is about 500 km (310 mi) from London to Berwick alone, so a whole England-sized kingdom fits on one A3 sheet only at about 1:2,000,000 (1 mm = 2 km). At that scale even a great city is a 1–2 mm symbol.

The sizes of places come from [How big to draw it](#) and [Footprints](#).

The feature filter: what to show and how many

RULE OF THUMB

Draw what a traveller at that scale would notice. A continent map shows the few great places, a kingdom map shows every town, and a local map shows every village and church.

The counts are for c. 1300 and an English-style realm unless the table says otherwise. They are averages: real features cluster on good land and along routes. For areas, 100 km² is 39 sq mi and 1,000 km² is 386 sq mi. The local column fits maps up to about 20 × 20 km (12 × 12 mi). On a bigger local map, such as a 50 × 50 km (30 × 30 mi) county, draw villages as dots and leave out mills and hamlets.

Feature	Continent (1:10,000,000)	Kingdom (1:1,000,000)	Local (1:50,000–1:100,000)
Realms and fiefs	All realms and great fiefs; at least three kinds of state	The realm border, 4–8 duchies or great fiefs, 1–3 special zones (a march, a palatinate, a church territory)	Only a border that crosses the map
Counties and districts	—	Counties 50–70 km (30–45 mi) across: 30–40 in an England-sized realm (England had 39 shires)	Hundreds (districts within a county) 12–15 km (7–9 mi) across; parish boundaries
Great cities (50,000+)	3–8 on the whole continent, almost all on the sea or a navigable river	All, as large symbols	Full outline with suburbs, 20–80 mm across
Cities and towns (2,000–50,000)	Capitals and the main cities only, as dots	Every town of 2,000+, as a symbol	Outline with walls, gates and market place
Market towns (500–2,000)	—	The most important third or so: 30–40 of about 90 in a 60,000 km ² realm of 1.2 million	All, 9–16 km (6–10 mi) apart
Villages	—	Only as texture in crowded lowland	All: 1.5–4 km (1–2.5 mi) apart, about 7 per 100 km ² in lowland
Hamlets and farmsteads	—	—	0–5 per 100 km ² in open-field lowland; 20+ in dispersed uplands
Castles and forts	The greatest fortresses, citadels and naval arsenals: about 1 per 20,000–50,000 km ² (8,000–19,000 sq mi)	Royal and major castles: about 1 per 1,000 km ² on average, many more on a contested border	All: in England about 12 castle sites and 4 active castles per 1,000 km ² , so about 5 sites and 1–2 active castles on a 20 × 20 km map; 3–5 times more in a march
Routes	Sea lanes, river spines and long trade routes; 1–3 main passes per mountain range	Main roads: about 30 km per 1,000 km ² (48 mi per 1,000 sq mi); an inn or village every 15–30 km (10–19 mi); bridges about 20 km (12 mi) apart on a lowland river; minor passes and hospices	All roads, lanes and footpaths: 2–3 roads out of each village
Church	1–2 great shrines on the whole continent; archbishops' seats; universities (only about 20 in all Europe c. 1300)	Cathedrals: 5–10 per 50,000 km ² in an English-style church, 25–100 in an Italian-style one; great abbeys; 1–3 shrines; 0–2 universities	Every parish church (about 7 per 100 km ² , or 73 per 1,000 km ²); abbeys, friaries, hospitals, leper houses and gal-lows
Industry and resources	Mining-district symbols on 3–6 mountain areas; one or two great fair regions	1–3 mining districts, 1–3 salt sources, one cloth region, 1–2 fair towns and several ports	Shafts, quarries and salt pans; mills, about 8–11 per 100 km ² (roughly one per village)
Manors and parks	—	Royal forests, as green dash-dot boundaries	Moated sites (4–5 per 100 km ²), deer parks (2–3 per 100 km ²), fishponds and warrens
Ruins and older layers	Only famous ruins and the old empire's roads	Big ruins and long straight old roads	All: hillforts (1–3 per 100 km ² in hilly country), deserted villages and earth-works

Example: an England-sized kingdom (130,000 km², 50,000 sq mi) on one sheet. Draw about 6 cities of 10,000+, roughly 50–60 towns of 2,000–10,000, 30–40 counties, 13–26 cathedrals in the English style (England had 17), about 130 royal and major castles and about 3,900 km

(2,400 mi) of main road. Leave the 9,500 villages and parish churches for local maps.

Read: Map symbols and labels; How many to draw; How much main road to draw; What to show at each scale; How

many features: England c. 1300 averages; Mining districts on the map; Drawing it all: symbols, density and scale

Check: Multiply each “per area” count by the area of your map. Is every count close to the target, or can you say why not (a march, a desert, a crowded Flanders-like plain)?

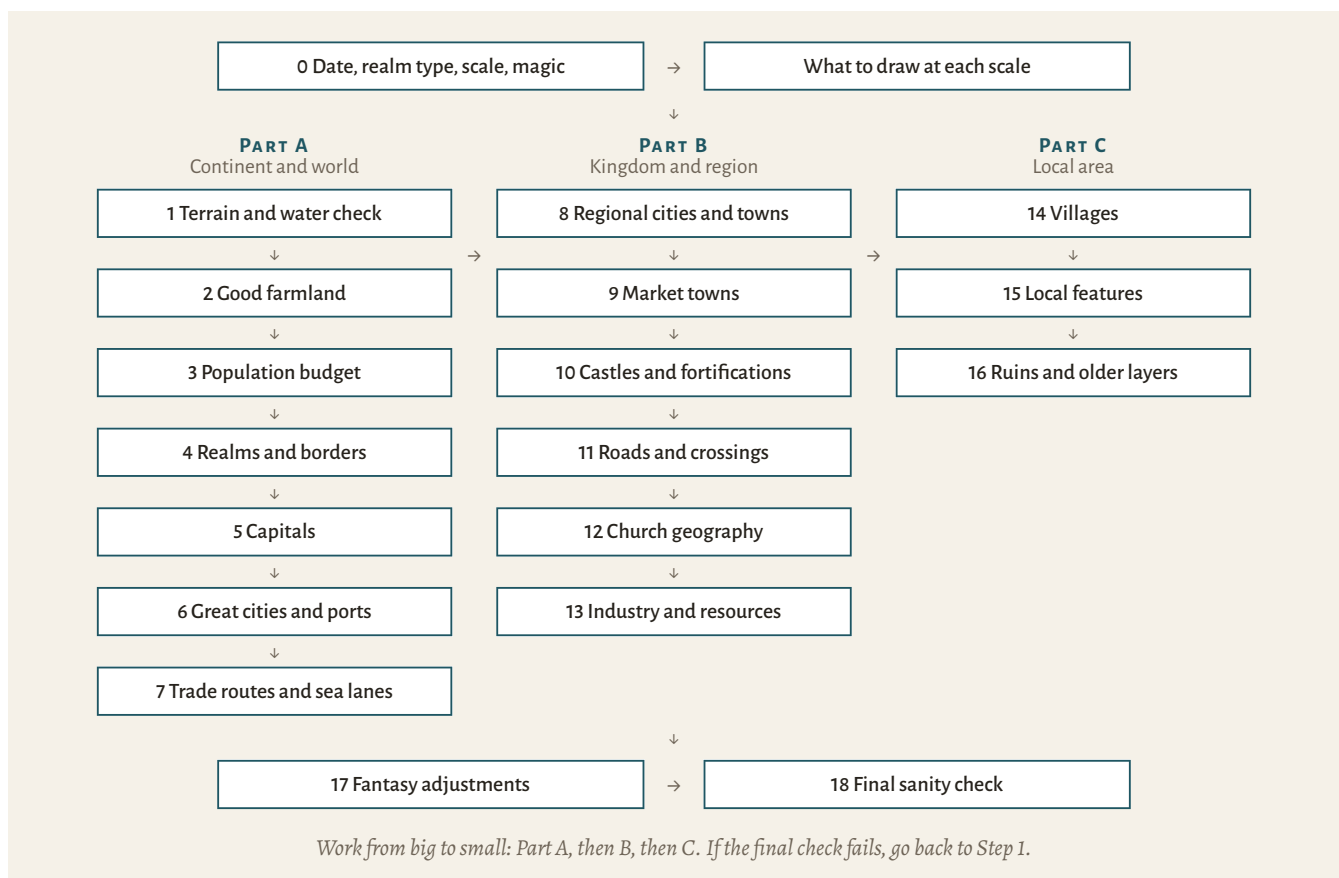
MAP TIP

Copy the column for your scale into the margin of your sketch, write the target number next to each feature and tick them off as you place them. A count far from the target is not always wrong, but it needs a reason you could write on the map.

The order of work

RULE OF THUMB

Work from big to small, and from nature to people: water and food first, then people, then power, then everything that serves them.



Each step says what to **place**, gives the key **rules**, links to the section to **read** and ends with a **check**. Fix a failed

check before you go on: early mistakes spread into every later step.

MAP TIP

Work in pencil or on separate layers, so you can redraw an early step without erasing everything after it.

Part A: Continent and world scale

Here you decide where the people are, who rules them and how goods move.

Step 1: Check the terrain and water

Place: rivers, springs, marshes, fords, and on each big river the tidal limit (where the tide stops) and the head of navigation (the highest point boats can reach).

Rules:

- Mark which stretches are **navigable** (deep enough for boats) and where they stop: falls, rapids or shallows.
- Where the lowest bridge, the tidal limit and the highest point for sea ships come together is the best spot on the map for a big town.

Step 2: Find the good farmland

Place: light shading for crowded lowland, ordinary farmland, thin hill country and empty land (mountain, marsh, forest, desert).

Rules:

- A person needs about 1 ha (2.5 acres) of ploughland, or 2–3 ha (5–7 acres) of all land once pasture, meadow and woodland are counted.
- A village needs arable (ploughed fields), meadow (hay), pasture and woodland, so the best land is where these meet: river valleys, plains and gentle hills.

Step 3: Work out the population budget

Place: nothing yet. Work out the numbers per region now, and add up each realm after Step 4.

Rules:

1. **Measure** each zone. A hex's area is about $0.866 \times (\text{width across the flat sides})^2$; a 30 km hex is about 780 km².

Per 1 million people	Frontier (Poland, Hungary, Scotland, Scandinavia)	Average (England, France, Germany c. 1300)	Highly urban (Flanders, Lombardy, Tuscany)
Cities of 10,000+	0–1	1–2	5–10
Towns of 2,000–10,000	8–12	12–18	30–40
Small market towns	40–60	60–90	80–100
Villages	3,000–4,500	2,500–3,500	2,000–2,500

4. **Compare** with real realms: a medium European kingdom c. 1300 had 1–5 million people, and a realm the size of England (130,000 km², 50,000 sq mi) should have 2–5 million. After the plague, use a third to a half fewer.

Step 4: Draw realms and borders

Place: realm outlines, great fiefs (duchies and large counties) and frontier zones.

- Rivers join as they flow downhill. They almost never split, except in deltas and marshes.
- Mark strong sites too: hills, river loops, islands and spurs (ridge ends with cliffs on two or three sides).

Read: [Placing settlements on your map: a method](#)

Check: Can you point to the head of navigation on every big river, and does no river split outside a delta or marsh?

- Good lowland holds 30–50 people per km² (80–130 per sq mi) at its peak. Hill country (farmed valleys with upland grazing) holds 8–15 (21–39 per sq mi) at its peak; high uplands, moors and mountains hold only 2–5 (5–13 per sq mi).

Read: [Population density](#)

Check: Can you name the reason for every empty zone: mountain, marsh, heath, forest or desert?

2. **Multiply by a density:** rich lowland 30–40 people per km², average land 15–25, hill country 8–15, high uplands, mountains, forest and marsh 2–5, desert 0. (Multiply by 2.59 for people per sq mi.)
3. **Split the total** by an urban profile:

Read: [Step-by-step population calculator](#)

Check: Is each realm's total within the range of a real kingdom of the same size?

Rules:

- Use at least three kinds of state: one or two feudal kingdoms, a fragmented zone (city-states or a loose empire)

and one unusual kind (a town league, church state, sea empire or nomads).

- A kingdom covers 10,000–450,000 km² (4,000–175,000 sq mi). Direct rule reaches roughly 500–700 km (300–450 mi) from the capital, 8–12 days for a messenger. Farther out, a realm needs viceroys, great vassals, sea routes or a relay post, or it splits.
- Borders are lines in settled farmland and wide bands in mountains, forests, marshes and war zones. Draw a

march (a militarised border province) as hatching 10–50 km (6–30 mi) wide. Add an enclave or a disputed strip.

Read: [Borders and frontiers](#); [How big can a realm be](#)

Check: Can you say what every border segment follows: a crest, a river, a forest, an old district line or a treaty?

Step 5: Place the capitals

Place: one capital per realm, or a set of royal residences if the court travels.

Rules:

- A capital needs a rich core region, river or sea access, a defensible site, loyal land and legitimacy (an old royal or sacred site). Centrality matters less: Paris, London and Kraków all lay off-centre.
- In a strong, centralised kingdom the capital holds about 1–2% of the realm's people and is 3–7 times the size of the second city. In a loose realm the largest city holds about 0.5% or less.

- Before about 1200 most western kings had no fixed capital: draw 5–15 royal residences about a day's travel apart instead.

- Around a fixed capital, draw 3–6 residences within about 60 km (37 mi), a royal forest and a burial church, and perhaps a coronation town 50–150 km (30–90 mi) away.

Read: [What makes a capital](#); [Around the capital: the royal landscape](#)

Check: Can you say why each capital is where it is, with a better reason than “it is in the middle”?

LATER ERA (1500s+)

Fixed capitals win everywhere (Madrid from 1561). For a 1600-style map, draw the capital 2–4 times bigger than its medieval size. See [Population and bigger cities](#).

Step 6: Place the great cities and ports

Place: the few great cities (50,000+ people) and the main ports.

Rules:

- Around 1300, Europe, the Middle East and North Africa together had only about 22 cities of 50,000+ and 8 of 100,000+. Mark only 3–8 great cities on a continent map.
- A city of 50,000 almost always needs river or sea supply (rare exception: Granada on its irrigated plain). A city of 100,000 is only possible by water.

- Best sites: the lowest bridging point, the head of navigation, a confluence, an estuary or a superb harbour with shelter, deep water and a route inland. On tidal coasts, big ports often sat 50–120 km (30–75 mi) up an estuary.

Read: [The biggest cities](#); [What makes a good harbour](#)

Check: Is every great city on the sea or a navigable river (a rare exception, such as Granada, sits in a rich irrigated plain), with a clear reason to be great (a court, a strait, a delta, an export trade)?

FANTASY TWIST

A griffon carries only a pack horse's load (100–120 kg). Only magic that moves bulk, such as a portal passing shiploads of grain every day, frees great cities from the water rule. See [Teleport circles and portals](#).

Step 7: Draw the trade routes and sea lanes

Place: sea lanes, river routes, main mountain passes and long land routes.

Rules:

- Land : river : sea transport cost about 8 : 4 : 1 (England c. 1300). Bulk goods (grain, timber, stone, salt, wine) fol-

low water; they go more than 30–50 km (19–31 mi) by land only if valuable, or in war or famine.

- A continent usually has one or two bulk sea networks, several river spines and one or two long luxury land routes. Copy a real one (the Hanse, Venice and Genoa, the Silk Roads).
- Sea lanes follow coasts and island chains. On a galley coast, put a watering port every 100–300 km (60–190 mi).

MAP TIP

A continent map shows realms and great fiefs, 3–8 great cities, sea lanes, main passes, great shrines, mining districts in 3–6 mountain areas and only the greatest fortresses (about 1 per 20,000–50,000 km², 8,000–19,000 sq mi). Leave counties, villages and ordinary castles for the kingdom map.

- Give each mountain range 1–3 main passes and 2–5 minor ones. Desert routes need water every 30–40 km (19–25 mi).

Read: [Real trade networks to copy](#); [Mountain passes](#)

Check: For each region, can you write “exports ...”, imports ...” and trace each export to water or a market?

Part B: Kingdom and region scale

Now zoom in to one realm and fill it in.

Step 8: Place regional cities and towns

Place: counties and their seats, cities, and towns of 2,000–10,000 people.

Rules:

- Divide the realm into 4–8 duchies or great fiefs, then into counties 50–70 km (30–45 mi) across, each with a seat near its middle. An England-sized kingdom has 30–40 counties.
- Put towns at confluences, heads of navigation, gaps in hill ranges, the feet of passes, harbours and crossings of main roads.
- Cities of 10,000+ stand 60–90 km (37–56 mi) apart in very urban regions, 130–170 km (80–105 mi) in average

ones and 260–400 km (160–250 mi) in thinly urban ones. For each, expect about 10 towns and 50–100 market towns.

- A town of 10,000 needs farmland within 21–27 km (13–17 mi) on good land. Draw that circle faintly; if two circles overlap heavily, shrink one town or put it on water.

Read: [How far apart: the numbers](#); [Political units and their sizes](#)

Check: Can you write a “site” note (the ground it stands on) and a “situation” note (its place among routes and regions) for every town?

Step 9: Fill in the market towns

Place: small market towns of 500–2,000 people, each with a weekly market and a yearly fair.

Rules:

- Space them about 9–16 km (6–10 mi) apart, so a peasant can walk to market and home in one day (6–10 km, 4–6 mi, each way). In thinly settled land they can be up to about 25 km (15 mi) apart.

- Put them on rivers and road junctions. Two market towns 3 km (2 mi) apart compete; make one a village.
- Christaller’s central places form neat hexagons only on a flat, even plain. Stretch the pattern along rivers and roads, and squeeze it on rich land.

Read: [Central place theory, made simple](#)

Check: Can every village reach a market town and get home the same day?

Step 10: Place castles and fortifications

Place: castles, town walls, watchtowers and beacon chains.

Rules:

- Every fortification controls something. Write its job in one word (*road, bridge, town, border, estate, refuge or watch*); the job gives the site.

- Peaceful core: a working castle every 20–30 km (12–19 mi). Contested border: one every 4–6 km

(2.5–4 mi). Conquest chain: one every day's march, 20–40 km (12–25 mi), each reachable by water.

Zone, per 1,000 km ² (about 390 sq mi), c. 1300	All castle sites ever built	Active castles
Peaceful lowland core	4–8	1–3
England and Wales average	~12	~4
March or contested border	35–60	12–20

- Put the castle at the edge of its town, with the market at its gate. Wall cities, regional capitals, frontier towns and raided ports; leave most market towns open.
- Peacetime garrisons are only 5–20 men. Beacon chains run from the border to the capital, about 5–20 km (3–12 mi) apart in hilly country.

Read: [How many castles and how far apart](#); [How many to draw](#)

Check: Can you name what each castle controls, and is the map sparse in the core and dense on the border?

LATER ERA (1500S+)

Low, thick star forts with angled bastions spread beyond Italy in the 1530s–1540s. They were so expensive that only frontiers, capitals, main ports and key crossings get them. See [Gunpowder and the new fortifications](#).

Step 11: Draw roads and crossings

Place: bridges, fords, ferries, roads, inns, hospices and tolls.

Rules:

- Crossings first: roads bend to reach them and towns grow at them. A great river gets only a few bridges, often tens of km apart, with ferries and fords between.
- Main roads link cities through crossings and passes, along valleys and dry ridges; only an older empire's roads run straight. Village tracks run to the market town like wheel spokes.

- Put an inn or village every 15–30 km (10–19 mi) on main roads, a caravanserai (walled roadside inn) every 30–40 km (19–25 mi) in desert, and a hospice at each main pass. Tolls sit at bridges, gates, gorges, straits and borders.
- Walkers cover 25–35 km (15–22 mi) a day, ox carts 15–25 km (10–15 mi) and armies with baggage 13–20 km (8–12 mi).

Read: [Drawing the route network](#)

Check: For each road, can you say where it goes, what it carries and why it bends where it does?

LATER ERA (1500S+)

Post stations every 20–40 km (12–25 mi) carry letters up to about 150 km (95 mi) a day, and canals with pound locks begin to climb hills. See [Posts, roads and canals](#).

Step 12: Lay out the church geography

Place: cathedrals, abbeys, friaries, shrines and pilgrim roads, hospitals and universities.

Rules:

- **Dioceses** (one bishop's area): in an English-style realm, one per 5,000–10,000 km² (2,000–4,000 sq mi), so a cathedral means a main city. In an Italian-style land, one per 500–2,000 km² (190–770 sq mi), many in small towns.
- **Monasteries:** about 1 religious house per 150 km² (60 sq mi). Cistercians in empty, well-watered valleys; canons at town edges; friars inside towns of about

3,000–5,000 and up. Give big abbeys granges (outlying farms), most within about 25 km (15 mi).

- **Friaries measure towns:** usually none at 2,000–3,000 people, about 2 at 10,000, 4 at 20,000–40,000.
- **Pilgrim roads** follow existing roads, with stops about 20 km (12 mi) apart and a hospice at every pass.
- **Hospitals and universities:** a hospital in every market town; a leper house 1–2 km (about 1 mi) outside each town of 2,000+; one or two universities in a kingdom of 2–5 million.

Read: [Monasteries, friaries and military orders](#); [Drawing it all: symbols, density and scale](#)

Check: Is every Cistercian abbey in an empty valley, and every friary in a town big enough to feed it?

Step 13: Place industry and resource sites

Place: mines, salt works, quarries, cloth districts, fisheries and fair towns.

Rules:

- Industry goes to its heaviest input: the ore, the fuel or the water power, not the customer.
- Per kingdom, aim for 1–3 mining districts, 1–3 salt sources, one cloth region, one or two fair towns and several ports. Every realm needs salt, or a route that brings it in.

- A mining district holds 2–7 mining towns 10–40 km (6–25 mi) apart, up side valleys, plus a mint town. Most had 1,000–5,000 people; rare boom towns reached 10,000–20,000.
- Ironworks need charcoal and fast streams: draw many small forges, not one big town.

Read: [The basic rules: what pulls industry to a place; Special town types and how to spot them](#)

Check: For every industry, can you point to the resource, the fuel and the way out?

MAP TIP

A kingdom map shows the capital, cities, towns and the main market towns, with villages only as texture. Add county borders, the main castles, cathedrals, great abbeys and mining districts. Leave mills, hamlets and gallows for the local map.

Part C: Local area scale

A local map covers roughly 5 × 5 km to 50 × 50 km (3 × 3 to 30 × 30 mi). Almost every square kilometre has a human feature.

Step 14: Place the villages

Place: villages, hamlets and farmsteads, each village with its parish church.

Rules:

- Open-field land gives tight (nucleated) villages. Hills, forest and marsh give hamlets and lone farms. A road, valley stream or dyke gives villages in a line.
- Lowland villages stand 1.5–4 km (1–2.5 mi) apart: on terraces above the floods, along spring lines and on the edge between two kinds of land. In hills, parish churches

stand about 7–10 km (4–6 mi) apart, with hamlets between.

- A typical village has 30–60 households (150–300 people), with fields within about 2 km (1.3 mi). Land use forms rings: gardens, then arable, then pasture, then wood and waste at the parish edge. Meadow follows the streams.

Read: [Sizes, territory and walking distances](#)

Check: Does every village touch water and have its fields within about 2 km (1.3 mi)?

FANTASY TWIST

Where monsters roam, people crowd into fewer, walled villages of 300–1,000, with fields within 4–5 km (2.5–3 mi) of the walls. See [Dangerous wilderness and monsters](#).

Step 15: Add the local features

Place: mills, manor houses, parks and forests, meeting places, gallows and boundaries.

Rules:

- About one mill per village: watermills on streams with a weir and a leat (a channel to the wheel); windmills on high ground in flat, dry country.

- The manor house stands beside the church, often moated, with a dovecote and fishponds.
- A royal forest is a legal hunting area: draw a green dash-dot boundary and a large italic name, with villages and fields inside, not solid trees. Deer parks are rounded enclosures 0.5–1 km (0.3–0.6 mi) across.
- Hundreds (districts within a county) about 12–15 km (7–9 mi) across meet at a mound, stone or ford. A gal-

lows stands on a hill by the main road outside each town.

On a 20 × 20 km (12 × 12 mi) lowland map, England c. 1300	Number
Villages with parish churches	~25–30
Mills	~30–45
Moated sites	~15–20
Deer parks	~10
Market places (many tiny)	~5

Read: [Drawing a local-area map](#)

Check: Does every patch of land have a use, and is any empty space labelled as marsh, moor or forest?

Step 16: Add ruins and older layers

Place: old roads, hillforts, castle ruins, deserted villages and moved towns.

Rules:

- Ruins near living towns are quarried away; they survive in empty country.
- Draw a few long, straight “old roads” linking cities, and in hilly country 1–3 hillforts per 100 km² (39 sq mi).

- In long-settled land, draw one or two castle ruins or earthworks for every active castle; on a new frontier, almost none.
- England has more than 3,000 known deserted villages, at least 1,500 abandoned c. 1350–1520. Draw them as a grey lone church labelled “(lost village)”.

Read: [Ancient and ruined layers](#); [How settlements change over time](#)

Check: Is there at least one older layer, and is every feature possible by the date in your title box?

MAP TIP

Use three styles: working features, old features still in use (like an ancient road), and ruins drawn with the feature’s own symbol, broken or grey, labelled “(ruin)”. These styles match the guide’s [master legend](#). For the fine detail of Steps 14–16, follow the [method for a local map](#) in chapter 06.

Finishing: fantasy and the final check

Step 17: Make the fantasy adjustments

Place: the places that magic, monsters and other peoples create, and the places they destroy.

Rules:

- Real rules first. Then ask which input each element changes (water, food, defence, travel cost, threats or resources) and write: “X exists, so Y changes.”
- Add magic in order: resources (a crystal is ore; a ley line is a river), then towers and boom towns, density changes, new hubs (portals, aeries) and finally fortresses.
- Danger concentrates people: walled hilltop villages, refuge forts about a day’s walk (24–32 km, 15–20 mi)

apart, and a frontier band of 30–100 km (20–60 mi) between the safe core and the wild.

- Strong fertility magic (about twice the yield) gives density like Flanders: 50–75 people per km² (130–190 per sq mi).
- Magic can give a town its reason to exist, but not its water and food. Use no more than three or four fantasy elements per region.

Read: [The method: change the inputs, not the rules](#); [Consistency checklist](#)

Check: Does every magical city still show where its water and food come from?

LATER ERA (1500s+)

For a 1500–1650 map, keep the villages and market towns. Grow the capital and ocean ports, turn interior castles into ruins or palaces, and add star forts where they pay, post roads, a drained polder and faith areas (soft colour tints, never a new border line). See [Drawing a world in transition](#).

Step 18: Run the final sanity check

Place: nothing new. Fix what fails.

Rules:

- Run the “why is it here?” test on every dot. A village needs water, mixed land nearby and safety from floods. A town must also stand where routes meet. A city must also be on navigable water and be the best-connected place in a rich region.
- Count symbols: far more villages than towns, far more towns than cities.

- Every empty area needs a reason; every crowded area needs good land.
- Place labels last, biggest first, with 2–3 typefaces at most.

Read: The “why is it here?” test; Sanity check: 20 questions for a finished map

Check: Can you answer “yes” to all 20 sanity-check questions?

MAP TIP

Keep a short list beside your map: name, rank, site type and a one-line situation (“Kingsford: town; ford and confluence; where the north road meets the Silverwater”). It takes a few minutes and catches most mistakes.

Printable checklist**Step 0: Before you start**

- ☐ Date chosen (c. 1100, 1300, 1450 or 1600) and written in the title box
- ☐ Realm type chosen: centralised, fragmented or in between
- ☐ Scale chosen, scale bar drawn, target counts for that scale written in the margin
- ☐ Amount of magic decided

Part A: Continent and world

- ☐ 1. Navigable stretches, heads of navigation, fords and tidal limits marked
- ☐ 2. Land shaded by quality; every empty zone has a reason
- ☐ 3. Population = area × density, checked against real kingdoms
- ☐ 4. At least three kinds of state; every border follows something
- ☐ 5. Capitals in rich, defensible cores on navigable water
- ☐ 6. Only 3–8 great cities, almost all on the sea or a navigable river
- ☐ 7. Sea lanes, river spines and passes drawn; exports and imports noted

Part B: Kingdom and region

- ☐ 8. Counties 50–70 km (30–45 mi) across; towns at route nodes
- ☐ 9. Market towns about 9–16 km (6–10 mi) apart
- ☐ 10. Every castle has a one-word job; sparse core, dense border
- ☐ 11. Few bridges on big rivers; inns every 15–30 km (10–19 mi)
- ☐ 12. Dioceses, monasteries, friaries, shrines and hospitals placed
- ☐ 13. Every industry has a resource, fuel and a way out; every realm has salt

Part C: Local area

- ☐ 14. Villages 1.5–4 km (1–2.5 mi) apart in lowland, each on water
- ☐ 15. Mills, manors, parks, meeting places and gallows added
- ☐ 16. Ruins and older layers added; nothing too new for the date

Finishing

- ☐ 17. Each fantasy element has an “X exists, so Y changes” line

- 18. “Why is it here?” test and the 20 sanity questions passed; labels placed last

Quick summary

- Decide the date, the realm type, the scale and the amount of magic first; date the map and draw a scale bar.
- At 1:N, 1 mm on paper is $N \div 1,000$ metres. On a kingdom map (1:1,000,000) a great city is a 2–4 mm symbol; on a local map (1:50,000–1:100,000) a village is a 2.5–9 mm outline.
- Each scale has its own filter: a continent map shows 3–8 great cities and the greatest fortresses; a kingdom map shows every town of 2,000+, the counties and the major castles; a local map shows every village, church, mill and moated site.
- Work from big to small and from nature to people: continent first, then kingdom, then local area.
- Population is area $\times$ density. Good lowland holds 30–50 people per km² (80–130 per sq mi).
- Capitals sit in rich, defensible cores on navigable water. Great cities of 50,000+ are rare and almost always fed by river or sea.
- Bulk goods follow water (land : river : sea $\approx$ 8 : 4 : 1). Roads bend to crossings and passes, with a stop every 15–30 km (10–19 mi).
- Spacing: villages 1.5–4 km (1–2.5 mi), market towns 9–16 km (6–10 mi), working castles 20–30 km (12–19 mi) in a peaceful core and 4–6 km (2.5–4 mi) on a contested border.
- Churches form a carpet, monasteries a scatter (about 1 per 150 km², 60 sq mi), cathedrals and universities a handful. Industry sits on its heaviest input.
- Finish with ruins, then fantasy changes, then the “why is it here?” test and the 20 sanity questions.

Sources and further reading

Almost every historical number in this chapter comes from the chapter section linked in each step or table. Each chapter lists its own sources at the end, and the shared figures are in [research/baseline-numbers.md](#).

The scale arithmetic in [What to draw at each scale](#) (sizes on paper, what one sheet covers, the England-sized example) is this guide’s own calculation. It uses these references:

- Ordnance Survey, [A Beginner’s Guide to Understanding Map Scales](#) (what a scale such as 1:25,000 or 1:50,000 means; “large scale” and “small scale”; what each OS scale shows).
- Wikipedia, [ISO 216](#) (A3 paper is 297 $\times$ 420 mm; A4 is 210 $\times$ 297 mm).

- *Encyclopaedia Britannica*, [Mediterranean Sea](#) (west–east extent from Gibraltar to the Gulf of Iskenderun about 4,000 km, 2,500 mi).
- Wikipedia, [Cartographic generalization](#) (selecting features and replacing them with symbols as the scale gets smaller).

For the date table in Step 0:

- Mate, Mavis. “The Rise and Fall of Markets in Southeast England”. *Canadian Journal of History* 31:2 (1996). Summary at [medievalists.net](#): market revenues began to fall in the 1390s and fell further in the mid-15th century, with empty market stalls common in the Weald.

CHAPTER 1

WHERE SETTLEMENTS ARE BUILT (AND WHY)

Real people did not put towns in random places. Every village, town and city answered the same few questions: is there water, can we grow food, can we defend it, and can goods and people reach it? This chapter explains those questions, shows the classic places where settlements grow, and gives you numbers for spacing and a method for naming places, so that every dot on your map has a reason to be there.

The baseline period is the High and Late Middle Ages in Europe (c. 1000–1500). Population numbers for each settlement type are in [Population and Settlement Sizes](#). Roads and ports are in [Trade Routes, Roads and Transport](#).

IN THIS CHAPTER

The six things every settlement needs

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Patterns and spacing

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The six things every settlement needs

RULE OF THUMB

Water, food, fuel, safety, access, health. A village needs the first three close by. A town also needs access. A city needs access by water.

Need	What people looked for	How close it must be	What it looks like on a map
Fresh water	River, stream, spring, reliable well, lake	Inside or next to the settlement	Every village touches blue: a river, a stream or a spring line
Food	A mix of land: arable (ploughed fields), meadow (hay), pasture (grazing), woodland	Fields within a 15–30 minute walk (roughly 1–2.5 km, 0.6–1.5 mi)	Villages sit where two kinds of land meet
Fuel and building material	Wood, peat or coal; stone, clay or timber	Within a few km for daily fuel; farther for building stone (by water)	Woods near most villages; quarries near stone towns
Defence	Hills, river bends, islands, spurs (rock tongues), marsh around a dry spot	The site itself	Towns on hills, in river loops, on islands
Communications	Fords, bridges, confluences, harbours, the mouth of a mountain pass, crossroads	The site itself	Bigger places at river crossings and route junctions
Shelter and health	Sunny slopes, out of the wind, above floods, away from “fever marshes”	The site itself	Villages on the sunny side of valleys and on the edge of marshes, not in them

Fresh water

Rule: No settlement is far from water. Villages need it every day for people, animals, brewing and washing.

Why: Water is heavy. Carrying it uphill every day is hard work. Deep wells were expensive to dig.

Real examples:

- Old Sarum is an Iron Age hill fort near modern Salisbury, England, with a Norman castle and cathedral

inside it. The clergy complained about three things: water was scarce and expensive (and the castle guards sometimes controlled access to it), the wind was fierce, and they quarrelled with the castle garrison. Pope Honorius III gave permission to move (1217–18), and in 1220 the bishop laid the foundation stone of a new cathedral down in the river valley. New Salisbury grew by the river. Old Sarum slowly emptied.

- On chalk and limestone hills, water sinks into the rock. It comes out as springs where the porous rock sits on clay. Villages form a line along these springs (see [spring-line villages](#) below).

Flood risk. People wanted water close, but not in the house. The usual answer was a **river terrace**: a flat step of gravel a

few metres above the flood plain. Roman and medieval London stood on two low hills (Cornhill and Ludgate Hill) on the north bank of the Thames. The higher one, Cornhill, rises only to about 18 m (58 ft) above sea level. Across the river, on the south bank, was a patch of dry ground at Southwark. The flood plain itself was left as meadow.

Food: a mix of land

Rule: A farming village needs four kinds of land, not one.

Land type	Use	Where it is
Arable	Grain for bread, ale and pottage (a thick stew of grain and vegetables)	Well-drained, fairly flat land
Meadow	Hay to feed animals in winter	Flood plains and wet valley floors
Pasture	Summer grazing for oxen, sheep, cows	Hills, heaths, rough ground, fallow fields
Woodland	Fuel, building timber, pigs (acorns)	Steeper or poorer land, parish edges

Why: Ploughs need oxen. Oxen need hay in winter. Hay needs meadow. Fields need manure from animals. So a village with only one kind of land fails.

Numbers:

- Meadow was precious. Medieval English land surveys usually valued an acre of meadow well above an acre of arable, often several times as much.
- A person needs about 1 ha (2.5 acres) of arable land, or 2–3 ha (5–7 acres) of all land including pasture, meadow and woodland (see the shared baseline in [Population and Settlement Sizes](#)).

- So a village of 200 people needs about 2 km² (0.8 sq mi) of fields and 4–6 km² (1.5–2.3 sq mi) of land in total. Add common waste (unfarmed rough land shared by the village) and woods, and you get the typical English lowland **parish** (the area served by one village church) of about 14 km² (5.4 sq mi).

Map result: Villages love **edges**. They sit where a river valley meets a hillside, where a fen meets dry land, or where downland meets a clay vale. From one spot the village can reach meadow, fields and pasture.

RULE OF THUMB

Put villages on the boundary between two kinds of land, not in the middle of one.

Fuel and building material

Every household burned fuel every day for cooking, heating, baking and brewing. Most villages had woodland or a fuel source within a few kilometres. Where wood was scarce, people burned peat (fens, the Netherlands, Ireland, the Scottish islands) or dung (steppe). Building material

was local: timber and wattle in forest country, stone where there was stone, clay brick in the stone-poor plains of the Low Countries and the Baltic coast. Fine building stone for cathedrals and castles moved long distances, but almost always by water.

Defence

Rule: The more dangerous the times, the higher and more enclosed the site.

Defensive site	How it helps	Real examples
Hilltop or plateau	Attackers must climb	Orvieto (Italy, on a volcanic rock plateau); Laon (France); Old Sarum (England)
Spur or promontory	Cliffs on three sides; wall only one side	Luxembourg (the Bock rock, acquired by Count Siegfried in 963); Edinburgh (castle rock with a ridge behind it)
River bend (meander)	Water on three sides	Durham (founded 995 in a loop of the River Wear); Bern (founded 1191 in a loop of the Aare); Shrewsbury (Severn)

Defensive site	How it helps	Real examples
Island	Water all round	Paris (Île de la Cité); Stockholm (first recorded 1252); Venice; Lindau (Lake Constance)
Dry spot in a marsh	Marsh slows attackers	Ely (England, on an “isle” in the Fens)

Mediterranean example: In Latium, central Italy, many scattered farms were regrouped into fortified hilltop villages called *castelli*. This happened mainly in the 10th and early 11th centuries. Historians call it *incastellamento* (“castle-building”), a term made famous by Pierre Toubert (1973). Scholars still argue about exactly when and why it happened, but the result is clear: central Italy is full of hill-

top villages. Raided coasts show the same effect (see [Coasts](#)).

The trade-off: A strong site is often a bad site for trade and water. Old Sarum lost to Salisbury for exactly this reason. In peaceful times, towns move down to the river.

Communications

A place that people **must pass through** has a reason to grow: a ford, a bridge, a harbour, the foot of a pass. These

are explained in [Classic site types](#). The routes themselves are in [Trade Routes, Roads and Transport](#).

Shelter and health

- **Sunny slopes.** In the Alps the sunny side of a valley (*adret* in French) has a longer growing season, less snow and higher tree and crop limits than the shady side (*ubac*). Villages and fields are on the adret; the ubac is left to forest.
- **Out of the wind.** At Old Sarum, the writer Peter of Blois said the wind was sometimes so strong that the church services could not be heard.
- **Away from fever marshes.** Malaria (“ague”, “marsh fever”) existed in Europe’s coastal marshes, including

Kent, Essex, the Fens, the Italian Maremma and the Pontine Marshes. Mary Dobson (1997) showed that in the 17th and 18th centuries, marsh parishes in Kent and Essex often buried more people than they baptised.

- **Healthy air.** The Abbasid caliph al-Mansur founded Baghdad in 762. Historians list his reasons as its place on trade routes, its water supply and its mild climate. When he later built his Khuld Palace there, he put it on the highest part of the Tigris bank, for fresh air and away from the insects of the riverside.

MAP TIP

Before you draw any settlement, mark on your terrain map: rivers and springs, flood plains, marshes, sunny valley sides, good soil, and defensible spots (hills, river loops, islands).

Most villages will fall on the edges between these zones. A dot with none of these features nearby needs a special reason, such as a mine or a monastery.

Site and situation

These two words come from school geography, and they are the most useful tools in this chapter.

	Site	Situation
Meaning	The exact spot: the ground the settlement stands on	Its place in the wider world: routes, neighbours, resources, region
Questions	Is it dry? Is there water? Can it be defended? Is there room to grow?	Is it on a river people sail? At a crossing? Between two regions that trade? In rich farmland?
Decides	Whether the place survives	Whether the place grows big

Rule: Thousands of places have a good site. Only a few have a great situation. Situation decides which village stays a village and which becomes a city.

Real examples:

- **London.** The *site* was modest: two low gravel hills by a marshy river. The *situation* was superb: the lowest point where the Thames could be bridged, near the old tidal

limit (the highest point the tide reaches up a river), where sea ships, river boats and roads all met. That is why it became the largest city in England.

- **Venice.** The *site* was terrible: mud islands in a lagoon, no farmland, and little fresh water (rain cisterns, plus water brought by boat). The *situation* was excellent: safe from land armies, at the head of the Adriatic, close to the passes into Germany. Venice became one of Europe's largest cities (see [Population and Settlement Sizes](#)).
- **Old Sarum and Salisbury.** Same situation, different site. The better site (water, flat ground by the river) won.

RULE OF THUMB

Good site = the village survives. Good situation = the village becomes a town. Great situation on water = it becomes a city.

Why situation matters so much: Moving bulk goods over-land was very expensive. In England around 1300, carrying goods by land cost about 8 times as much as by sea and 2 times as much as by river (the shared baseline gives land : river : sea $\approx 8 : 4 : 1$). So towns that needed to import grain, timber or stone had to sit on **navigable** water (water deep enough for boats). Nearly every town above about 10,000 people sits on a navigable river or the coast. The few exceptions had something special. Granada sat in an unusually rich irrigated plain (the *Vega*). Siena lived on banking and on the main pilgrim road to Rome.

MAP TIP

For each large settlement, write two short notes: "site: ..." and "situation: ...". If you cannot write the situation note for a city, it should probably be a town.

Classic site types

These are the standard "reasons to be here." Most real towns combine two or three of them.

Site type	What it is	Why a town grows there	Real examples
Ford town	Where a river is shallow enough to wade or drive a cart across	Roads must meet here; travellers stop	Oxford ("oxen ford", first recorded 912); Frankfurt ("ford of the Franks", first recorded 794); Hertford; Stratford
Bridge town	Where a bridge replaced or added to a ford	Tolls, markets, controls traffic	Cambridge; Innsbruck ("bridge over the Inn"); Saarbrücken
Lowest bridging point	The bridge nearest the sea	Roads on both banks AND ships from the sea; often also the tidal limit	London (Thames); York (Ouse); Exeter (Exe); Gloucester (Severn); Stirling (Forth: a map by Matthew Paris from the 1240s–1250s seems to show only one bridge in all Britain, at Stirling)
Head of navigation / break of bulk	The highest point big ships can reach, or a point where cargo must change boats ("break of bulk")	Goods must be unloaded, stored, sold, reloaded	Cologne (ships from the lower Rhine were too big for the river above, so cargo usually changed boats here; from 1259 a <i>staple right</i> , a town's legal right to make passing merchants stop, forced them to unload and offer their goods for sale for 3 days); Schaffhausen (the name probably means "boat houses": goods were unloaded above the Rhine Falls and carried around them); Seville (sea port about 80 km (50 mi) up the Guadalquivir from the Atlantic)
Confluence town	Where two rivers join	Two river routes meet; a defensible tongue of land between them	Koblenz (Latin <i>Confluentes</i> : Rhine and Moselle); Lyon (Rhône and Saône); Passau (Danube, Inn and Ilz); Belgrade (Sava and Danube)
River-bend (meander) town	Inside a tight loop of a river	Water on three sides; only one side to wall	Durham (995); Bern (1191); Shrewsbury; Besançon (Doubs); Toledo (Tagus)
Hilltop town	On a hill, ridge or plateau	Defence, health, view	Italian <i>castelli</i> ; Orvieto; Carcassonne; Shaftesbury
Island town	On a river, lake or sea island	Defence; easy river crossing in two short bridges	Paris (Île de la Cité); Stockholm; Venice; Lindau; Mont-Saint-Michel (tidal island)

Site type	What it is	Why a town grows there	Real examples
Spring-line village	A row of villages where springs come out at the foot (or near the top) of a hill scarp (a steep slope along the edge of a ridge)	Reliable water plus fields below and pasture above	Lincoln Cliff (a limestone ridge over 80 km (50 mi) long, with more than 40 villages and towns along it: roughly one every 2 km (1.2 mi)); the foot of the South Downs (Fulking, Poynings, Edburton); the edges of the Mendips (Wells, Wookey Hole, Blagdon, Compton Martin)
Gap town	Where a river cuts a gap through a line of hills	All routes across the hills funnel through the gap	Lincoln (River Witham through the limestone ridge); Guildford (River Wey through the North Downs); Lewes (River Ouse through the South Downs)
Pass town	At the foot of a mountain pass	Travellers rest, change animals, hire guides; customs posts	Altdorf and Bellinzona (Gotthard: the first bridge across the Schöllenen gorge, built around 1220, opened the route); Chur (several Alpine passes); Susa (Mont Cenis); Aosta (Great St Bernard)
Harbour town	A sheltered bay, deep water, protection from storms	Sea trade, fishing, shipbuilding	Genoa; Bergen; Dubrovnik (Ragusa); Brest
Estuary town	Up a tidal river mouth, usually at the first good crossing	Sheltered from storms and raiders; reaches inland by river	Bordeaux (Gironde/Garonne, about 100 km inland); Hamburg (Elbe, about 110 km); Antwerp (Scheldt, about 80 km); Rouen (Seine) — all roughly 80–120 km (50–75 mi) from the open sea
Oasis town	Around a desert spring, river or underground channel	The only water for days of travel; caravans must stop	Sijilmasa (Morocco, founded 757, northern end of a trans-Saharan route, abandoned 1393); Damascus (watered by the Barada river and the Ghouta gardens); Yazd and Kerman (Iran, on <i>qanats</i> , underground water channels: see Desert)
Wet-point settlement	A spot chosen for water in a dry land	Water is the scarce thing	Spring-line villages on chalk; oasis villages; wells on dry plateaus
Dry-point settlement	A spot chosen for dry ground in a wet land	Dry land is the scarce thing	Ely (on a clay “isle” about 26 m (85 ft) high in the Fens; abbey founded 673); Glastonbury (Somerset Levels); Frisian <i>terpen</i> (man-made dwelling mounds)

Notes on the most important types

Lowest bridging point and head of navigation. On many rivers the lowest bridge, the old tidal limit and the highest point sea ships could reach were all at the same place, or close together. That place usually became the region's chief town (London, York, Exeter, Gloucester). It is the single best spot on your map for a big town.

Tidal coasts: ports sit inland. On the tidal Atlantic and North Sea coasts, the big medieval ports often sat 50–120 km (30–75 mi) up an estuary (a wide tidal river mouth). There they were safe from storms and raiders, they

had the first bridge, and the tide helped ships upstream. Later, as ships grew bigger or rivers silted up, small “out-ports” grew nearer the sea: Damme and Sluis for Bruges, Topsham for Exeter, Sanlúcar (at the river mouth) for Seville. In the almost tideless Mediterranean, ports more often sat directly on a sheltered bay (Genoa, Marseille, Barcelona, Dubrovnik).

Fords. A ford needs firm gravel and water shallow enough for carts at normal levels. On big rivers fords are rare, so a crossing that serves several roads is a strong candidate for a town.

Reading place names

Many real names tell you the site type. You can use the same trick for your fantasy names. (The everyday village endings, and how to build a whole region's names, are in [Naming settlements realistically](#).)

Name element	Meaning	Examples
-ford, -furt, -voorde	Ford	Oxford, Frankfurt, Erfurt
-bridge, -brück, -bruck	Bridge	Cambridge, Innsbruck, Saarbrücken
-mouth, -münde, Aber-	River mouth	Plymouth, Monmouth, Warnemünde, Aberdeen
Confluentes, Coblenz, Condé	Confluence	Koblenz, Condé-sur-l'Escaut
-wick, -wich, -vik	Trading place, bay	Ipswich, Reykjavík

Name element	Meaning	Examples
-ley, -sart, -rode, -roth	Forest clearing	Many English villages; “Les Essarts” in France; Wernigerode
-ey, -holm	Island, or dry ground in wet land	Chertsey, Romsey, Stockholm

FANTASY TWIST

If griffon riders or teleport circles move people and letters, crossings matter less for **travellers**. They still matter for **goods**: grain, timber and stone still go by boat and cart. Keep the big towns at the crossings. See [Fantasy Variants](#).

MAP TIP

Give each town one or two site types and draw them clearly: the river loop around the meander town, the bridge symbol at the lowest crossing, the gap in the hills at the gap town. Then name it so the name matches the site, like “Oakford” at a ford or “Highbridge” at a bridge. Readers notice when geography and names agree.

Settlement by terrain and biome

The same rules apply everywhere, but the answer changes with the land. The first eleven rows below cover Europe and its nearest neighbours, the steppe and the desert. The last

four cover other climates, with examples from Asia, Africa and the far north.

Terrain	Where settlements go	Typical pattern	Density (see chapter 02)
Fertile river valleys and plains	River terraces, valley edges, crossings	Many nucleated (tightly grouped) villages; towns on the river	Highest: 30–50 per km ² (80–130 per sq mi) at peak
Rolling hills	Valley floors, spring lines, sheltered slopes	Villages in valleys; hilltops empty or pasture	Medium
Hill country (farmed valleys with upland grazing)	Valley floors and lower slopes, spring lines	Small villages and hamlets in the valleys; scattered farms; open grazing on the hills above	Lower: 8–15 per km ² (21–39 per sq mi) at peak
High uplands, moors and mountains	Sheltered valley floors, sunny terraces, pass feet	Hamlets and farmsteads; seasonal huts higher up	Low: 2–5 per km ² (5–13 per sq mi)
Coasts	Sheltered bays, estuaries, river mouths, slightly inland on raided coasts	Chains of fishing villages and ports	Varies
Marsh and fens	Islands of dry ground, fen edges, man-made mounds	Villages on the edge; few inside	Low inside, high on the edge
Forest	Clearings, river valleys through the forest	Strings of new villages pushing in	Low, rising as land is cleared
Steppe and grassland	River valleys, steppe edges	Nomad camps; a few trade towns on rivers	Very low: under 1–2 per km ² (under 3–5 per sq mi)
Desert	Oases, wadis, <i>qanat</i> lines	Isolated clusters; caravan stops	Near zero between oases
Islands	Sheltered side, inland on raided islands	One main port; villages on fertile patches	Varies
Cold north	Sheltered fjords, coastal lowlands	Scattered farms; very few towns	Very low
Tropical forest and monsoon uplands	Ridges and river banks; towns only on rivers and coasts	Small villages that move their fields every few years (shifting cultivation); paths, not roads	Low: about 6 per km ² (14 per sq mi) as a 1950s world average; about 12–35 per km ² (30–90 per sq mi) in modern northern Laos and Thailand; at most about 55–56 per km ² (about 145 per sq mi)

Terrain	Where settlements go	Typical pattern	Density (see chapter 02)
Wet-rice lowlands and deltas	Mounds, dikes, river levees and canal banks between the paddies	Continuous rice fields with villages packed along the water lines	Very high: Song China averaged roughly 30–40 per km ² (80–100 per sq mi) over the whole empire; Greater Angkor about 230–300 per km ² (600–780 per sq mi) over its core (derived)
Savanna and Sahel (dry grassland south of the Sahara)	Desert-edge river ports, wells, seasonal rivers	Trade towns where caravans meet boats; farming villages by water; herders moving with the rains	Low, higher along rivers
Tundra and northern forest (taiga)	Winter villages by rivers and lakes; fishing waters, hunting grounds and pastures in summer	A few winter villages; families scatter to seasonal camps	Extremely low

Fertile river valleys and plains

This is where most people lived. Villages sit on terraces just above the flood plain, every 1.5–4 km (1–2.5 mi). The flood plain is meadow. Towns grow at crossings and confluences.

Rolling hills

Valleys hold the villages; hilltops hold sheep pasture and sometimes old hill forts. On chalk and limestone, the spring line decides everything. Parishes here often became

In England's Midlands (the "Central Province" of Roberts and Wrathmell 2000), large nucleated villages with open fields filled almost all the land.

strip parishes: long, thin parishes running from the hilltop pasture, through the village at the spring line, down to the meadows in the valley. Each village gets a slice of every land type.

Uplands and mountains

- **Two kinds of upland.** Farmed **hill country** (valleys with upland grazing above them) held about 8–15 people per km² (21–39 per sq mi) at the peak around 1300. **High uplands, moors and mountains** held only about 2–5 per km² (5–13 per sq mi). See chapter 02.
- **Valley floors and terraces** hold permanent villages, usually on the sunny side and on alluvial fans (fan-shaped slopes of gravel where side streams enter the valley) that are safe from the main river's floods.
- **Altitude bands.** In the Valais (Switzerland), hamlets sit on sunny terraces at about 1,000 m (3,300 ft). Higher hay barns and chalets (*mayens*) sit at 1,500–1,600 m

(4,900–5,200 ft). Summer pastures start around 2,000 m (6,600 ft).

- **Seasonal settlement.** Herds moved up in summer and down in winter (*transhumance*). This is very old in the Alps: archaeology traces it back to about 3000 BC, and written records mention summer pastures by 1204. In northern England and Scotland the summer huts used for the same purpose are called *shielings*.
- **The upper limit.** Permanent villages above about 2,000 m (6,600 ft) were very rare. Juf in Switzerland, at 2,126 m (6,975 ft), was settled by Walser colonists in 1292 and is usually called the highest permanent village in western Europe.
- **Passes** create towns at both ends (see the table above).

Coasts

Fishing villages sit in every sheltered cove. Real ports need shelter plus a route inland, usually a river. On coasts with raiders, villages sit a few kilometres inland and uphill, with only boat sheds and a watchtower on the shore. In

Sardinia, the village of Posada moved to the foot of its castle hill in the late 15th or early 16th century to escape coastal raids. On many Aegean islands, villages came back down to the shore only after the pirate threat ended, often in the 19th century.

Marsh and fens

People lived **around** marshes, not in them. The fen edge was rich: fish, eels, reeds, peat, salt and summer grazing. Inside the marsh, settlements sat on islands of higher

ground (Ely, Glastonbury) or on man-made mounds. In Friesland and Groningen alone there were about 1,200 *terpen* (dwelling mounds), the first built around 500 BC.

Forest

In 1000–1300 Europe cleared huge areas of forest. Clearing land for farming was called **assarting** in England (from French *essarter*, to grub out). Clearings show up in place names (-ley, -field, -royd, -sart, -rode). In Germany, new forest villages were often laid out as **Waldhufendörfer** (“forest-strip villages”): a line of farms along a road or stream, each with a long strip of land running back into the

forest. This form appeared around 1000 in the northern Black Forest and spread east with German settlers in the 12th–13th centuries.

A “forest” was not always a forest. A medieval English “forest” was a **legal** area for royal hunting that could include villages and fields; clearing land in it needed the king’s permission. See [Royal forests](#).

Steppe and grassland

RULE OF THUMB

On the steppe, people live in small camps of a few tents that move several times a year. Label pasture zones and draw routes, not village dots. Fixed towns are rare and sit on rivers or at the steppe edge.

Why: Grass is thin, so herds must keep moving to fresh pasture. A big crowd of animals in one place would eat the grass bare in days.

Camp sizes. Nomad camps came in three sizes:

Camp	What it is	Size	Example
Herding camp	A few round felt tents (<i>ger</i> or <i>yurt</i>), one family in each, sharing the herding work	Usually 2–10 families	The Mongolian <i>khot ail</i> (“camp group”) today
Ring camp	A circle of tents with the wagons and animals inside and the chief’s tent in the middle; used by chiefs and at big seasonal gatherings	A whole clan or tribe	The Mongol <i>güre’en</i> (also written <i>kuriyen</i>). The word later meant any fortified camp, and then the Mongol army unit of “a thousand”
The ruler’s camp (<i>ordu</i> , the origin of the word “horde”)	A moving town around the khan’s great tent, with officials, guards and traders	Many thousands of people and animals	Ibn Battuta saw Khan Uzbek’s camp in 1332: “a vast city on the move with its inhabitants, with mosques and bazaars in it, and the smoke of the kitchens rising in the air”

Even a side branch of the court was large. When one of Uzbek’s wives travelled to Constantinople in 1332, Ibn Battuta counted an escort of 5,000 troops under an amir (a commander), her own 500 horsemen, about 200 maidservants, about 400 carts, 2,000 horses, 300 oxen and 200 camels.

How often and how far they move. Medieval figures are scarce, so modern Mongolia is the best guide. (Many modern herders move by truck, which may make some moves longer than in the past.)

Measure	Typical value	Source
Camps used in a year	Usually about 4–10; GPS loggers on 142 households found 2–14 camps in 9 months	Teickner et al. 2020; an earlier survey (Ganbold 2015) found 4–8
Distance from one camp to the next	Typically about 12 km (7 mi); some families move about 36 km (22 mi) or more each time	Teickner et al. 2020
Total distance in a year	Most families under 170 km (105 mi); a few over 300 km (190 mi)	Teickner et al. 2020
Dry desert pasture (Dzungarian Gobi)	About 9 moves a year; 70–123 km (43–76 mi) between summer and winter camps	Michler et al. 2022

Where the camps go:

- **Winter camps** sit in sheltered places out of the wind, such as valleys and the foot of hills, where the snow is

not too deep for the animals to graze. Each family returns to its own winter site year after year, so these sites have stone pens for the animals. They are the most “fixed” points of nomad life.

- **Summer camps** spread out over open pasture near water: rivers, lakes and wells.
- **Spring and autumn camps** sit between the two, often along the migration route.

MAP TIP

On a nomad region, do not colour the pastures: a soft tint means a language or faith area. Instead, write italic area labels such as “Summer pastures” and “Winter pastures” across the land they cover. Mark a few winter camps in sheltered river valleys with the nomad-camp symbol: 2–3 tiny **open** half-circles (round tents). Do not use dots, because a tiny dot means a farmstead (or, on a kingdom

The whole system needs a lot of land: steppe nomads lived at under 1–2 people per km² (under 3–5 per sq mi) (see [Population and Settlement Sizes](#)).

map, a village). Draw the seasonal migration routes between the pastures as dotted lines with arrows, with wells and fords on the way. Add one or two fixed towns (a Sarai or a Karakorum) on a river, and frontier markets where the steppe meets farmland. If the ruler’s camp matters to your story, draw it as a large ring of the same open half-circles with a label such as “Khan’s Ordu (summer)”.

Desert

Water decides everything. Settlements form islands and lines: green islands at oases, lines along wadis (dry river-beds that flood sometimes) and along *qanats*. A *qanat* is a gently sloping tunnel that brings groundwater from the foot of the mountains to the fields. They are typical of Iran.

Most Iranian qanats are under 5 km (3 mi) long, but some near Kerman measure about 70 km (43 mi). Desert cities such as Yazd and Kerman sit at the foot of mountains where qanats can be dug. Caravan stops sit one day’s march apart, about 30–40 km (19–25 mi) (see [Trade Routes, Roads and Transport](#)).

Islands

An island usually has one main port on the sheltered side, facing the nearest mainland or trade route. Villages sit on fertile patches. On raided islands the main village (the *chora*

on Greek islands) sits inland on a hill, often under a castle, with a small harbour below.

Cold north

Growing grain is hard, so people lived on scattered farms with hay meadows and animals, along sheltered fjords and coastal strips. Towns were very few. Norway had only a handful, such as Bergen, Trondheim and Oslo, all on the coast or a fjord. Medieval Iceland had no towns at all: foreign traders came in summer to seasonal trading places such as Gásir. At the end of the 11th century, Bishop Gizur Ísleifsson counted about 4,560 farmers in Iceland who were

rich enough to pay the assembly tax (the count is recorded in Ari’s *Book of the Icelanders*). That gives you a sense of scale: a whole country of a few thousand farms, and not one town. Norse Greenland had around 500 farms in its Eastern Settlement and maybe 100 in its Western Settlement, deep in the fjords. Estimates of its peak population vary: recent ones say about 2,000–3,000, while some older ones were higher.

Tropical forest and monsoon uplands

Rule: In hot, wet forest, farmers usually practise **shifting cultivation** (also called *swidden* or slash-and-burn): they clear and burn a patch of forest, farm it for a year or two, then let it grow back for many years while they clear another patch.

Why: Tropical forest soils lose their fertility fast once cleared. A long fallow (rest period) lets the forest rebuild it. So each village needs a large area of forest of different ages around it.

Numbers: Population stays low.

Measure	Density	Source
World average, 1950s	About 6 per km ² (14 per sq mi), for very roughly 200 million shifting cultivators	FAO (the UN's food and farming agency), 1957
Modern examples in Southeast Asia	About 12–35 per km ² (30–90 per sq mi): 12 in northern Laos, 35 in northern Thailand	Warner 1991 (FAO)
Ceiling for traditional systems	About 55–56 per km ² (about 145 per sq mi); above that, fallows get too short and the soil fails	Ruthenberg 1980, cited by FAO

Pattern: Small villages on ridges or river banks, linked by footpaths. Some villages move every few years to follow the

fields. Towns grow only where boats can reach them: on big rivers and coasts.

Wet-rice lowlands and deltas

Rule: Where monsoon rain or rivers can flood level fields, **wet rice** (rice grown in flooded fields called paddies) feeds far more people per km² than European grain land. These are the most crowded farming landscapes of the medieval world.

Why: A flooded paddy keeps its fertility year after year and can give one or even two harvests a year. But it needs water control: dikes, canals, reservoirs and terraces. Whoever builds and runs the water works shapes the map (for local water features, see [Villages, Farms and the Countryside](#)).

Numbers and examples:

- **Song China** (960–1279) held about 3.1 million km² (1.2 million sq mi) in 980. A census of 1102 recorded about 20.5 million households, and estimates of the population around 1100 range from about 90 to over 120 mil-

lion. That is roughly 30–40 people per km² (80–100 per sq mi) over the whole empire (derived), with the densest crowds in the rice lands of the Yangzi valley and the south.

- **Greater Angkor** (Cambodia, peak in the 13th century) had about 700,000–900,000 people spread over about 3,000 km² (1,160 sq mi) of houses, ponds, embankments and rice fields (Klassen et al. 2021). That is about 230–300 people per km² (600–780 per sq mi) (derived). Lidar (laser scanning from the air) shows a grid of small house mounds and little ponds between the fields.

Pattern: An unbroken carpet of paddies. Villages sit on anything slightly higher than the fields: river levees, dikes, canal banks and man-made mounds. Rivers and canals are the roads.

Savanna and Sahel

The savanna is open grassland with scattered trees; the Sahel is its dry northern edge, next to the Sahara. Rain falls in one short season, so water decides everything.

- **Towns** grow where the desert meets a great river, because there camel caravans from the Sahara meet river boats. **Timbuktu** (Mali) was founded as a seasonal Tuareg camp around 1100 and grew into a great trading and scholarly town about 13–15 km (8–9 mi) north of the Niger, with a river port at Kabara. **Gao** sits on the Niger

itself. See the trans-Saharan routes in [Trade Routes, Roads and Transport](#).

- **Farming villages** sit by wells, seasonal rivers and the river flood plain.
- **Herders** move north into the short-lived Sahel pastures in the rainy season, and back south to moister land, rivers and harvested fields in the dry season (a pattern still followed today). It works like the steppe migrations above, but the timing follows rain, not snow.

Tundra and northern forest

North of the grain line (the tundra, or treeless arctic plain, and the taiga, the great northern conifer forest) nobody farms. In northern Scandinavia, the Sámi lived by hunting wild reindeer, fishing and trapping, and kept small tame herds for transport. Large-scale reindeer herding came

later, as many communities shifted to it between the 16th and 19th centuries.

- **The siida.** The traditional Sámi community was the *siida*, a group of up to about 12 families who shared a territory. It moved between several seasonal sites during the year, with a different task at each: fishing, hunting, trapping or grazing.

- **The winter village.** The centre of each *siida* was its winter village. Families and extended families gathered there for the winter, then split into smaller groups for the snowless part of the year. Archaeologists have excavated a chain of eight such winter villages along the small River Nukkumajoki near Lake Inari (northern Finland), dated to the late 16th and early 17th centuries. One of them (Nukkumajoki 5) had 8 huts (*goahti*) and 7 hearths in a line (Halinen 2019).
- **Summer.** The small groups moved to their own seasonal sites: fishing waters, hunting grounds and pastures.

FANTASY TWIST

Dwarves change the mountain rule. A dwarf city can be large and deep inside a mountain, but it still needs food, so expect a busy human or dwarf market town at the mountain gate, on the nearest valley floor. Treat it like a pass town. See [Fantasy Variants](#).

MAP TIP

Draw shifting fields as scattered clearings of different ages around small villages, with forest in between. Draw wet-rice land as a continuous fine grid along rivers and deltas, with villages strung along the dikes and canals. In the savanna, put a trade town where a desert route meets a river. In the arctic, mark a few winter villages by lakes and rivers with the village symbol, with dotted seasonal routes out to the summer camps (2–3 tiny open half-circles, as for nomads); leave the rest empty.

MAP TIP

Before placing settlements, lightly shade your map by terrain type, in pencil or on a layer you hide later. Then use the table above as a checklist. Plains get dense villages. Hill country gets small villages and hamlets in the valleys; high uplands get only a few. Marshes get a ring of villages around the edge. Deserts get a few oases, each drawn as a green belt of gardens with small palm marks around the town or well, joined by caravan routes. Rice deltas get the densest carpet of all. Leave the high mountains, deep forest, open steppe and tundra mostly empty. These symbols match the guide's [master legend](#).

Patterns and spacing

Nucleated, dispersed and linear

Pattern	What it looks like	Where and why	Real examples
Nucleated	Houses grouped tightly around a church, green or crossroads; fields all around	Good arable land farmed in common open fields; or a need for defence	English Midlands "Central Province"; Italian <i>castelli</i> ; much of northern France
Dispersed	Single farms and small hamlets scattered over the land	Uplands, pastoral farming, forest clearings, wet or broken land	Wales, Devon, Cornwall, the Alps, Scandinavia, much of south-east England
Linear	Houses in a line along a road, stream, dyke, spring line or shore	Each farm needs access to one line feature	German <i>Waldhufendorf</i> and <i>Straßendorf</i> ("street village"); marsh villages along dykes; spring-line rows

England is a useful model: Roberts and Wrathmell (2000) mapped a central belt of nucleated villages (their "Central Province") running from the north-east down through the Midlands to the south coast, with more dispersed settle-

ment on both sides. A real kingdom usually has more than one pattern. Village layouts in detail are in [Villages, Farms and the Countryside](#).

Central place theory, made simple

In 1933 the German geographer Walter Christaller studied towns in southern Germany. He noticed that towns of the same rank were spaced fairly evenly, and he built a model to explain why. The key ideas:

- **Central place:** a settlement that sells goods and services to the people around it.

- **Threshold:** the number of customers a service needs to survive. A smith needs few customers; a cathedral or university needs many.
- **Range:** how far people will travel for it. People walk a short way for bread and a long way for a lawyer.
- **Result:** many small centres close together, a few big centres far apart. Each market area is roughly a **hexagon**, because hexagons fill the land with no gaps.

In his main model (called **K = 3**), each step up the hierarchy is spaced $\sqrt{3}$ (about 1.7) times farther apart than the step below, and serves about 3 times the area.

Christaller's level (1930s Germany, as his table is usually reproduced)	Spacing between places of this level	Model population (1930s)
Market hamlet (<i>Markort</i>)	~7 km (4.5 mi)	~1,000
Township centre (<i>Amtsort</i>)	~12 km (7.5 mi)	~2,000
County seat (<i>Kreisstadt</i>)	~21 km (13 mi)	~4,000
District city (<i>Bezirksstadt</i>)	~36 km (22 mi)	~10,000
Small state capital (<i>Gaustadt</i>)	~62 km (39 mi)	~30,000
Provincial capital (<i>Provinzstadt</i>)	~108 km (67 mi)	~100,000
Regional capital (<i>Landstadt</i>)	~186 km (116 mi)	~500,000

Careful: These are model numbers for the 1930s. Medieval places at each level were much smaller (see [chapter 02](#)), but the **spacing pattern** is a good guide. The medieval English evidence (below) shows market places 9–16 km (6–10 mi) apart, close to Christaller's lower levels.

Christaller had two other versions. In the **transport version** (**K = 4**), smaller towns sit on the straight roads between bigger towns. In the **administrative version** (**K = 7**), each big centre's territory fully contains its smaller centres, as in a county or diocese. Real landscapes mix all three. Outside

Europe, G. William Skinner (1964) found a similar pattern in rural China: each basic market town served about 18 villages.

What it looks like on a map: Put a city in the middle. Around it, at roughly equal distances, put six towns. Between and around those, put market towns. Then **distort** the pattern: stretch it along rivers and roads, squeeze it together on rich land, spread it out on poor land, and break it at mountains and coasts. Real maps never show perfect hexagons, but you can usually see the idea underneath.

How far apart: the numbers

These spacings come from the shared baseline used across this guide.

Feature	Typical spacing	Notes
Villages, lowland	1.5–4 km (1–2.5 mi)	England c. 1300: one parish per ~14 km ² (5.4 sq mi), about 4 km (2.5 mi) church to church. Upland parishes cover 50–100+ km ² (20–40+ sq mi).
Market towns, legal rule	6½ mi (10.7 km)	The lawyer Bracton (13th century): a new market should not be closer than this to an old one. It was rarely enforced.
Market towns, in practice	9–16 km (6–10 mi)	1,746 English markets were recorded by 1300 (one per 75 km ² (29 sq mi), ~9 km (6 mi) apart), but many were tiny. In the late 16th century there were ~650 working markets in England and Wales, ~16 km (10 mi) apart.
Peasant walk to market	6–10 km (4–6 mi) one way	There and back in one day, with time to trade.
Bridge towns on a lowland river	About every 20 km (12 mi) of river; gaps 10–35 km (6–22 mi)	Thames, London to Oxford, c. 1300: 9 road bridges in about 180 km (112 mi) of river. Great rivers had far fewer bridges. See How often is a river bridged?
Head ports on a coast	About one per 100–150 km (60–95 mi) of coast	A rough figure from late medieval England's 15 customs head ports; smaller ports and landing places sit between them. See Port ranks
Towns of 5,000+	Flanders 35–45 km (22–28 mi); northern Italy 50–60 km (31–37 mi); France, Germany, Iberia 80–95 km (50–59 mi); Britain 130+ km (80+ mi); Poland 180–250 km (110–155 mi)	An upper bound derived from city data.
Cities of 10,000+	Flanders and Italy 60–90 km (37–56 mi); France, Germany, Iberia 130–170 km (80–105 mi); Britain and Poland 260–400 km (160–250 mi)	Same basis.

How many in one area? If places are spaced d apart in a rough hexagon pattern, each one “owns” about $0.87 \times d^2$ of land. Using that rule on the table above, here is what you

would expect in a square of 100×100 km (62×62 mi), which is $10,000$ km² ($3,860$ sq mi):

Feature	Region type	Land per place	Number in 100×100 km
Parish villages	Lowland farmland	~14 km ²	~700
Parish villages	Upland	50–100+ km ²	100–200
Market places	England at the c. 1300 peak (many tiny)	~75 km ²	~130
Market places	Busy markets only (England and Wales, late 1500s)	~230 km ²	~45
Towns of 5,000+	Very urban (Flanders)	1,100–1,750 km ²	6–9
Towns of 5,000+	Northern Italy	2,200–3,100 km ²	3–5
Towns of 5,000+	France, Germany, Iberia	5,500–7,800 km ²	1–2
Towns of 5,000+	Britain / Poland	15,000+ / 28,000–54,000 km ²	Under 1 / 0–1
Cities of 10,000+	Flanders, Italy	3,100–7,000 km ²	1–3
Cities of 10,000+	France, Germany, Iberia	15,000–25,000 km ²	Usually 0–1
Cities of 10,000+	Britain, Poland	59,000–139,000 km ²	Usually none

These are rough, derived numbers. Use them as a sanity check, not a rule. Note that England had many markets but

few big towns: market density and city density are separate questions.

Chains along rivers, coasts and roads

Settlements line up along anything that carries traffic:

- **Rivers.** On the Rhine between Mainz and Koblenz (about 90 km (56 mi) by river), small towns follow each other every 5–20 km (3–12 mi): Bingen, Bacharach, Oberwesel, St Goar, Boppard. Many had toll castles above them. Bridges are a separate question: this stretch of the Rhine had ferries, not bridges, but on a settled lowland river such as the Thames a bridge town came about every 20 km (12 mi) (see [How often is a river bridged?](#)).
- **Coasts.** On the southern Baltic, Hanseatic ports (members of the Hanse, a league of north German trading

towns) such as Wismar, Rostock and Stralsund sit about 50–70 km (30–45 mi) apart in a straight line, each on its own bay or river mouth. Not every port is equal: late medieval England had about one big customs **head port** per 100–150 km (60–95 mi) of coast, with smaller ports and fishing landings between (see [Port ranks](#)).

- **Roads.** On main roads, inns and stopping places appear every 15–30 km (10–19 mi), so a traveller always finds one within a day’s travel. Roman road stations (*mansiones*) were about 25–30 km (16–19 mi) apart, and many later towns grew on them.

Clusters and empty spaces

Settlement is never even. In England in 1086, Suffolk had about 23–25 people per km² (60–65 per sq mi), while much of the north and west had under 5 per km² (under 13 per

sq mi). Good land is crowded; poor land is nearly empty. A map with evenly spaced dots looks false.

MAP TIP

First decide which pattern each region has: nucleated, dispersed or linear. Then use the spacing table. Draw villages close together on good lowland, in lines along springs and rivers, and widely spaced in uplands. Put market towns about 9–16 km (6–10 mi) apart, so that every village can

walk to a market and back in one day. Space larger towns by the regional row in the table. Finally, check the “how many” table. If your 100×100 km (62×62 mi) farming region has only five villages, it is far too empty. If it has three cities, it is too full, unless it is your Flanders.

How settlements change over time

A real map is a snapshot of a long story. Showing that story makes a map feel alive.

Reuse of ancient and Roman sites

Old sites were reused because they already had walls, roads, a bridge or a bishop. In Italy, southern France and the Rhineland, most Roman cities stayed cities: Cologne, Trier, Lyon, Milan. In Britain many Roman towns shrank to almost nothing after 400 and later revived (London, York, Winchester). Some never came back:

- **Silchester** (Roman *Calleva*) was abandoned sometime between about 550 and 650 and never became a town

again. A small medieval village grew up beside it and was deserted around 1400. Today its Roman walls still stand around open fields, with only a church and a farm inside.

- **Wroxeter** (Roman *Viroconium*, probably the fourth-largest town in Roman Britain) faded out in the late 7th or early 8th century. **Shrewsbury**, 8 km (5 mi) away in a strong river loop, became the main town of the area instead.

Planted “new towns” and bastides

In the High Middle Ages lords founded towns on purpose, to earn rents, tolls and market dues and to control land. Planted towns are easy to spot: a **grid** or a regular street plan, a big **central market place**, and often a name with “New”, “Ville-”, “-ville”, “Villa Nova” or “Bastide”.

Region	How many	Examples
England (1066–1370)	About 170 planted towns (Beresford 1967)	Bury St Edmunds (grid laid out c. 1080 by Abbot Baldwin; Domesday Book records 342 houses built on former ploughland by 1086); New Salisbury (1220s); New Winchelsea (charter 1281, laid out from 1288)
Wales	More than 80 (Beresford)	Edward I's castle towns in North Wales (see chapter 04)
Gascony and southwest France	About 125 in English Gascony (Beresford); almost 700 bastides (planned new towns of southern France) between 1222 and 1372	Cordes (1222, usually called the first); Monpazier (1284, Edward I); Villeneuve-sur-Lot
German lands	Central Europe had only a few hundred towns around 1100. Very many were founded from 1120 onwards, mostly in the next 250 years. Estimates vary with the definition of “town”, but by the late Middle Ages there were several thousand, most of them small	Freiburg im Breisgau (1120); Bern (1191); Lübeck (founded 1143, refounded 1159)

Many planted towns failed. A lord could grant a charter, but he could not force traders to come if the situation was poor.

Castle-gate and monastic towns

Castles and abbeys were big customers. They needed food, crafts, labour and services, so a town often grew at the gate.

- **Castle-gate town:** Ludlow (castle begun c. 1085; a planned town laid out in the 12th century). See [Castles, Forts and Military Outposts](#).

- **Monastic town:** Bury St Edmunds; St Albans; Ely. See [Religious, Cultural, Legal and Ancient Sites](#).

Why some places grow and others shrink

Cause	What happens	Real example
Silting	The harbour fills with mud; ships cannot reach the town	Bruges: the Zwin inlet began silting in the 13th century. By the 14th century big ships could no longer reach Damme and used Sluis instead. In the 15th century trade suffered badly, and it moved to Antwerp (the Hanseatic merchants moved their office there in 1520)
The sea	Storms and erosion destroy the town	Dunwich (England): storms in 1286–87, 1328, 1347 and 1362 destroyed much of it; a leading port became a village. Old Winchelsea was destroyed by storms in 1287–88 and rebuilt on a hill
A blocked river	A weir (a low dam across a river) or a bridge cuts the town off from the sea	Exeter: weirs built in the 1270s–80s and in 1317 cut off its port; ships had to unload at Topsham
A better site nearby	People move a short distance	Old Sarum to Salisbury (1220s)
A new route	A new bridge, road or pass shifts traffic	The first Schöllenen bridge (around 1220) opened the Gotthard pass and brought trade to central Switzerland
Plague and low prices	Too few people to farm all the land; the worst land is abandoned	The Black Death (1347–51) and later plagues
Conversion to pasture	Lords turn arable land into sheep runs	Many deserted villages in the English Midlands, mostly in the 15th century
Politics	A ruler moves the court, a bishop moves his see (the seat of his diocese), a war ruins a region	See Capitals, Realms and Borders

Deserted medieval villages

England has **more than 3,000** known deserted medieval villages, counting all dates and causes (estimates vary). Most were deserted between about 1350 and 1520, especially in the 15th century, when landlords turned arable land into sheep pasture. Others were lost to floods, sand, the sea, or later to parks around great houses. The plague usually

weakened a village; sheep or a landlord finished it. The best-studied is **Wharham Percy** in Yorkshire, on the western edge of the chalk Wolds. It shrank after the Black Death, and its last villagers were evicted for sheep pasture in the early 1500s. Archaeologists excavated it every year from 1950 to 1990. Today such sites show as bumps in fields: house platforms, sunken lanes and a lonely church.

LATER ERA (1500s+)

New planned towns appear again, now for war and for princely show: Valletta (Malta, 1566, after the Great Siege of 1565), Palmanova (Italy, 1593, a star-shaped fortress town), Freudenstadt (Germany, 1599), Charleville (France, 1606). Engineers also make new land and new routes. The Exeter Canal (1564–66) bypassed the blocking weirs and brought small ships back to the city. Ocean trade makes Atlantic ports such as Seville, Lisbon, Antwerp and then Amsterdam grow fast. Seville's river was hard for big ships, and silting made it worse by the 1620s, so more and more of the American trade moved downstream to Sanlúcar and the Bay of Cádiz. See [What Changes After 1500](#).

MAP TIP

Add history to your map. Mark a few ruins, each drawn with its own symbol broken or grey and labelled "(ruin)": for example an "Old Sarum"-style hill fort above a newer valley town. Draw one or two planted towns with a grid plan and names like "Newbridge" or "Villeneuve". Leave a deserted village or two in sheep country, drawn as a grey church alone in a field and labelled "(lost village)", as in the [master legend](#). If your map is set after a plague or war, show fewer villages on poor land and more sheep pasture.

Where people do not build

Empty space is as realistic as settled space. Here is where medieval people avoided building, and what you find there instead.

Place	Why people avoid it	What you find there instead
Active flood plain	Floods every winter	Hay meadows, mills at the edge, causeways
Fever marsh	Malaria; no firm ground	Fishing huts, salt pans, summer grazing; villages on the edge
High mountains (above ~2,000 m (6,600 ft) in the Alps)	Too cold for crops; snow for months	Summer huts and pastures only
Shady valley sides	Short growing season; long snow cover	Forest
Raided coasts	Pirates and slave raiders	Watchtowers, beacons, boat sheds; villages a few km inland
Royal forests and hunting parks	Laws limit clearing and building	Hunting lodges, keepers' houses, a few old villages
Poor soils (sandy heath, thin chalk, bare rock)	Crops fail	Sheep walks, rabbit warrens, a few hamlets
Contested borders	War and raids	Castles and empty "march" land (see Capitals, Realms and Borders)
Deep old forest	Clearing takes generations	Charcoal burners, swineherds, hermits, slowly growing clearings
Desert away from water	No water at all	Nothing, except caravan stops at wells
Right next to an existing market	Competition and legal rules	Villages, but not another market town

RULE OF THUMB

On a realistic map, a lot of land is empty or only lightly used. Mountains, marsh cores, heaths, forests and borders should look thin.

LATER ERA (1500S+)

Two changes. First, drainage begins to turn empty marshes and shallow lakes into farmland with new farms and villages (the Beemster lake in Holland, drained by 1612; the English Fens, 1630s–1650s). Second, Barbary corsair (North African pirate) raids make some Mediterranean coasts **more** dangerous: Genoa built about 85–90 watchtowers around the coast of Corsica between 1530 and 1620 (a 1617 list counts 86), and coastal villages stayed inland and uphill. See [What Changes After 1500](#).

FANTASY TWIST

Monsters push people together. Where wolves, trolls or raiding beasts are real dangers, expect fewer lone farms, more walled villages, and settlements on defensible sites even in peaceful kingdoms. The "empty" lands become truly empty: no charcoal burners in the cursed forest. See [Fantasy Variants](#).

MAP TIP

After you place settlements, look for "holes" in the pattern. Each hole should have a reason you could explain: marsh, mountain, forest, heath, a royal forest, a dangerous border. If a hole has no reason, add villages. If a crowded area sits on bad land, remove some.

Naming settlements realistically

Names are the first thing a reader sees on a map. Real names follow patterns, and those patterns tell the story of who settled the land.

RULE OF THUMB

Most village names are plain. They join a common ending ("farm", "homestead", "clearing") to a person's name, a tree, an animal or a direction. Each region uses a small set of endings again and again. Save dramatic names for the few places that earned them.

Why: The first settlers simply described the place or its owner: "Acca's farm", "the oak clearing", "the north farm".

Once a name was in use and written down, it rarely changed. Domesday Book (1086) already names 13,418 places, covering most of England except the far north, so

most English village names are older than the castles and abbeys on your map.

Common English endings

These endings come from Old English, the language of the Anglo-Saxons.

Ending	Meaning	Notes	Examples
-ton	Farmstead, estate, village (Old English <i>tūn</i>)	The commonest of all: over 4,000 places	Acton ("oak farm"), Wootton ("wood farm"), Newton ("new farm"), Sutton ("south farm")
-ham	Homestead, village (<i>hām</i>)	Often among the older names	Nottingham, Dagenham
-wick, -wich	Specialist farm (often dairy), or trading place (<i>wīc</i> , from Latin <i>vicus</i>)	Inland usually a farm; on coasts and rivers often a port	Keswick ("cheese farm"), Ipswich, Norwich
-worth	Enclosure	—	Tamworth, Kenilworth
-stead	Place, site	—	Hampstead
-cot, -cote	Cottage, hut	Small places	Didcot, Ascot
-bury, -borough, -burgh	Fortified place (<i>burh</i>); later a town	Use it at old forts, earthworks or fortified towns	Canterbury, Peterborough
-chester, -caster, -cester	Roman fort or town (<i>ceaster</i> , from Latin <i>castra</i>)	Only on Roman sites	Chester, Lancaster, Gloucester
-ley, -leigh	Wood clearing, woodland pasture (<i>lēah</i>)	Common in once-wooded country	Bradley ("broad clearing"), Henley
-field	Open land without trees	—	Sheffield, Lichfield
-den, -don	Valley (<i>denu</i>); hill (<i>dūn</i>)	Easy to confuse; check the terrain you drew	Croydon (a valley name)
-hurst	Wooded hill	Wooded country	Lyndhurst, Sandhurst
-stow	Meeting place, holy place	—	Bristol ("place at the bridge"), Padstow
-minster	Mother church, monastery	One per large district (see chapter 08)	Westminster, Axminster

The Norse (Viking) layer. Danish and Norwegian settlers of the 9th–11th centuries added their own endings in northern and eastern England:

Ending	Meaning	How many	Examples
-by	Farm, village	More than 600 places in England (Baugh and Cable); Domesday already records 303 in the East Midlands alone (Cameron)	Grimsby, Whitby, Derby
-thorpe, -thorp	Secondary settlement, outlying farm	About 300	Scunthorpe, Mablethorpe
-thwaite	Clearing, meadow	About 300; mostly in the north-west (Cumberland, Westmorland, Lancashire), where Norwegians settled, and in Yorkshire	Braithwaite, Applethwaite
-toft	Building plot, homestead	—	Lowestoft
-holm(e)	Island, dry ground in wet land	—	Axholme
Kirk-	Church	—	Kirkby

The Normans, by contrast, gave few new village names in England, because the land was already named. They named their new castles and abbeys instead: Richmond in Yorkshire ("strong hill", founded 1071 by Alan Rufus, probably

after a Richemont in Normandy), Belvoir ("fine view"), Beaulieu ("fine place").

Names in other lands

Region and language	Common elements (meaning)	Examples
Celtic: Wales, Scotland, Ireland, Cornwall, Brittany	Aber- (Welsh) and Inver- (Gaelic): river mouth; Llan- and Kil-: church; Caer- and Dun-: fort; Pen-: head, hilltop, end; Tre- and Bally-: farm, settlement; Strath-: broad valley	Aberystwyth, Inverness, Llandaff, Kilkenny, Caernarfon, Dundee, Penzance, Tredegar, Ballymena, Strathclyde
German lands	-ingen ("people of"; mostly 6th–9th centuries), -heim (home), -dorf (village), -hausen (houses), -hof (farm), -weiler (hamlet), -stadt or -stedt (place, town), -burg (fortress); clearing names -rode, -rath, -reuth, -ried (high-medieval clearance); -hagen (hedged settlement)	Tübingen, Mannheim, Düsseldorf, Mühlhausen, Darmstadt, Magdeburg, Wernigerode, Bayreuth
East of the Elbe (Slavic names, later Germanized)	-itz, -ow, -in	Chemnitz, Güstrow, Schwerin
France	Gallo-Roman <i>-acum</i> ("estate of") became -ac in the south and -y or -ay in the north; -court (farm with a yard) and -ville (farm, later village) from the early Middle Ages; in Normandy, Viking-age -tot (house site; over 300 places), -beuf (booth, shed) and -fleur (stream running to the sea)	Cognac, Bergerac, Vitry; Harcourt, Abbeville; Yvetot, Elbeuf, Honfleur
Italy	-ano (a Roman estate named after its owner, Latin <i>-anum</i>), Castel- and Rocca (fortified village, rock fortress), Borgo (market settlement or new town), Pieve (rural mother church)	Castelfranco, Rocca di Papa, Borgo San Sepolcro (today Sansepolcro), Pieve di Cadore
Iberia, Arabic layer	Al- ("the"); Medina- (<i>madina</i> , city); Alcalá (<i>al-qaḥa</i> , castle); Alcázar (<i>al-qasr</i> , palace-fortress); Guad- (<i>wadi</i> , river valley); Almadén (<i>al-māḍin</i> , "the mine")	Medinaceli, Medina del Campo, Alcalá de Henares, Guadalquivir (<i>al-wadi al-kabir</i> , "the great river"), Almadén
Iberia, Christian resettlement (<i>repoblación</i>)	Villanueva ("new town"), Villafranca ("free town"), Castro- (castle)	Villanueva de la Serena, Villafranca del Bierzo, Castrojeriz
Slavic lands	-grad, -gorod, -gród (fort, town); -ice, -owice ("people of", from a founder's name); -ów, -ov (possessive: "X's place")	Novgorod ("new town"), Belgrade ("white town"), Kraków ("Krak's town")

How names pile up in layers

- 1. Rivers and big hills keep the oldest names.** In England the oldest place names are river names, many of them Brittonic (the Celtic language spoken before English). *Avon* simply means "river". Use your oldest language for rivers and mountains.
- 2. Villages carry the language of the people who founded them.** Danish settlers left -by and -thorpe in eastern England; Norwegians left -thwaite in the north-west. Where the older language survived longer, as in Cornwall and Cumbria, many village names stayed Brittonic. So a border between two kinds of names is often an old settlement frontier, not a political border.
- 3. Later settlements fill the gaps.** Clearing names (-ley, -thwaite, -rode, -reuth) mark land taken from the forest; in Germany the -rode and -reuth names belong mainly to the great clearances of the High Middle Ages (see [Forest](#)). Planted towns get "new" names: Newport, Villeneuve, Neustadt, Villanueva, Novgorod.

Endings change at language lines. Where two languages meet, the endings switch within a few villages, even inside one realm. In Belgium, north of the Romance–Germanic

- 4. Same-name villages get extra words.** Two neighbouring Suttons or Stokes need telling apart. Common extras:

- **the lord's family:** Stoke Mandeville (the suffix is first recorded in 1284, when the Mandeville family held the manor); Sutton Courtenay (after the Courtenay family took the manor in the 1170s);
- **size or position:** Great and Little, Upper and Lower, North and South;
- **a market:** Market Harborough, Chipping Norton (*chipping* = market);
- **a church or royal owner:** Bishop's Stortford, Abbots Langley, King's Lynn;
- **a river:** Stratford-upon-Avon, Newcastle upon Tyne.

- 5. Owners change, names follow.** King's Lynn was "Bishop's Lynn" until 1537, when the town passed from the Bishop of Norwich to Henry VIII.

line, many Flemish village names end in Germanic -gem ("home", like German -heim) or -zele ("house, settlement", as in Brussels; spelled -zeele across the border in French

Flanders): Wevelgem, Zottegem, Herzele, Bollezeele. South of the line, Romance names such as Soignies and Ottignies (“Otto’s lands”: a Frankish owner’s name with a Romance ending) take over; in Lorraine, -ange (the French spelling of German -ingen: Hayange is German *Hayingen*) on the Ger-

man-speaking side faces -ville and -court on the French side, and two villages called Audun were told apart as Audun-le-Tiche (“German Audun”) and Audun-le-Roman (“Romance Audun”). See [Language lines are not borders](#).

A method for naming a region

1. **Write the region’s history in two lines.** Who lived here first? Who came later? Who rules now? Choose 1–3 language layers.
2. **Name rivers and mountains** from the oldest layer.
3. **Pick 4–8 common endings** from the main settler layer, and build most village names from them. Join them to personal names, trees, animals, colours and directions. (This is a mapmaking suggestion, not a measured rule, but real regions work this way: in England, -ton alone names over 4,000 places.)
4. **Use site words only where the map shows the site:** -ford at a ford, -bridge at a bridge, -mouth at a river mouth, -ey at an island (see [Reading place names](#)).

FANTASY TWIST

Give each people its own set of endings, and let border zones mix them. A valley where dwarf names sit under older elf river names, with human “-ton” villages on the valley floor, tells the reader three chapters of history before they read a word of your book. See [Fantasy Variants](#).

5. **Save special elements for special places.** Fort words (-bury, Caer-, Dun-, -grad, Alcalá) go on places with a fort or an old earthwork. Market words (Market-, Chipping-) go on market towns. Church words (Kirk-, Llan-, -minster, Pieve) go on places with an important church. Roman words (-chester) go only on Roman sites.
6. **Show history with patches.** Put a band of second-layer names along the coast where raiders settled. Use clearing names in old forest. Mix both sides’ names in a march.
7. **Split duplicates with extra words** (Great/Little, a lord’s name, a river).

MAP TIP

Keep a name sheet for each region: its 4–8 endings, its old river words and 10–20 first parts. Reuse them. Real maps repeat themselves (England has many Suttons, Nortons and Newtons), and that repetition is what makes a region feel like one place. Readers notice when the names in one corner of the map share an “accent”.

Placing settlements on your map: a method

Work from big to small, and from nature to people. Start with a finished terrain map: coasts, mountains, rivers, forests, marshes.

1. **Mark the water.** Draw all rivers. Decide which stretches are **navigable** (deep enough for boats) and where they stop: falls, rapids, shallows. Mark fords and the tidal limit on each major river.
2. **Mark the land quality.** Lightly shade good farmland (river valleys, plains, gentle hills), pasture land (uplands, heath), marsh, forest, mountain and desert, in pencil or on a layer you hide later.
3. **Place the great city or cities.** Look for the best situation in the richest region: the lowest bridging point or estuary of a big navigable river, or a superb harbour. Remember that a city of 50,000+ must be fed by water.
4. **Place regional towns.** Put them at the next-best nodes: confluences, heads of navigation, gap towns, pass towns, harbours, crossings of major roads. Space them

by the regional row in the spacing table. Cities of 10,000+ are 60–90 km (37–56 mi) apart in a very urban region, 130–170 km (80–105 mi) in an average one, and 260–400 km (160–250 mi) in a thinly urban one.

5. **Place market towns.** Fill the farmland with market towns about 9–16 km (6–10 mi) apart, on rivers and road junctions where possible. Spread them more widely on poor land.
6. **Place villages.** Fill good land with villages 1.5–4 km (1–2.5 mi) apart, on terraces, spring lines and land edges. In uplands, use hamlets and single farms in valleys. Leave marsh cores, mountains and heaths nearly empty.
7. **Add special sites.** Castles ([chapter 04](#)), monasteries and shrines ([chapter 08](#)), mines, salt works and other special towns ([chapter 07](#)). These are the main reasons for a settlement in an “odd” place.

8. **Connect them.** Draw roads and river routes ([chapter 05](#)). Small adjustments are fine: shift a village to the road, or a town to the bridge.

9. **Add time.** Mark ruins, deserted villages, moved towns and planted towns.

10. **Run the test below on every settlement.**

The “why is it here?” test

Ask these questions about every dot. A **village** must pass 1–3. A **town** must also pass 4. A **city** must pass all six.

#	Question	If the answer is “no”
1	Is there fresh water on or next to the site?	Move it to a river, stream, spring or lake shore.
2	Is there a mix of farmland, meadow, pasture and wood within a few km?	Move it to an edge between land types, or make it a special site (mine, monastery).
3	Is it safe from floods (on a terrace, rise or island)?	Move it up a few metres onto a terrace or rise.
4	Do routes meet here (crossing, confluence, junction, harbour, gap, pass)?	Make it a village, or move it to the crossing.
5	Is it on navigable water or the coast?	Make it a town under ~10,000 people.
6	Is it the best-connected place in a rich region?	Make it a regional town, not a great city.

Also check:

- **Defence:** In a dangerous region, is there a hill, a river loop or a wall? If not, why not?
- **Spacing:** Is it too close to a place of the same rank? Two market towns 3 km (2 mi) apart compete; one should be a village.

- **Name:** Does the name fit the site (ford, bridge, mouth, hill) and the region's language layer? See [Naming settlements realistically](#).

FANTASY TWIST

A magical reason, such as a ley-line nexus, a dragon's peace treaty or a portal, can replace test 4 or 6. It cannot replace tests 1–3. Even a wizard's city needs water and food, or a very good explanation of where they come from.

MAP TIP

Keep a short list next to your map: settlement name, rank, site type, and one-line situation (“Kingsford: town; ford and confluence; where the north road meets the Silverwater”). It takes a few minutes and catches most mistakes. More common errors are in [Common Mistakes and How to Fix Them](#). The full workflow is in [Step-by-step](#).

Quick summary

- Every settlement needs **water, food, fuel, safety, access and health**. Villages need the first three nearby; towns also need access; cities need access **by water**.
- Villages sit on **edges** between land types (valley and hill, fen and dry land, downland and clay), on **river terraces** above the flood plain, and along **spring lines**.
- **Site** (the spot) decides if a place survives. **Situation** (its place in the network) decides if it grows big.
- The best town sites are **lowest bridging points, heads of navigation, confluences, gaps, pass feet, sheltered harbours and estuaries**. On tidal coasts, big ports often sat 50–120 km (30–75 mi) up an estuary.
- In dangerous times and places, people choose **hills, river loops, islands and spurs**.
- Spacing: villages **1.5–4 km (1–2.5 mi)**, market towns **9–16 km (6–10 mi)**, bridge towns on a lowland river **about 20 km (12 mi)**, head ports **about one per 100–150 km (60–95 mi) of coast**, towns and cities much farther apart, depending on how urban the region is.
- Christaller's **central place** model: many small places close together, a few big ones far apart, in rough hexagons. Distort it along rivers and roads.
- Settlement is **clustered on good land and thin on bad land**. Empty areas need a reason, and so do crowded ones.
- Outside Europe: **shifting cultivators** live thinly (about 6 per km² as a 1950s world average, 12–35 in modern northern Laos and Thailand, at most about 55–56), **wet-rice lands** are the most crowded farmland of all (Angkor about 230–300 per km² over its core), and **steppe nomads** use small camps of 2–10 families that move about 4–10 times a year, with fixed winter sites.
- Places change: Roman sites reused or abandoned, **planted towns** with grids, towns moved, harbours silted, and **more than 3,000 deserted villages** in England alone.
- **Names** come in layers: rivers keep the oldest names, villages use a few common endings of their founders' language (-ton, -by, -dorf, -ac), extra words (Great/Little, a lord's name, a river) split duplicates, and the endings switch at language lines (-gem and -ange on one side, -ville and -court on the other).
- Use the **"why is it here?" test** on every dot.

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CHAPTER 2

POPULATION AND SETTLEMENT SIZES

This chapter gives you the numbers: how many people live in a village, a town or a city, how many of each your realm should have, and how crowded the land can be. All population and size figures used elsewhere in this guide come from here. The baseline is Europe in the High and Late Middle Ages (c. 1000–1500), with the Mediterranean and Middle East added where they help, plus a short section on Asia, Africa and the Americas. The numbers are real historical estimates, given as ranges because scholars disagree.

IN THIS CHAPTER

- The settlement ladder
- How many of each kind
- Population density
- Carrying capacity: how much land feeds people
- Realm populations and areas
- The biggest cities
- Beyond Europe: Asia, Africa and the Americas
- Step-by-step population calculator
- Change over time: growth, famine and plague
- Who lives where
- Gaming rules of thumb vs history
- Quick summary
- Sources and further reading

The settlement ladder

RULE OF THUMB

Almost everyone lives in villages. A “big city” in medieval Europe is anything above 10,000 people. A place of 50,000 is one of the great cities of the known world.

Why: Before modern farming, most families had to farm just to feed themselves. Across Europe 85–95% of people lived in the countryside. So only about one family in ten (at best one in seven) could live mainly from a craft, trade or

service. Towns therefore stayed small and few. A place grew big only when it had something special: a ruler’s court, a bishop, a great port, a river junction, or an export industry (see [Where Settlements Are Built](#)).

Numbers. Population means people, not households. A household is about 4.5–5 people. Medieval tax records often count *hearths* (households) instead of people, so multiply by about 4.5 to get people.

Tier	People	Households	What it usually has
Isolated farmstead	5–15	1–2	Its own fields and pasture. No church. The main pattern in uplands, Wales, Devon, the Alps, Scandinavia and forest clearings.
Hamlet	10–75	2–15	A few houses. A chapel at most. Belongs to a nearby village’s parish (the area served by one church).
Village	75–500 (English lowland: usually 150–300)	15–100	Parish church, manor house (the local lord’s house), a watermill or windmill, a smithy, open fields and common pasture. Sometimes an alehouse.
Small market town	500–2,000	100–400	A chartered weekly market and a yearly fair, an open market place, burgage plots (long narrow house plots rented by townspeople), inns, specialist craftsmen, one parish church, often a hospital (a charity house for the sick, old or poor).
Town	2,000–10,000	400–2,000	Usually walled. Several parishes, merchant and craft guilds (associations that controlled trade and crafts), friaries (houses of preaching friars, found in towns from the 1220s), hospitals, often a castle, a borough charter (a written grant of town rights) and its own council.

Tier	People	Households	What it usually has
City	10,000–50,000	2,000–10,000	Cathedral (a bishop's main church), many parishes and friaries, suburbs outside the walls, strong guilds, law courts, a mint (where coins are made); sometimes a university.
Great city	50,000+	10,000+	A capital or a major port. Several markets, a university, quarters of foreign merchants, an organised water supply. Its grain must arrive by river or sea.

Real examples:

- **England in 1377** (after the plague). These estimates come from the 1377 poll tax, a tax on every person over 14.
 - London had about 35,000 people; York about 11,000; Bristol about 9,500; Coventry about 7,000; Norwich about 6,000.
 - About 10 more towns had roughly 3,500–5,500 (for example Lincoln, Salisbury, King's Lynn, Colchester and Boston).
 - Outside London, only York passed 10,000. About 5 towns had 5,000–10,000, about 24 had 2,000–5,000, and many more had 1,000–2,000.
 - Before the Black Death, all of these were about 1.5–2 times larger.
- **Universities were rare:** about 20 in all of Europe c. 1300 and about 60–80 by 1500. (Over 80 had been founded by 1500, but not all of them lasted.)

Regional differences: The ladder above fits England, northern France and Germany best. Elsewhere:

- **Italy:** *castelli* were fortified hilltop villages of 200–1,000 people. Italy also had hundreds of small bishoprics, so a cathedral does not prove a big city there. England had only 17 bishoprics (sees), so there a cathedral does mean an important city.
- **Southern Spain, Sicily and Apulia:** *agro-towns* had 2,000–10,000 people, but most of them were farmers who walked out to their fields every day. They are large in population but not real towns in function.
- **Germany and the Low Countries:** many places with a town charter (“legal towns”) had fewer than 2,000 people.
- **Byzantine lands:** most villages were small, usually well under 100 households (estimate). Even Lampsakos, a small town on the Dardanelles, had only about 173 households in one medieval record, and about a third of them did town work rather than farming.

How big to draw it

Medieval towns were small on the ground. Historians often assume about 100–200 people per hectare (40–80 per acre) inside a town. (A hectare is 100 × 100 m, about 2.5 acres.)

One major city database uses 150 per hectare, and 75 for spacious “garden cities” such as Baghdad, Basra and Sanaa (Bosker et al. 2013). Bigger cities were denser than small ones (Cesaretti et al. 2016).

Place	People	Built-up area	Width if roughly round
Village	150–300	5–15 ha (12–37 acres), houses spread out with gardens	250–450 m (0.15–0.3 mi)
Small market town	500–2,000	5–20 ha (12–50 acres)	250–500 m (0.15–0.3 mi)
Town	2,000–10,000	20–80 ha (50–200 acres)	0.5–1 km (0.3–0.6 mi)
City	10,000–50,000	80–300 ha (200–740 acres)	1–2 km (0.6–1.2 mi)
Great city	50,000–200,000	300–1,000+ ha (740–2,500+ acres)	2–4 km (1.2–2.5 mi)
Packed island city (Tenochtitlan, Mexico, 1519)	200,000–250,000 (some estimates up to 400,000)	800–1,350 ha (8–13.5 km ² , 3.1–5.2 sq mi)	3–4 km (2–2.5 mi)
Dispersed “garden” city (Greater Angkor, Cambodia, 13th century)	700,000–900,000	About 3,000 km ² (1,160 sq mi) of house mounds, ponds, embankments and rice fields around a walled core of 9 km ² (Angkor Thom)	About 60 km (37 mi) for the whole zone; the walled core is 3 × 3 km (2 × 2 mi)

Checks against real places:

- **Bristol c. 1300:** about 130 ha including its suburbs, but only 55 ha inside the walls.

- **Paris:** the wall of Philip Augustus (built 1190–1213) enclosed 253 ha.
- **Florence:** its last medieval wall (finished 1333) enclosed about 430 ha for roughly 100,000 people.

The village row is a rough estimate.

The two extremes. The last two rows show how far city density can stretch. Tenochtitlan packed about 150–300 people per hectare onto its island (derived; Smith 2005 estimates 212,500 people on 13.5 km², about 160 per ha). That is at or above the top of the European range. Greater Angkor averaged only about 2–3 people per hectare over its

whole zone: about 50 per ha inside the walls of Angkor Thom, but only about 1.5 per ha in the wide outer zone of houses, ponds and fields (Klassen et al. 2021). Most medieval cities sit between these two: 100–200 per ha for walled towns and about 75 per ha for Islamic “garden cities” such as Baghdad. See [Beyond Europe](#) for more on both cities.

Map symbols and labels

Tier	Continent map	Kingdom map	Local map	Label size
Farmstead	—	—	Tiny dot	None
Hamlet	—	—	2–5 small house marks, no church	6–7 pt italic, or none
Village	—	A tiny dot, only for notable villages	House marks around a church	7–8 pt
Small market town	—	Small open circle	Houses around a market square	8–9 pt
Town	Only the important ones, as a dot	Circle with a ring; battlements on the ring if walled	Real outline with wall and gates	9–10 pt
City	Dot or circle	Circle with a thick battlemented ring (a larger walled symbol)	Full outline	11–12 pt, bold or small capitals
Great city	Large circle with a ring	Large walled symbol	Full outline with suburbs	13–14 pt CAPITALS

These symbols match the guide’s [master legend](#). A realm capital takes a filled star in place of its tier symbol, so do not use a star for any other place.

FANTASY TWIST

Where monsters roam the countryside, people give up isolated farmsteads and crowd into fewer, larger, walled villages. Italy did this for human reasons in the 10th–12th centuries (*incastellamento*, the move into hilltop *castelli*). Shift some of your hamlet population into fortified villages of 300–1,000 people. See [Fantasy Variants](#).

LATER ERA (1500S+)

The ladder stays the same, but the top grows taller. By 1600 Istanbul had perhaps 400,000–700,000 people, and Paris, Naples and London had 200,000–300,000 each. Garrison towns built around artillery forts become a new type. See [What Changes After 1500](#).

MAP TIP

On a kingdom map at 1 cm = 10 km (about 1 inch = 16 mi), even a great city is only 2–4 mm across, so use a symbol, not an outline. Draw real outlines only on local maps (1 cm = 1 km, about 1 inch = 1.6 mi, or larger). Make label size follow the tier, not the story importance. If you want a small place to look important, give it a special icon (castle, cathedral, shrine) instead of a big label. Capital markings are covered in [Capitals, Realms and Borders](#).

How many of each kind

RULE OF THUMB

85–95% of people are rural almost everywhere. For every city of 10,000+, expect about 10 towns, about 50–100 small market towns, and a few thousand villages.

Why: Each town needs a ring of farmland around it to feed it and to buy its goods. Small towns serve a few villages each. Bigger towns serve the small towns. Only a few places serve a whole region.

Urbanization by region

Urbanization means the share of people who live in towns. The number depends on where you draw the line: “towns of 10,000+” gives a low figure, while “all towns” gives a higher one.

Region	c. 1300	c. 1500	Notes
Europe	~5% in cities of 10k+; ~9.5% in all towns (broad definition)	5.6% in 10k+	—
England	~2% in 10k+ (only London); ~10% in towns of 2k+; 15–20% counting small boroughs	3.1% in 10k+	Dyer argues for ~20% in all towns by 1300.
France	~3–4% in 10k+; ~7–9% in 2k+	~4.2% in 10k+	Paris dominated everything.
Low Countries	Flanders ~33–40% in towns of all sizes	Belgium 21%, Netherlands 16% in 10k+	Holland ~45% urban by 1500.
Northern Italy	~18% in 10k+ (Centre-North); ~20% in 5k+	~15–16% in 10k+	Italy had 48–79 cities of 10k+ in 1300 (estimates differ).
Iberia	~8–10% in 10k+	Spain ~6%, Portugal ~3% in 10k+	Inflated by agro-towns.
Germany	~2–3% in 10k+	3.2% in 10k+	Many small towns of 1,000–2,000.
Denmark	~10% in all towns	~14% in all towns	About 80 towns c. 1350, most of them small.
Poland, Hungary	under 2% in 10k+	~1–2% in 10k+	—
Arab Middle East and North Africa	~7% in 10k+	~8% in 10k+	Stayed roughly level (about 6–10%) from 800 to 1800. It was clearly more urban than Europe around 800–1000; Europe only caught up in share c. 1700–1800. (Rough figures from the Bosker et al. city data.)

Counting the cities

Year	Cities of 10k+	Cities of 50k+	Cities of 100k+
c. 1000	~130 (Europe 81, Middle East and North Africa 48)	~11	4
c. 1300	~230 (Europe 172, Middle East and North Africa 60)	~22	8
c. 1500	~250 (Europe 195, Middle East and North Africa 52)	~31	8

(Bosker, Buringh & van Zanden 2013 dataset; it excludes Russia and Iran.)

Looking further down the ladder, the historian Paul Bairoch counted about 1,450 towns and cities in 14th-century Europe. He sorted them by what they did:

- **About 60%** were small towns of 2,000–6,000 people with purely local roles: a church or two, a local market, and craftsmen serving nearby villages.

- **About 300–330** were “regional centres” of 4,000–12,000. They usually had walls, trade, and often a bishop.
- **About 210** were large cities of more than about 8,000–12,000.

(Bairoch 1988, as summarised by Jedwab, Johnson & Koyama 2020.)

England c. 1300 as a model kingdom (4.75 million people on 130,000 km²):

Tier	Approximate number	One per ... people
Parishes (each with a village church)	~9,500	~500
Places with a market grant	~1,750, but many were village markets and only 39% still had a market c. 1600	~2,700
Boroughs (places with town status)	about 600 (Beresford & Finberg 1973), many of them tiny	~8,000

Tier	Approximate number	One per ... people
Towns of 2,000–10,000	roughly 50–60 (estimate)	~80,000–100,000
Cities of 10,000+	~6: London (60–100k) plus about five provincial cities of 10–15k. The Bosker et al. data list Winchester, Norwich, Coventry, Plymouth and Bristol; other historians would include York.	~800,000

How many people per city?

Region c. 1300	Cities of 10k+	One city per ... people
Italy	48–79	130,000–260,000
Iberia	~24	250,000–375,000
Egypt	~12	350,000–650,000
Germany (modern borders)	~24	roughly 400,000–500,000
France (modern borders)	~34	~600,000
England	~6	~800,000
Poland	~2	~1,000,000

(City counts from the Bosker, Buringh & van Zanden data, using modern borders. Italy's higher count, 79, comes from Malanima.)

Population totals are from the realm table below. Treat every figure as a rough estimate.)

The rank-size rule and primate cities

The **rank-size rule** (also called Zipf's law) says: the 2nd city has about $\frac{1}{2}$ the population of the largest, the 3rd about $\frac{1}{3}$, the 4th about $\frac{1}{4}$, and so on. It describes many modern countries well. A **primate city** is a capital far bigger than the rule predicts, often 3 or more times the size of the 2nd city.

Medieval reality did not follow the rule neatly:

- **Primate realms (strong kings, one capital):**
 - **England, 1377:** London was 3–4 times the size of York or Bristol, and 8–9 times the size of Colchester or Canterbury. The rule predicts York at about 17,000; it had about 11,000.
 - **France, c. 1300:** Paris had perhaps 200,000–250,000 (estimates vary widely). The next cities (Rouen, Montpellier, Saint-Omer) had about 35,000 each, so Paris was roughly 6–7 times larger.
- **Flat-topped regions (many independent cities):** In Italy c. 1300, Venice (~110k), Milan (~100k), Genoa (~100k) and Florence (~95k) were all about the same size. The Low Countries were similar, with Ghent and Bruges at about 40,000 each. (Some estimates put Ghent higher, at 50,000–65,000.)
- **Overall:**
 - Economic historians find that European city sizes only came to fit the rank-size rule between 1500 and 1800 (Dittmar 2008).

- Before that, the pattern was bent: in most regions the biggest cities were *smaller* than the rule predicts. Primate kingdoms such as England and France were the exception.
- In the Arab world, by contrast, the largest city always held 12–38% of all city-dwellers. In Western and Central Europe it never held more than 12% (Bosker et al. 2008).

RULE OF THUMB

A strong, centralised kingdom has one capital 3–7 times bigger than its second city. A land of city-states or weak kings has several rival cities of similar size.

LATER ERA (1500S+)

In 1500–1650 some capitals pulled further ahead and reached 5–10 times the size of the second city. London was the extreme case: about 200,000 people by 1603, over 13 times the size of Norwich, the next-largest English city (Mortimer 2012, via Wikipedia). See [What Changes After 1500](#).

MAP TIP

Decide first what kind of realm you have, centralised (one strong king) or fragmented (many rival lords or city-states), then size the top cities. For a centralised kingdom, draw one great capital and then a gap down to a handful of towns. For a fragmented region, draw 3–5 rival cities of similar size, each with its own little territory. Do not give every country a 100,000-person capital: around 1300, only 8 cities in all of Europe, North Africa and the Middle East reached that size. (A rice-growing empire with canals, like Song China, is different; see [Beyond Europe](#).)

Population density

RULE OF THUMB

Good lowland farmland supports 30–50 people per km² (80–130 per sq mi) at its peak. A whole realm, uplands included, averages about 7–40 per km² (18–100 per sq mi): 10–40 in settled western kingdoms, 7–15 in young eastern ones. Europe as a whole c. 1300 averaged about 20 per km² (52 per sq mi).

Why: Density depends on how much food the land can grow (see the next section). Above about 50 per km² (130 per sq mi), a region needs towns, specialised commercial farming or imported grain.

By type of land

Land type	per km ²	per sq mi	Real examples
Rich lowland, peak times	30–50	80–130	Northern Italy 34–40 c. 1300; England's <i>whole-country</i> average was already ~36 in 1290, so its best lowland was higher
Very urban, rich lowland	50–75	130–190	Flanders and Holland in the 15th century (estimate). Exceptional districts went far higher: Norfolk, where most people lived in villages (Norwich was its only city), had 486,920 people in 1290 (Broadberry, Campbell & van Leeuwen 2011, Table 8B) on about 5,400 km ² (2,100 sq mi). That is roughly 90 per km ² (230 per sq mi) (derived).
Average mixed farmland	15–30	40–80	Suffolk ~23–25 in 1086; England ~19 in 1377
Poorer land, young kingdoms	7–15	18–40	Hungary ~7 (c. 1200) and ~9–11 (c. 1495); Crown of Poland 8.6 (1370) and 13 (1500)
Hill country (farmed valleys with upland grazing), peak times	8–15	21–39	Cumberland and Westmorland about 15–17 in 1290, as whole counties that include the Eden valley and the Solway plain, so the fells themselves held fewer. The same two counties had about 6–7 in 1377, and northern and western England had under 5 in 1086. (Derived from county estimates in Broadberry, Campbell & van Leeuwen 2011.)
High uplands, moors, mountains	2–5	5–13	Scottish Highlands, high Alps, Pyrenees, northern Pennines (estimate; few direct counts exist). Upland parishes covered 50–100+ km ² (20–40+ sq mi) each.
Forest or steppe frontier	1–5	3–13	Lithuania, the Rus' forest zone (estimate)
Shifting (swidden) farming in tropical forest	about 6 on average; 12–35 in busier districts; at most about 55–56	about 14 on average; 30–90; at most about 145	World average about 6 in the 1950s (FAO); modern Southeast Asia from 12 in northern Laos to 35 in northern Thailand (Warner 1991, FAO); ceiling about 55–56 (Ruthenberg 1980)
Steppe nomads	under 1–2	under 3–5	Herders need huge pastures
Desert	close to 0	close to 0	Except oases and irrigated valleys

Land type	per km ²	per sq mi	Real examples
Irrigated river valley	100+ on the farmed strip	260+	Egypt c. 1340: perhaps 4–8 million people, almost all on the Nile floodplain and delta
Wet-rice or lake-garden basin (outside Europe)	80–125 over a whole region; 230–300 in a city-and-rice-field zone	210–325; 600–780	Yangzi delta and Zhejiang ~80 (1102); Valley of Mexico ~100–125 (1519); Greater Angkor ~230–300 (13th century). All derived; see <i>Beyond Europe</i>

By realm and date

Realm and date	per km ²	per sq mi
England 1086 (1.71m on 130,000 km ²)	~13	~34
England 1290 (4.75m)	~36	~94
England 1377 / 1450	~19 / ~15	~50 / ~38
France c. 1328	~35–40	~90–105
Byzantium 1025 (about 12m on 1.675m km ²)	~7	~19
Poland-Lithuania 1500	6.6	17

How to read this: A realm's average hides big differences. In 1086 Suffolk was five times denser than northern Eng-

land. Even at the 1290 peak, much of England's upland was nearly empty while East Anglia was crowded.

MAP TIP

Before placing villages, lightly shade your map in pencil (or on a layer you hide later) into three or four density zones: crowded lowland, ordinary farmland, thin hill country and empty land (mountain, marsh, forest, desert). Then place settlements to match. Lowland villages sit 1.5–4 km

(1–2.5 mi) apart. In uplands, parish churches may be 7–10 km (4–6 mi) apart, with scattered farms and hamlets between them. Leave real gaps: empty land makes the full land look full.

Carrying capacity: how much land feeds people

Carrying capacity means how many people a piece of land can feed.

RULE OF THUMB

Each person needs about 1 hectare (2.5 acres) of plough-land, or 2–3 hectares (5–7 acres) of all land once pasture, meadow and woodland are counted.

Why: Medieval harvests were small, and a lot of each harvest went back into the ground:

- **Seed-to-yield ratio:** Typically 3–5 : 1, with a range of 2–7 : 1 in the 14th century. You sowed 1 sack and harvested 3–5. Modern farms get 30 : 1 or more.
- **Seed:** A quarter to a third of every crop had to be kept as next year's seed.
- **Fallow:** Land left unplanted for a year so the soil could recover. It took ⅓ of the arable land under the three-field system and ½ under the two-field system.

- **Wheat yields:** English demesne wheat (grown on the lord's own farm) yielded about 8–12 bushels per acre (0.55–0.8 tonnes per hectare). A bushel is an old measure of volume, about 36 litres.

- **Result:** About 250–350 kg of food grain per hectare of arable land per year. A person ate about 200–250 kg of grain a year, plus barley for ale and oats for horses.

Check against history: England c. 1300 had about 11.5 million acres of arable, counting the fallow field. About 7.4 million acres of it were sown each year (Broadberry, Campbell et al. 2010). For 4.75 million people, that is about 2.4 acres (1.0 ha) of arable per person, of which only about 1.6 acres (0.6 ha) carried a crop in any one year. Counting all land, it is about 2.7 ha per person. Even so, in 1300–1319 the main crops and animal products supplied only about 1,300–1,350 kcal per person a day. That is after seed, fodder, storage losses and the waste of milling and brewing are taken out. Grain gave about 1,200–1,250 kcal of it, and

meat and dairy about 120 (Broadberry, Campbell & van Leeuwen 2011). Grain calories rose to about 1,800 a day by the 1380s, because the plague had cut the number of mouths. The same authors reckon a well-fed population

needs about 2,000 kcal a day, with at least 1,500 of it from these main crops and animal products. Garden vegetables, poultry, fish and game filled only part of the gap. England was at its limit.

Village territory

• Family holdings:

- A peasant family needed very roughly 10–15 acres (4–6 ha) of arable to live from its land alone (estimates vary).
- The standard English holding, the *virgate* or *yardland*, was in theory about 30 acres (12 ha), but its real size varied a lot from place to place. Many families held only half a virgate, and many held less, so they also worked for wages.

• **One example parish:** Milton Ernest (Bedfordshire) in the 14th century had about 1,600 acres (650 ha, or 6.5 km²): 1,405 acres of arable, 81 of meadow, 84 under the village itself and 28 of roads (Hutchings 1969).

• **Average size:** England c. 1300 had about 14 km² (5.4 sq mi) per parish. Upland parishes covered 50–100+ km² (20–40+ sq mi).

Village size	Arable needed	Total territory (lowland)	Territory (upland)
100 people	~100 ha (250 acres)	2–4 km ² (0.8–1.5 sq mi)	10–30 km ² (4–12 sq mi)
250 people	~250 ha (620 acres)	5–10 km ² (2–4 sq mi)	25–60 km ² (10–23 sq mi)
500 people	~500 ha (1,240 acres)	10–20 km ² (4–8 sq mi)	rare in uplands

For how villages and their fields are laid out, see [Villages, Farms and the Countryside](#).

Feeding towns: the hinterland

A **hinterland** is the area of countryside that feeds a town. The model below assumes rural density of 30 per km² on good land (10 on marginal land). It also assumes each

country person grows enough surplus to feed 0.15–0.25 townspeople (0.1 on marginal land). Real hinterlands were bigger, because towns competed for the same farms.

Town size	Overland radius, good lowland	Overland radius, marginal land	With water transport
1,000	7–8 km (4–5 mi)	~18 km (11 mi)	Not needed
10,000	21–27 km (13–17 mi)	~55 km (35 mi)	A river or coast lets food come from 2–10 times farther along the waterway
50,000	46–59 km (29–37 mi)	~125 km (78 mi)	Almost always needs a river or sea supply (rare exceptions: Granada)
100,000	65–84 km (40–52 mi)	~180 km (110 mi)	Only possible by water

Why big cities need water: Moving bulk goods overland was very expensive. In England around 1300, land transport cost roughly 8 times as much as sea transport, and river transport about 4 times as much (see [Trade Routes, Roads and Transport](#)). Real examples:

- **London c. 1300** (60,000–100,000 people) drew its grain mainly along the Thames, through river ports such as Henley, and along the Thames estuary. In bad years it

reached up the coast as far as the Humber. Firewood was worth carting overland only 18–20 km (11–12 mi). Charcoal, which is more valuable for its weight, came from 40+ km (25+ mi) away (Galloway 1999).

- **Constantinople, Cairo, Venice and Genoa** were all fed by ship or by the Nile. Venice and Genoa brought grain from Sicily and the Black Sea.

FANTASY TWIST

Magic that raises harvests changes everything downstream. If priests can bless fields from a 4 : 1 to an 8 : 1 yield, the same land feeds about twice as many people. You can then double densities, enlarge towns and shrink hinterlands. Cheap magical transport (portals, flying freight) would also free great cities from the “must be on water” rule. See [Fantasy Variants](#).

MAP TIP

Draw a faint pencil circle around each town at the overland radius from the table, and erase it when you are done (on the finished map, rings around a place belong to the town symbol). If two towns' circles overlap heavily, one of them should be smaller or should be on water. Put most places over 10,000 people on a navigable river or a coast. Inland exceptions existed, but they sat in very rich farmland with good roads: Coventry in England (about 12,000 c. 1300) and hilltop Siena in Italy (perhaps 50,000) had no navigable river. Places over 50,000 almost always need water transport.

Realm populations and areas

RULE OF THUMB

A medium European kingdom c. 1300 had 1–5 million people. Only France had more than 10 million people under one king (the Holy Roman Empire had more people, but it was loosely ruled). Europe as a whole had about 70–80 million.

Realm	c. 1000	c. 1300 (peak)	c. 1500	Area	Density c. 1300
England	1.5–2.0m (1.71m in 1086)	4.75m (1290); 4.81m (1348)	2.2–2.4m (low point 1.9m in 1450)	130,000 km ²	~36/km ² (94/sq mi)
Scotland	—	0.5–1m	—	~79,000 km ²	~6–13/km ²
France	5–6.5m	16–17m in the kingdom (1328); ~20m within modern borders	15–16m (modern borders)	551,000 km ² (modern)	~35–40/km ²
Iberia (all realms)	4.6–7m	6–9m	7.8–9m	583,000 km ²	~10–15/km ²
German lands / Holy Roman Empire	3–4m	11–14m	~16m in the HRE; Germany proper 9–13m	0.8–1.1m km ²	~10–17/km ²
Italy	~5m	10–12.5m	9–11m	301,000 km ²	~33–41/km ²
Denmark (medieval)	—	~0.95m (before 1348)	~0.6m	—	—
Poland	1.0–1.25m	~2m (1370)	Crown 3.4m; 7.5m with Lithuania	~240,000 km ² (1370)	~8.6/km ²
Hungary	1–1.5m	~2m (1200) to ~3m (14th century)	3m+ (1495)	280,000–325,000 km ²	~7–11/km ²
Byzantium	~12m (1025; some estimates up to 18m)	3–5m (1282), 2m (1312)	Gone (fell 1453)	1.675m km ² (1025)	~7/km ² (1025)
Egypt	—	perhaps 4–8m	—	Farmed strip only a few tens of thousands of km ²	very high on the farmed strip
Europe total	32–40m (Russell: 38.5m)	70–80m (Russell: 73.5m in 1340; some say up to 100m)	70–90m	—	~20/km ²

Notes on uncertainty:

- England's 1300 figure was once put anywhere from 3.7 to 6 million. The 4.75–4.81 million series (Broadberry, Campbell and van Leeuwen) is now standard.

- Europe totals for 1000 depend on what you count as Europe. Russell's 38.5 million includes Russia. Other series put it higher.

- Scotland's figure is little more than a guess: it may have grown from about half a million to a million by the mid-1300s.

LATER ERA (1500S+)

Europe reached about 100 million by 1600. England had about 4.1 million in 1600 and 5.3 million in 1650, finally passing its 1348 peak. See [What Changes After 1500](#).

- Byzantium's 1282 figure is often given as 5 million; some historians put it as low as 3 million.

MAP TIP

Measure your realm's area (count grid squares or hexes), then compare it with this table. A realm the size of England (130,000 km², 50,000 sq mi) should have 2–5 million people. A realm the size of Hungary should have 2–3 million. If your numbers are ten times bigger, either your land is magically fertile or you have made a mistake.

The biggest cities

RULE OF THUMB

In any period, only about 4–8 cities in Europe, North Africa and the Middle East had 100,000+ people. Nearly all of them were capitals or great ports, and almost all were on the sea or a big river. (Granada, fed by its rich irrigated plain, is a rare exception.) China was in a different league: its Song capitals Kaifeng and Hangzhou were each probably bigger than any city in this table (see [Beyond Europe](#)).

Ranges are given in thousands. Where scholars disagree a lot, the central figure from Bosker, Buringh & van Zanden (2013) is in brackets.

Rank	c. 1000	c. 1300	c. 1500
1	Constantinople 150–400 (300)	Cairo 220–600 (220)	Cairo 150–400 (180)
2	Baghdad 125–500 (300), declining	Paris 75–270 (250); ~200 is the most common figure	Istanbul 100–280 (280); a 1478 count suggests ~80–100
3	Cairo-Fustat 135–200	Granada 50–150 (150)	Paris 95–250 (200)
4	Córdoba 100–450 (100)	Venice 90–120	Naples 100–150 (125)
5	Seville 70–90	Milan 100–150	Edirne ~100–127
6	Tinnis and Damietta ~80 each	Genoa 85–100	Venice 100–150
7	Palermo 60–75	Constantinople ~100	Milan ~100
8	Alexandria ~60	Florence 95–120	Tunis ~65–100
9	Kairouan, Salerno ~50	Baghdad ~95 (after the 1258 sack)	Aleppo 65–80
10	Venice 17–45	Seville ~90	Granada ~70; Valencia ~75
11	Antioch ~45; Thessaloniki, Regensburg, Damascus ~40	London 60–100	Lisbon, Palermo, Fez, Damascus 55–65
12	Rome 30–35; Kiev 20–45	Fez, Marrakesh, Tunis, Naples, Aleppo, Córdoba 55–70; Ghent 40–65	Genoa, Rome, Florence, Bursa ~55–60; London ~50

How uncertain are these? Very.

- An older standard database (Bairoch, Batou & Chèvre 1988) gave Córdoba 450,000 people c. 1000 and Palermo 350,000. Later scholars cut these to about 110,000 (Glick) and 60,000 (Epstein and Johns).
- For many Middle Eastern cities there are no counts at all. Historians estimate the population from the walled

area (about 150 people per hectare), or from the number of mosques or public bathhouses (about 1,000 people each) (Bosker et al. 2013, data appendix).

Treat every number above 100,000 as a rough guess.

Things to notice:

- Around 1000, the biggest cities were in the Byzantine and Islamic worlds. Before about 1000, the Middle East's total city population was larger than Europe's.

LATER ERA (1500s+)

c. 1600: Istanbul 400–700k; Paris ~220–300k; Naples ~250–280k; London ~200k (roughly 300–400k by the 1650s–60s; estimates vary); Venice, Seville, Lisbon and Milan 110–150k.

- By 1300, Italian and Flemish trading cities had caught up. The great cities of the Middle East held steady or shrank.
- Cities could fall fast: Baghdad after the Mongol sack of 1258; Constantinople after 1204, and again by 1453.

MAP TIP

On a continent-sized map of a “medieval Europe” world, mark only 3–8 great cities of 50,000+. Put almost all of them on coasts, at river mouths or on big navigable rivers. Give at least one of them a clear reason: an emperor's court, a strait, a delta or an export trade.

Beyond Europe: Asia, Africa and the Americas

All the tiers and densities above come from Europe, North Africa and the Middle East. Other ways of farming gave very different numbers. Use this section if your realm grows rice, farms by clearing forest, feeds a city from lake gardens, or lives from cattle and long-distance trade.

RULE OF THUMB

What people farm sets the numbers. Wet rice and lake gardens can feed a much denser countryside and much bigger cities than European grain fields. Shifting forest farming feeds far fewer people. From the 11th to the 13th century, the biggest cities in the world were probably in China, not in Europe.

Why:

- **Wet rice** grows in flooded fields (paddies). A paddy can be farmed every year with no fallow year, so the same land carries more people. A rice empire with good canals can also move grain cheaply to its capital.
- **Shifting (swidden) cultivation** means clearing and burning a patch of forest, cropping it for one to three years, then leaving it to grow back for many years. At any time, most of the land is regrowing forest, so few people can live on it.
- **Lake gardens, reservoirs and canals** (Mexico, Angkor) let a city grow much of its food at its own edge or inside its own zone. In Mexico, canoes on the lakes did the job that rivers and the sea did for Europe's great cities.

Density beyond Europe

Place and date	People	Area	Density	Notes
Song China, c. 1100	90 to over 120 million (estimates vary); the 1103 census listed about 20.5 million households	About 3 million km ² (1.2 million sq mi)	~30–40 per km ² (80–105 per sq mi) on average (derived)	Like England or France c. 1300 on average, but packed in the rice lands of the south
Liangzhe circuit (Yangzi delta and modern Zhejiang), 1102	~10 million (households × 5)	Roughly 120,000–130,000 km ²	~80 per km ² (~210 per sq mi) (derived)	Denser than Europe's most crowded large regions (Flanders and Holland 50–75), even with the Zhejiang hills included. Only single districts in Europe went higher, e.g. Norfolk, about 90 in 1290 on about 5,400 km ² (derived)
Valley of Mexico, 1519	1–1.2 million	9,600 km ² (3,700 sq mi), including about 1,500 km ² of lakes	~100–125 per km ² (260–325 per sq mi) (derived)	Lake-bed gardens (chinampas) and irrigated fields

Place and date	People	Area	Density	Notes
Greater Angkor, 13th century	700,000–900,000	About 3,000 km ² (1,160 sq mi)	~230–300 per km ² (600–780 per sq mi) averaged over the zone (derived)	A whole city-and-rice-field zone, not a walled town
Shifting cultivation in tropical forest (world in the 1950s; modern Southeast Asia)	About 200 million people worldwide in the 1950s (very rough FAO estimate)	About 36 million km ² (14 million sq mi) worldwide	About 6 per km ² (14 per sq mi) as a 1950s world average (FAO 1957); about 12–35 per km ² (30–90 per sq mi) in modern examples	12 in northern Laos, 35 in northern Thailand (Warner 1991, FAO). Ruthenberg (1980, cited by FAO) calculates a ceiling of about 55–56 per km ² (about 145 per sq mi).

How Song China grew. Registered households roughly tripled between 980 and 1101, from 6.2 to 17.5 million (Deng 2013). In 1012 the emperor had 30,000 bushels of seed of quick-ripening Champa rice (first brought from Champa, in today's central Vietnam) sent from Fujian to farmers in

the lower Yangzi and Huai valleys (Ho 1956). Historians still argue about how much this new rice mattered. Deng (2013) points out that growing two rice crops a year on the same field was rare under the Song and became common only under the Ming (1368–1644).

Cities beyond Europe

City	Date	People	What it was like
Kaifeng (Northern Song capital)	c. 1100	600,000 to over 1 million (estimates vary)	Three rings of walls; a trade and industry centre where four major canals met
Hangzhou (Southern Song capital)	1270s	A count of 1270 listed about 186,000 families; Gernet estimated well over 1 million by 1276	Probably the largest city in the world at the time
Greater Angkor (Khmer capital, Cambodia)	13th century	700,000–900,000 in the whole zone	A walled temple-city core (Angkor Thom, 3 × 3 km) inside a huge spread of house mounds, ponds, embankments and rice fields (Klassen et al. 2021)
Tenochtitlan (Aztec capital, Mexico)	1519	200,000–250,000 (estimates run from 200,000 to 400,000)	An island city of 8–13.5 km ² in a shallow lake, joined to the shore by three causeways and fed partly by chinampas in Lakes Xochimilco and Chalco
Great Zimbabwe (southern Africa)	13th–15th centuries	Under 10,000 (older estimates 18,000–20,000)	Capital of a state rich in cattle and gold trade; stone-walled royal enclosures. Chirikure et al. (2017) argue that it never passed 10,000 people.
Kilwa Kisiwani (East African coast)	14th century	At least 10,000, perhaps 20,000	Island port that controlled the coast's gold trade; a great mosque and palace built of coral stone

Most cities in the Americas were far smaller than Tenochtitlan. Of 87 Mesoamerican cities of the last centuries before the Spanish conquest with good survey data, the median covered only 90 ha (Smith 2005): less than 1 km², the area of a large European town.

What this means for your map:

- A **rice empire** can have a whole-realm average like France's (30–40 per km²), with deltas and river basins at 80 per km² or more, and one or two capitals of several hundred thousand people or more. The “only 4–8 cities over 100,000” rule applies to Europe and the Middle East, not to Song China.

- A **lake or canal capital** can be packed (Tenochtitlan) or spread out over a huge farmed zone (Angkor). Decide which before you draw it.
- A **tropical forest with shifting farming** has small villages, about 6 people per km² on average (12–35 in busier districts, never much above 55), and real towns only on rivers and coasts.
- A **trading or herding state** (Great Zimbabwe, Kilwa) can be rich and powerful with a capital of only 10,000–20,000 people.

LATER ERA (1500S+)

After falling during the Mongol conquest, China's population grew strongly under the Ming: estimates for c. 1600 run from about 160 million to 200 million. In Mexico the opposite happened. After the Spanish conquest (1519–21), epidemics and war cut the native population by at least half in the 16th century, and perhaps by as much as nine-tenths over wide areas (McCaa 2000; estimates vary widely). Tenochtitlan was rebuilt as Mexico City. See [What Changes After 1500](#).

MAP TIP

For a rice empire, shade the deltas and river basins as your densest zone, link the capital to them by canals or rivers, and draw many small villages on dikes and mounds among the paddies (see [Villages, Farms and the Countryside](#)). For an Angkor-type capital, draw a walled square core with large rectangular reservoirs, then a wide zone of tiny house-mound dots, small ponds and rice fields with no outer wall. (Avoid stipple here: the master legend keeps it for commons and heath.) For a Tenochtitlan-type capital, draw a dense island city, causeways to the shore and a band of narrow garden plots in the shallow lake. For shifting farmers, scatter small villages far apart with clearings of different ages around them. Terrain and biome patterns are in [Where Settlements Are Built](#), and canal and causeway transport in [Trade Routes, Roads and Transport](#).

Step-by-step population calculator

Use this to get believable numbers for any realm.

Step 1: Measure the area. Count grid squares or hexes and multiply by the area of one square or hex. A hex's area is about $0.866 \times (\text{width across the flat sides})^2$. For example, a 30 km hex is about 780 km²; a 30 mi hex is about 780 sq mi.

Step 2: Split the land by quality and apply a density (from the [Population density](#) tables). Typical values: rich lowland 30–40 per km², average land 15–25, hill country 8–15, high

uplands, mountains, forest and marsh 2–5, desert 0. For rice basins and tropical forest farming, use the figures in [Beyond Europe](#).

Step 3: Add up to get the total population.

Step 4: Choose an urban profile and split the people into tiers.

Share of people	Frontier (Poland, Hungary, Scotland, Scandinavia)	Average (England, France, Germany c. 1300)	Highly urban (Flanders, Lombardy, Tuscany)
Cities 10,000+	~1%	~3%	~15%
Towns 2,000–10,000	~3%	~5%	~12%
Small market towns 500–2,000	~4%	~6%	~8%
Villages, hamlets, farms	~92%	~86%	~65%

Step 5: Turn shares into places. Divide by a typical size: about 3,000–4,000 for towns and about 800 for market towns. For villages, use about 250 in lowland and 100–150 in hill country. Size the top cities using the rank-size section above. The capital of a strong kingdom usually holds about 1–2% of all its people (London c. 1300 $\approx$ 1.3–2%; Paris $\approx$ 1.25%). In a fragmented realm, the largest city holds about 0.5% or less (Cologne, Germany's largest city c. 1300, held about 0.4–0.5%).

Step 6: Check spacing and food.

- Villages should end up 1.5–4 km (1–2.5 mi) apart in lowland.
- Market centres should be about 9–16 km (6–10 mi) apart in a crowded realm, and up to about 25 km (15 mi) apart in a thinly settled one.
- Most places over 10,000 should be on navigable water or a coast. An inland one needs very rich farmland around it.

If the spacing is wrong, adjust the densities or the shares.

Quick ratios per 1 million people (from the profiles above):

Per 1 million people	Frontier	Average	Highly urban
Cities 10,000+	0–1 (largest often 8–15k)	1–2 (largest 15–30k)	5–10
Towns 2,000–10,000	8–12	12–18	30–40
Small market towns	40–60	60–90	80–100
Villages	3,000–4,500, plus many hamlets and farms	2,500–3,500	2,000–2,500

Real check: Denmark before 1348 had about 950,000 people and about 80 mostly small towns, holding ~10% of the people. That fits the frontier profile.

Worked mini-example

A kingdom of **60,000 km² (about 23,000 sq mi)**: a fertile river plain, hilly country and a mountain border.

Land	Area	Density	People
Fertile lowland (40%)	24,000 km ²	35 per km ²	840,000
Hills and mixed land (35%)	21,000 km ²	15 per km ²	315,000
Mountains, forest, marsh (25%)	15,000 km ²	4 per km ²	60,000
Total	60,000 km²	~20 per km² (52 per sq mi)	~1.2 million

Using the **average** profile:

Tier	Share	People	Places
Cities 10,000+	3%	~36,000	Capital of ~22,000 (1.8% of the realm) on the main river; a second city of ~11,000 at the river mouth or on the coast
Towns 2,000–10,000	5%	~60,000	~16 towns: 2 of 6,000–7,000, 4 of about 4,000, 10 of 2,500–3,000
Small market towns	6%	~72,000	~90 market towns of about 800, plus 50–100 large villages with a weekly market
Rural	86%	~1,030,000	~3,000–3,500 villages of about 250 (mostly in the plain), plus thousands of hamlets and farmsteads (mostly in the hills)

Checks:

- **Villages:** about 2,900 villages in the 24,000 km² plain means one per ~8 km² (3 sq mi), or about 3 km (2 mi) apart. ✓
- **Market centres:** about 110 market centres on 60,000 km² means one per ~550 km² (210 sq mi). That is about 25 km (15 mi) apart on average, closer in the plain. This is wider than in England, because this realm is less crowded. ✓

- **Capital:** a city of 22,000 would need an overland hinterland 30–40 km (19–25 mi) in radius, so it goes on the navigable river. ✓
- **Big places:** 4 places of 5,000+ in 60,000 km² means they are about 130 km (80 mi) apart. That is like medieval Britain. ✓

MAP TIP

You do not have to draw all 3,000 villages. On a kingdom map, draw the capital, the city, all 16 towns and the 30–40 most important market towns. Then add villages only as texture in the plain, so the viewer *feels* the crowding. Save full village detail for local maps.

Change over time: growth, famine and plague

RULE OF THUMB

The date you choose changes the map. 1300 is the crowded peak. 1350–1450 has a third to a half fewer people, with deserted villages and half-empty towns. 1500 is in the middle of recovery.

Date	Europe	England	What your map shows
c. 1000	32–40m	~1.5–2m	Few towns, much forest and waste, castles spreading
c. 1200	growing fast	~3–4m	New towns founded, markets granted, forests cleared
c. 1300–1348	70–80m (peak)	4.75–4.81m	Every good field farmed; villages climbing into marginal hills; towns at their largest
c. 1350–1450	~50–60m	2.5m (1377), 1.9m (1450)	Deserted and shrunken villages; marginal land returned to pasture and woodland; towns with empty plots inside the walls
c. 1500	70–90m	2.2–2.4m	Recovery under way in the south, slower in the north; sheep pastures where villages once stood

The three big events:

1. **Growth, 1000–1300:** Europe's population roughly doubled to tripled. England grew about 2.7 times between 1086 and 1315. In England, 57% of all markets that existed by 1516 were created in the 13th century.
2. **Great Famine, 1315–22:** Northern Europe lost about 10% of its people (range 5–15%). England fell 12%, from 4.69 million in 1315 to 4.12 million by 1325. Cattle plague then killed up to 80% of livestock in places. The Mediterranean was largely spared. Some regions, such as parts of Tuscany, were already shrinking before 1348.
3. **Black Death, 1347–51, and later plagues:**
 - Europe lost 30–60% of its people.
 - England fell from 4.81 million in 1348 to 2.6 million in 1351.
 - Florence fell from about 100,000–120,000 c. 1300 to 37,000 in the 1427 tax survey.
 - Denmark fell from about 950,000 to 550,000.
 - Egypt lost roughly half its people once the later outbreaks are counted (Borsch 2005).
 - Plague kept returning (1361–62, 1369, 1374–75 and later), so recovery took until c. 1500–1550 in much of Europe. England only regained its 1348 peak c. 1600–1630.

Effects you can draw:

- **Deserted villages:**
 - England: Beresford and Hurst's 1971 survey listed well over 2,000 deserted medieval villages, and later estimates exceed 3,000.
 - Most were abandoned or shrank between c. 1350 and 1520, especially in the 15th century.

- Plague alone rarely emptied a village. More often, the survivors drifted away and lords turned the fields into sheep pasture.
- Germany: the number of settlements fell from about 170,000 to 130,000 between 1300 and 1500 (–23.5%). In one district of northern Thuringia, only 33 of 179 earlier settlements survived to the mid-16th century.

- **Shrunken towns:** Walls built for the 1300 population enclosed gardens, orchards and ruins after 1350. Florence did not regain its 1300 population until the 19th century, so for centuries its 1333 walls held gardens and fields as well as houses.

- **Abandoned marginal land:** Fields high on hillsides, in heath or at forest edges went back to rough pasture or woods. You can still see their old ridges.

FANTASY TWIST

A magical plague, a dragon's long rampage or an undead war works on a map exactly like the Black Death. Expect empty villages, towns too small for their walls, and good land going back to wild. A ruin-filled landscape needs a disaster in its recent past. See [Fantasy Variants](#).

LATER ERA (1500S+)

Population grew again after c. 1500, and big towns grew fastest. Italy suffered new plagues in 1629–31 and 1656–57, each killing 20% or more of the people in the regions they hit. The Thirty Years' War (1618–48) devastated parts of Germany. See [What Changes After 1500](#).

MAP TIP

For a post-plague setting, mark a few deserted villages in the lowland: a grey church alone in a field, labelled “(lost village)”. Sometimes a manor house (small open square) survives beside it in sheep pasture. Show towns with open green space inside their walls, and push the hill-farm line downhill. For a 1300 setting, do the opposite: farms right up the valley sides and suburbs spilling outside the town gates.

Who lives where

RULE OF THUMB

Out of every 100 people, about 85 farm, 10–15 live in towns, 1–2 are clergy, about 1 is noble in Western Europe (many more in Poland and Hungary), and fewer than 1 is a full-time soldier.

Group	Share of people	Where they are on the map
Peasants and rural workers	80–90% (65% in Flanders or Tuscany)	Villages, hamlets, farmsteads; also rural craftsmen such as smiths, millers and carpenters
Townspeople	5–20% (England 15–20% counting all small towns; Flanders 33–40%)	Market towns, towns, cities
Clergy (priests, monks, nuns, friars)	~1–2% (estimates vary). In England in the 1530s, monks, canons, friars and nuns alone came to about 1 adult man in 50, before counting parish priests.	Every parish has a priest. Monks and nuns live in monasteries, often in the countryside; friars live in towns; cathedral clergy live in cities. England had nearly 900 religious houses with about 12,000 monks, canons, friars and nuns in the 1530s.
Nobility and gentry	About 1% in France and Italy; in England titled nobles were far fewer, but the gentry (untitled landowners) added more. ~4–5% in Hungary and ~8–10% in Poland (early modern figures). Perhaps ~10% in Castile, concentrated in the north (rough estimate).	Manor houses and castles across the countryside; town houses in capitals. Where nobles are 10% of the population, many are poor and live like farmers.
Full-time soldiers	Well under 1% in peacetime	Castle garrisons (5–20 men in peace; see Castles, Forts and Military Outposts), royal guards, mercenaries. Even 500–600 garrisoned castles (England's number after the Conquest) with 5–20 men each add up to only about 2,500–12,000 men, well under 1% of the people (estimate). Big wartime armies existed for one season only. Edward I's English army at Falkirk in 1298 had about 15,000 men (some older estimates are higher).

Towns needed newcomers. Medieval towns had more deaths than births. Death rates could be as high as 6% a year even in normal years. Towns grew only by drawing people in from the countryside, and a quarter to a half of a

growing town's people were migrants from nearby villages (Nicholas 2014, via Jedwab et al. 2020). This is why towns sit inside a ring of villages that feed them both food and people.

MAP TIP

Put a manor house (small open square) in most lowland villages (some English villages had two or more). Place castles much more sparsely. Religious houses were common but mostly small: England had nearly 900 in the 1530s, roughly one per 145 km² (56 sq mi). Friaries sit in towns, while

abbeys (a church with a square cloister beside it) often sit in remote valleys (see [Religious, Cultural, Legal and Ancient Sites](#)). Specialists are rare: a university, a great cathedral school or a famous library belongs to only one or two cities in a kingdom.

Gaming rules of thumb vs history

Many worldbuilders use S. John Ross's free article *Medieval Demographics Made Easy* (1993–2018) or the size table in the *Dungeons & Dragons 3.5 Dungeon Master's Guide* (2003). Both are useful starting points. Ross says plainly that his num-

bers are simplified and not meant to recreate any real country. Here is how they compare with the historical data in this chapter.

Topic	Gaming rule	History	Verdict
Density of a settled realm	Ross: 30–120 per sq mi (12–46 per km ²)	7–40 per km ² for whole realms; Europe c. 1300 ~20 per km ² (52 per sq mi); Flanders up to 75 per km ²	Good for crowded western kingdoms. Too high for frontier, upland or young realms, which can be 7–15 per km ² (18–40 per sq mi).
Village size	Ross: 20–1,000, typically 50–300. D&D: thorp 20–80, hamlet 81–400, village 401–900	Hamlet 10–75; village 75–500, typically 150–300	Ross matches well. D&D's "villages" are big villages or small market towns.
Town and city size	Ross: town 1,000–8,000; city 8,000–12,000; big city 12,000–100,000. D&D: metro-polis 25,001+	Town 2,000–10,000; city 10,000+; 50,000+ was very rare	Broadly right. Remember that 25,000 was a first-rank city almost everywhere in Europe.
Largest city	Ross: $\sqrt{(\text{population})} \times 15$ on average	Strong kingdoms: 1–2% of realm population	Too small for centralised capitals: it gives England c. 1300 about 33,000 (real London 60–100k), France about 60,000 (real Paris ~200,000) and Byzantium 1025 about 52,000 (real Constantinople ~300,000). Fair for fragmented realms.
Second city	Ross: 20–80% of the largest	England 1377 ~30%; France c. 1300 ~15–20%; Italy ~90–100%	Real realms fall outside both ends of the range. Choose by the type of realm.
Towns per city	Ross: number of cities $\times 2d8$ (average 9)	England c. 1300: ~50–60 towns of 2k+ to ~6 cities of 10k+, about 9–10 to 1	Very close.
Farmland	Ross: 180 people per sq mi of cultivated land (≈ 1.4 ha each, pasture and roads included)	~1 ha arable per person; 2–3 ha of all land	Consistent. Ross's "cultivated land" includes pasture.
Urban density	Ross: ~150 per hectare	100–200 per ha; 150 is a standard assumption	Agrees.
Castles	Ross: 1 active castle per 50,000 people	England had 500–600 occupied at once after 1066, when it had only about 1.7–3 million people; the German lands had ~14,000 castles in all	Far too few for most of medieval Europe. See Castles, Forts and Military Outposts .
Universities	Ross: 1 per 27.3 million people	~20 in Europe c. 1300 (≈ 1 per 3.5–4 million); 60–80 by 1500	Far too few. Use about 1 per 3–4 million for 1300 and 1 per ~1 million for 1500.
Clergy	Ross: 1 clergyman per 40 people (2.5%)	~1–2% (estimates vary)	Somewhat high but close.

In short:

- Ross's village sizes, towns-per-city ratio, farmland and town density all match history well.
- Ross underestimates big capitals, castles and universities.

- Ross's minimum density is too high for thinly settled lands.
- D&D's size names put the labels "village" and "town" on larger places than historians would.

MAP TIP

Use a gaming formula as a first draft. Then run the checks from this chapter: Are most 10,000+ cities on water, and does each inland one have rich farmland around it? Does the capital fit the type of realm? Are villages 1.5–4 km (1–2.5 mi)

apart in lowland? Is the realm's total within the range of a real kingdom of the same size? For more pitfalls, see [Common Mistakes and How to Fix Them](#).

Quick summary

- Population means people, not households. A household is about 4.5–5 people.
- Tiers: farmstead 5–15; hamlet 10–75; village 75–500 (typically 150–300); small market town 500–2,000; town 2,000–10,000; city 10,000–50,000; great city 50,000+.
- 85–95% of people are rural. A typical kingdom c. 1300 had about 3% of its people in cities of 10,000+. Northern Italy had about 18% in cities of 10,000+, and Flanders had 33–40% in towns of all sizes.
- For every city of 10,000+, expect about 10 towns, 50–100 market towns and a few thousand villages.
- Good lowland supports 30–50 people per km² (80–130 per sq mi). Whole realms average 7–40 per km² (10–40 in settled western kingdoms). Europe c. 1300 averaged about 20 per km² (52 per sq mi).
- Each person needs about 1 ha (2.5 acres) of ploughland, or 2–3 ha (5–7 acres) of all land.
- A town of 10,000 needs a farmland radius of 21–27 km (13–17 mi) overland on good land. Most cities above 10,000, and almost all above 50,000, are fed by river or sea.
- Strong kingdoms have one primate capital (1–2% of the realm's people, 3–7 times bigger than the second city; 5–10 times in some realms after 1500). Fragmented regions have several rival cities of similar size.
- Only about 8 cities in Europe, North Africa and the Middle East had 100,000+ people c. 1300 or c. 1500.
- Outside Europe the numbers change with the way of farming: Song China averaged ~30–40 per km² with capitals of perhaps a million; Angkor spread 700,000–900,000 people over ~3,000 km²; Tenochtitlan packed 200,000–250,000 onto an island of 8–13.5 km²; shifting forest farming supports only about 6 per km² on average (12–35 in busier districts, at most about 55).
- 1300 is the crowded peak. After the Black Death (1347–51), 30–60% of people are gone, villages are deserted and towns are half-empty for 100–150 years.
- Gaming formulas are a good first draft, but they shrink great capitals and make castles and universities far too rare.

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CHAPTER 3

CAPITALS, REALMS AND BORDERS

A map is not only towns and roads. It is also a pattern of power: who rules where, from which city, and where one ruler's land stops and another's begins. This chapter explains where capitals were built and why, what you find in and around them, how big kingdoms, duchies and counties were, and what borders really looked like in the Middle Ages, including why language lines rarely match them. It ends with practical advice for drawing realms, capitals and borders on your map.

The baseline period is the High and Late Middle Ages in Europe (c. 1000–1500). Castles and their garrisons are covered in [Castles, Forts and Military Outposts](#). City sizes and the “one giant capital” pattern are in [Population and Settlement Sizes](#).

IN THIS CHAPTER

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What makes a capital

RULE OF THUMB

A capital sits in the richest, most loyal region of the realm, on navigable water, at a place that is easy to defend. Centrality matters less than you think.

The five things a capital needs

Need	Why	Real examples
A rich core region	The court, officials and garrison must be fed. The ruler's own estates (the <i>royal domain</i>) supply food and money.	Paris sits in the grain lands of the Île-de-France. London sits in the Thames valley. Prague sits in the middle of the fertile Bohemian basin.
River or sea access	A capital grows into a big city. Big cities need grain by water (see Population and Settlement Sizes).	Paris (Seine), London (Thames), Prague (Vltava), Kraków (Vistula), Buda (Danube), Cairo (Nile), Constantinople (Bosporus).
A defensible core site	The ruler's family, treasure and archives must be safe.	Kraków's Wawel hill, Prague Castle on its ridge, Buda's Castle Hill, Paris's island (Île de la Cité), Toledo in a deep loop of the Tagus river.
Loyalty	The capital is in the dynasty's home region, where the ruler owns land and the local nobles are relatives or clients.	The Capetian kings of France ruled from the Paris–Orléans region, their old family land.
Legitimacy	People must accept the ruler. Old sacred places, saints' tombs and old royal sites give that acceptance.	Westminster Abbey (coronations since 1066); Reims (French coronations); Aachen (Charlemagne's church).

Centre, fortress or frontier?

A ruler choosing a capital balances three pulls:

Type	Where	Strength	Weakness	Examples
Central capital	Near the middle of the realm	Fair to all regions; short messenger times	Often not on the best river; no natural defences	Madrid, where Philip II settled his court in 1561. Its position near the centre of the Iberian peninsula is often given as one reason, though historians debate the others.
Core-region capital	In the dynasty's richest home region, which may be off-centre	Rich, loyal, easy to feed	Far from outer provinces	Paris, London, Kraków
Forward capital	Close to a frontier the ruler wants to push or defend	The ruler is near the army and the conquests	Exposed to attack	Ottoman capitals: Bursa (taken 1326), then Edirne (1360s), then Constantinople (1453), each closer to the conquests in Europe

“Forward capital” is a modern geographers’ term, but the pattern is old. The Castilian kings moved their main residences south as they conquered: Toledo (taken in 1085), then Seville (taken in 1248). The Babenberg rulers of Austria

moved their seat east to Vienna in 1145, closer to the frontier with Hungary. (Austria became a duchy in 1156.)

RULE OF THUMB

Expanding realms move their capital toward the frontier. Defensive or settled realms keep it in the old core.

Two capitals in one: the palace and the merchant city

Many capitals were really two towns side by side: a palace quarter and a merchant city.

- **London and Westminster:** London was the merchant city. Westminster, about 3 km (2 mi) upstream, had the royal palace, the abbey and the law courts.
- **Prague, Kraków, Buda:** a royal castle on a hill (Hradčany, Wawel, Castle Hill) above or across the river from the

merchant town (Prague’s Old Town, Kraków’s market town, Pest).

- **Islamic capitals:** rulers often built a new palace city beside the old one. The Fatimids founded al-Qahira (Cairo) in 969 next to the older city of Fustat. The caliph Abd al-Rahman III of Córdoba founded the palace city of Madinat al-Zahra in 936 (main building work perhaps from 940), about 5 km (3 mi) west of Córdoba.

MAP TIP

On a local or regional map, draw the capital as two linked parts: a fortified palace on the high ground and a larger merchant town by the river, often with a bridge between them. On a kingdom map, one capital symbol (a filled star) is enough.

Capital is not always the biggest city

The capital is usually the biggest city in a strong, centralised kingdom (London, Paris). In other realms it is not:

- The Holy Roman Empire (HRE) had no capital at all for most of the Middle Ages.

- The Valois dukes of Burgundy (1363–1477) moved their court between Dijon, Brussels, Lille and Bruges. Dijon, the old ducal capital, was much smaller than the great Flemish cloth cities of Ghent and Bruges.
- In Castile, the court moved between several cities, and the biggest city, Seville, was only one of its stops.

MAP TIP

In a strong, centralised kingdom, make the capital the biggest city on the map. In a loose realm (an elective empire, a group of duchies under one ruler), put the court in a middling town and let a trading port or cloth city be bigger. That contrast tells the viewer how the realm works.

FANTASY TWIST

A capital can also be chosen for magic: a ley-line crossing, a dragon-proof mountain, the only place a sacred tree grows. Keep the other four needs, though. Even a magical capital needs food, water transport and loyal land around it, unless you explain how magic replaces them. See [Fantasy Variants](#).

Travelling courts and fixed capitals

RULE OF THUMB

Before about 1200, most western kings had no fixed capital. The court travelled. Fixed capitals appear when government needs paper, courts and coin more than it needs to eat the king's food on the spot.

Why courts travelled

- **The court went to the food.** Taxes and rents were often paid in food. Carting food was expensive (see [Trade Routes, Roads and Transport](#)), so the king and several hundred followers moved from one royal estate to the next and ate what was stored there.

- **Presence was power.** A king who was seen, held court, judged disputes and received oaths in each region kept the loyalty of the nobles there.
- **Elections, not inheritance (in Germany).** HRE kings were elected. They came from different regions, so no single city became the permanent base.

How it worked

Feature	Numbers	Source or example
Spacing of German royal palaces (<i>Pfalzen</i>) and royal manors	About 30 km (19 mi), one day's travel for the royal train	Wikipedia <i>Kaiserpfalz</i>
Spacing of royal farms on routes	24–31 km (15–19 mi)	Wikipedia <i>Itinerant court</i>
Length of stay	Rarely more than a few weeks; a few days at transit stops	German royal itineraries
Late example	Emperor Charles V (reigned 1519–56) made 40 journeys and spent about a quarter of his reign on the road	Wikipedia <i>Itinerant court</i>

Famous Pfalzen include Aachen (Charlemagne's favourite, 790s), Ingelheim (c. 790), Paderborn and Goslar (built

1030–1055 near the silver mines and hunting forests of the Harz).

When and why capitals became fixed

Realm	Fixed capital	Date	What made it fixed
Byzantium	Constantinople	From 330	Salaried officials, coin taxes, huge walls. Never itinerant in the western sense.
Islamic caliphates	Baghdad (founded 762), Córdoba, Cairo (969)	8th–10th centuries	Same: paid bureaucracy and coin taxes. New capitals were often planned from scratch.
France	Paris	From the 1190s, firmly by the mid-1300s	In July 1194 Philip II lost his travelling archives, seal and treasure to Richard the Lionheart at the battle of Fréteval. After that, the king kept his records in a fixed archive in Paris (the <i>Trésor des Chartes</i>) instead of carrying them around. The Parlement (high court, 1302) and the Chambre des comptes (audit office, c. 1303) sat in the royal palace on the Île de la Cité.
England	Westminster / London	From the 12th century, firmly by the mid-1300s	The Exchequer (treasury office) was based at Westminster by the reign of Henry II (1154–89). Magna Carta (1215), clause 17, said common pleas (ordinary lawsuits) must be held "in some fixed place". In practice that place became Westminster Hall. The Chancery (writing office) followed.
Poland	Kraków	Seat of government from 1038; coronations from 1320	Royal castle and cathedral on Wawel hill.
Bohemia	Prague	Always the main seat; imperial capital under Charles IV (1346–78)	Archbishopric 1344, New Town and university 1348, Charles Bridge begun 1357.

Realm	Fixed capital	Date	What made it fixed
Hungary	Buda (with moves)	Buda until 1315; Temesvár 1315–23; Visegrád 1323–1408; Buda again from 1408	The seat followed the king's power base and his building projects.
Castile / Spain	Madrid	1561 (Valladolid briefly 1601–06)	Before 1561 the court moved between Toledo, Valladolid, Burgos, Seville and others.
Holy Roman Empire	None	—	Remained itinerant. Dynasties used their own home cities (Prague, Vienna).

The forces that fixed capitals were:

1. **Paperwork.** Archives, account rolls and registers became too bulky and too valuable to move.
2. **Law courts.** People with lawsuits needed a court they could find.
3. **Money taxes.** When taxes came in coin, food could be bought anywhere, so the court no longer had to travel to eat.
4. **A bigger staff.** Clerks, judges and accountants needed offices and houses.

LATER ERA (1500S+)

Fixed capitals win everywhere. Philip II fixed Madrid in 1561 and built the Escorial palace-monastery nearby (1563–84). Sigismund III moved the Polish court from Kraków to Warsaw in 1596, closer to Lithuania and the Baltic. German princes settled into fixed *Residenzstädte* (residence towns). See [What Changes After 1500](#).

MAP TIP

If your kingdom is in an “early medieval” stage, do not draw one capital. Draw 5–15 royal residences (small crowns) scattered across the royal lands, about a day's travel apart along the main routes, and mark one or two as favourites. If your kingdom is “late medieval”, draw one capital (a filled star) and a ring of royal residences around it (see the next section).

FANTASY TWIST

Fast flying mounts, teleport circles or magical messages reduce the need to travel. A realm with them can have a fixed capital much earlier and govern a bigger area. A realm of nomadic or migrating people (or a king who must personally renew a magical oath in every province) stays itinerant.

What is in and around a capital

RULE OF THUMB

A capital is not one building. It is a cluster: palace, great church, treasury, writing office, law courts, mint, and later a university, surrounded within a day's ride by royal residences and hunting forests.

Inside the capital

Institution	What it is	Real examples
Palace or royal castle	Where the ruler lives and holds court; often on high ground or an island	Palais de la Cité and the Louvre (Paris); Westminster Palace; Prague Castle; Wawel (Kraków); Buda Castle (first royal residence built 1247–65)
Cathedral or great abbey	The ruler's main church, often a bishop's or arch-bishop's seat	Notre-Dame (Paris); St Vitus (Prague); Wawel Cathedral; Hagia Sophia (Constantinople)
Treasury	Stores coin, plate and crown jewels	The Paris house of the Knights Templar (the Temple) managed the French royal money from the 12th century until Philip IV moved it to the Louvre in 1307. Charles IV kept the imperial regalia (crown, sceptre and other symbols of rule) in Karlštejn castle, founded in 1348 about 30 km (19 mi) from Prague.
Chancery and archives	Writes and stores charters, letters and laws	Trésor des Chartes in Paris (set up after 1194)
Law courts	The high court of the realm	Westminster Hall; the Parlement in the Palais de la Cité
Mint	Makes the royal coinage	The Tower of London mint; the Paris mint

Institution	What it is	Real examples
University (later)	Trains lawyers, clerks and priests for government and church	Many capitals founded one: Prague 1348, Kraków 1364, Vienna 1365
Royal chapel or relic church	Holds relics that give the king holy status	Sainte-Chapelle, Paris (1248), built for the Crown of Thorns

How many mints?

RULE OF THUMB

Early medieval kingdoms struck coins in many towns. Later, strong kings closed most mints and kept a few they could control. In loose realms, the opposite happened: more and more lords got the right to mint.

A mint was often a small workshop of a few moneyers (coin makers), not a big building. England is the best-recorded case:

England	Number of mints	Notes
Æthelred II (978–1016)	About 70 (one standard catalogue lists 73)	Mostly in boroughs (fortified towns, <i>burhs</i>), from London and York down to small places such as Lydford in Devon. A single coin type was struck at 60 or more places.
Henry II, 1154–58	Nearly 50	—
Henry II's recoinage of 1158	30 during the recoinage, then only 12 kept open	A <i>recoinage</i> is when old coins are called in and struck again as new ones.
Henry II's death, 1189	9	—
Henry III (1216–72), between recoinages	4: London, Canterbury, Bury St Edmunds and Durham	The capital plus three old church centres. For a big recoinage (1247), many old mints were briefly reopened.

The Holy Roman Empire went the other way. From the 10th century, kings granted the right to mint to bishops, from the 11th century to lay princes, and later to towns. By the 12th century the empire was split into many regional coin areas, each with its own penny (*Pfennig*).

- **Early medieval or newly united realm:** a mint in most fortified towns.

- **Centralised kingdom of the High or Late Middle Ages:** a mint in the capital, one to three church or regional centres, and one at each big silver mine (see [Industry, Resources and Special Towns](#)).
- **Loose realm (an empire of princes, a land of city-states):** a mint in every princely seat, many bishops' cities and the big free towns.

MAP TIP

Mints are too small for a symbol on a kingdom map, but they are good detail for a town description or a regional map. Use the number of mints to show what kind of realm it is: many mints for an old or loose realm, three or four for a strong, centralised one.

Ceremonial cities: coronation and burial

The city where a ruler is **crowned** or **buried** is often not the city where he governs. The ceremonial city is usually an older place, where the dynasty or the church first became strong. It keeps that role for centuries, even after the government has moved elsewhere.

Realm	Coronation	Burial	Government
France	Reims (nearly all kings; about 130 km / 80 mi from Paris)	Saint-Denis abbey, just north of Paris	Paris
England	Westminster Abbey (since 1066)	Westminster Abbey (rebuilt by Henry III from 1245)	Westminster / London (Winchester was an older royal and treasury centre)

Realm	Coronation	Burial	Government
Holy Roman Empire	Aachen (about 30 kings crowned 936–1531), then Rome for the imperial crown	Speyer cathedral (Salian emperors) and others	No capital. The Golden Bull (1356) fixed elections at Frankfurt, coronation at Aachen and the first diet (assembly) of each reign at Nuremberg.
Hungary	Székesfehérvár (about 38 monarchs crowned, 1038–1527)	Székesfehérvár	Esztergom, Visegrád, Buda (about 60 km / 37 mi from Székesfehérvár)
Poland	Gniezno (early), then Kraków's Wawel (from 1320)	Wawel Cathedral	Kraków
Scotland	Scone	Dunfermline abbey	Edinburgh, from the 15th century
Aragon	Zaragoza	Poblet monastery	Barcelona, Zaragoza, Valencia (itinerant)

RULE OF THUMB

Governments move; sacred sites stay. A small town with a coronation church or royal tombs can be more important on your map than a large trading city.

Legitimacy can also be borrowed. After the Mongols destroyed Baghdad in 1258, the Mamluk sultans of Egypt brought a member of the Abbasid family to Cairo (1261) and kept him there as a “shadow caliph” to bless their rule.

Around the capital: the royal landscape

Within about one day's ride of a fixed capital you usually find:

Feature	Distance from capital	Real examples
A ring of royal residences: smaller palaces, manors and castles	5–60 km (3–37 mi)	London: Eltham and Sheen (~13 km / 8 mi), Windsor (~35 km / 22 mi). Paris: Vincennes (~7 km / 4 mi), Saint-Germain-en-Laye (~20 km / 12 mi), Fontainebleau (~55 km / 34 mi). Buda: Visegrád (~35 km / 22 mi). Prague: Karlštejn (~30 km / 19 mi). Straight-line distances, rounded.
Royal hunting forests and lodges	10–100 km (6–60 mi)	England: the New Forest (c. 1079); Windsor Forest; Woodstock and Clarendon hunting palaces. Kraków: Niepołomice hunting castle (built by Casimir the Great in the mid-1300s).
The ruler's favourite abbeys and burial church	0–20 km (0–12 mi)	Westminster, Saint-Denis
Estates of courtiers and officials	Within 1–2 days' ride	Nobles' town houses in the capital and country manors in the home counties

Royal forests were legal hunting zones that could contain villages, fields and heath as well as woods (see [Royal](#)

[forests](#)). Kings used them for hunting and to raise money from fines and licences.

MAP TIP

Around your capital, draw 3–6 small royal residences (small crowns) within about 60 km (37 mi), at least one big royal forest with a hunting lodge, a great abbey or burial church close by, and, if you want, a separate coronation town within 50–150 km (30–90 mi), marked with a small crown

beside its town symbol. Draw each royal forest as a green dash-dot boundary with its name in large italics, and leave its villages and fields visible inside. These details make a capital feel real.

Political units and their sizes

RULE OF THUMB

Kingdom > duchy (or principality) > county > barony or lordship > manor. (A *fief* is any land held from a lord in return for loyalty and service; a *vassal* is the person who holds it.) Each level is usually several times to a few dozen

times smaller in area than the one above. But medieval units overlapped, were scattered and changed constantly. Do not expect a neat pyramid.

The units, top to bottom

Unit	Typical area	Typical population (c. 1300)	Seat	Notes and examples
Empire	0.8–1.7 million km ² (300,000–650,000 sq mi)	5–15 million	One great capital, or none (HRE)	HRE 0.8–1.1m km ² ; Carolingian Empire c. 800 ~1.2m km ² ; Byzantium in 1025, 1.675m km ²
Kingdom	10,000–450,000 km ² (4,000–175,000 sq mi)	0.1–17 million	Capital or itinerant court	Navarre ~10,000–12,000 km ² (~100,000 people c. 1300); Scotland ~79,000; Portugal ~89,000; England 130,000; Hungary 280,000–325,000; France was the giant
Duchy or principality (great fief)	10,000–50,000 km ² (4,000–19,000 sq mi)	0.5–2 million (estimates)	A ducal city with castle, cathedral and court	Rough areas, based on the matching modern regions: Normandy ~30,000 km ² ; Brittany ~34,000 km ² ; Flanders (a county, but as powerful as a duchy) ~10,000 km ²
County (shire)	1,000–10,000 km ² (400–4,000 sq mi); England averaged ~3,300 km ²	20,000–300,000; England averaged ~120,000	County town with castle, sheriff and county court	39 English shires: Rutland 394 km ² (152 sq mi) to Yorkshire ~15,700 km ² (6,067 sq mi). Hungary had ~70 counties in the 15th century, averaging ~4,000–4,500 km ² .
Hundred or wapentake (English sub-county district)	Average ~130–150 km ² (50–60 sq mi) in 1086, ~150–200 km ² (60–80 sq mi) by 1316; range 4.6–982 km ² (2–380 sq mi)	2,000–10,000	A meeting place, often open-air: a mound, a ford, a crossroads, a stone	812 hundreds can be traced in Domesday Book (1086), from Worth in Kent (4.6 km ²) to Salford in Lancashire (982 km ²); roughly 10–12 parishes each in 1086; after mergers about 630 named hundreds remained by 1316, with 10–15 parishes each
Barony, lordship, honour (a great lord's group of estates)	Very variable: a few manors to 100+	—	The <i>caput</i> (head manor), usually with a castle	A great baron's manors were often scattered across several counties
Knight's fee (land expected to support one knight)	1–2 manors; no fixed size	—	A manor house	England had only about 5,000–6,000 knights' fees (1166 survey)
Manor	2–10 km ² (one village, part of one, or several)	50–500	Manor house, often moated	The village of Elton (Huntingdonshire) had 37 recorded households in 1086 (roughly 170–190 people), split among three manors
Parish	~14 km ² in the lowlands; 50–100+ km ² in uplands	75–500	Parish church	~9,500 parishes in England c. 1300

Units do not nest neatly

- **Manors and villages often differed.** In a 1279 survey of 134 Leicestershire villages, only 37 had a single lord. 63 villages (47%) were split among three or more lords.
- **Baronies were scattered.** In 1086 the people who held land directly from the king were called *tenants-in-chief*. A small group of great barons held a large share of the country; one common estimate is that about a dozen

men held around a quarter of it. Their manors lay in many counties.

- **Counties had detached pieces.** Many English counties had small “islands” of land inside neighbouring counties. An Act of 1844 removed many of them, but some survived into the 20th century.

- **Church lands cut across everything.** Bishops and abbots held roughly a quarter of England in 1086 (estimates vary), and a bishop's lands did not match his diocese.

Who sits where

Lord	Seat	What the seat looks like
King	Capital or main palaces	See above
Duke or great count	Main city of the duchy	Big castle or palace, cathedral, mint, the duke's own abbey and burial church. Example: Rouen for Normandy, Dijon for Burgundy, Ghent and Bruges for Flanders.
Count or sheriff (royal officer)	County town	Royal castle with the sheriff's office and gaol, the county court, a market. In England many shires are named after their county town: Warwickshire, Oxfordshire, Bedfordshire. In Hungary each county head (<i>ispán</i>) sat in a royal castle.
Baron	Caput of the barony	A castle (see Castles, Forts and Military Outposts), often with a small borough and a priory founded by the family
Knight or manor lord	Manor	Manor house with hall, barns and often a moat, next to the parish church

How many units in one kingdom?

Realm	Count	Source
England c. 1300	1 kingdom; 39 shires (plus special zones such as the Palatinate of Durham); about 630 hundreds (628 named in a 1316 inquest; 812 can be traced in Domesday Book, 1086); ~9,500 parishes; 5,000–6,000 knights' fees	Domesday; 1166 survey; 1316 inquest; baseline
France c. 1200	Royal domain plus 6 great lay fiefs: the duchies of Burgundy, Normandy and Aquitaine and the counties of Flanders, Champagne and Toulouse (the six lay "peers of France", fixed c. 1180), plus many lesser counties	Peerage of France
Hungary	~37 counties named for 1038 (many of them disputed), ~45–50 (c. 1074), ~70 (late 14th and 15th centuries)	Wikipedia <i>Administrative divisions of the Kingdom of Hungary</i>
Carolingian Empire c. 800	Somewhere between about 110 and 600 counties (estimates vary widely)	Wikipedia <i>Carolingian Empire</i>
Holy Roman Empire 1521	About 400 <i>imperial estates</i> (territories and cities with a place in the imperial diet). About 53 were ecclesiastical principalities (archbishops and bishops, three of the archbishops being electors) and 87 were free and imperial cities. The rest were secular princes, counts and lords, abbots and abbesses.	Worms register of 1521
China (for comparison)	About 1,230 counties (<i>xian</i>) under the Song (960–1279); about 1,385 under the Ming (1368–1644)	Skinner 1977

MAP TIP

For a kingdom the size of England (130,000 km², 50,000 sq mi), draw 30–40 counties, each about 50–70 km (30–45 mi) across, each with a county town near its middle. Group them into 4–8 bigger regions held by great lords or

old duchies. Draw duchy borders as medium dash-dot lines and county borders as thin ones. Leave 1–3 special zones: a march (hatching or a fading colour band), or a palatinate or church territory (its own colour).

A different model: the bureaucratic empire (China)

RULE OF THUMB

An empire run by salaried officials gives a much more regular map than feudal Europe: county seats of similar size, spread fairly evenly, each one a walled government town, with market towns between them and few private castles.

Why it is different. In imperial China the basic unit was the county (*xian*). A magistrate ran it. He was an official appointed by the central government, not a lord who inherited the land. A magistrate could not serve in his home county (the "rule of avoidance", strictly enforced from the 6th century), and he was moved to a new post after a few years (in Ming times usually two or three). Nobody owned a county, so counties were not split by inheritance, scattered

by marriage or turned into private lordships. Their number also stayed surprisingly steady from one dynasty to the next.

Feature	Numbers	Source
Number of counties	Song: about 1,230. Ming: about 1,385. A recent study argues that the Song also had hundreds of “administrative towns” with officials doing county-like work.	Skinner 1977; Han and Sng 2023
Average county size	About 2,000–2,500 km ² (770–970 sq mi) in Song China: about 1,230 counties on 2.5–3.1 million km ² (derived; area estimates vary). That is a little smaller than an average English shire (~3,300 km ²).	Derived
Spacing of county seats	About 45–55 km (28–34 mi) apart on average (derived from the area above). Closer on crowded river plains, much farther apart in hills and on the frontier.	Derived
City walls	Of all administrative cities (county seats and above): 62% walled in 1368, 70% in 1393, 80% in 1469, 90% in 1540 and 95% in 1576	Xue et al. 2021
Below the county seat	Periodic market towns. In Skinner’s model, each “standard market town” served about 18 villages (6 close by, 12 farther out) and roughly 6,000–12,000 people. Markets met on fixed days of a ten-day cycle, for example days 1, 4 and 7. A bigger market nearby used other days (for example 2, 5 and 8), so traders could visit both.	Skinner 1964–65

Be careful with Skinner’s market figures. They come from the 19th and 20th centuries, when China was far more

crowded than medieval Europe. Use them for the pattern, not for exact numbers in 1000–1500.

MAP TIP

For a bureaucratic empire, draw county seats as walled towns of similar size (the walled-town symbol with the county-town dot in the centre), spread fairly evenly about 45–55 km (28–34 mi) apart on good farmland, closer on crowded plains and farther apart in the hills. Give each one a regular wall, often a rectangle on flat land. Group every few counties into a prefecture, with a bigger city as its seat. Fill the gaps with unwalled market towns, each in the middle of its ring of villages. Draw few or no private castles: in this kind of state, the walls belong to the government’s towns.

FANTASY TWIST

Other peoples may divide land differently. Dwarf holds might be counted by mountain or mine, not area; elves by forest; a sea people by islands or harbours. Keep the same rule: one seat per unit, one unit per day or two of travel.

Borders and frontiers

RULE OF THUMB

In settled farmland, borders were known lines, walked and argued over field by field. In empty land (mountains, forest, marsh, desert, steppe), borders were wide zones. Draw lines where people live and bands where they do not.

What borders follow

Feature	How often used	Why	Real examples
Old boundaries	Very often	Borders are “sticky”. Once fixed, they last for centuries.	Many French dioceses follow Roman city territories. Many English shire borders lasted from the 10th or 11th century until 1974.
Watersheds and mountain crests	Often, but not always	Mountains are empty, so a crest line divides little.	Much of the Pyrenees. But the Val d’Aran, on the French side of the crest, belonged to Aragon, and Lower Navarre lay north of the mountains.
Pass states (an exception)	In the Alps	Whoever holds a pass wants both sides of it.	Savoy, Tyrol and the Swiss canton of Uri straddled the main Alpine passes instead of stopping at the crest.

Feature	How often used	Why	Real examples
Rivers	Some-times	Easy to name in a treaty and easy to see. But a river valley is one economic unit, so peoples on both banks are often the same.	Tweed (England–Scotland, Treaty of York 1237); Scheldt (France–Empire, from the partition of 843); the Danube (old Roman frontier)
Forests, marshes, heaths	Often in early times	Thin population; hard to cross.	The “Wildnis” (wilderness) left between the Teutonic Order’s Prussia and Lithuania
Empty wasteland	Frontier wars	Raiding emptied the land between enemies.	The Castile–Granada frontier (1246–1492); the steppe “Wild Fields” north of the Black Sea
Man-made dykes and walls	Rare but striking	Mark a claim or block raids	Offa’s Dyke on the Welsh border (traditionally credited to King Offa of Mercia, who reigned 757–796, though recent dating suggests parts are older); the Danevirke in Denmark; the Scots’ Dike (1552)

Frontier zones vs border lines

- **Line borders** existed in settled land. Parish and manor bounds were marked with stones, crosses, ditches and old trees, and villagers walked them every year (“beating the bounds”). Treaties named rivers and landmarks. The Anglo-Scottish border fixed by the Treaty of York (1237)

changed very little afterwards; the only later changes were Berwick and the Debatable Lands.

- **Zone frontiers** were the norm on the edges of Christian Europe, in mountains and in war zones. Here power faded out over 20–100+ km (12–60+ mi). People paid tribute to both sides, or to neither.

Marches: militarised border provinces

A **march** (German *Mark*, Italian *marca*) was a border province given to a strong lord to defend. The **margrave** (march count) or marcher lord had more military power than an ordinary count.

March	Dates	Features
Spanish March (Marca Hispanica)	From c. 795; Barcelona taken 801	A ring of counties (Barcelona, Girona, Urgell, Cerdanya and others) facing al-Andalus. The count of Barcelona was effectively independent by the late 980s; the march became Catalonia.
Welsh March	From the 1070s to 1536	Dozens of small Norman lordships between England and Wales. Marcher lords ruled “like kings”: own courts, own castles, private war, their own markets and boroughs. Britain’s densest cluster of motte-and-bailey castles (castles on a man-made earth mound with a fenced yard). The lords’ special powers were abolished in 1536 by the first of the Laws in Wales Acts (1535–42).
Anglo-Scottish Marches	13th century to 1603	Three marches (East, Middle, West) on each side, each under a royal Warden. A special border law, first written down in 1249 by six English and six Scottish knights. Raiding clans (“reivers”).
Brandenburg	From 1134	Albert the Bear was made margrave of the Northern March in 1134 and called himself Margrave of Brandenburg from 1157. It grew into a major principality.
Austria	From the 10th century	The eastern march of Bavaria; Vienna became its seat in 1145, and Austria became a duchy in 1156.

Other frontier words that survive as place names: *Extremadura* (often explained as “the far edge of the Duero”

lands), *Marche* (France and Italy), *Mark*, and *Krajina* (Slavic for “borderland”).

RULE OF THUMB

A march is bigger, emptier and more fortified than an inland county of the same importance. Expect noticeably more castles per area (the Welsh March has Britain's densest cluster of earth-and-timber castles) and fewer, smaller towns.

Buffer states, enclaves, exclaves and contested land

Thing	What it is	Real examples
Buffer state	A small realm between two big ones. Each big neighbour prefers it independent rather than in the other's hands.	Navarre between France, Castile and Aragon (its southern part was conquered by Ferdinand of Aragon in 1512 and later joined to Castile); Savoy on the Alpine passes; Wallachia and Moldavia between Hungary, Poland and the Ottomans; Andorra (co-ruled since 1278 by the bishop of Urgell and the count of Foix; 468 km ² / 181 sq mi)
Enclave / exclave	A piece of one realm surrounded by another (an enclave from the outside view, an exclave from the owner's view)	Baarle (Low Countries): medieval land deals between the dukes of Brabant and the lords of Breda left dozens of tiny pockets that still exist (22 Belgian pieces inside the Netherlands today). Avignon and the Comtat Venaissin: papal land inside France from 1274 (Venaissin) and 1348 (Avignon) until 1791.
Overseas bridgehead	A defended exclave across a sea	The Pale of Calais, English from 1347 to 1558. It ran roughly from Gravelines to Wissant, about 35 km (22 mi) of coast, with marshy, often-changing borders inland. It was supplied by sea.
Palatinate	A border county where a great lord holds royal powers	The prince-bishops of Durham guarded England against Scotland
Contested land	Claimed by both sides, ruled by neither	The Debatable Lands on the Anglo-Scottish border: about 16 × 6.4 km (10 × 4 mi), disputed for about three centuries and a refuge for raiders, split in 1552 by the Scots' Dike. Berwick-upon-Tweed changed hands more than a dozen times before becoming English for good in 1482.

Border markers and crossings

- **Markers:** boundary stones, crosses, ditches, banks, marked trees and posts. Treaty commissioners rode the line and ordered markers to be set up. For the Scots' Dike (1552), an earth bank a little over 5 km (3 mi) long, they ordered a stone with a cross at each end.
- **Crossings:** bridges, fords and passes on main roads, with a toll station, often a border castle on each side,

and an inn or hospice. Tolls and customs could be collected at sea straits too: Denmark's Sound Toll at Helsingør (from 1429).

- **Meeting places:** on the Anglo-Scottish border, wardens met on fixed "truce days" at traditional spots on the line to settle complaints.

MAP TIP

Draw dash-dot border lines where settled farmland meets, and a hatched band (a "march") across mountains, forests and war zones. Put paired castles (small castle icons) facing each other where a main road crosses the border, with a toll mark (a bar and a "T") on the road between them. Add one or two odd features: an enclave, a tiny buffer state, a disputed strip. These make borders look historical rather than planned. Castle spacing along frontiers is covered in [Castles, Forts and Military Outposts](#).

FANTASY TWIST

Monsters behave like an extra enemy. A monster-haunted forest or mountain becomes a frontier zone even between friendly realms, with marcher lords, watchtowers and wide empty bands. A magical barrier (a ward line, a cursed river) can create a sharp line border even in wilderness.

Language lines are not borders

RULE OF THUMB

A language line is not a political border. It is usually older and it moves much more slowly. Land changes hands across a language line many times while the line itself stays put. When a language line does move, it creeps a few villages at a time over centuries.

Why they differ

- **Villagers stay put.** When the lord or king changes, the villagers keep speaking what their grandparents spoke. Language follows who settled and farmed the land long ago, not who rules it now.
- **Realms are built across language lines.** Marriage, inheritance and war join lands with different languages into

one realm, and split lands with one language between several.

- **Lines move by slow pressure.** A language spreads when towns, lords, the church and trade use it. Its rival survives longest in remote villages. A language can also jump in all at once, when a ruler brings in settlers.

Real examples

Example	What happened	What it shows
The Romance–Germanic line in today's Belgium	It took shape as Frankish settlers moved in, roughly between the 4th and 10th centuries. After about the 10th century it hardly moved in what is now Belgium. (Farther west, the coast from Boulogne to Dunkirk became French-speaking only in the later Middle Ages.) Medieval states sat across it: the County of Flanders held Romance-speaking (Picard) Lille and Douai, and the Prince-Bishopric of Liège held Romance-speaking (Walloon) Liège and the Dutch-speaking County of Loon. The line became a legal boundary only in 1962–63.	A stable line, with realms on both sides of it
Breton in Brittany	Breton reached its farthest east in the 9th century, roughly from the bay of Mont-Saint-Michel to Donges on the Loire (the “Loth line”, named after the linguist Joseph Loth). Even then the region's chief cities, Rennes and Nantes, stayed Romance-speaking. After that the line moved west: to about Dinan–Pornichet by the 12th century, and to about Binic–Guérande by 1588. That is roughly 30–80 km (20–50 mi) in seven centuries (derived from these places).	A slow retreat inside one realm, with the capital on the “wrong” side of the line
Transylvanian Saxons (a language island)	King Géza II of Hungary (1141–62) began bringing German-speaking settlers to the thinly settled south-eastern edge of his kingdom, to farm it and defend it. The <i>Diploma Andreanum</i> (1224) gave them self-government on their own “Royal Land”. Their chief towns included Hermannstadt (Sibiu), Kronstadt (Braşov) and Bistritz (Bistriţa), and they built about 300 villages with fortified churches. In 1848 the Royal Land still held 271 villages, market towns and towns.	A language island planted by a ruler, lasting for centuries inside another language area
Mudéjar villages in Valencia	After James I of Aragon conquered Valencia (1230s–40s), whole villages of Muslims (<i>Mudéjars</i>) kept their Arabic speech and their faith under Christian lords. In the late 1300s they may still have been a majority of the countryside. See Religious, Cultural, Legal and Ancient Sites .	The border moved at once; the language of the villages did not

MAP TIP

Draw languages or cultures as a separate layer: a soft colour tint only, never a line, so nobody mistakes a language edge for a border or a track. Let it cross borders, and let one realm hold two or three language areas. Sometimes put the capital on the “wrong” side of the line, like Rennes. Change the place-name endings across the line, for example German-style *-ingen* and *-heim* on one side and French-style *-ville* and *-court* on the other (see [Naming settlements realistically](#)). Add one to three language islands where a ruler planted settlers: on an empty frontier, in a mining district or around a new town.

FANTASY TWIST

Long-lived peoples change their speech slowly, so elf or dwarf language lines can be even older and more stable than human ones. A dwarf hold or an elven wood inside a human kingdom is a language island: give it its own place names, even if a human king rules the land around it.

Kinds of states and how they look on a map

RULE OF THUMB

Do not colour all of Europe-like land in big, solid blocks. Medieval maps were patchworks: kingdoms, duchies, free cities, church lands, leagues and empires overlapped and interlocked.

Kind of state	How it works	How it looks on a map	Real examples
Feudal kingdom	A king over dukes, counts and barons who hold land in return for loyalty and service. The royal domain is only one part.	One outline, but inside it a patchwork of great fiefs. The royal domain is the core around the capital. Some fiefs are nearly independent.	France c. 1200: royal domain in the Île-de-France, surrounded by Normandy, Flanders, Champagne, Burgundy, Aquitaine and Toulouse
City-state and its <i>contado</i>	A self-governing city rules the countryside around it (the <i>contado</i> , often the same area as its diocese) and later conquers neighbouring cities.	A city with a territory of roughly 20–50 km (12–30 mi) radius, many such states side by side, with borders that move often.	Florence (took Arezzo in 1384, conquered Pisa in 1406). In 1427 the Florentine state had roughly 260,000 people, of whom the city held only about 37,000–38,000 (the <i>catasto</i> tax survey, studied by Herlihy and Klapisch-Zuber). Also Milan, Siena, Genoa.
League of towns	Towns cooperate for trade and defence but keep their own small territories.	A network, not an area: dots linked by sea routes, plus trading posts in foreign cities.	Hanseatic League: between about 70 and 170 member towns at different times, and nearly 200 places linked to it overall. Its power peaked around 1370 (Peace of Stralsund). Main trading posts (<i>Kontore</i>) in Novgorod, Bruges, London (the Steelyard) and Bergen; meetings (diets) usually in Lübeck
Confederation	Rural valleys and towns ally for defence and rule conquered land together.	A cluster of small units; some areas shown as joint subject lands.	Swiss Confederacy: 8 cantons by 1353, 13 by 1513. Aargau (1415) and Thurgau (1460) were ruled jointly as subject lands.
Composite empire	Many realms and hundreds of lords under one elected ruler with weak central power.	A big outer border with hundreds of small internal units.	Holy Roman Empire: ~400 imperial estates in 1521, including 87 imperial cities and ~50 prince-bishops and archbishops
Church state	A bishop, abbot or the Pope rules land as a prince. The territory is not the same as the diocese.	Separate colour; often scattered pieces.	Papal States in central Italy; prince-bishoprics of Cologne, Mainz, Trier, Liège, Münster, Würzburg, Salzburg; Durham in England
Military-order state	A crusading order of knights rules conquered land.	A frontier territory with many castles and commanderies (houses of the order).	Teutonic Order in Prussia, seat at Marienburg (Malbork) from 1309
Maritime empire	A trading city holds ports and islands along a sea route, later also land behind it.	A home city plus a chain of coastal dots and islands, not a solid area.	Venice: Crete (c. 1210–1669), Negroponte (to 1470), Modon and Coron (1207–1500), Corfu (from 1386), Cyprus (1489–1570), Dalmatian ports; mainland expansion from 1337–39 (Mestre, Treviso) to Brescia and Bergamo (1420s)
Nomadic empire	Herders on seasonal pastures; settled peoples nearby pay tribute.	A huge, vague area: core pastures, a few capital cities, and a ring of tributary states around them. Borders are wide zones.	Mongols: Karakorum walled in 1235; the Golden Horde's Sarai on the lower Volga (founded by Batu in the 1240s–1250s), destroyed by Timur in 1395–96. Khans also kept moving camps and tent palaces. Rus' princes paid tribute but kept their own cities.

Drawing the unusual kinds

- **City-states:** many small territories, each 1–3 days across, centred on rival walled cities about 50–90 km (30–55 mi) apart.
- **Leagues:** do not colour an area. Underline each member town's name in the league's colour (members can sit inside other rulers' kingdoms) and draw dotted sea lanes between them.

- **Nomadic empires:** show summer and winter pastures, one or two built capitals on rivers or trade routes, and the tributary ring in a lighter tint of the empire's colour.
- **Maritime empires:** colour only the islands and harbour towns along the sea route; the land behind them keeps the local ruler's colour.

MAP TIP

For a continent map, use at least three kinds of state: one or two big feudal kingdoms, one fragmented zone (city-states or a composite empire) and one unusual kind (a league, a church state, a maritime empire or nomads). This mix is what makes a medieval map look medieval.

FANTASY TWIST

Different peoples suggest different state types. A dwarven realm may be a league of mountain holds linked by tunnels (a network, like the Hanse). An elven forest realm may look like a nomadic or church state with vague borders. A wizards' city may rule a contado like Florence.

How big can a realm be

RULE OF THUMB

A ruler can govern directly up to roughly 500–700 km (300–450 mi) from the capital. At a messenger's normal pace of about 50–65 km (30–40 mi) a day, that is about 8–12 days' travel. Beyond that, the ruler needs viceroys (governors

who rule in the king's name), great vassals, sea routes or a relay post, or the realm splits. This is a guide's heuristic, not a measured law.

Why distance limits power

- Orders, tax money and news travel at the speed of a horse (see the travel table in [Trade Routes, Roads and Transport](#)).

- A rebellion far away can grow for weeks before the king even hears of it.
- Feudal armies often owed only about 40 days of service, so a host raised in the core could not campaign for long on a far frontier.

Distance from the capital and what it means

Speeds from the shared baseline: royal messenger 50–90 km (30–56 mi) a day; army with baggage 13–20 km (8–12 mi) a day.

Distance from capital	Royal messenger	Army	What it means
100 km (60 mi)	1–2 days	5–8 days	Core: royal domain, residences, hunting forests. Direct control.
300 km (190 mi)	3–6 days	15–23 days	Normal kingdom: counties ruled by royal officers (sheriffs, bailiffs, <i>ispáns</i>).
600 km (370 mi)	7–12 days	30–45 days	Edge of a large kingdom: needs great lords, marcher lords or a royal governor.
1,000+ km (600+ mi)	11–20 days	50–75+ days	Only possible for empires with sea routes, relay posts or semi-independent governors.

A real measure comes from news reaching Venice around 1500 (after Sardella and Braudel). Venice had one of the best news networks in Europe. Typical times were:

From	Time for news to reach Venice (c. 1500)
Milan, Genoa, Florence	About 1 week
Paris, Lyon, Marseille, Vienna, Buda, Naples	About 2 weeks
Antwerp, Cologne, Barcelona	About 3 weeks
London	3–4 weeks
Castile (Burgos, Valladolid), Constantinople	4–5 weeks
Lisbon	About 6 weeks

Realistic realm sizes

Realm	Area	Population	Farthest point from the capital (approx.)
Andorra (micro-state)	468 km ²	tiny	Under 1 day
Navarre	~10,000–12,000 km ²	~100,000 (c. 1300)	2–3 days

Realm	Area	Population	Farthest point from the capital (approx.)
Scotland	~79,000 km ²	0.5–1m (c. 1300; estimates vary)	5–8 days
Portugal (border almost unchanged since 1297)	~89,000 km ²	~1.0m (1500)	4–7 days
England	130,000 km ²	4.75m (1290)	London to Berwick ~500 km (310 mi) in a straight line, ~550 km (340 mi) by road: 6–11 days
Poland (1370)	~240,000 km ²	~2m	6–10 days
Hungary	280,000–325,000 km ²	2–3m	8–12 days
France (1328)	smaller than modern France's 551,000 km ²	16–17m	10–14 days to the Mediterranean or the Pyrenees
Holy Roman Empire	0.8–1.1m km ²	11–14m (c. 1300)	Weeks; no single capital
Byzantium (1025)	1.675m km ²	~12m	Weeks, but sea routes helped

Around 1500, Europe held roughly 500 more or less independent political units (Tilly 1975), from tiny lordships to great kingdoms. Most were small.

Realms that grew too big had three choices. They split (the Carolingian Empire at Verdun in 843; the Mongol Empire into separate khanates by the 1260s). They were run through regents and governors (Charles V). Or they held their distant parts by sea (Venice, Byzantium).

MAP TIP

Measure from your capital along the roads, not in a straight line. Mark the 5-day and 12-day messenger lines in pencil (about 250–300 km and 600–700 km, or 150–190 mi and 370–430 mi, at a normal messenger pace). Put direct royal counties inside the first line, great vassals and marches between the lines, and tributary or semi-independent lands beyond the second. If your kingdom is much bigger than that, add a second-tier capital (a viceroy's seat) or make the outer parts nearly independent.

LATER ERA (1500S+)

Organised postal relays (Taxis, from about 1500, up to ~150 km or 95 mi a day) and faster shipping roughly double the range a capital can control. Empires like Spain and the Ottomans govern huge areas through viceroys and provincial governors. Borders are increasingly surveyed, mapped and marked by boundary commissions. See [What Changes After 1500](#).

FANTASY TWIST

Every faster messenger extends the realm. A griffon post at 300 km (190 mi) a day is 3–6 times faster than a royal messenger, so orders reach 3–6 times farther in the same time. Instant magical messages remove the communication limit but not the army limit. Armies with baggage still march at 13–20 km (8–12 mi) a day unless magic moves them too.

Drawing realms, borders and capitals

Step by step

1. **Choose the realm type** and **find the core**: the best farmland on a navigable river, with a defensible hill or island for the capital.
2. **Add the royal landscape**: 3–6 residences, a royal forest, the burial church, perhaps a separate coronation town.
3. **Draw the outer border**: dash-dot lines in settled land, march bands across mountains, forests, marshes and war zones.
4. **Divide** into 4–8 duchies or great fiefs, then counties 50–70 km (30–45 mi) across, each with a seat near its middle. Add a few oddities (church land, an enclave, a free city, a disputed strip).
5. **Check distance**: anything more than about 12 days' messenger travel (roughly 700 km or 430 mi by road) from the capital needs a viceroy or becomes semi-independent.

Symbols and line styles

RULE OF THUMB

One symbol, one meaning. A filled star means the capital and nothing else. Borders are always dash-dot lines (a line of dashes with dots between them); roads, tracks and sea lanes are always solid or dotted lines. Then nobody mistakes a border for a road.

These symbols match the guide's [master legend](#). The “town symbol” below means the normal symbol for the place's size (village, town, city and so on), from [Map symbols and labels](#).

Feature	Symbol	Scale
Realm capital	Filled star in place of the town symbol, bigger for a bigger city; name in bold capitals	Continent, kingdom
Coronation or royal burial town	Small crown beside the town symbol, labelled “(coronations)” or “(royal tombs)”	Kingdom
Duchy or regional seat	Town symbol with a small pennant (flag) on top	Kingdom
County town	Town symbol with a dot in the centre	Kingdom
Royal residence or hunting lodge	Small crown on its own	Kingdom, local
Royal forest	Green dash-dot boundary with a large italic name; villages and fields inside	Kingdom, local
Moot (open-air meeting place of a hundred)	Small lone tree, labelled “Moot”	Local
Toll at a border crossing	One bar across the road or river, with a “T”	Kingdom, local
Realm border	Thick dash-dot line, or a coloured band along the inside edge	Continent, kingdom
Duchy or great fief border	Medium dash-dot line	Continent, kingdom
County border	Thin dash-dot line	Kingdom
Hundred, parish or manor bounds	Thin dash-dot-dot line	Local
Boundary stone	The letters “BS” on the boundary line, as on Ordnance Survey maps	Local
Disputed border	Two parallel dash-dot lines, or alternating colours	Kingdom
March or frontier zone	Hatching, or a fading colour band 10–50 km (6–30 mi) wide	Continent, kingdom
Enclave	Same colour as its owner, with a thin outline	Kingdom
League member town	Town symbol as usual; name underlined in the league's colour	Continent, kingdom
Language or culture area	Soft colour tint only, never a line	Continent, kingdom

Castles, beacons and watchtowers on the border use the symbols in [Castles, Forts and Military Outposts](#).

Colour and labels

- Use pale fills for realms, so roads and towns stay readable. Give vassal duchies a lighter shade of their king's colour.
- Give church lands and free cities their own colours, and underline league towns' names in the league's colour. Do not merge them into the surrounding kingdom.

- Write realm names in widely spaced capitals across the whole territory. Write duchies in smaller italic capitals and counties in lower case.
- Label marches and frontier zones by name (“the West March”, “the Debatable Land”). Named empty spaces make a map feel lived-in.

MAP TIP

On a continent map, show only realms and great fiefs. On a kingdom map, add counties and the capital's royal landscape. On a local map, add hundreds, manors and parish bounds (thin dash-dot-dot lines) and boundary stones (“BS”). Do not show every level at every scale.

Quick summary

- A capital needs a rich core region, navigable water, a defensible site, loyal land around it and a source of legitimacy. Centrality matters less.
- Expanding realms move their capital toward the frontier (“forward capital”); settled realms keep the old core.
- Before c. 1200 most western courts travelled. German kings moved between royal palaces and manors about 30 km (19 mi) apart. Fixed capitals came with archives, law courts and money taxes: Paris and Westminster in the 12th–14th centuries, Madrid only in 1561. The Holy Roman Empire never had one.
- Coronation and burial cities are often different from the governing city, and keep their role for centuries.
- Around a capital, draw 3–6 royal residences and hunting forests within about 60 km (37 mi). At their peak (late 12th and early 13th centuries), royal forests covered about one-third of southern England.
- Mints shrink as kings get stronger: about 70 in late Anglo-Saxon England, 9 in 1189, normally 4 under Henry III. Loose realms like the Holy Roman Empire had more and more.
- Typical sizes: kingdom 10,000–450,000 km²; duchy 10,000–50,000 km²; county 1,000–10,000 km² (England averaged ~3,300 km² and ~120,000 people); hundred ~130–150 km² in 1086 (~150–200 km² by 1316); manor 2–10 km².
- A bureaucratic empire (China) looks more regular: about 1,230–1,385 counties, each with a walled seat roughly 45–55 km (28–34 mi) from the next, and periodic market towns between them.
- Borders are lines in settled land and wide zones (marches) in empty land. They follow old boundaries, crests, rivers, forests and wasteland. Marches, buffer states, enclaves and contested strips were normal: add some. Draw every border as a dash-dot line and keep the filled star for the capital only.
- Language lines are older and slower than borders. Draw them as a separate layer that crosses borders, and add a few language islands of planted settlers.
- Medieval political maps are patchworks: kingdoms, city-states, leagues, church states, maritime chains and nomadic empires side by side. Europe c. 1500 had about 500 political units.
- Direct rule reaches roughly 500–700 km (300–450 mi), about 8–12 days’ messenger travel. Beyond that, rulers need viceroys, vassals, sea routes or relays.

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- Wikipedia: Itinerant court; Kaiserpfalz; Court of Common Pleas (England); Battle of Fréteval; Trésor des Chartes; Royal forest; Golden Bull of 1356; Timeline of Aachen; Timeline of Kraków; Coronation of the Hungarian monarch; Kingdom of Hungary (1301–1526); Administrative divisions of the Kingdom of Hungary; Buda Castle; Peerage of France; Knight-service; List of ancient counties of England by area in 1891; Welsh

- More Wikipedia pages used in fact-checking: [Exchequer](#); [Chambre des comptes](#); [Parlement of Paris](#); [Sainte-Chapelle](#); [Aachen Cathedral](#); [Karlštejn Castle](#); [Niepolomice Castle](#); [Madinat al-Zahra](#); [Bursa](#); [Henry II, Duke of Austria](#); [Margraviate of Brandenburg](#); [Leges Marchiarum](#); [Treaty of York](#); [Berwick-upon-Tweed](#); [Scots' Dike](#); [Offa's Dyke](#); [Sound Dues](#); [Baarle-Hertog](#); [Kingdom of Navarre](#); [Carolingian Empire](#); [Counties \(Detached Parts\) Act 1844](#); [Malbork Castle](#); [Arezzo](#); [History of Madrid](#); [Philip II of Spain](#); [Warsaw](#).
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the distance of the Breton retreat) are worked out by this guide from area, density or map distances and are not measured values.

CHAPTER 4

CASTLES, FORTS AND MILITARY OUTPOSTS

Castles are the most “fantasy” feature on a medieval map, and also the easiest to get wrong. Real castles were not scattered at random. Each one controlled something: a road, a river crossing, a town, a border or an estate. This chapter explains what fortifications were for, the main types and their dates, where they were built, how many there were and how far apart, how big their garrisons and armies were, how armies moved, camped and were fed, and how to draw all of this on your map.

The baseline period is the High and Late Middle Ages in Europe (c. 1000–1500). Borders and marches as political units are covered in [Capitals, Realms and Borders](#). Roads, river transport and travel speeds are covered in [Trade Routes, Roads and Transport](#).

IN THIS CHAPTER

What fortifications are for
Types of fortification
Where to put a castle
How many castles and how far apart
Castles and towns
Garrisons and armies
Military geography and logistics
Camps and siege towns
Outposts in wild and contested land
Watchtowers, beacons and coastal warning
Where battles happen
Drawing military sites on your map
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Sources and further reading

What fortifications are for

RULE OF THUMB

Every fortification answers one question: “What does it control?” If you cannot name the road, crossing, town, border or estate it watches, move it or delete it.

A castle was not only a military building. It was the lord’s house, his office, his court, his prison, his store for rents paid in grain, and a barracks. Many historians (for example Charles Coulson) stress that walls and battlements were also a **status symbol**. A castle said “a lord rules here”.

Purpose	What it looks like	Typical site	Real example
Control a route	A castle or tower right beside a pass, ford, bridge or narrow river	The narrowest point of the route	Krak des Chevaliers (Syria) guarded the Homs Gap between the coast at Tripoli and inland Homs. Château Gaillard (1196–98) guarded the Seine valley road to Rouen.
Protect a town or port	Town walls and gates, often with a castle or citadel at one corner	Around the town; the castle on the highest point or by the river	Carcassonne (two walls, about 3 km / 1.9 mi); Ávila (about 2.5 km / 1.6 mi); Visby (3.4 km / 2.1 mi)
Hold a border	A chain of castles and towers, much denser than inland	Along the border, at every crossing	The Welsh March; the Arab–Byzantine frontier (the <i>thughur</i>); the Christian–Muslim frontier in Spain
Dominate conquered land	Big new castles in or beside the main towns, often with new settler towns	Inside existing towns, at ports, on rivers	William I’s castles in English towns after 1066; Edward I’s castles in North Wales (1277–1295); the Teutonic Order in Prussia (from 1231)
Lord’s home and admin centre	A castle at the head manor (the <i>caput</i>) of a barony	Near the best farmland, a village and a church	Most of the 500–600 castles occupied at any one time in post-Conquest England
Refuge	Fortified churches, tower houses, walled villages, old hillforts	Inside the village	Over 150 Saxon villages in Transylvania with fortified churches (13th–16th centuries)
Watch and signal	Watchtowers, beacons, coastal towers	Hilltops and headlands with long views	The Byzantine beacon line across Asia Minor (9th century)

MAP TIP

Before you place a castle, write down its job in one word: *road, bridge, town, border, estate, refuge* or *watch*. The job tells you the site. A “road” castle sits on the road; an “estate” castle sits in the middle of good farmland; a “border” castle sits on the border.

Types of fortification

Castles: from earth to stone

Type	Dates	What it is	Real examples
Ringwork	10th–12th centuries	A ring of earth bank and ditch with a timber fence (palisade) on top. No mound.	Common in south-west England and south Wales after 1066
Motte-and-bailey	From c. 1000 in the Loire valley and Normandy; in England after 1066	A motte (a man-made earth mound) with a timber tower on top, and a bailey (a fenced yard) beside it for halls, stables and stores	About 80% of Norman castles in England were of this type. Mottes were about 3–30 m (10–100 ft) high and 30–90 m (100–300 ft) across at the base. A bailey usually covered about 0.15–0.7 ha (0.4–1.7 acres): The Mount, Ilketshall St John (Suffolk), has a bailey platform of 53 × 30 m; Ewyas Harold (Herefordshire) has a bailey of about 0.7 ha. A large baronial castle such as Castle Acre (Norfolk) covered about 3 ha (7.5 acres) to the outer edge of its defences. See Footprints . An average motte, 5 m (16 ft) high and 15 m (49 ft) across at the top, is estimated to have taken 50 people about 40 working days. The biggest mottes took up to about 24,000 working days.
Stone keep (great tower, <i>donjon</i>)	From c. 950 in France; in England from the 1070s	A massive stone tower holding the lord's hall and rooms	Doué-la-Fontaine (France, c. 950, often called the oldest known keep); the White Tower, London (c. 1075–1100); Dover's great tower (1180s)
Shell keep	Mostly 11th–12th centuries	A stone ring wall replacing the timber fence on top of a motte, because a mound cannot carry a heavy tower	Windsor (Henry II); Restormel (1270s–80s); Lewes; Launceston
Curtain-wall castle	Late 12th–13th centuries	A strong stone wall (the curtain) with towers along it that let defenders shoot along the wall face	Framlingham (late 1100s, finished by 1213; 13 wall towers and no keep); Château Gaillard (1196–98, taken 1204)
Concentric castle	Byzantine examples from the 11th century; crusader examples from the 1160s; Wales 1268–1295	Two or more rings of walls. The inner wall is higher, so both rings can shoot at once.	Belvoir (Hospitallers, from 1168); Krak des Chevaliers (Hospitaller from 1142; its outer ring of walls was added in the early 1200s); Caerphilly (begun 1268); Harlech (1282–89, cost £8,190); Beaumaris (begun 1295, never finished)

Stone castles were slow and costly. Stone needed skilled masons, and they had to be paid. A keep's walls usually rose no more than about 3.7 m (12 ft) a year, so a great tower took around ten years (the keep at Scarborough is a typical case). Edward I spent at least £80,000 on his Welsh castles between 1277 and 1304. For Beaumaris alone his master

mason asked for 400 masons, 2,000 labourers, 100 carts, 60 wagons and 30 boats. Earth-and-timber castles, by contrast, could go up in a few weeks or months, within a single building season, using unpaid local labour. That is why invaders built them first.

Small fortified homes

Type	Dates	What it is	Real examples and numbers
Tower house	14th–17th centuries	A tall stone house of 3–6 storeys for one family and its servants, often with a walled yard (<i>barmkin</i> in Scotland, <i>bawn</i> in Ireland). Built where raiding was common and the king was far away.	Ireland: over 3,000 built (more than 2,000 still stand). In 1429 the English crown offered £10 to any man in the Pale (the English-ruled area around Dublin) who built one, though recent studies doubt this made much difference. Scotland about 800. England over 250, mostly near the Scottish border. Also common in the Basque lands, Italy and the Caucasus.
Peel tower and bastle house	Peels 14th–17th centuries; bastles mostly 16th–17th centuries	A peel is a small tower house on the Anglo-Scottish border. A bastle is a fortified farmhouse: animals below, family above.	Northumberland, Cumbria and the Scottish Borders
Fortified manor house	13th–15th centuries	A manor house with a moat, gatehouse and battlements. Strong enough to stop a raid, not an army.	England has about 460 surviving licences to crenellate (royal permission to add battlements) for just over 500 sites, mostly for knights. Stokesay (1291), Cooling (1381), Bodiam (1385), Herstmonceux (1441).

Community defences

Type	Dates	What it is	Real examples and numbers
Town walls and gates	All periods; a big wave in the 12th–14th centuries	Stone wall or earth bank and ditch, with towers and gatehouses. Gates also collected tolls.	Carcassonne about 3 km (1.9 mi); Ávila about 2.5 km (1.6 mi); Visby 3.4 km (2.1 mi). England paid for walls with murage (a toll on goods entering a town), first granted to Shrewsbury in 1220; about 1,470 murage grants are recorded in England, Wales, Scotland and France.
Citadel	All periods	A fortress inside or at the edge of a city. It defends the city, and also watches it.	Tower of London; Cairo Citadel (begun by Saladin in 1176); Aleppo Citadel (rebuilt c. 1200)
Fortified bridge	12th–14th centuries	A bridge with a gate tower on it, to stop attackers and collect tolls	Pont Valentré, Cahors (1308–78; 138 m / 453 ft long, three towers); Monnow Bridge, Monmouth (bridge traditionally dated 1272, gate tower added c. 1300), the only British river bridge with its gate tower still standing on it
Fortified church or monastery	13th–17th centuries	A church with thick walls, a strong tower and often a ring wall. Villagers took refuge inside.	Transylvania: over 150 villages (13th–16th centuries); Slovenia: about 300 churches fortified against Ottoman raids (late 15th–17th centuries); Albi Cathedral, a brick fortress-church (1282–1480)
Reused hillfort or Roman fort	Whenever old walls were handy	Old earthworks or Roman walls given a new castle inside	Old Sarum: an Iron Age hillfort about 400 × 360 m (1,300 × 1,180 ft), a Norman castle by 1069 and a cathedral in 1092, abandoned for New Salisbury from 1220. Pevensey and Portchester: Norman castles built inside Roman forts.

Frontier and special types

Type	Dates	What it is	Real examples
Crusader castle	1100–1291	Large stone castles on spurs and hills, held by small numbers of Western knights among a hostile population	Kerak (1142); Krak des Chevaliers (Hospitaller 1142–1271); Chastel Pèlerin at Atlit (Templar, 1218–1291); Montfort (Teutonic Order, 1220s–1271)
Islamic <i>ribat</i>	8th–11th centuries	A fortified lodge for volunteer frontier fighters, with a watch and signal tower. Built in chains along coasts and borders.	Monastir (founded 796) and Sousse (rebuilt or enlarged 821) in Tunisia
Islamic frontier fortress	8th–15th centuries	Big walled fortresses and fortified towns in the frontier zone	Gormaz, Spain (built or rebuilt 965–66 for the caliph of Córdoba; about 1 km / 0.6 mi of walls with 24–27 towers); Tarsus (main Arab base facing Byzantium, 4,000–5,000 troops in its garrison)

Type	Dates	What it is	Real examples
Military-order castle	12th–15th centuries	A castle that is also a monastery for knight-monks (an <i>Ordensburg</i> in German)	Malbork/Marienburg (begun c. 1274, built over about 130 years; Teutonic Order headquarters from 1309; 21 ha / 52 acres inside its outer walls, the largest castle by area); Calatrava (Order of Calatrava, from 1158)
Siege castle (counter-castle)	All periods	A fort built to block an enemy castle or town: either a temporary siege fort close by, housing the besiegers, or a permanent counter-castle within a day's ride	Malregard at Antioch (1097); Mons Peregrinus at Tripoli (1102–03; the city fell only in 1109); four castles ringing Fatimid Ashkelon (Castrum Arnaldi, Beth Gibelin, Ibelin and Blanchegarde, 1133–42; the city fell in 1153). In 12th-century England, siege castles stood about 180–270 m (200–300 yd) from the target, just beyond bow range.

MAP TIP

Match the castle type to your map's date. Around 1100, almost every castle is earth and timber. By 1300, the important ones are stone and the small ones are either rebuilt in stone or abandoned. From 1350, add tower houses and fortified manors, and let old castles decay.

LATER ERA (1500S+)

Cannon broke tall thin walls (the Ottomans breached Constantinople's walls in 1453). From the late 1400s Italian engineers built walls **lower and thicker**, often of earth faced with brick. They added angled **bastions** (projecting platforms for guns). This *trace italienne* ("Italian outline"), or "star fort", spread across Europe in the 1530s–1540s. A famous example is Palmanova (1593), a whole star-shaped new town. Star forts were very expensive: Siena bankrupted itself on its walls in 1544. Henry VIII's **Device Forts** (1539–47, total cost £376,000) were low, round artillery forts guarding harbours and anchorages, such as Deal and Pendennis. Their garrisons were small: from 9 men (West Tilbury) to 29 (Camber). Draw one with the guide's coastal artillery fort symbol: a small **rosette** outline (rounded lobes, like the plan of Deal Castle, 1539–40, a round tower ringed by six rounded bastions). Never give it points: pointed outlines mean star forts. See [What Changes After 1500](#).

Where to put a castle

RULE OF THUMB

Put the castle where a few men can stop many: at the narrowest point of a route, on the strongest ground next to it, with water nearby.

Site	Why it works	Real example
River crossing (ford or bridge)	Every road and every boat must pass	Stirling (lowest crossing of the Forth); Chepstow (begun 1067, on a cliff above the Wye)
Mountain pass or gap	The only easy way between two regions	Krak des Chevaliers (Homs Gap); Loulon at the northern exit of the Cilician Gates (Byzantine–Arab frontier)
Spur or promontory	A ridge end with cliffs on two or three sides. Cut one ditch across the neck and the castle is strong on all sides.	Château Gaillard above the Seine; Kerak; Montfort
River bend (meander)	Water on three sides, a short neck to defend	Durham (castle begun 1072 in a loop of the Wear); Toledo in a loop of the Tagus
Confluence of two rivers	Controls traffic on both rivers	Koblenz, where the Moselle meets the Rhine
Island or lake shore	Water as a moat; controls river or lakeside road	Pfalzgrafenstein, an island in the Rhine (1326–27); Chillon, on Lake Geneva beside the road to the Great St Bernard Pass
Coastal headland or harbour mouth	Sea on several sides; supply by ship; watches the harbour	Dunstanburgh (begun 1313); Harlech, with a water gate down to the shore

Site	Why it works	Real example
Hill above a town	Overlooks and protects the town, and watches it	Edinburgh; Prague; Carcassonne
Flat wet lowland	No hills, so dig a moat or flood the ground	Caerphilly (artificial lakes); Bodiam (moat); Muiderslot near Amsterdam (c. 1285). These are “water castles” (German <i>Wasserburg</i>).

Hills or lowlands? It depends on why the castle was built.

- In the German lands about two-thirds of castles were on high ground (*Höhenburg*, “height castle”). Many lords there built for safety and display on wooded spurs.

MAP TIP

Draw the terrain first, then look for the “choke points”: fords, bridges, passes, narrow valleys, river bends and harbour mouths. Put a castle at the most important ones. Put estate castles in the middle of good farmland, beside a village. Do not put a castle on a remote peak far from any road or village. Nobody would pay for it.

- In England most castles stood on low ground, next to villages and towns. Norman castles were built to control people and farmland, so they sat where the people were.

FANTASY TWIST

If your world has flying mounts or flying monsters, high spurs lose some value and roofs matter more. Expect covered courtyards, roofed towers, heavy crossbows aimed upward, and aeries (bases for flying mounts) inside the strongest castles. See [Fantasy Variants](#).

How many castles and how far apart

Counts and densities

The numbers below count **every castle site ever built** in the period, including small mottes that were used for only a few decades. The column “typical gap” is the square root of the area per castle. It is a rough guide to the distance from one castle to the next.

Region	Castles	Area per castle	Typical gap to next castle
England and Wales overall	1,761 sites (Gatehouse gazetteer)	~86 km ² (33 sq mi)	~9 km (6 mi)
Peaceful eastern England (Norfolk, Essex, Kent, Suffolk, Lincolnshire)	22–45 per county	130–250 km ² (50–95 sq mi)	11–16 km (7–10 mi)
Welsh March (Herefordshire, Shropshire, Monmouthshire, Pembrokeshire, Glamorgan)	60–127 per county	17–28 km ² (7–11 sq mi)	4–6 km (2.5–4 mi)
North-west Wales (Caernarfonshire), conquered 1283	21	~71 km ² (27 sq mi)	~8 km (5 mi), but a few very large royal castles
Anglo-Scottish border (Northumberland, Cumberland)	32–46 per county	110–130 km ² (42–50 sq mi)	~11 km (7 mi), plus hundreds of tower houses not counted here
German-speaking lands	~14,000	~35–40 km ² (14–15 sq mi) (estimate)	~6 km (4 mi)
Switzerland	~4,000 (many very small)	~10 km ² (4 sq mi)	~3 km (2 mi)
Teutonic Prussia	Several dozen brick convent castles plus smaller ones	~370 km ² (143 sq mi) (estimate)	~19 km (12 mi)
Livonia (modern Latvia and Estonia)	—	~790 km ² (305 sq mi) (estimate)	~28 km (17 mi)
Western Europe as a whole	75,000–100,000 built	—	—

Notes:

- County figures are from the Gatehouse gazetteer (England and Wales; its total area is about 151,000 km² / 58,000 sq mi). The German, Swiss and Western Europe totals follow the guide’s shared baseline. The German

area per castle assumes German-speaking lands of roughly 500,000–550,000 km², so it is only a rough guide. The Prussia and Livonia figures are uncited estimates quoted on Wikipedia (its Livonia figure also covers Courland and Estonia); treat them as rough.

- **Not all at once.** In England only 500–600 castles were in use at any one time after the Conquest. That is about one-third of all sites. Active castles were therefore about one per 250–300 km² (about 95–115 sq mi), or roughly **16–17 km (10–11 mi) apart**.
- **Castle booms happen in times of weak rule.** In Provence there were 12 castles in 950, 30 by 1000 and over 100 by

1030. England had a civil war in 1139–53 (the “Anarchy”). The chronicler Robert of Torigni claimed that 1,115 unlicensed castles were built during it, but elsewhere he gave a figure of only 126. Expect exaggeration in your own world’s chronicles too.

Control radius

- **Direct military reach:** only about 400 m (1,300 ft). This was a castle’s immediate area of influence, roughly the reach of the defenders’ bows and crossbows.
- **Practical control:** a castle’s mounted men could ride out, patrol and return in one day. That gives a radius of about

15–30 km (10–20 mi). This is a working heuristic, not a measured figure.

- **Why it still matters:** an enemy cannot safely march past an untaken castle. Its garrison can cut his supply line, attack foragers and retake the land behind him. That is why so much medieval war was sieges.

Borderlands and conquered land

Borders and newly conquered land have far more castles than the peaceful core of a realm. The Welsh March had roughly **5–10 times** as many castle sites per km² as eastern England. (A **march** is a border zone held by local lords with special powers to defend it.)

Situation	Spacing	Real examples
Peaceful core of a kingdom	All sites 11–16 km (7–10 mi) apart; active castles about 20–30 km (12–19 mi) apart (estimate): one per county town plus scattered baronial seats	Eastern England (Norfolk, Essex, Suffolk)
March or contested border (many small lords)	4–6 km (2.5–4 mi) apart; small castles, many short-lived	Welsh March, 1070s–1280s
Conquest chain along a coast or river	20–40 km (12–25 mi) apart, about one day’s march; each castle reachable by water	Edward I’s coastal castles in Wales, from Flint to Aberystwyth: about 18–50 km (11–31 mi) apart in a straight line, most about 18–25 km (11–16 mi), all supplied by sea. The Teutonic Order’s castle-towns along the Vistula (Toruń 1231, Chełmno, Grudziądz, Kwidzyn, Malbork, Elbląg 1237): 25–40 km (15–25 mi) apart.
Ring around a hostile city	About 25–50 km (15–30 mi) from the city and 10–25 km (6–15 mi) from each other, all within a day’s ride of it	The four crusader castles around Fatimid Ashkelon (1133–42), which blocked its raiders until the city fell in 1153. Three of them stood about 25–32 km (15–20 mi) from the city; the fourth, Castrum Arnaldi, was nearer 45 km (28 mi) away.
Thinly settled frontier zone	Fortress towns 25–100 km (15–60 mi) apart, with open land between	The Arab <i>thughur</i> facing Byzantium (Tarsus, Adana, Mopsuestia, Malatya); the Duero frontier in 10th-century Spain

RULE OF THUMB

Peaceful core: a working castle every 20–30 km (12–19 mi). Contested border: one every 4–6 km (2.5–4 mi). Conquest chain: one every day’s march (20–40 km, 12–25 mi) along the main road, river or coast.

MAP TIP

Do not spread castles evenly. Cluster them along borders, at river crossings and around rich towns. Leave the safe middle of the kingdom fairly empty: a castle in each county town and a few baronial seats. On a frontier, draw a line of castles along the road or river the invaders used, about a day’s march apart.

Castles and towns

RULE OF THUMB

Castles and towns come in pairs. A lord who builds a castle wants a market and rents beside it; a town wants protection. Where one exists, look for the other.

Three common patterns:

1. **Castle planted in an existing town.** After 1066 the Normans built most royal castles inside or beside English towns, to control them. They demolished houses to make room: 166 at Lincoln, 113 at Norwich, 27 at Cambridge (Domesday Book).
2. **New town planted beside a new castle.** This is the conqueror's method. Edward I built Conwy, Caernarfon, Harlech and Beaumaris each with a new English borough beside it. At Conwy a royal charter of 1284 made the castle's constable (the officer in charge of the castle) also the town's mayor. The Teutonic Order did the same in Prussia: castle first, then a chartered town under its

walls. Many of these towns used Kulm law, a model town charter first issued for Chelmno (Kulm) in 1233.

3. **Village gathered around a castle.** In Italy many villages moved into fortified hilltop sites (*incastellamento*). In south-west France nearly 700 planned new towns (*bastides*) were founded in 1222–1372, many walled later. See [Where Settlements Are Built](#).

How they sit together on the ground:

- The castle is at the **edge** of the town, not the middle: usually at the highest point, at a corner of the town wall, or by the river or bridge.
- The castle has its own gate to the outside. The lord could bring in troops even if the town rebelled.
- The town grows along the road leading from the castle gate, with a market place in front of the gate.
- In a capital, the pattern is often “castle on the hill, merchant town below” (see [Capitals, Realms and Borders](#)).

MAP TIP

On a local map, draw the castle on the high ground or river bank at one corner of the town wall, with the market street running away from its gate. On a kingdom map, nearly every town of 2,000+ people should have a castle, a citadel or at least walls (see [Population and Settlement Sizes](#)).

Garrisons and armies

Garrisons

RULE OF THUMB

Garrisons were tiny. The walls did the work. A castle with 30–60 men could often hold off an army ten or more times bigger for weeks or months, if it had food and water.

Situation	Garrison	Real examples
Peacetime caretaker	5–20	A constable (the officer in charge), a porter, watchmen and a few men-at-arms (armoured professional soldiers)
New frontier castle in peacetime	30–60	Harlech 1284: 36 (a constable, 30 soldiers of whom 10 crossbowmen, a chaplain, a smith, a carpenter and a mason). Conwy 1284: 30 soldiers, 15 of them crossbowmen, plus a chaplain and craftsmen. Caernarfon 1284: 40.
Small castle under siege	20–150	Caernarfon in 1403–04: a garrison of only about 30 men held out against Welsh besiegers of Owain Glyndŵr, who had French help.
Great castle in a major war	150 to about 1,200	Dover 1216: 140 knights plus many more soldiers held out against the French prince Louis. Kenilworth 1266: a garrison usually estimated at about 1,200 held out for about six months (June to December).
Great crusader fortress	1,500–2,200	Krak des Chevaliers: about 2,000 at its peak, but only about 60 knights. Safed (rebuilt by the Templars 1240–43, described c. 1260): 1,700 in peace and 2,200 in war, of whom only 50 were knights.
Frontier fortress city	4,000–5,000	Tarsus, the main Arab base facing Byzantium. In early Abbasid times (late 700s and early 800s) the whole Arab frontier zone held about 25,000 troops.

Most people in a big garrison were not knights. They were crossbowmen, sergeants, servants, craftsmen and (in the East) slaves and workers.

Field armies and population

Campaign	Army (modern estimates)	Realm population	Share of population
Hastings, 1066	William: 7,000–8,000 (some say 10,000–12,000). Harold: 5,000–13,000, most say 7,000–8,000.	England 1.5–2.0 million	~0.4% (Harold)
Falkirk, 1298	Edward I: about 15,000 (about 2,000–2,500 cavalry and about 12,000 paid foot soldiers, many of them Welsh). Some estimates are lower.	England and Wales ~5 million	~0.3%
Crécy, 1346	English: 7,000–15,000 (Ayton: ~14,000)	England ~4.8 million	~0.3%
Siege of Calais, 1347	English: a peak of about 32,000 in July 1347, the largest English army sent overseas before 1600	England ~4.8 million	~0.7%, for a few weeks
Agincourt, 1415	English: about 6,000–9,000 (Curry: about 8,500–9,000). French: about 12,000 (Curry) or 14,000–15,000 (Rogers, Barker, Sumption); up to ~24,000 if armed servants are counted	England ~2–2.5 million; France well over 10 million	English ~0.3–0.4%; French under 0.3%

RULE OF THUMB

A medieval realm puts about 0.2–0.5% of its people into a field army. About 1% is an all-out effort that cannot last. A kingdom of 4 million people fields 10,000–20,000 soldiers, not 100,000. Chroniclers often claimed much bigger numbers; modern counts from pay records are usually several times smaller.

LATER ERA (1500s+)

Armies grew fast. Gunpowder forts needed bigger garrisons, and states began to keep permanent troops in **garrison towns**. Field armies of 20,000–50,000 became common in major wars of the 1500s and 1600s.

MAP TIP

You do not draw garrisons, but they tell you what a castle is. On a kingdom map, an ordinary castle is a post of a few dozen men, not a barracks for thousands. Put big troop numbers only in two or three great frontier fortresses and the capital's citadel. If you label armies or invasion routes, keep the sizes realistic: about 0.2–0.5% of the realm's people. A kingdom of 2 million fields about 4,000–10,000 soldiers.

Military geography and logistics

Speed and season

Army speeds follow the guide's shared baseline (details in [Trade Routes, Roads and Transport](#)):

Mode	Per day
Large army with baggage	13–20 km (8–12 mi)
Army living off the land	10–26 km (6–16 mi)
Forced march, a few days only	30–55 km (20–35 mi). Harold's march north to Yorkshire in 1066 covered about 300 km (185 mi) in perhaps 5–7 days (often said to be only four), about 43–60 km (27–37 mi) a day, probably partly mounted.
Mounted force with spare horses	55–65 km (35–40 mi)

Campaign season. Most campaigns ran from late spring, when there was grass for the horses, until after the harvest in autumn. Winter campaigns were rare. A vassal usually owed only **40 days** of military service a year, so feudal

armies melted away during long sieges. There were exceptions: the Teutonic Order often raided Lithuania in winter, because then the rivers and marshes were frozen and could be crossed. Galley fleets mostly stayed in port in winter.

Food and fodder

- A soldier needs about 1–1.5 kg (2–3 lb) of food a day, mostly bread or grain.
- A horse needs about 4.5 kg (10 lb) of grain a day if it can also graze. Without grazing it needs roughly double that, about 9–10 kg (20–22 lb) of grain and hay. On summer grass it can graze for much of its needs. Otherwise, hay and oats must be carried.
- An army of 10,000 men and 4,000 horses therefore needs about 10–15 tonnes of food and 18–40 tonnes of fodder **every day**. That is very roughly 30–100 cartloads a day (an estimate).
- A moving army can take only about 5–15% of the food in the country it passes through and still keep moving (Devereaux's estimate). So armies **keep moving**. An army that stops, for example for a siege, eats out the surrounding land within weeks.

RULE OF THUMB

Armies follow rivers and coasts. Bulk supply came by boat, so invasion routes, siege camps and frontier castles cluster along navigable water. Edward I placed every one of his royal castles in North Wales where ships could reach it. In 1294–95 Harlech was resupplied from Ireland by sea through its water gate.

Military sites behind the front

Site	What it is	Where to put it	Real examples
Muster point	Where the army gathers before a campaign	A royal town or castle near the frontier, on a main road or river, with grazing land around it	English armies gathered at Chester for North Wales, at Newcastle, Carlisle or Berwick for Scotland, and at Portsmouth or Southampton for France
Armoury and royal store	Weapons, armour, arrows, siege engines, tents	In the capital's citadel, or a major castle on the main route to the frontier	The Tower of London was the English crown's main armoury and store
Arsenal and naval base	A state shipyard that builds, stores and fits out war galleys	A sheltered harbour or river port in or next to the capital or main port	Venice Arsenal (traditionally founded in 1104, though the exact date is uncertain; enlarged from 1320 and again in 1473; about 45 ha / 110 acres, roughly 15% of the city; up to 16,000 workers claimed for the early 1500s). Barcelona's royal shipyard, the Drassanes (ordered in 1285, enlarged 1328–90). The French royal galley yard at Rouen, the Clos des Galées (founded by Philip IV from c. 1292–94).
Invasion port	Where a fleet gathers and sails, and where it lands	The nearest good harbour to the enemy coast, with roads behind it to bring troops and food	1066: William's fleet sailed from Saint-Valéry-sur-Somme and landed at Pevensey. 1346: Edward III sailed from Portsmouth and landed at Saint-Vaast-la-Hougue. 1415: Henry V sailed from Southampton to Harfleur. Calais, taken in 1347, stayed an English bridge-head until 1558.

MAP TIP

For each realm, mark one or two muster towns near each threatened border, one main armoury (usually the capital), and one naval base if it has a coast. Draw the obvious invasion route between two rival realms: the shortest sea crossing or the easiest river valley. Then fortify both ends of it.

FANTASY TWIST

Magic that feeds armies (conjured food, bottomless wagons) removes the river rule. Armies could then march straight across dry land and besiege for months. That one change would make castles far less useful. If you allow it, make it rare and costly. See [Fantasy Variants](#).

Camps and siege towns

RULE OF THUMB

An army that stays in one place for more than a few weeks turns its camp into a town, with streets, a market, workshops and a defended edge, all fed by boat or by the main road. Most of these camps vanish when the war ends. A few become real towns.

Why camps matter. A field army of 10,000–30,000 people (see [Garrisons and armies](#)) is bigger than almost every town in its own kingdom. When it stops for a long siege or for the winter, it needs shelter, food markets, smiths, carpenters and stables. It also needs a defended edge, because it can be attacked from the besieged town and by a relief army from outside. Merchants, servants and other camp followers come with it. So a long siege camp looks and

works like a town of timber, thatch and canvas. Like the castles in [Military geography and logistics](#), it sits where boats or a main road can supply it.

Kinds of camp

Kind	How long it lasts	What it looks like	Size	Real examples
Overnight marching camp	One night to a few days	Roman armies: rows of tents inside a bank and ditch. Medieval armies: tents and rough shelters, usually with only light defences	Roman camps from under 1 ha to 67 ha (165 acres); most 10–18 ha (25–45 acres)	Roman camps in Scotland, a day's march apart along the campaign roads
Winter camp	One winter, several months	Naturally strong ground beside a river, with boats drawn up, workshops and trading	Torksey: at least 55 ha (136 acres)	The Viking "Great Army" at Torksey, England, winter 872–73
Siege camp-town	Months, up to about a year	Timber houses and huts around the besieged town, a market, stables, a fenced or ditched edge, and the commander's own quarters	For an army of about 10,000–30,000	Calais 1346–47; Neuss 1474–75
Permanent siege town	Built to last	A small walled town on a planned grid, built to show the enemy that the besiegers will not leave	Santa Fe: about 335 × 260 m (1,100 × 850 ft), about 9 ha (22 acres)	Santa Fe near Granada, 1491

Camps that became towns (or nearly did)

- **Torksey, 872–73 (England; earlier than this guide's baseline period).** The Viking Great Army spent the winter on higher ground beside the River Trent, partly surrounded by marsh. The camp covered at least 55 ha (136 acres), bigger than York at the time, and held several thousand people, including warriors, craftworkers and traders. They repaired boats, melted looted silver and gold into ingots, made and traded goods. After the army left, Torksey grew into an important Anglo-Saxon *burh* (fortified town) with a pottery industry. Archaeologists see the camp as a trigger for the later town.
- **Calais, 1346–47 (northern France).** Edward III besieged Calais for 11 months (4 September 1346 to 3 August 1347). His army built a temporary town on firm ground in the marsh, along the causeway road about 800 m (half a mile) from the walls. They called it *Villeneuve-la-Hardie* ("Bold New Town"). It had timber halls for the king and nobles, market halls, stables and thousands of huts of brushwood and thatch. Markets were held twice a week, and observers said its meat and cloth markets looked like those of Arras or Amiens. At its height it held more people than any English town except London. Supplies came by sea: 853 ships with 24,000 sailors served the siege at one time or another. It did not survive as a town: once Calais fell, Edward expelled the French townspeople and settled English colonists inside the captured city instead.
- **Neuss, 1474–75 (Rhineland, Germany).** Charles the Bold, Duke of Burgundy, surrounded Neuss, a fortified town of about 4,000 people, with an army of about 14,000 (estimates run from 13,000 to 20,000). He expected a

siege of a few days or weeks. It lasted about ten months, from 29 July 1474 until a relief army sent by the Emperor and peace talks ended it in late May and June 1475. By mid-August 1474 the town was completely shut in. Charles kept his court in the camp in great luxury. His courtier Olivier de La Marche is quoted as calling the camp "like a city", with craftsmen, merchants, barbers, inns and taverns, and even windmills. Disease still spread through it.

- **Santa Fe, 1491 (near Granada, Spain).** In the last campaign against the Muslim kingdom of Granada, Ferdinand and Isabella set up their royal camp in April 1491 on the open plain (the *Vega*), about 10 km (6 mi) west of the city and within sight of its walls. They began building a real walled town almost at once. In July 1491 a fire destroyed much of the tent camp. Older accounts say the town was built only after the fire, but recent research shows building had already begun. Chroniclers say the work took about eighty days. In October 1491 the royal court's historian, Peter Martyr, described it as nearly rectangular, about 400 by 312 paces (about 335 × 260 m, 1,100 × 850 ft), with battlemented walls, moats and towers, a square in the middle and a gate in each of the four sides. Two main streets crossed at that square, and the moat was filled from the plain's irrigation channels. Granada agreed to surrender in November 1491. In 1492 the camp became a town (*villa*), and its land was shared out among about 200 settler households. Santa Fe still exists; some of its gates still stand, and the old grid can still be seen.

Roman marching camps as old earthworks

Roman armies on campaign built a defended camp at the end of each day's march. On a medieval map these camps are part of the older layer of the landscape: low banks in moorland or marks in fields (see [Religious, Cultural, Legal and Ancient Sites](#)).

- **How many:** nearly 500 are recorded in Britain, almost half of them in Scotland. No other Roman province has so many known, mostly because of aerial photography.
- **Shape:** the classic “playing-card” plan, a rectangle with rounded corners, made of one earth bank (**rampart**) and an outer ditch. Many real camps are less regular.
- **Gates:** most often 4 or 6, sometimes fewer (the Roman writer Hyginus describes a model camp with six). Each gate was a gap in the bank, usually guarded by a short bank and ditch set in front of it (a *titulus*), so attackers could not rush straight in.
- **Where:** beside Roman roads and forts, often just after a river crossing. Near some frontier forts, at river crossings and road junctions, are “gathering grounds” where armies camped again and again before a campaign.
- **Spacing:** camps built on the same campaign lie about a day's march apart: in Scotland mostly 11–24 km (7–15 mi), less across hills.

Camp size	Area	Roughly how wide (estimate)	Examples (Scotland)
Small	1–5 ha (2.5–12 acres)	100–220 m (330–720 ft)	Pennymuir II, 3.9 ha (9.7 acres)
Common	10–18 ha (25–45 acres)	320–420 m (1,050–1,400 ft)	Pennymuir I: almost 18 ha (44 acres), 520 × 343 m (1,710 × 1,125 ft), probably six gates
Large campaign series	25 ha (63 acres)	about 500 m (1,650 ft)	About 20 camps in two lines north of the Forth, from the Stirling area towards Stracathro
Very large campaign series	45–67 ha (110–165 acres)	670–820 m (0.4–0.5 mi)	Camps north of the Forth (45–54 ha) and on Dere Street, the main Roman road north through southern Scotland (67 ha)

How many soldiers? Pennymuir I is said to have had room for two full legions, roughly 8,000–12,000 men. That is very roughly 500–700 soldiers per hectare (estimate). So an army of 10,000 in a tight, ordered camp needs about 15–20 ha (35–50 acres). A looser medieval camp, with baggage, horses and camp followers, spread far wider: the Viking camp at Torksey covered at least 55 ha for several thousand people.

MAP TIP

On a local map of a siege, draw the besieged town first. Then draw the camp as a rectangle or ring of huts and tents on dry ground beside the main road or the river landing, a few hundred metres to about 1 km (0.6 mi) from the walls, with a market square and the commander's quarters inside. Give it a name, as real soldiers did (*Villeneuve-la-Hardie*). For 10,000 soldiers, allow about 15–20 ha, roughly 400 × 450 m (1,300 × 1,500 ft), and more for horses and followers. On a kingdom map a siege camp is only a symbol (a closed dashed rectangle with rounded corners), and only on a map with a date. On a later map, leave old camps as the same outline drawn grey and labelled “Camp” or “Roman Camp”, or, if the besiegers won and stayed, as a small grid-planned town with a religious or military name (*Santa Fe* means “Holy Faith”).

LATER ERA (1500s+)

Santa Fe's plan, two main streets crossing at a central square with a gate at each end, is often seen as a forerunner of the grid towns that Spain planted in the Americas. See [What Changes After 1500](#).

FANTASY TWIST

Where monsters roam the wilds, every army halt needs a defended camp, as the Romans built every evening. Draw chains of earthwork camps (closed dashed rectangles) a day's march apart along old campaign roads, and let a few of them grow into towns.

Outposts in wild and contested land

Frontier posts

Real frontiers were zones, not lines (see [Capitals, Realms and Borders](#)). Their military layout was usually:

- **Forward line:** small castles, towers and fortified villages at fords, passes and wells, held by settlers or military orders.
- **Rear line:** big fortress towns a day or two behind, where the army and stores were kept. The Arabs facing Byzantium called the forward zone the *thughur* (“openings”) and the rear zone the *awasim* (“the defences”).
- **No man’s land between:** thinly settled, raided every summer.

LATER ERA (1500S+)

The best-known early modern frontier belt is the Habsburg **Military Frontier** (German *Militärgrenze*, Croatian *Vojna krajina*), facing the Ottomans in Croatia and Hungary. Austrian troops began to garrison Croatian border forts in 1522. From the 1550s (1553 is the usual founding date) it became a permanent system, partly paid for by the Austrian and Bohemian lands. By the end of the 1560s the Habsburg border-fort system had about 100–120 forts of all sizes along a border of almost 1,000 km (620 mi), from the Adriatic to Transylvania. In the later 1500s about 20,000–22,000 paid soldiers held them. That is very roughly one fort every 8–10 km (5–6 mi) (estimate). Between the forts were small outposts and palisades at river crossings: in 1601–03 the local lord in Međimurje (northern Croatia), György Zrínyi,

Examples:

- **Iberia:** the Christian–Muslim frontier moved south over centuries, from the Duero valley (10th century) to the Tagus (Toledo taken 1085) and finally Granada (1492). Castile’s own name comes from its castles. Fortress towns, castles and military orders (Calatrava from 1158, Santiago from 1170) held each new frontier. The Granada frontier lasted about 250 years (1240s–1492). Both sides lined it with castles and *atalayas* (watchtowers).
- **Prussia and the Baltic:** the Teutonic Order advanced from 1231 along the Vistula and the coast, building a castle and a town together at each step.

set up nine outposts along the Mura to guard its crossings, several of them with 100 guards each. Behind the forts the Habsburgs settled Christian refugees from Ottoman lands as **armed peasants**: free land and no feudal dues in return for military service. The first came in the 1530s (to Žumberak), a large wave came in 1597–1600, and the *Statuta Valachorum* (“Vlach Statutes”) of 1630 obliged every man over 16 to serve. New star forts anchored the belt, such as Karlovac (begun 1579 as a six-pointed star), the headquarters of the Croatian frontier. The Military Frontier lasted until 1881. On a 1500s–1600s map, draw such a frontier as a line of forts and posts at the crossings, with a band of soldier-villages behind it. See [What Changes After 1500](#).

Toll castles on rivers

- **Where:** narrow, busy stretches of navigable river, especially in gorges where ships cannot avoid the bank.
- **Numbers:** the Rhine Gorge (between Bingen/Rüdesheim and Koblenz) has about 40 castles in 65 km (40 mi), one every 1.5–2 km (about 1 mi). Many of them collected tolls. Along the whole navigable Rhine, recorded toll stations grew from 19 at the end of the 1100s to 44 by 1300 and 62 by 1400.

Robber-knight castles

When royal power was weak, small lords used their castles to rob travellers, demand illegal tolls and hold merchants for ransom. Germans later called them *Raubritter* (“robber knights”).

- During the German Interregnum (1250–73), toll stations multiplied.

- **How it worked:** Pfalzgrafenstein (built 1326–27 by King Louis IV) stands on an island in the Rhine at Kaub. A chain could block the channel. Ships paid at the customs house in Kaub. Ships that refused were seized, and their crews were held in the castle’s dungeon until a ransom was paid.
- **Who owned them:** the archbishops of Mainz, Cologne and Trier, the Count Palatine and noble families. Tolls were a pillar of their power.

- In 1254 Rhine cities and princes formed the Rhenish League. It took and destroyed robber castles: about 10–12 strongholds within three years.
- In 1282 King Rudolf of Habsburg destroyed more robber castles, among them Sooneck.

Military-order houses

- **Frontier castles:** the orders held the hardest frontier castles: Krak des Chevaliers, Safed, Montfort, Calatrava, Malbork.
- **Commanderies in the home countries:** most of the orders' houses in western Europe were not forts. A com-

MAP TIP

On a frontier, mark the march with hatching or a fading colour band, the guide's symbol for a frontier zone (see [Capitals, Realms and Borders](#)). Put small forts at every ford, pass and well along its front edge, and two or three fortress towns a day or two's march behind them. The whole military zone, from the forward forts back to the fortress towns, is then about 30–80 km (20–50 mi) deep. Leave the land in front nearly empty. On a busy river gorge, crowd the cliffs with toll castles (small castle icons, with a toll bar and "T" across the river). Put one military-order commandery in the countryside of each inland county, drawn as a small church with a shield, not as a castle.

mandery (or **preceptory**) was a manor farm with a chapel and a small community, run to send money, horses and food to the frontier. The Templars had nearly 1,000 commanderies and fortifications at their height, and 15,000–20,000 members, only about one-tenth of them knights.

FANTASY TWIST

A wilderness full of monsters works like a hostile neighbour. Its edge becomes a frontier zone even inside one realm, with forward posts, refuge towers and fortified villages, and a rear line of strong towns. Where dwarves or others hold mountain passes, the pass fort may belong to them, not to the human kingdoms on either side.

Watchtowers, beacons and coastal warning

RULE OF THUMB

Warning beats walls. A chain of fires on hilltops could carry a message hundreds of kilometres in an hour, faster than any rider.

Beacons. A beacon is a pile of fuel or an iron fire basket on a hill or tower, kept ready to light.

- **Byzantine beacon line (9th century, under Emperor Theophilos):** about 720 km (450 mi) from the fortress of Loulon near the Cilician Gates to Constantinople, with about nine main stations. Stations stood about 55–100 km (35–60 mi) apart on high peaks: closer in rugged country, farther in open country. Modern experiments suggest a message could cross the whole line in about one hour. Each of the 12 hours of the day stood for a set message. The hour at which the first fire was lit told the capital which message was meant.

LATER ERA (1500s+)

Dense chains of coastal towers are mostly a 16th-century answer to Barbary and Ottoman raids. Genoa built about 85–90 towers around Corsica in 1530–1620 (a 1617 list counts 86). Each was usually about 12 m (39 ft) high and held 2–6 guards. Smoke, fire or a conch horn could put the whole island on alert within a few hours. Spain and the Kingdom of Naples built similar chains.

- **Scottish Borders (Act of 1455):** one bale (fire) meant the English were coming; two bales together meant they were "coming indeed". Watches were kept at fords and passes.
- **Islamic coasts:** ribats were built at intervals along the North African shore so they could signal to each other. From Monastir's watchtower, signals were exchanged at night with neighbouring ribats. Sousse, the next big ribat up the coast, is about 20 km (12 mi) away.

Spacing. Beacon intervals depend on height and haze. In ordinary hilly country, plan for about **5–20 km (3–12 mi)** between beacons. Between high mountain peaks in dry air, 55–100 km (35–60 mi) is possible. These spacings are estimates.

Watchtowers stood at passes, fords, wells and frontier hilltops. They held a handful of men whose job was to see, signal and run, not to fight.

MAP TIP

Draw beacon chains as flame dots on hilltops leading from the border or coast to the nearest castle and on to the capital. Label a few hills "Beacon Hill" or "Watch Hill"; such names are common in Britain. On a raided coast, add a watchtower (a small triangle) on every headland that can see the next one.

Where battles happen

RULE OF THUMB

Battles happen where armies must pass or where one side must save a besieged place. Pitched battles were rare and risky. Sieges and raids were normal war.

Historians such as John Gillingham point out that many successful commanders avoided pitched battle almost entirely. When battles did happen, they clustered in a few kinds of place:

Where	Why	Real examples
River crossings	The attacker must cross; the defender waits on the far bank	Stirling Bridge (1297); the fight at the Blanchetaque ford before Crécy (1346)
Near a besieged castle or town	The relieving army must come to the siege, and the besiegers must stop it	Lincoln (1141 and 1217); Muret (1213); Bannockburn (1314, near besieged Stirling Castle)
On the main road to a capital or a port	One army blocks the other's line of march	Hastings (1066, on the road to London); Agincourt (1415, on the road to Calais)
Near water in dry country	Armies must drink	Hattin (1187), where the crusader army ran out of water

The defender usually picks ground with a slope in front and protected flanks: woods, villages, marsh or a river. At Crécy the English stood on a slope with their flanks resting on

two villages. At Agincourt woods on either side narrowed the field.

MAP TIP

Name a battlefield after the nearest village and mark it with crossed swords and the year. Put it near a ford, a bridge, a road junction or a besieged castle, never in the middle of nowhere. Three or four famous battlefields in a kingdom's history are plenty.

Drawing military sites on your map

Suggested symbols

These symbols match the guide's [master legend](#).

Feature	Symbol	Notes
Great royal castle or fortress	Large castle icon (three towers)	Label by name
Ordinary castle	Small castle icon (one tower with battlements)	
Motte or small earthwork castle	Small mound with a tower tick on top	Local maps only. A whole motte-and-bailey is only about 100–150 m (330–490 ft) long (estimate), about 1–1.5 mm at 1 cm = 1 km, so use a symbol, not a plan (see Footprints)
Tower house, peel or bastle	Small solid square	Many in raided borderlands
Fortified or moated manor house	Small open square; double outline if moated	An open square, so it does not look like a tower house
Walled town	Town symbol (a circle with a ring) with battlements on the ring	A city has a thick battlemented ring
Citadel	Small castle icon at the edge of the town ring	Not a star: a filled star means a capital
Fortified bridge	Bridge symbol with a small tower	
Fortified church	Church with battlements	
Watchtower or coastal tower	Small triangle	Draw chains of them; on a raided coast, one on each headland (coastal chains are mostly 1500s+)

Feature	Symbol	Notes
Beacon	Flame dot	Draw chains of them from the border or coast to the capital
Military-order commandery	Small church with a shield	A manor farm, not a fort. Use a castle icon only for the orders' real frontier castles
Arsenal or naval base	Anchor plus a castle tower	
Siege castle	Small dashed castle near its target	Short-lived; show only for a dated map
Army or siege camp	Closed dashed rectangle with rounded corners, labelled with its name or "camp"	Dated maps only. Always a closed shape, never an open line. An old Roman or abandoned camp: the same outline drawn grey, labelled "Camp"
Battlefield	Crossed swords and the year	
Star fort or fortress town (1500s+)	Five- or six-pointed star outline, or a star-edged town outline; never filled	Later era only
Coastal artillery fort (1500s+)	Small round outline with rounded lobes (a rosette, like the plan of Deal Castle, 1539–40); never pointed	Later era only. At harbour mouths, anchorages and river mouths. Pointed outlines mean star forts: a fort with angled bastions takes the star outline
Ruin	The feature's own symbol, broken or grey, labelled "(ruin)"	

Ruined and active castles

- In England only about one-third of all castle sites were in use at any one time. The rest were abandoned, never finished, or replaced.
- So a map dated 1300 should show many **abandoned mottes and ringworks** from the 1070s–1150s, especially in old border zones that are now peaceful.

- By 1450, many 12th-century castles were half-ruined. The crown kept up only the important ones.
- Old hillforts and Roman forts are often reused (Old Sarum, Pevensey). Others stay as grassy earthworks with names like "Castle Hill" or "Camp" (old army camps are covered in [Camps and siege towns](#)). Ancient ruins in general are covered in [Religious, Cultural, Legal and Ancient Sites](#).

RULE OF THUMB

For every active castle, draw one or two ruins or earthworks in long-settled land. In a peaceful core, more ruins; in a newly conquered frontier, almost none (everything there is new).

How many to draw

Per 1,000 km² (about 390 sq mi) of land, at about 1300:

Zone	All sites ever built	Active castles	Show on a kingdom map	Show on a local map
Peaceful lowland core	4–8	1–3	The county town's castle, plus 1 major baronial castle per 2,000–3,000 km ²	All active castles plus ruins
Average (England and Wales overall)	~12	~4	Royal and major castles only, about 1 per 1,000 km ²	All
Hilly lands with many small lords (German type)	~25	~8–12 (estimate)	Major castles and toll castles	All, clustered on spurs and river cliffs
March or contested border	35–60	12–20	All major castles; draw the border as a hatched band studded with castle icons	All, including mottes and towers
Raided border of the late Middle Ages	8–10 castles plus many tower houses	Most of them	Castles and the main tower houses	Every tower house and bastle
Thin conquered frontier (Prussia type)	1–3	1–3	All of them, each paired with a town	All

The “active” column assumes about one-third of sites are in use at once, as in England. It is an estimate for other regions.

Map scale guide:

- **Continent map:** show only capitals’ citadels, the greatest frontier fortresses, famous fortress cities and naval

MAP TIP

Check your castles with three questions. (1) What does each one control? (2) Can it be supplied by water or from rich farmland nearby? (3) Is the density right for the zone: sparse in the core, dense on the border? Common errors are covered in [Common Mistakes and How to Fix Them](#).

arsenals. As a suggestion, about 1 per 20,000–50,000 km² (8,000–19,000 sq mi).

- **Kingdom map:** add royal castles, county-town castles, border fortresses and the main border chain.
- **Local map:** show everything: castles, ruins, mottes, tower houses, fortified churches, beacons and watchtowers.

FANTASY TWIST

If mages can break walls like cannon, castles should evolve early into low, thick, star-shaped forts. If enemies can teleport or fly over borders, strong citadels inside cities matter more than border chains. Long-lived races (elves, dwarves) may keep very old fortresses in use, so their lands show fewer ruins and older styles.

Quick summary

- Every fortification controls something: a route, a crossing, a town, a border, an estate, a refuge or a lookout. If it controls nothing, delete it.
- Types change with time: earth-and-timber mottes and ringworks (c. 1000–1150), stone keeps and curtain walls (1150–1250), concentric castles and town walls (1250–1350), then tower houses and fortified manors (1350–1500). From the 1500s come star forts and artillery forts.
- Best sites: fords, bridges, passes, spurs, river bends, confluences, headlands, and hills above towns. German castles were mostly on high ground; English castles mostly beside villages and towns.
- England and Wales had about 1,761 castle sites (one per ~86 km², or 33 sq mi), but only 500–600 were in use at one time: an active castle every 16–17 km (10–11 mi).
- Border zones are roughly 5–10× denser: the Welsh March had one castle site per 17–28 km² (7–11 sq mi), a gap of 4–6 km (2.5–4 mi). In a peaceful core, working castles stand about 20–30 km (12–19 mi) apart. Conquest chains put a castle about every day’s march (20–40 km, 12–25 mi) along a coast or river.
- Castles pair with towns: castle at the edge of town, on the high point or by the river, with the market at its gate.
- Garrisons were tiny: 5–20 in peace, 30–60 for a new frontier castle, a few hundred (rarely up to about 1,200) in a great siege. Only giant frontier fortresses held 2,000 or more.
- Field armies were about 0.2–0.5% of a realm’s population: 10,000–15,000 for England in the 1300s, briefly 32,000 at most.
- Armies march 13–20 km (8–12 mi) a day, campaign from late spring to autumn, and follow rivers and coasts for supply. A stationary army eats out the land within weeks.
- Long sieges and winter quarters turn camps into temporary towns with markets (Calais 1346–47, Neuss 1474–75); a few became real towns (Torksey, Santa Fe 1491). Roman marching camps (nearly 500 known in Britain, mostly 10–25 ha, a day’s march apart) survive as rectangular “camp” earthworks.
- Add muster towns near borders, an armoury in the capital, an arsenal at the main port, toll castles on river gorges and beacon chains from the border to the capital.
- Battles happen at crossings, near sieges and on roads to capitals and ports, and are named after the nearest village.

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Population figures, travel speeds, total castle counts, the control-radius heuristic and the Harlech and Safed garrisons follow the guide's shared baseline. Figures marked "estimate" (gaps between castles, active-castle densities outside England, cartloads per day, beacon spacing, army shares of population, distances between Edward I's and the Teutonic Order's castles, distances from Ashkelon to its ring of castles and from Santa Fe to Granada, measured in straight lines on a modern map; the widths of Roman camps, worked out from their areas; soldiers per hectare and camp

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CHAPTER 5

TRADE ROUTES, ROADS AND TRANSPORT

Roads, rivers and sea lanes are the blood vessels of your map. In the Middle Ages, moving goods over land was slow and expensive, and moving them by water was cheap. That one fact explains where cities grow, where roads go, which places become rich, and which stay poor. This chapter gives you travel speeds and costs, the spacing of inns, tolls and ports, and real trade networks you can copy.

The baseline period is the High and Late Middle Ages (c. 1000–1500), mainly in Europe. Site types such as ford towns, bridge towns and head-of-navigation towns are explained in [Where Settlements Are Built](#). How much land a town needs to feed itself is in [Population and Settlement Sizes](#). Castles that guard roads are in [Castles, Forts and Military Outposts](#). Pilgrim shrines are in [Religious, Cultural, Legal and Ancient Sites](#).

IN THIS CHAPTER

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 Roads
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 Mountain passes
 Inns, hospices and caravanserais
 Roads without wheels: the Andes and Mexico
 Tolls and customs
 Rivers as roads
 The sea: ships, harbours and sea lanes
 Trade nodes: markets, fairs, staples and entrepôts
 What is traded
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How fast and how expensive: travel and transport

RULE OF THUMB

Over land, a traveller covers about 30 km (19 mi) a day and a loaded ox cart about 20 km (12 mi). By water, goods move at least as fast for a small part of the cost. Heavy, cheap goods go by water or stay at home.

Speed per day

These are the guide's shared numbers. Use them for every journey on your map.

Mode	km per day	mi per day	Notes
Walking traveller	25–35	15–22	Archbishop Sigeric's journey from Rome to Canterbury (c. 990): about 1,700 km (1,060 mi) in 80 stages, so an average stage was about 20 km (12 mi). A pilgrim's stage was often shorter than a hurried traveller's day
Peasant to market	6–10 one way	4–6	Out, trade and home in one day
Ox cart	15–25	10–15	Slow, strong and cheap to feed
Horse cart or wagon	30–40	20–25	Horses can double the speed of oxen
Pack horse or mule train	20–40	12–25	Loads: horse 100–120 kg, mule 150–180 kg, donkey 80–100 kg
Camel caravan	25–40	15–25	About 230 kg per camel
Cattle or sheep drove	16–30	10–20	About 3 km/h (2 mph), grazing on the way: about 16 km (10 mi) a day in hills, up to 30 km (20 mi) in lowlands. Figures mostly from the 1700s–1800s; see Drove roads
Army with baggage	13–20	8–12	Forced march 30–55 km (20–35 mi) for a few days
Rider, many days	30–50	20–30	—

Mode	km per day	mi per day	Notes
Royal messenger with fresh horses	50–90	30–56	30–40 mi normal, 52–56 mi when urgent
River boat, downstream	40–100	25–60	Estimate; travels by daylight
River boat, upstream, towed	10–20	6–12	Danube, Vienna to Passau: about 300 km (190 mi) in 20–31 days
Galley along a coast	about 55 (voyage average with stops)	about 30–35	Venice to Jaffa pilgrim galley: 48 days (1480)
Sailing ship (cog or round ship)	40–100 voyage average; 90–220 while under way	25–60; 55–135	Candia (Crete) to Alexandria, about 600 km (370 mi), in 4 days (1323)

How long does it take to cross your map?

These are rough ranges from the table above. Add rest days, Sundays, bad weather and waiting for wind.

Distance	On foot	Ox cart	Courier with fresh horses	Galley	Sailing ship
100 km (62 mi)	3–4 days	4–7 days	1–2 days	about 2 days	1–3 days
300 km (190 mi)	9–12 days	2–3 weeks	3–6 days	5–6 days	3–8 days
1,000 km (620 mi)	4–6 weeks	6–10 weeks	1.5–3 weeks	2.5–3 weeks	1.5–3.5 weeks

A sailing ship can be faster than anything on land, or it can sit in port for two weeks waiting for a fair wind. Land travel is slower but more predictable.

Cost: why water beats land

Rule: Per tonne and per kilometre, land transport cost several times more than river transport, and river transport cost more than sea transport.

Why: A pack horse carries about 100–120 kg. It needs a driver, food, a stable and a smith to shoe it. The Bremen cog of 1380 (a cog is the main northern cargo ship) carried about 90–130 tonnes. A rule of thumb gave one sailor per 10

tonnes of cargo, so it needed a crew of only about 10–13 sailors, perhaps a few more in practice. One small ship moved as much as about 1,000 pack horses. For comparison, one modern estimate puts the goods that crossed the Brenner Pass, a main Alpine route, at about 3,000 tonnes a year in the 14th century. Treat that figure as a rough order of size. If it is right, a whole year's traffic over a main pass was the cargo of only 25–30 cogs.

Numbers:

Comparison	Ratio	Source and place
Land : river : sea, per ton-mile (the cost of moving one ton one mile)	about 8 : 4 : 1	England, 1290s–1340s (Masschaele)
Water against land	Water costs about one-tenth of land	England (Campbell et al.; Clark)
Sea : river : land	about 1 : 5 : 28–56	Roman Empire, Diocletian's Price Edict, 301 AD

• **Grain by cart, England c. 1300:** each 10 mi (16 km) of carting added about 3.3% to the value of wheat. The cost doubled the price after about 480 km (300 mi). This is the best case: horses, carts and flat land.

• **Grain by wagon, Roman Empire:** carting wheat 100 Roman miles added 55% to its price. The cost doubled the price after about 270 km (170 mi).

- **By sea:** shipping a quarter (about 220 kg) of grain from Exeter to London, about 560 km (340 mi) by sea, cost the same as carting it 65–70 km (40–45 mi).

Map advice:

- Bulk goods (grain, timber, stone, salt, wine) follow water.

FANTASY TWIST

A griffon or a teleport circle that carries 100 kg is a pack horse with wings. It changes the speed of news, letters and luxuries, not the price of bread. Only magic that moves *bulk* cheaply (wind-calling for ships, golem wagons, river spirits that tow barges) lets big cities grow away from water. See [Fantasy Variants](#).

- Bulk goods travel more than 30–50 km (19–31 mi) over land only when they are valuable, or in war or famine.
- Most towns above 10,000 people sit on navigable water or the coast (inland exceptions: see [chapter 02](#)). See [feeding towns](#).

LATER ERA (1500S+)

Organized relay posts with fresh horses at every stage made messages much faster. The Taxis family set up their first permanent relay line in 1490, from Innsbruck towards Mechelen in the Low Countries, with horse stations roughly every 37 km (23 mi). By about 1500 such posts could cover up to about 150 km (95 mi) a day. English royal post stages were about 32 km (20 mi) apart. Goods still moved at the old speeds.

Roads

RULE OF THUMB

A medieval “road” was a right to pass, not a built thing. Only old Roman roads, causeways and bridges were engineered. Everything else was a strip of earth worn by use.

Why: In England, the “king’s highway” was a legally protected route on which everyone had the right to travel. It did not mean a paved surface. Kings protected roads by law, not by building them.

Kinds of roads

Type	What it is	Where	How to draw it
Old Roman road	A straight, engineered road on a raised bank (an <i>agger</i>). Still used, often decayed	Former Roman lands: Italy, Gaul, Spain, Britain, the Rhineland	Straight thick solid line, labelled “Street” or “(Roman)”; grey where abandoned. Real names: “Street”, “Strata”, “Chaussée”
Royal highway (king’s highway)	A main route between cities and towns, under the ruler’s protection	Between major towns	Thick solid line
Local road or lane	Village to market town, village to village, field access	Everywhere	Thin solid line
Hollow way (holloway, sunken lane)	A lane worn below the land around it by feet, hooves, wheels and rain. Some are 5 m (16 ft) deep	Soft rock (chalk, sandstone, loess) on slopes near towns and fords	Two thin parallel solid lines (local maps only)
Causeway	A raised road across marsh or a flood plain, often with small bridges	Fens, flood plains, approaches to bridges	Solid road line on a bank across the marsh
Pack-horse track or bridle way	A narrow track for animals, not carts; narrow pack-horse bridges	Hills and mountains	Thin dotted line
Drove road	A wide grassy route for walking cattle or sheep to distant markets	From uplands to towns	Broad dotted line (two rows of dots)
Caravan route	A track from well to well and from caravanserai to caravanserai	Desert and steppe	Dotted line with a larger dot at each stage

The Roman legacy

At its height the Roman road network had more than 400,000 km (250,000 mi) of roads, of which more than 80,500 km (50,000 mi) were stone-paved. Official lodges

(*mansiones*) stood about 25–30 km (16–19 mi) apart. After Rome fell, surfaces decayed and many bridges collapsed, but the **routes** stayed in use because they linked the old Roman towns.

In 12th-century England, law books placed **four highways** under the king's special peace: Watling Street, Ermine Street and the Fosse Way (all Roman) and the Icknield Way (prehistoric). In Italy the Via Aemilia still formed the spine of the Po plain.

Medieval people did draw road maps. The Gough Map of Britain shows about 190 thin red lines between towns, nine out of ten marked with a distance in Roman numerals. Its date is uncertain. It is conventionally dated c. 1360–70, but

Who maintained roads

- **Landholders.** Repairs usually fell on local lords, landholders and communities, not on the king. England's Statute of Winchester (1285) added a security rule. On highways between market towns, the landholders had to clear ditches, trees and bushes for **200 feet (about 61 m) on each side**, so that robbers could not hide.
- **Towns.** Towns paid for paving their streets and the roads just outside the gates, often with a special toll called *pavage*.

Drove roads

RULE OF THUMB

A drove moves at the speed of a grazing cow: about 16–30 km (10–20 mi) a day, with a field or pasture for the night at the end of each day's stage.

Why: Live animals walk themselves to market, so they can cross a kingdom over land far more cheaply than grain. But they must eat and drink on the way, or they arrive thin and sell for less. So drovers move slowly, let the animals graze, and keep to open hill tracks and broad green lanes. They avoid towns, crops and, where they can, tolls. A 19th-cen-

a study of its handwriting (Smallwood 2010) suggests it was made one or two decades after 1400, and parts were redone or added later in the 15th century (Solopova 2012; Delano-Smith et al. 2017; both summed up by Oksanen & Brookes 2025). It is a good model for a “period” style map: towns, rivers, and the main routes between them with distances. It also shows how much main road to draw on a kingdom map (see [How much main road to draw](#)).

- **Charity.** Building or mending roads and bridges counted as a pious work. For example, in 1474 Maud Heath gave land and houses to pay for a raised footway (a causeway) from Wick Hill across the flood plain of the River Avon to Chippenham in Wiltshire. A charity still looks after it from her gift, more than 500 years later.
- **Lords who took tolls** were supposed to keep the road or bridge in repair. Many did not.

tury saying about Welsh drovers was that “they’d lose a day goin’ round sooner’n they’d pass a gate”: they would rather lose a day going round than pay at a toll gate.

When: Cattle were driven from Wales to England from at least the 13th century. There are records of cattle driven from Wales to London, and sheep from Lincolnshire to York, in the early 14th century. Scottish drovers were being licensed by 1359. The great age of droving came later, in the 1600s–1800s, and most of the numbers below come from that time. Use them as the best guide we have for the Middle Ages too.

Item	Number	Where and when
Speed	About 3 km/h (2 mph). About 16 km (10 mi) a day in steep hill country, up to 30 km (20 mi) a day in the lowlands	Scotland and Wales, 18th–19th c.
Cattle herd	Usually 100–400 head, with 4–8 drovers and their dogs	Welsh droves, 19th c.
Sheep flock	1,500–2,000 sheep, about 19–23 km (12–14 mi) a day	Welsh flocks driven to southern England
Journey time	North Wales to Kent: about three weeks	Same
Overnight halts (<i>stances</i>)	About 16 km (10 mi) apart on average	Scotland, evidence in a court case of 1848
Width of a drove road	12–27 m (40–90 ft), much wider than a pack-horse track	Britain
Hoof care	Cattle were shod before the long trek, and re-shod on the way	Wales, 19th c.

A Mediterranean example. In Castile, the Mesta (a guild of sheep owners, with a royal charter from 1273) moved huge flocks between summer pastures in the north and winter pastures in the south. From 1436 to 1549 more than 2.5 mil-

lion sheep made this journey each year. Flocks from León and Old Castile walked 550–750 km (340–470 mi) each way, along legally protected sheep walks (*cañadas*) that no one could farm, build on or block.

MAP TIP

Draw a drove road as a broad dotted line (two rows of dots) from upland breeding grounds (hills, moors, mountain pastures) to a lowland fair or a big city. Put a halt with a pasture and water every 16–25 km (10–15 mi), and a large livestock

fair near the end. Keep the route off the main highway where you can: along ridges, across commons and around towns. Inns along it can have names like “Drovers’ Arms”.

Road quality by season

- **Summer:** dry, hard roads; carts move well.
- **Autumn to spring:** clay roads turn to deep mud. Carts get stuck, so many carriers switch to pack animals. Floods close fords for days.

- **Northern winter:** in Russia and Scandinavia, winter was often the *best* time for land travel. Sledges ran fast over frozen rivers, lakes and marshes.

MAP TIP

Do not draw roads as straight lines between cities. Real roads follow valleys and dry ridges, bend to reach fords and bridges, and avoid marshes and steep slopes. Only an older empire’s roads run straight across country. If your world had one, draw its roads straight and let medieval roads reuse some of them and wander elsewhere.

LATER ERA (1500S+)

England’s Highways Act of 1555 made each parish choose unpaid surveyors of highways, and made parishioners give 4 days of labour a year to the roads (6 days from 1563). In France, Henry IV made the Duke of Sully *grand voyer* (chief officer of roads) in 1599. Main roads got better, but most were still dirt until the 1700s.

Crossing rivers: bridges, fords and ferries

RULE OF THUMB

Every crossing is a magnet. Roads bend to reach it, a town grows at it, and someone takes a toll there.

Fords

A ford needs firm gravel and water shallow enough for carts at normal levels. Floods close it. Place names show fords: English *-ford* (Oxford), German *-furt* (Frankfurt), Dutch *-*

voorde. On big lowland rivers good fords are rare, so a ford that serves several roads is a strong place for a town (see [ford towns](#)).

Ferries

Where a river is too deep or wide for a ford and too expensive to bridge, there is a ferry. Ferry rights belonged to a lord, a town or a monastery and earned money. For example, in 1330 Edward III gave the monks of Birkenhead Priory the right to run a ferry across the Mersey.

On very big rivers, ferries were normal. On the Rhine below Basel there were only a handful of fixed bridges in the Middle Ages. Cologne, the biggest city on the river, had no fixed Rhine bridge from Roman times until 1859.

Bridges

When were stone bridges built? Early medieval bridges were mostly timber. A boom in stone bridges began in the 12th century.

Bridge	Date	Notes
Stone Bridge, Regensburg (Danube)	1135–1146	A model for later stone bridges
Judith Bridge, Prague	1158–1172	Badly damaged by a flood in 1342; replaced by the Charles Bridge, begun in 1357
Old London Bridge (Thames)	1176–1209	19 arches and a drawbridge, about 282 m (926 ft) long, a chapel, and houses along it
Pont Saint-Bénézet, Avignon (Rhône)	First bridge 1177–1185 (probably timber); rebuilt in stone from 1234	About 900 m (2,950 ft) and 22 arches when complete
Middle Bridge, Basel (Rhine)	c. 1225	Part timber, part stone; the oldest bridge site still in use on the Rhine between Lake Constance and the sea
Krämerbrücke, Erfurt	1325 (in stone)	Lined with shops and houses
Ponte Vecchio, Florence	1345	Shops on the bridge

How many? For England, the historian David Harrison found that there were almost as many bridges in 1250 as in 1750. By the end of the Middle Ages most were solid, well-built structures that were repaired regularly, and most important crossings in a settled lowland country had a bridge. On the biggest rivers, bridges stayed rare. London Bridge was the only Thames bridge in London until Putney Bridge opened in 1729; the next bridge upstream was at Kingston, about 33 km (20 mi) up the river (see [the Thames example](#) below).

unloaded *below* the bridge, so the port grew downstream of it.

Who paid?

- **Old duties** on certain lands or villages to repair a bridge.
- **Tolls.** In England a *pontage* grant let someone take a toll for a few years to repair a bridge. About 370 such grants were made between 1228 and the 1440s.
- **Rents.** Houses on London Bridge paid for its upkeep.
- **Charity and trusts.** Gifts and wills paid for bridges. The Rochester Bridge Trust (founded 1399) still owns and maintains bridges there.
- **Bridge chapels.** A small chapel on or beside the bridge collected gifts for repairs, and travellers prayed there for a safe journey. They were not uncommon in the Middle Ages, but most were lost after the Reformation, when an Act of 1547 dissolved chantries (endowed chapels for prayers). Only six survive in England: Wakefield (1342–56), St Ives (1426), Rotherham (1483), Bradford-on-Avon, Derby and Rochester. The last two stand beside their bridges, not on them.

A bridge blocks ships. Old London Bridge had such thick piers that the water level could differ by up to 1.8 m (6 ft) from one side to the other, making rapids. Sea ships

How often is a river bridged? The Thames example

RULE OF THUMB

On a navigable lowland river in a settled kingdom, put a bridge about every 20 km (12 mi) of river, with gaps of 10–35 km (6–22 mi). Each bridge has a town. Put ferries and fords in the gaps.

Why: A bridge is expensive, so it is built only where main roads must cross. Each one pulls the roads of a whole district towards it. A second bridge close by would split the traffic and the tolls, so bridges spread out along a river.

Numbers: The River Thames from London to Oxford is about 180 km (112 mi) long by river. By about 1300 it had nine road bridges, and a tenth (Marlow) was added in the 14th century. That is one bridge every 20–22 km (12–14 mi) on average. In the gaps, people used ferries: in 1300 a ferrywoman called Sibille was paid three shillings to carry King Edward I across the Thames at Chertsey. After a bridge at Chertsey was licensed in 1410 and Abingdon's bridges were built in 1416–22, there were twelve, about one every 16 km (10 mi).

Bridge town	Distance from London Bridge by river (approx.)	First record of a bridge
London	0	Stone bridge 1176–1209 (earlier bridges since Roman times)
Kingston	33 km (20 mi)	12th century: a timber bridge on stone approaches
Chertsey	51 km (32 mi)	A ferry in 1300; a bridge licensed in 1410
Staines	57 km (35 mi)	Repairs with wood from Windsor Forest ordered in 1222
Windsor	69 km (43 mi)	Probably 12th century; tolls on boats passing under it recorded by 1172; oaks granted for a new bridge in 1242
Maidenhead	80 km (50 mi)	A wooden bridge c. 1280; "almost broken down" by 1297, when a toll (pontage) was granted for repairs
Marlow	91 km (57 mi)	Perhaps 1309; certainly by the reign of Edward III (1327–77)
Henley	104 km (65 mi)	Recorded 1232; timber on stone piers, with chapels and granaries
Reading (Caversham)	120 km (75 mi)	Recorded 1231, with a chapel of St Anne on it
Wallingford	145 km (90 mi)	Mentioned 1141; first stone bridge in the 13th century
Abingdon	166 km (103 mi)	Built 1416–22, with a second bridge at Culham and a causeway between them
Oxford (Grandpont, now Folly Bridge)	179 km (111 mi)	Stone bridge c. 1085

Distances are measured on today's river and rounded, so treat them as rough. Around 1300 the gaps between bridges ranged from about 11 km (7 mi), from Windsor to Maidenhead, to about 34 km (21 mi), from Wallingford to Oxford before Abingdon had its bridge.

A new bridge can make one town and break another.

Abingdon's bridges (1416–22) were paid for by a town guild and two of its members, a London merchant and his wife.

They replaced a ferry, so traffic that had crossed the Thames at Wallingford could now cross at Abingdon instead. The completion of the Abingdon bridge badly damaged trade at Wallingford, 21 km (13 mi) downstream.

On the greatest rivers, bridges are much rarer. Compare the Rhine below Basel, where Cologne had no fixed bridge for centuries (see [Ferries](#)).

MAP TIP

Use three symbols: a bridge (two short curved lines “)(” with the road running through), a ford (a short dashed line across the river) and a ferry (a short dotted line across the river with “F”). Put a settlement at almost every bridge on a main road. Put the main port of a river just **downstream** of the lowest bridge. On a settled lowland river, draw a bridge town about every 20 km (12 mi) of river, with ferries and fords between. On a great river, draw only a few fixed bridges, often many tens of km apart, and ferries in between.

FANTASY TWIST

A troll that charges travellers is just a strict toll collector. Traffic keeps using the bridge as long as the toll is cheaper than the detour. If the detour is short, the troll starves.

Mountain passes

RULE OF THUMB

A few passes carry almost all the traffic across a mountain range. The lowest, gentlest one takes carts; the high ones take only mules. Each pass has a town at the foot on both sides and a hospice near the top.

Why: Snow, cold and steep climbs cost time and animals. Traders choose the pass with the lowest summit, the shortest climb, a road for wheels and a safe valley.

Pass	Height	Route	Medieval notes
Brenner	1,370 m (4,495 ft)	Augsburg – Innsbruck – Bolzano – Verona – Venice	The lowest main Alpine pass. Mule trains and carts by the 12th century. Emperors (for example Frederick Barbarossa) used it to reach Italy. Perhaps about 3,000 tonnes of goods a year in the 14th century (one estimate)
Great St Bernard	2,469 m (8,100 ft)	Martigny – Aosta – Italy	The pilgrim road to Rome (Via Francigena). Hospice founded c. 1050 by St Bernard of Menthon to shelter pilgrims and protect them from bandits
St Gotthard	2,106 m (6,909 ft)	Lucerne – Uri – Ticino – Milan	Became a through route after the first wooden bridge over the Schöllenen gorge, c. 1220. Only for people on foot and pack animals: the first carriage crossed in 1775
Mont Cenis	2,085 m (6,841 ft)	Lyon – Savoy – Susa – Turin	A main France–Italy route. Emperor Henry IV crossed it in the winter of 1076–77 on his way to Canossa, a famous dangerous winter crossing

Seasons. High passes are snowbound for much of the year. The historic road over the Great St Bernard is open only from June to September, and winter snow there can be up to 10 m (33 ft) deep. Medieval travellers did cross in winter, but with local guides and real danger.

Organization. Valley villages organized the carrying trade. Local carriers took goods over their own stretch and handed them on to the next village’s carriers. Lords and towns took tolls at gorges and narrows on the approach roads.

MAP TIP

For each mountain range, choose 1–3 main passes and 2–5 minor ones. Draw a chain of villages up each valley, a toll (one bar across the road with a “T”) at the narrowest gorge, a hospice (a small house with a cross) near the summit, and a market town at the foot on each side, often where a lake or navigable river begins. Draw minor passes as thin dotted lines, label them “mule path” and give them no towns.

FANTASY TWIST

A dwarven tunnel through a range is a pass with no snow season. It would take most of the traffic from the passes nearby, and the towns at its two ends would grow rich. Whoever holds the tunnel holds the toll.

Inns, hospices and caravanserais

RULE OF THUMB

Places to sleep sit one day's stage apart: 15–30 km (10–19 mi) on main roads in Europe, 30–40 km (19–25 mi) between caravanserais on plains and in deserts, and 10 km (6 mi) or less in mountains.

Why: A traveller must reach shelter by nightfall. Animals need water, fodder and rest. On busy roads, inns appear in every town and large village that is one stage from the last.

Type of stop	Spacing	Example
Roman official lodge (<i>mansio</i>)	25–30 km (16–19 mi)	Roman roads across the empire
Pilgrim stage	about 20 km (12 mi) on average	Sigeric, c. 990: 80 stages from Rome to Canterbury
Inn town on a main road	15–30 km (10–19 mi)	Most European main roads
Caravanserai on plains and desert	30–40 km (19–25 mi)	Seljuk Anatolia and Iran
Caravanserai in mountains	10 km (6 mi) or less	Iran
Mongol relay station (<i>yam</i>)	32–64 km (20–40 mi)	Mongol Empire, 13th–14th centuries
Inca way station (<i>tambo</i>)	About one day's walk; from under 10 km to nearly 45 km (6–28 mi)	Andes, 15th century (see Roads without wheels)
Pass hospice	One near the top of each main pass	Great St Bernard

Kinds of lodging

- **Town inns** clustered near the town gates, the market place and the bridge. In Chaucer's *Canterbury Tales* (c. 1390s) the pilgrims set off from the Tabard Inn in Southwark, on the main road leading south from London Bridge.
- **Monastic guest houses.** The Rule of St Benedict told monks to receive all guests as Christ. Monasteries on main roads ran large guest houses, and some built inns. The New Inn in Gloucester (mid-15th century) was built by the abbey there for pilgrims.
- **Hospitals and hospices** sheltered pilgrims and the poor, at town gates, river crossings and mountain passes.

MAP TIP

On a main road, put an inn (a small house on the road) or a village or town every 15–30 km (10–19 mi). In a desert, put a caravanserai (the same small house, labelled “Han” or “Caravanserai”) or a well every 30–40 km (19–25 mi). Mark a hospice with a small house with a cross. If two towns on a road are more than 40 km (25 mi) apart, add a village or an inn between them. Label them: “Inn of the Seven Wells”, “Han of ...”, “Hospice of ...”.

Roncesvalles, at the Spanish foot of a Pyrenees pass on the pilgrim road to Santiago, had a large pilgrim hospital.

- **Caravanserais** (Persian, “caravan palace”) were fortified rural inns with one gate, a central courtyard, rooms, stables, water and often a mosque. Depending on the region, the same kind of building is called a *khan*, a *wikala* (Egypt) or a *funduq* (North Africa). A famous example is Sultan Han (1229). The Seljuk sultan Kayqubad I built it about 40 km (25 mi) west of Aksaray, on the road to Konya. Rulers and pious endowments (*waqf*) paid for such buildings. In Seljuk Anatolia they gave travellers free food, fodder and lodging.

FANTASY TWIST

Where monsters roam at night, every stop becomes a small fort with walls, a gate and guards, like a caravanserai. Stops may sit closer together than a full day, so that travellers always arrive before dark.

Roads without wheels: the Andes and Mexico

RULE OF THUMB

Where there are no carts and no horses, roads can be narrow, steep and stepped. Goods move on human backs, on llamas and in canoes. Way stations still sit about one day's walk apart, but news travels by relay runners.

Why: Before the Spanish arrived in the 1500s, the Americas had no horses, oxen or carts. The Inca of the Andes had llama-

mas as pack animals; the Aztecs of Mexico had no pack animals at all. A road for walkers and llamas does not need gentle slopes, so it can climb straight up a mountainside on stairs. Without carts, a state that wants to move food and news must organize people instead: way stations, storehouses and relay runners.

The Inca road system (Andes, mainly 15th century)

The Inca road network (in Quechua, *Qhapaq Ñan*) ran from today's Colombia to Chile and Argentina. Much of it was built or rebuilt in the 15th century, after the Inca conquests began (c. 1438), but many routes were older roads that the Incas took over, some built centuries earlier by the Wari and Tiwanaku cultures.

Feature	Number	Notes
Total length	Estimates vary: 23,000–40,000 km (14,000–25,000 mi), some up to 60,000 km (37,000 mi)	Two main north–south roads, one in the mountains and one along the coast, joined by more than twenty cross roads. The mountain road alone, from Quito through Cusco to Mendoza, was about 5,650 km (3,500 mi) long
Road width	Usually 1–4 m (3–13 ft)	The Cusco–Quito highway was always over 4 m (13 ft) wide, in places up to 16 m (52 ft)
Way stations (<i>tambos</i>)	2,000 or more; usually about one day's walk apart, but from under 10 km to nearly 45 km (6–28 mi)	Lodging, kitchens and storehouses; sited by water and good ground, away from marsh and steep slopes
Relay posts for runners (<i>chaski</i>)	About 2.5 km (1.6 mi) apart in one account; other sources give up to 7.5 km (4.7 mi). Four to six runners waited at each post	A message could travel up to 240–300 km (150–190 mi) a day
Llama	Carries about 30 kg (66 lb) for about 20 km (12 mi) a day	Up to 45 kg (99 lb) on short trips
Rope suspension bridges	The greatest spanned about 45 m (150 ft), over the Apurímac canyon	Made of woven grass; local villagers renewed them every one to three years (accounts differ) as labour service. One, the Q'eswachaka, is still rebuilt every June
State storehouses (<i>qollqa</i>)	Huánuco Pampa, a provincial centre, had 497 storehouses, about 37,100 m ³ in all	Rows of storehouses on hillsides near provincial centres and along the roads

Tenochtitlan: a city of causeways and canoes (Mexico, 1325–1521)

The Aztec capital Tenochtitlan stood on an island in Lake Texcoco. Its traditional founding date is 1325. Three main causeways (raised roads across the water) tied it to the shore: north to Tepeyac, south to Iztapalapa and west to Tlacopan. The western causeway was about 3 km (2 mi) long, by one classic account. The causeways had gaps crossed by bridges. Canoes could pass through the gaps,

and the bridges could be removed to defend the city. Inside the city, a network of canals meant that every part could be reached on foot or by canoe. Two aqueducts, each over 4 km (2.5 mi) long, brought fresh water from the springs of Chapultepec.

Without pack animals, goods came across the lakes by canoe or on the backs of professional porters (*tlamemeh*). A porter carried about 23 kg (50 lb) for about 21–25 km (13–16 mi) a day, then handed the load on to the next porter.

MAP TIP

For a people without carts or horses, draw thin roads that climb straight up slopes (mark the steep parts with steps or small chevrons “>”, as modern road atlases mark steep hills). Put a way station (the inn symbol: a small house on the road) about every day’s walk, about 20–25 km (12–16 mi), closer in mountains. Add small relay huts every few km along the trunk roads (on a local map only), the bridge symbol labelled “rope bridge” at gorges, and a labelled group of storehouses near each provincial town. A lake capital can sit on an island, joined to the shore by two to four straight causeways with gaps for boats, and fed by canoe traffic across the lake.

FANTASY TWIST

This is a good model for any people without draft animals: dwarves in mountain halls, forest elves or a hill people who travel on foot. Their roads can be narrow and stepped, and their relay runners can carry news faster than a medieval royal messenger with fresh horses (see [Speed per day](#)). A flying mount does not change the model for bulk goods: it is still one small load per animal.

Tolls and customs

RULE OF THUMB

Tolls are taken where traffic cannot avoid them: bridges, town gates, river narrows, mountain gorges, straits and borders.

Why: A lord or town needs money, and a narrow point makes it easy to stop every cart or boat. Merchants paid as long as the toll cost less than the detour.

Numbers and examples:

- **The Rhine.** In 1250 there were 12 toll stations on the Rhine between Mainz and Cologne alone, about one every 15 km (9 mi). Stations seem to have been at least 5 km (3 mi) apart. Between about 800 and 1800, 79 different places took tolls on the Rhine and its tributaries. The number of toll stations shot up during the Great Interregnum (1250–1273), when there was no emperor to

stop lords from adding new ones. The Pfalzgrafenstein (1326–27) is a toll castle built on an island in the middle of the Rhine near Kaub.

- **Straits.** From 1429 Denmark took the Sound Dues from every foreign ship passing Helsingør, at the entrance to the Baltic. The strait there is only about 4 km (2.5 mi) wide, so no ship could slip past. In 1479, 795 ships passed. By 1583 it was 5,400.
- **Bridges and towns.** Pontage (bridge tolls), murage (tolls to pay for town walls) and pavage (tolls for paving) were common in England.
- **Export customs.** From 1275 the English crown taxed every sack of exported wool. From 1363 all wool for export had to go through the wool staple at Calais (see [staple towns](#)).

MAP TIP

Mark a toll with one bar across the road or river and a “T” (two bars close together mean a lock). Put toll castles (the small castle icon) on river cliffs and islands at narrow points

(see [Castles, Forts and Military Outposts](#)). Too many tolls on one route is a story: merchants will look for another road, and a rival town on that road will grow.

Rivers as roads

RULE OF THUMB

A navigable river is a highway with a one-way discount. Downstream is fast and cheap. Upstream is slow, hard work.

How far up can boats go?

Every river has a **head of navigation**: the highest point boats can reach. Big sea ships stop lower down; small river boats go higher. Where cargo must move from one kind of boat to another, or onto carts, a town grows (see [head of navigation towns](#)). Around 1300, grain for London was

loaded onto boats on the Thames as far upstream as Henley.

How far up English rivers were really navigable is debated. In 1991 Edwards and Hindle argued that rivers were navigable much farther upstream than once thought. On their map, most of England lay within about 15 miles (24 km) of

navigable water. Other historians were more cautious. Langdon (1993) argued that some of their evidence showed only occasional river use, not regular traffic. Jones (2000)

Going upstream: towpaths

Boats went upstream by sail when the wind allowed, by poles and oars, or were **towed** by men or horses walking

Mills, weirs and locks

Mills need **weirs** (low dams) to raise the water. Fish traps were also built across rivers. Both blocked boats, so millers and boatmen fought for centuries.

- **Magna Carta (1215), clause 33**, ordered all fish-weirs removed from the Thames, the Medway and all of England except the sea coast.
- **Exeter.** In the 1270s or 1280s Isabella de Fortibus, Countess of Devon, built a weir for her mills that cut Exeter off from the sea. Trade was restored in 1290. Then in 1317 her cousin Hugh de Courtenay built another weir and a quay at Topsham, so boats had to unload there and pay his tolls. Exeter petitioned the king for about 250 years. In the end the city built its own canal (1564–66) around the weirs. It had the first pound locks in Britain (see below).

Portages

A **portage** is a place where boats or cargo are carried or dragged overland between two rivers, or around rapids. On the Russian route “from the Varangians to the Greeks” (Baltic to Constantinople), merchants crossed watersheds between river systems by portage. On the

Staple rights

Some river towns forced passing merchants to stop and sell. From 1259 Cologne made every merchant passing on the Rhine unload and offer his goods for sale for three days.

River ports

A river port needs a quay, warehouses, a crane and a customs post. The great treadwheel crane of Gdańsk (1442–44), wrecked in 1945 and rebuilt, can still be seen on the water-

front. Rivers freeze in northern winters, run low in summer droughts and flood in spring, so river traffic has seasons too.

found that the navigable network shrank in the later Middle Ages, as weirs and other obstructions grew.

along a **towpath** on the bank. That is why upstream travel was only 10–20 km (6–12 mi) a day.

Flash locks were a gap or gate in a weir. It was opened to let boats shoot down on a rush of water, or to winch them up against it. Slow and dangerous.

Pound locks are a chamber with a gate at each end, as in modern canals. China had one by 984. In Europe the first appeared in the Low Countries in the late 14th century: a large lock basin at Vreeswijk (dated 1373 or 1385; sources differ) and the first true pound lock at Damme near Bruges (1396). Near Milan, Bertola da Novate built 18 pound locks on the Bereguardo canal in 1452–58.

The Stecknitz Canal (1391–98), the first canal in Europe to cross a watershed (the higher ground between two river systems), linked the Elbe at Lauenburg with Lübeck on the Baltic, so that salt from Lüneburg could travel by water. It ran 97 km (60 mi) along two small rivers, with 11.5 km (7 mi) of dug canal across the watershed. It had 13 locks at first (17 later), most of them simple flash locks. Its barges carried about 7.5 tonnes of salt each and needed at least 10 days for the trip one way.

Dnieper they also had to carry or drag their boats past seven rapids (later counts give nine), where the river fell 50 m (165 ft) in 66 km (41 mi), while nomads attacked. Russian place names keep the word *volok* (portage): Volokolamsk means “portage on the Lama”.

Vienna got a similar right on the Danube in 1221. Staple rights made river towns rich and annoyed everyone else.

front. Rivers freeze in northern winters, run low in summer droughts and flood in spring, so river traffic has seasons too.

MAP TIP

Draw navigable stretches of rivers a little thicker, from the head of navigation down to the sea, with a small boat at the head of navigation. Put a town at the head of navigation, at confluences and at every portage end. Draw portages as short dotted lines between two rivers, labelled with their length. Add weirs and mills (a wheel on the stream) on small rivers, and a toll castle at a narrow gorge on big ones.

LATER ERA (1500S+)

Pound locks spread, and more canals crossed watersheds. The Canal de Briare (1604–42) linked the Loire and the Seine; it was the first canal in Europe to cross a watershed using pound locks. In the Netherlands, horse-drawn passenger canal boats (*trekschuiten*) ran on timetables from 1632 (Amsterdam–Haarlem).

The sea: ships, harbours and sea lanes

RULE OF THUMB

Sailing ships carry bulk cheaply; galleys carry people and luxuries quickly and safely. Both prefer to sail along coasts and island chains, and both stay in port in winter.

Ship types

Ship	Where and when	Size	Crew	Best for
Cog	North Sea and Baltic, 12th–14th c.	15–25 m (49–82 ft) long, 30–200 t; Bremen cog (1380): about 23–24 m (75–79 ft), 90–130 t	About 1 sailor per 10 t (a rule of thumb; real crews were often larger)	Bulk cargo: grain, timber, salt, fish, beer; the Hanseatic ship
Hulk	Northern Europe, 14th–15th c.	By the 1300s carried as much as a cog	Small	Bulk cargo; became the main Hanseatic ship in the 14th century
Merchant great galley	Mediterranean (Venice, also Genoa and Florence), c. 1290s–1530s	Up to about 46 m (150 ft); the Flanders galley about 37 m (121 ft) long; about 140–250 t of cargo	Up to about 250, including 150–180 oarsmen	Spices, silk, money and pilgrims; sailed in state convoys
Round ship (<i>nave</i> , <i>nef</i>)	Mediterranean	Often hundreds of tonnes	Small	Bulk: grain, salt, wine, cotton, alum
Carrack	Mediterranean and Atlantic, 14th–16th c.	Large; 16th-century Portuguese carracks were often over 1,000 t	Large	Long voyages with big cargoes
Caravel	Portugal and Spain, 15th c.	12–18 m (39–59 ft), 50–60 t	Small	Exploration and speed; too small for cheap cargo

Why galleys carried only valuable goods: a galley with 200 men aboard must stop for water and food every few days and pay all those wages. A cog of the same cargo needs

about 10 men. So galleys carried spices, silk and passengers, and round ships carried grain and salt.

Coastal hopping and open sea

Ships stayed near coasts when they could, for shelter, landmarks, water and trade at each port. They crossed open water on short, well-known lanes: across the Channel and the North Sea, across the Bay of Biscay, or from Crete to Egypt.

In the Mediterranean, the historian John Pryor showed that the main **trunk routes** (the main long-distance sea lanes) ran mostly along the northern coasts and island chains (the Adriatic coast, the Ionian islands, Crete, Rhodes, Cyprus). These routes hardly changed from antiquity to the 16th cen-

tury. Whoever held the islands and harbours along them controlled the sea. Venice held Crete (1211–1669) and the ports of Modon and Coron in southern Greece (1207–1500) as stations for its ships.

How far apart were galley stops? The Venice–Jaffa pilgrim galleys usually called at Poreč, Zadar, Korčula or Hvar, Dubrovnik, Corfu, Modon, Crete, Rhodes and Cyprus. Measured on a modern map, the stops were roughly 100–450 km (60–280 mi) apart, which is about two to eight days of travel. The whole voyage took 48 days in 1480, and could be done in as little as 29 days.

RULE OF THUMB

On a galley coast, put a watering port every 100–300 km (60–190 mi). Big ships can cross wider gaps, but they still prefer a friendly harbour every week or so.

Sailing seasons

Sea	Season	Real example
Mediterranean	Main season roughly April to October; winter sailing avoided	Romans treated the sea as “closed” (<i>mare clausum</i>) from November to March, though bans were probably never enforced. 14th-century Genoese statutes banned winter sailing in the Black Sea, with heavy fines. Venetian convoys: Flanders galleys left in March–April and returned about eight months later; Alexandria and Beirut galleys left in late August or September and returned in November
North Sea and Baltic	Spring to autumn; Baltic ports freeze	Hanseatic rules discouraged or banned most winter sailing, roughly from mid-November to late February
Indian Ocean	One voyage each way per year, timed to the monsoon	The south-west monsoon (summer) carries ships from Arabia and East Africa to India; the north-east monsoon (winter) brings them back. Merchants waited months in port between seasons

What makes a good harbour

Feature	Why it matters	Example
Shelter from storms and the prevailing wind	Ships at anchor survive	Bergen (deep fjord); Venice (lagoon behind sand bars)
Deep water at low tide	Big ships can enter	Atlantic ports often sat up estuaries for depth and shelter (see chapter 01)
Narrow, defensible entrance	A chain or tower closes it in war	Constantinople's Golden Horn was closed by a chain
Fresh water and food	Crews need supplies	Every galley stop
A river or road into the land behind	Goods can go on inland	Bordeaux (Garonne), Rouen (Seine), Hamburg (Elbe)
No silting	Harbours that fill with mud die	Bruges lost its sea access as the Zwin inlet silted; it used outports at Damme and Sluis, and Antwerp took its place by about 1500

Port ranks: how many ports on a coast?**RULE OF THUMB**

A kingdom's coast has a few big head ports, several smaller ports under each of them, a landing place in almost every sheltered cove or creek, and one or two famous anchorages where fleets wait for the wind.

Why: Overseas trade needs merchants, warehouses, money and a customs house, and these gather in a few places. Rulers made this stronger on purpose: it is easier to tax trade that must pass through a few ports. But fishing boats and small coasters (boats that carry goods along the coast)

need only a beach or a creek, so small landing places are everywhere.

Numbers (late medieval England): England's customs began with a tax on wool exports in 1275, and by the 1400s all foreign trade was taxed. It had to pass through designated **head ports**. After some changes in the 1200s and early 1300s their number settled at **15**, from Newcastle in the north-east round to Bristol in the south-west. That is roughly one head port for every 100–150 km (60–95 mi) of coast (a rough figure, measured on a modern map without the small inlets). London was by far the busiest: 462 ships arrived there in nine months in 1390. Hull, a big regional port, had about 80–90 arrivals a year in the late 1300s.

Rank	What it is	How many	Real example
Head port	A big port with a customs house, quays, warehouses and resident merchants. All foreign trade for its stretch of coast must clear customs here	15 for all of England	London, Hull, Boston, King's Lynn, Southampton, Bristol, Newcastle
Member port	A smaller port inside a head port's district. It ships goods, but customs are run from the head port	Several per head port	In 1353–54 the customs district of Hull covered "all ports from Whitby inclusive to Grimsby", more than 100 km (60 mi) of coast
Creek or landing place	A beach, creek or river landing used by fishing boats and small coasters. "Creek" was a customs word for such places	Every few km of sheltered shore or tidal river (an estimate)	Gatcombe, Newnham, Berkeley and Tewkesbury on the Severn, named as creeks of Gloucester in 1580
Anchorage (a "road" or roadstead)	Sheltered open water off the coast where ships anchor to wait for a fair wind or to gather into a fleet	One or two on a busy coast	The roadstead off Sluis in the Zwin, Flanders, where a French fleet of over 200 ships lay at anchor in 1340. The Downs, off Kent, which later held as many as 800 sailing ships at once

Promotion and decline. Ranks changed. Throughout the Middle Ages, Gloucester and the whole Severn down to Avonmouth were under Bristol for customs. Only in 1580 did Gloucester become a customs port of its own, with its own creeks. Boston and King's Lynn were great wool ports in the 1200s and 1300s, but as wool exports fell, Hull, Southampton and Bristol became the biggest regional ports of the 1400s.

How many ships? The historian Wendy Childs estimates that England's whole commercial fleet was perhaps

1,000–2,000 vessels at times. In the 1300s about 300 ships a year sailed from England to Bordeaux for wine; most loaded 100–150 tons, and a few over 300 tons. In the later 1300s and the 1400s, most major ports could muster about 20 large ships over a decade. Hundreds of small coastal and fishing boats, often under 10 tons and up to 20 tons, made up the rest. In war the king seized merchant ships as transports: fleets of 150–200 were common, and nearly 700 ships were used for the Crécy campaign (1346).

MAP TIP

On a kingdom map, draw a large anchor with a quay, with the name in capitals, for each head port, about one every 100–150 km (60–95 mi) of coast, at the best harbours and river mouths. Draw several small anchors for member ports between them, and leave most coves and creeks as unlabelled landing places. Add one named anchorage (an anchor drawn in open water, labelled "the Roads" or "the Downs") off the busiest stretch of coast or at the mouth of the main trade estuary.

LATER ERA (1500S+)

Anchorage became places to defend. Henry VIII guarded the Downs with three artillery forts, Deal, Walmer and Sandown, built in 1539–40 (see [The Later Era, 1500–1650](#)).

Lighthouses and sea marks

Lighthouses were **rare** in medieval Europe. Most coasts had none. Ships relied on church towers, beacons, landmarks and local pilots. Lights stood only at the entrances to rich ports and at very dangerous points:

- **Genoa:** a light tower by 1128; the present Lanterna dates from 1543.
- **Hook Head, Ireland (1201–1240):** built by William Marshal to guide ships into Waterford Harbour and up to his port of New Ross, about 30 km (19 mi) upriver. Monks kept the light.

- **St Catherine's Oratory, Isle of Wight (1328):** built by a lord as a penance for stealing wine from a wreck. It is Britain's only surviving medieval lighthouse.
- **Cordouan (Gironde estuary):** a 14th-century tower built under Edward the Black Prince guarded the Bordeaux wine trade and charged ships "lighthouse dues"; the present tower dates from 1584–1611.

MAP TIP

Draw sea lanes as dotted lines from port to port, following coasts and island chains, with short open-sea crossings between them. Add arrows where the wind decides the direction, and label each lane with its season (“galleys, April–October”). Put lighthouses (a small tower with short rays; never a star, which means a capital) only at the mouths of rich estuaries and on dangerous capes near big ports. A port without a river, road or rich hinterland (the farmland and towns behind it) stays small.

FANTASY TWIST

Sea monsters push ships even closer to the coast and into armed convoys, and make every harbour mouth worth a watchtower. Magic that calms storms would open the winter season and make sea trade even cheaper, so port cities grow larger still.

LATER ERA (1500S+)

Ocean routes open. The Portuguese reached India by sea in 1498, and Seville got a monopoly of trade with Spanish America in 1503. Carracks and then galleons crossed oceans. Atlantic ports (Lisbon, Seville, Antwerp, then Amsterdam and London) grew fast. Baltic grain fed the Netherlands, and traffic through the Danish Sound rose about sevenfold between 1479 and 1583. See [The Later Era, 1500–1650](#).

Trade nodes: markets, fairs, staples and entrepôts

RULE OF THUMB

Trade has a ladder. Every market town has a weekly market. Many have a yearly fair. A few towns in a large region host great fairs or force goods through a staple. Only a handful of great ports in a whole continent are true world markets.

Node	What happens there	How often	Spacing or number	Real examples
Weekly market	Peasants sell food and buy tools, salt and cloth	Weekly	Market towns 9–16 km (6–10 mi) apart in practice; England had 1,746 recorded markets by 1300	Any market town; English place names with <i>Chipping</i> (“market”), such as Chipping Norton
Yearly fair	Livestock, cloth, regional merchants	Once a year, a few days to some weeks	Many market towns had one	St Ives (Huntingdonshire); Stourbridge near Cambridge (charter 1199, first fair 1211; later one of the biggest in Europe)
International fair cycle	Merchants from distant lands meet, trade and settle debts	A cycle of fairs through the year	One or two fair regions per continent at any time	The Champagne fairs (12th–13th c.); later Geneva, Lyon, Frankfurt, Leipzig, Medina del Campo
Staple town	The law forces certain goods to be unloaded, offered for sale or taxed there	Permanent	One per river stretch or per export trade	Cologne (from 1259); Vienna (from 1221); the English wool staple at Calais (1363–1558)
Entrepôt	A great port where goods from many regions are stored and shipped on	Permanent	Only a handful in all of Europe and the Mediterranean c. 1300	Bruges, Venice, Genoa, Constantinople, Alexandria, Lübeck; Antwerp from c. 1500
Merchant colony (<i>Kontor</i> , <i>fondaco</i>)	Foreign merchants live and store goods in their own walled compound, under their own rules	Permanent	One per foreign nation in each big port	See below
Customs house	Goods are weighed, sealed and taxed	Permanent	Every major port and border	The Kaufhaus (warehouse and customs hall) at Konstanz, 1388

The Champagne fairs

From the 12th century, the Counts of Champagne protected merchants travelling to their lands, fixed weights and measures, and ran fair courts. Six fairs in four towns ran almost all year, and each lasted about six weeks: Lagny (from 2 January), Bar-sur-Aube (Lent), Provins (May), Troyes (“hot fair”, summer), Provins (September) and Troyes (“cold fair”, November). Each fair had set days for

cloth, then leather, then spices and goods sold by weight. It ended with a few days to settle accounts. Flemish cloth came south, and Italian silks, spices and money came north. In 1284 the heiress of Champagne married the future Philip IV of France, and Champagne became part of France. Royal wars and taxes then hurt the fairs, and Italian galleys began sailing directly to Flanders. By about 1350 the fairs were only regional markets.

Merchant colonies: kontors and fondaci

Colony	Place	Dates	Notes
Peterhof	Novgorod (Russia)	c. 1200–1494 (reopened 1514, then faded)	Palisaded yard with a stone church used as a warehouse; about 150–200 German merchants a season; furs and wax out, cloth, silver, salt and herring in
Steelyard (<i>Stalhof</i>)	London	Cologne merchants had a hall by the 1170s and royal privileges from 1175; closed 1598	Walled compound of warehouses and halls on the Thames
Bryggen (German Wharf)	Bergen (Norway)	c. 1350–1754	Timber warehouses on the quay; stockfish (dried cod) out, grain in
Hanseatic Kontor	Bruges	13th century to 16th	No compound: merchants lodged with local innkeepers who acted as brokers; own house only from 1442
Fondaco dei Tedeschi	Venice	1228; rebuilt 1505–08	The official house of the German merchants, by the Rialto Bridge, under Venetian supervision; the rebuilt house had 56 rooms for over 100 merchants
Venetian fondaci	Alexandria	13th–15th c.	Venetian warehouses and lodgings for the spice trade
Genoese colonies	Pera (Constantinople), Caffa (Crimea)	Pera from 1267; Caffa c. 1266–1475	Whole walled towns run by Genoa
Venetian quarter	Constantinople	From 1082	Quays, warehouses and churches, free of Byzantine customs

MAP TIP

In each great port, draw or label the foreign quarters: “Hanse Steelyard”, “Genoese quarter”, “Venetian fondaco”. At fair towns, add the fair dates to the label (“Provins: May and September fairs”). A fair town sits where two economic regions meet, often on a political border or on neutral ground under a strong protector.

FANTASY TWIST

If the roads are full of monsters, trade collapses into fewer, bigger events: one or two great fairs a year, reached by armed caravans. A neutral fair town protected by a powerful mage or temple could become the Champagne of your world.

LATER ERA (1500S+)

Permanent exchanges began to replace the great fairs. Antwerp opened a purpose-built exchange (bourse) in 1531 and Amsterdam in 1611, so merchants could trade every day instead of a few weeks a year. Antwerp, then Amsterdam, became the leading entrepot of northern Europe.

What is traded

RULE OF THUMB

The higher the value per kilogram, the farther a good can travel over land. Cheap, heavy goods stay local unless they can float.

Bulk goods and luxury goods

Good	Value per weight	How far over land	How far by water
Grain, flour	Very low	Usually under 30–50 km (19–31 mi); routine sales within about 17 km (10 mi)	Hundreds of km by river and sea, especially to big cities and in famine
Firewood, timber, stone, brick	Very low	A few km	Far by water: Caen stone from Normandy built English cathedrals
Salt	Low	About 100 km (62 mi) or more on “salt roads” (Lüneburg to Lübeck)	Across Europe: salt from the Bay of Bourgneuf (France) went to the Baltic
Wine	Medium	Short distances	Very far: Bordeaux shipped about 100,000 tuns (a tun is a big cask of about 950 litres) in the record year 1308–09, much of it to England (James 1971)
Salted herring, stockfish	Medium	Short to medium	Continent-wide
Wool	Medium to high	Carried by pack horse and cart to ports	England to Flanders and Italy
Cloth	High	Any distance	Any distance
Metals (copper, tin, silver, iron)	Medium to high	Long distances	Long distances
Furs, wax, amber	High	Long distances	Long distances
Spices, silk, dyes, gems, sugar	Very high	Across continents	Across oceans
Live animals	Walk themselves	Hundreds of km along drove roads (see Drove roads)	—

A cart journey that doubles the price of grain adds only a few per cent to the price of pepper, which was worth many

times more per kilogram. That is why spices crossed Asia by camel while grain rarely crossed a county by cart.

Typical goods by region type

Region type	Exports	Imports
Fertile lowland plain	Grain, malt, ale or beer, flax, wool	Salt, metal, wine, luxuries
Hills and uplands	Wool, sheep, cattle, cheese, hides, lead	Grain, salt, cloth
Mountains	Silver, copper, iron, rock salt, timber, cattle, cheese; transit tolls	Grain, wine, cloth
Northern forests	Furs, wax, honey, timber, tar and pitch, potash	Salt, cloth, metal goods, wine
Coasts and islands	Salted fish, sea salt, ships and sailors	Grain, timber, cloth
Wine country (in Europe mostly south of about 50° N, though small vineyards grew farther north)	Wine, brandy (later)	Grain, cloth, fish
Mediterranean dry lands	Olive oil, wine, dried fruit, sugar (Cyprus, Sicily; see Sugar estates), alum, cotton (Syria), mastic (Chios)	Grain, timber, metals, cloth
Cloth towns	Woollen and linen cloth	Wool, dyes, alum, food

Region type	Exports	Imports
Desert and oasis	Dates, salt; transit of gold and slaves	Grain, cloth, metal goods
The “Indies” (South and East Asia)	Pepper and spices, silk, cotton cloth, porcelain, gems	Silver, gold, horses, coral, cloth

For mines, quarries, salt works and industrial towns, see [Industry, Resources and Special Towns](#).

MAP TIP

For each region on your map, write one line: “exports ..., imports ...”. Then draw the route that carries each export to the nearest water or market. If two regions export the same thing to each other, one of them is wrong.

Real trade networks to copy

RULE OF THUMB

A continent usually has one or two **bulk sea networks**, several **river spines**, and one or two **long luxury routes** over land. Copy a real one, rename it, and adapt it to your geography.

Network	Peak	Shape on the map	Hubs	Main goods
Hanseatic League (Baltic and North Sea)	c. 1250–1500	A chain of ports along a cold sea, fed by rivers; 70–170 member towns at different times, and nearly 200 in all; four great kon-tors	Lübeck, Hamburg (linked to Lübeck by a short land route of about 60 km (37 mi) across the neck of the Jutland peninsula), Visby, Danzig, Riga, Reval, Bergen, Bruges, London, Novgorod	Grain, timber, furs, wax, fish, salt, beer, cloth
Venice and Genoa (Mediterranean and Black Sea)	c. 1100–1500	State convoys and chains of island and coastal bases	Venice, Genoa, Crete, Pera, Caffa, Chios, Alexandria, Beirut, Acre (until 1291)	Spices, silk, cotton, alum, grain, slaves, salt
Silk Roads	Revived under the Mongols, c. 1250–1350	A relay of oasis cities over about 6,400 km (4,000 mi); hardly anyone travelled the whole way	Tabriz, Samarkand, Bukhara, Kashgar, Sarai, Tana, Khanbaliq (Beijing)	Silk, porcelain, spices, horses, silver
Trans-Saharan	8th to early 17th c.; Mali’s peak in the 13th–14th c.	North–south lines from oasis to oasis; crossing took about two months	Sijilmasa, Taghaza (salt mine), Walata, Timbuktu, Gao, Agadez	Gold, salt, copper, cloth, slaves
Indian Ocean monsoon	Whole period	Seasonal sea lanes; merchant quarters in every port	Aden, Hormuz, Cambay, Calicut, Malacca (from c. 1400), Kilwa, Quanzhou	Pepper, spices, cotton cloth, horses, gold, ivory
Rhine corridor	Whole period	One river spine with many tolls and staples, linked to Italy by Alpine passes	Basel, Strasbourg, Mainz, Cologne, Dordrecht	Wine, cloth, metals, stone, fish
Russian rivers	9th–15th c.	River systems joined by portages	Novgorod, Smolensk, Kiev, Bulghar	Furs, wax, honey, slaves, silver
Amber Road	Prehistoric and Roman, Baltic to the Adriatic	A long north–south land route	Aquileia, Carnuntum; in the Middle Ages, amber came from Prussia, where the Teutonic Order controlled it	Amber (later sold to rosary-makers in Bruges and Lübeck)
Salt routes	12th–16th c.	Short land or canal routes from salt works to fish ports	Lüneburg to Lübeck (about 100 km, 62 mi), for salting Baltic herring	Salt

Network	Peak	Shape on the map	Hubs	Main goods
Wine trade	Gascony to England 1152–1453	Big seasonal wine fleets from one estuary	Bordeaux, La Rochelle; Rhine wine via Cologne	Wine
Pilgrim roads	Whole period	Chains of hospices and inns to a shrine	Rome, Santiago de Compostela, Canterbury; Jerusalem by galley from Venice	People and their money (see chapter 08)

Notes on using them:

- **Silk Roads:** around 1340 the Florentine merchant Pegolotti wrote that the road from Tana (on the Sea of Azov) to China was “perfectly safe, whether by day or by night”, according to merchants who had used it. A strong empire across the route makes long land trade possible. When it breaks up, the route breaks into dangerous pieces.

MAP TIP

Pick one template for each kind of sea or land on your continent: a Hanse-like network for a cold northern sea, a Venice-like network for a warm inland sea, a Silk-Road-like chain across a steppe or desert. Then join them at a few great entrepôts. Those meeting points are your richest cities.

- **Trans-Saharan:** in 1352 Ibn Battuta crossed from Sijilmasa to Walata in about two months. At Taghaza, 25 days out, the houses and mosque were built of blocks of salt. Near the end, water was brought out to the caravan from Walata, four days ahead. An average caravan may have had about 1,000 camels; some are said to have had as many as 12,000.

FANTASY TWIST

Elves who travel by forest paths or dwarves with underground roads create networks that do not follow rivers. Treat them like any route: work out what they carry, how fast, and at what cost. A dwarven road that moves bulk ore cheaply is a river in stone.

Drawing the route network

Step by step

1. **Water first.** Mark which rivers are navigable and how far (thicker lines). Choose head ports about every 100–150 km (60–95 mi) of coast (see [Port ranks](#)). Add coastal sea lanes and short open-sea crossings.
2. **Crossings.** Place fords, bridges and ferries: on a settled lowland river, a bridge town about every 20 km (12 mi) of river. Big rivers get few fixed bridges.
3. **Passes.** Choose the main passes over each range.
4. **Main roads.** Link cities and big towns through the crossings and passes. Follow valleys and dry ridges. Connect every main road to a river port or harbour. Aim for about 30 km of main road per 1,000 km² of settled land (see [How much main road to draw](#)).
5. **Secondary roads.** Link each market town to its neighbours, 9–16 km (6–10 mi) away, and to the nearest main road.
6. **Local tracks.** Link villages to their market town, 6–10 km (4–6 mi) away, like spokes of a wheel.
7. **Stops.** Add inns every 15–30 km (10–19 mi) on main roads, caravanserais every 30–40 km (19–25 mi) in deserts, hospices on passes, and watering ports every 100–300 km (60–190 mi) on galley coasts.
8. **Tolls.** Put tolls at bridges, gorges, straits and borders.
9. **Hubs.** Check that the biggest towns sit where routes meet (see below).

Which places become hubs?

- Where **land meets water:** ports, river ports at the head of navigation, and the lowest bridge on a river.
- **Confluences**, where two navigable rivers join.
- The **foot of a main pass**, on both sides.
- **Both ends of a portage.**

- **Border towns** between two economic regions, where goods change hands (fair towns, staple towns).
- **Crossroads** of two main roads, especially at a bridge.

As a rough guide (not a measured rule): a village has 2–3 roads, a market town 3–5, and a city 5–8 or more, plus water routes.

How much main road to draw

RULE OF THUMB

On a kingdom map of settled land, draw about 30 km of main road for every 1,000 km² (about 48 mi per 1,000 sq mi). A square of 100 × 100 km (62 × 62 mi) gets about 300 km (190 mi) of main road: for example three roads crossing it, or four or five shorter roads meeting at towns inside it.

Why: A kingdom map shows the main roads that a merchant or a royal messenger would plan a journey on. The lanes between villages are many times denser, but drawn at

this scale they turn the map into a spider's web. They belong on a local map.

Numbers: A good medieval measure is the Gough Map of Britain (conventionally dated c. 1360–70; a handwriting study suggests one or two decades after 1400, with additions later in the 15th century; see [The Roman legacy](#)). It shows over 600 cities, towns and smaller settlements. About 190 red lines join about 180 of them in England and Wales. A GIS study (a study with digital map software) by Eljas Oksanen and Stuart Brookes matched 99% of these lines to real historical routes.

Measure	Value
Red route lines on the map itself	About 190
Length of the real roads behind the map's red lines, as reconstructed by Oksanen & Brookes (2025)	About 4,540 km (2,820 mi), stored as 455 segments in their database
Area of England and Wales	About 151,000 km ² (58,000 sq mi)
Main road per area	About 30 km per 1,000 km ² (48 mi per 1,000 sq mi)
Share of the route length on Roman roads	35.5%
Share on routes first recorded before 1086 (not Roman)	13%
Share on routes first recorded after 1086	16%
Share known only from later (early modern) evidence	32.5%
Other (a line along the River Witham, or no known road)	about 3%

The Gough Map is a selection, not a full road atlas. It leaves out some important roads, such as the road from London to Colchester and East Anglia, and Watling Street between London and Canterbury. So 30 km per 1,000 km² is a good amount for a kingdom map, not the whole network. The village lanes (2–3 per village, see above) still exist, but draw them only on a local map.

Roman roads live on. About a third of the Gough Map's route length runs on Roman roads, nearly 1,000 years after Roman rule ended in Britain. Much of the rest first appears in the Middle Ages, when new roads grew up to serve new towns, markets and bridges.

MAP TIP

Spread the main roads unevenly: more around the capital and in rich, crowded lowland, fewer in hills, moors and forests. Put a town or an inn every 15–30 km (10–19 mi) along each main road. If your kingdom was once part of an

older empire, draw about a third of the main-road length as long straight stretches on the old empire's roads, and let the rest bend between medieval towns and bridges.

Symbols and labels

These symbols match the guide's [master legend](#). Routes are always solid or dotted lines. On the route network a short dashed line means a ford, and dash-dot lines are borders, never routes.

Route	Suggested line	Label example
Major sea lane	Dotted line at sea, with arrows where the wind sets the direction, and a season label	"Galley route to Alexandria (Aug–Nov)"
Navigable river	Thicker river line up to the head of navigation, with a small boat there	"Navigable to Westbridge"
Royal highway	Thick solid line	"The King's Road", "Via Aurelia"

Route	Suggested line	Label example
Old Roman or older-empire road	Straight thick solid line; grey where abandoned	"Ermine Street", "Fosse Way (Roman)"
Regional road	Medium solid line	"The Salt Road"
Local road	Thin solid line	Usually unlabelled
Pilgrim road	The ordinary road line with small scallop marks along it	"Pilgrims' road to the shrine"
Track, bridle way or mule path	Thin dotted line	"Mule path, closed in winter"
Drove road	Broad dotted line (two rows of dots)	"Drovers' road"
Caravan route	Dotted line with a larger dot at each stage	"Caravan road, 9 stages"
Portage	Short dotted line between two rivers, labelled with its length	"Portage, 12 km"

Point on the network	Symbol	Label example
Bridge	Two short curved lines ")(" with the road running through	Usually the town's name
Ford	Short dashed line across the river	"Ford"
Ferry	Short dotted line across the river with "F"	"Ferry"
Toll	One bar across the road or river with a "T"	"Toll"
Lock	Two short bars close together across the river or canal	"Lock"
Inn or caravanserai	Small house on the road	"Inn of the Seven Wells", "Han of ..."
Hospice	Small house with a cross	"Hospice of ..."
Head port	Large anchor with a quay	Name in capitals
Member port	Small anchor	Name in ordinary letters
Anchorage	Anchor in open water	"the Roads", "the Downs"
Lighthouse	Small tower with short rays	"Light"

Naming roads. Real roads were named after where they go ("the London road"), who built them ("Via Appia", after Appius Claudius), or what they carry ("Salt Road", "Amber

Road"). Give long distances in days ("12 days to the capital"), because that is how travellers thought.

MAP TIP

Check every road with three questions. Where does it go? What does it carry? Why does it bend here? If a road has no destination or cargo, delete it. Common errors (roads straight over mountains, too many bridges on a great river, big cities with no water access) are listed in [Common Mistakes](#).

Quick summary

- Over land: about 30 km (19 mi) a day on foot, 15–25 km (10–15 mi) by ox cart, 16–30 km (10–20 mi) for a cattle drove, 50–90 km (30–56 mi) for a royal messenger.
- Water is much cheaper: land : river : sea $\approx 8 : 4 : 1$ (England c. 1300). Bulk goods follow water; most towns above 10,000 people sit on it.
- Medieval roads were rights of way, not built roads. Roman roads were reused (about a third of the Gough Map's routes). Roads follow valleys, ridges, fords and bridges. On a kingdom map, draw about 30 km of main road per 1,000 km² (48 mi per 1,000 sq mi).
- Stone bridges spread from the 12th century. They were paid for by tolls (pontage), rents, gifts and bridge chapels. A settled lowland river has a bridge town about every 20 km (12 mi), like the Thames; the biggest rivers had few bridges and many ferries.
- A few passes carry most traffic across a range. The lowest takes carts (Brenner, 1,370 m or 4,495 ft). Hospices sit near the summits.
- Inns and stages are 15–30 km (10–19 mi) apart on main roads. Caravanserais are 30–40 km (19–25 mi) apart on plains and 10 km (6 mi) or less in mountains. Without wheels or horses (the Inca), way stations still sit a day's walk apart, and relay runners carry news 240–300 km (150–190 mi) a day.
- Tolls sit at choke points. The Rhine had 12 toll stations between Mainz and Cologne alone in 1250.
- Rivers have a head of navigation, towpaths, weirs, locks and portages. Millers and boatmen fought over the water.
- Cogs and hulks carry northern bulk. Galleys carry Mediterranean luxuries and pilgrims in convoys, with stops every 100–450 km (60–280 mi). Winter is the closed season.
- A coast has a few head ports (15 for all of late medieval England, roughly one per 100–150 km or 60–95 mi of coast), more member ports, landing places in most creeks, and one or two named anchorages.
- Trade nodes form a ladder: weekly markets, yearly fairs, great fairs and staples, then a few great entrepôts with foreign merchant colonies.
- Copy real networks (Hanse, Venice and Genoa, Silk Roads, trans-Saharan, Indian Ocean, rivers) and make hubs where land meets water.

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CHAPTER 6

VILLAGES, FARMS AND THE COUNTRYSIDE

Most people in the Middle Ages lived in villages, hamlets and farms. Between 85% and 95% of people lived in the country, so most of any realistic map is countryside. This chapter is for local-area maps: one parish, one valley, one barony or one county. It tells you what a village looks like and what is inside it. It shows how fields, woods and pastures are arranged, how much of a whole realm they cover, where mills, parks, ponds and barns go, and how far apart everything is.

The baseline is Europe c. 1000–1500, with England as the best-documented example. Where to put villages on a big map is covered in [Where Settlements Are Built \(and Why\)](#). How many people live in them is covered in [Population and Settlement Sizes](#).

IN THIS CHAPTER

Village forms and layouts
 Sizes, territory and walking distances
 Fields and farming systems
 Woods, forests and hunting grounds
 How much of the map is fields and woods
 Mills and water management
 Other rural features
 Reclaimed land: fens, polders and assarts
 Local administration: parish, manor, hundred
 Drawing a local-area map
 Quick summary
 Sources and further reading

Village forms and layouts

RULE OF THUMB

Good arable land with shared open fields gives **tight villages**. Hills, forest, marsh and pasture give **scattered farms and hamlets**. A line feature (a road, a valley stream or a dyke) gives **villages in a line**.

Why: People live as close as they can to the land they work most. On open-field land, every family has strips all over the parish, so the best place to live is in the middle, together. Where each family farms its own block of land (in hills, forest clearings or marsh), it is better to live on that block.

The main village forms

Form	What it looks like	Where (real examples)	How to draw it
Nucleated village (houses in one cluster)	20–80 houses packed along a few lanes around the church and manor house; fields all around	English Midlands; northern France; much of Germany (<i>Haufendorf</i> , “heap village”); Denmark	A tight cluster with a church, open fields around it
Green village	Two rows of houses facing each other across a long or square green	Northern England: in County Durham many villages, perhaps most, show signs of deliberate planning, often as two rows round a green; German <i>Angerdorf</i> (a lens-shaped green with church and pond) in the lands settled from the 12th century	Two parallel rows around an open space
Street (row) village	One or two rows of houses along one street	Planned villages everywhere; German <i>Straßendorf</i>	A line of houses along a road
Polyfocal village (several centres)	Two or three small clusters (church end, manor end, green end) that grew together	Common in England	Two or three knots joined by a lane
Forest strip village (<i>Waldhufendorf</i>)	Farms spaced along a valley road or stream. Behind each farm a long strip of land runs uphill to the forest	First in the northern Black Forest c. 1000; spread in the 12th–13th centuries to the Odenwald, Ore Mountains, Bohemia and Silesia. Chains of these villages run up to 25 km (15 mi) along valleys in Saxony	Farms like beads on a string; parallel strips behind them

Form	What it looks like	Where (real examples)	How to draw it
Marsh and dyke village (<i>Marschhufendorf</i>)	Farms in a line along a dyke or drainage ditch, with long narrow plots behind	North Sea marshes, Holland	A line along the dyke, thin strips at right angles
Round village (<i>Rundling</i>)	A horseshoe or circle of farms around a green, with one entrance	The German–Slavic border zone, mostly founded in the 12th century. Usually 5–7 farms at first; farms were divided until there were about 20 by the 16th century. The church stands outside the ring. About 93 survive, in the Wendland (Lower Saxony)	A small horseshoe of house marks round a green, with the church outside it (not a plain circle, which means a market town)
Hamlet	2–15 households, no church	Everywhere. Main form in Devon, Cornwall, Wales, Ireland (<i>clachan</i>) and Scotland (<i>fèrmtoun</i>)	2–5 small house marks with no church
Dispersed farmsteads	Single farms, each in its own fields	Uplands, Alps, Norway, forest clearings; Kent, Essex and the Weald	Scattered tiny dots joined by lanes
Mediterranean hill village	A dense, walled cluster of stone houses on a hilltop or spur, with castle and church at the top	Lazio and Tuscany (<i>incastellamento</i> , the move into fortified hill villages, 10th–12th centuries); Provence (<i>villages perchés</i>); Catalonia	A dense blob on a hill, with a wall
Agro-town	2,000–10,000 people, mostly farmers, in one big settlement	Sicily, Apulia, southern Spain	A town-sized symbol, but with few town functions

England is the classic case of mixed patterns. A central belt of nucleated villages ran from the north-east through the Midlands to the south coast, with more dispersed settlement on both sides (Roberts and Wrathmell 2000). In

France, the open-field villages of the north-east contrast with the *bocage* (small hedged fields and scattered hamlets) of the west.

What is in a village

Feature	How common	Where it sits	Notes
Parish church	One per parish	On a slight rise, often at one end of the village, next to the manor house	The biggest building, usually stone. Has a churchyard
Manor house	One per manor; a village could have two or three manors	Next to the church; often moated	Hall, barns, stables, often a dovecote and fishpond. England has about 6,000 known moated sites, most dug c. 1250–1350
Mill	About one per village (England c. 1300)	On the stream, often a little away from the houses, with its leat and pond; a windmill on rising ground	The lord's monopoly (see Mills)
Parsonage and glebe	One per parish	Near the church	The priest's house and his own land (glebe)
Smithy	Most villages	On the main street, the green or the road	Shoes horses and oxen, mends ploughs
Alehouse	Common	Any house that brewed for sale	Before c. 1400 usually an ordinary house showing an <i>ale-stake</i> (a pole with a bush). Inns are mainly in towns and on main roads
Green	Many villages, mainly in the north and east of England and in German <i>Angerdörfer</i>	In the middle	Grazing, gatherings, sometimes a pond and the pound
Pond	Most villages	On the green or at the lowest point	For animals, not drinking
Pound or pinfold	Most villages by the 16th century	At the edge of the village or green	A walled or hedged pen for stray animals, run by the <i>pinder</i>
Lord's oven and winepress	Common in France and Germany	Near the manor or in the centre	In France these were <i>banalités</i> : the lord's mill, oven and press, which peasants had to use and pay for
Tofts and crofts	All villages	Along the streets	A toft is the house plot with its yard; a croft is the enclosed garden or paddock behind it

Real example: Wharram Percy (Yorkshire, England). This deserted village on the chalk Wolds had about 40 peasant house plots in planned rows, two manor houses, a church and a green. It also had two watermills, and their mill-ponds were fished. At its height, in the late 13th and early

14th centuries, it had perhaps 200 people. About 30 houses were occupied around 1368 and at least 16 in 1436. Around 1500 the lord evicted the last four families and turned the land into sheep pasture.

MAP TIP

Pick one village form for each region of your map and stick to it. Draw the church (a small church with a tower) as the biggest symbol, the manor house (a small open square) beside it, and the mill (a wheel on the stream) a short way off. On a local map, show tofts as a row of narrow plot outlines along the street, each with a small house mark at the street end, and crofts as long narrow plots behind. Do not draw them as plain small rectangles: in the guide's legend, a chain of small rectangles means fish-ponds. Most villages need a green, a pond or a road junction as their centre.

FANTASY TWIST

If wolves, raiders or monsters roam the countryside at night, dispersed farmsteads become rare and dangerous. People crowd into nucleated or hilltop villages with a ditch and palisade, as Italian peasants did during *incastellamento*. Isolated farms survive only as fortified towers or inside a lord's protection. See [Fantasy Variants](#).

Sizes, territory and walking distances

RULE OF THUMB

A lowland village has 30–60 households (150–300 people) and owns about 5–15 km² (2–6 sq mi) of land. Its farthest fields are a 20–35 minute walk from the houses.

Why: A person needs about 1 ha (2.5 acres) of arable land and 2–3 ha (5–7 acres) of all land to eat (see [Population and Settlement Sizes](#)). Peasants walked to their fields and back every day, often with an ox team and a plough. Every extra kilometre wastes working time.

Village territory

Settlement	People	Households	Territory, lowland	Farthest regular fields
Hamlet	10–75	2–15	1–5 km ² (0.4–2 sq mi), part of a parish	under 1 km (0.6 mi)
Small village	75–150	15–30	2–7 km ² (0.8–2.7 sq mi)	1–1.5 km (0.6–1 mi)
Typical lowland village	150–300	30–60	5–15 km ² (2–6 sq mi)	1.5–2.5 km (1–1.5 mi)
Large village	300–500	60–100	10–20 km ² (4–8 sq mi)	2–3 km (1.2–1.9 mi)
Upland parish	100–500, in many hamlets and farms	20–100	50–100+ km ² (20–40+ sq mi)	Each farm's infield is under 1 km away; summer pastures much farther
Agro-town	2,000–10,000	400–2,000	~50–250 km ² (20–100 sq mi), derived	5–10 km (3–6 mi), an hour or more each way

These numbers are worked out from national totals. England c. 1300 had about 9,500 parishes on 130,000 km², or about 14 km² (5.4 sq mi) per parish. A circle of 14 km² has a radius of about 2.1 km (1.3 mi). Lowland villages are therefore 1.5–4 km (1–2.5 mi) apart.

Real example: Milton Ernest (Bedfordshire) in the 14th century, according to a field survey by Hutchings (1969), had about 1,600 acres (650 ha, 6.5 km²): 1,405 acres of arable, 81 of meadow, 84 under the village itself and 28 of roads. This

was a “champion” (open-field) parish with almost no wood or waste.

How far is too far? The geographer Michael Chisholm (1962) collected evidence from many farming societies. Costs rise noticeably once fields are more than about 1 km (0.6 mi) from the farmhouse. At 3–4 km (2–2.5 mi) people usually change the system: they build an outlying farm or field barn, or they turn the far land into pasture. Agro-

towns of southern Italy and Sicily are the famous exception, where farmers walked long distances every day.

Rings of land use around a village

Land use forms rough rings. The land that needs the most work and manure is closest to the houses.

Ring	Distance from the houses	What is there	Why
1. House plots	0–100 m (0–330 ft)	Tofts and crofts, vegetable gardens, orchards, hemp and flax, pigsties, the manor's dovecote and fishponds	Daily work; household waste is the manure
2. Arable	0.2–2 km (0.1–1.2 mi)	The open fields, or the infield in the uplands	Most of the year's labour
Meadow (not a ring)	Wherever the flood plain is	Hay	Meadow follows water, not distance
3. Pasture and commons	1–3 km (0.6–2 mi)	Common pasture, heath, rough grazing	Animals walk there themselves
4. Wood and waste	Parish edge, 2–4 km (1.2–2.5 mi)	Coppice woods, wood-pasture, moor, marsh	Visited now and then
5. Summer pasture	5–50+ km (3–30+ mi)	Shielings, alps, long-distance grazing	Seasonal only

In the Mediterranean the rings have different crops. In the hill villages of Lazio, studied by Pierre Toubert (1973), the order from the walls outwards was: vegetable gardens, then

vineyards (and olives), then grain fields, then woods and pasture.

MAP TIP

For each village, imagine a circle of about 2 km (1.3 mi) radius. Put the open fields inside it, a ribbon of meadow along the stream, and woods, heath or moor in the gaps between the circles of neighbouring villages. Parish boundaries run through this outer land, often along streams, ridges or the edges of woods.

Fields and farming systems

Open fields: two-field and three-field

RULE OF THUMB

In open-field country there are no hedges between fields. Two or three huge fields surround each village, cut into hundreds of long, thin strips.

How it worked: Each family held strips scattered across all the fields, so everyone shared good and bad soil. All the strips in one field followed the same crop plan.

- **Three-field system:** one field of winter grain (wheat or rye), one of spring crops (barley, oats, peas, beans) and one left **fallow** (resting, grazed by the village animals). It spread in the 12th–13th centuries.
- **Two-field system:** half crop, half fallow. It was older, and it stayed normal in the dry Mediterranean, where a biennial (every-other-year) rotation was the rule.

Element	What it is	Typical size
Field	One of the 2–3 great unhedged fields	Several hundred acres each
Furlong	A block of parallel strips all running the same way; each had its own name	A parish had dozens. A “furlong” as a length is 220 yards (201 m)
Strip (selion, “land”)	The unit ploughed by one holder	About 200 m (220 yd) long; 4.6–20 m (5–22 yd) wide; 0.1–0.4 ha (¼–1 acre)

Element	What it is	Typical size
Ridge and furrow	Strips ploughed into raised ridges with furrows between, for drainage	Ridges 3–20 m (3–22 yd) apart; today up to 0.6 m (2 ft) high, once higher. Older ones curve in a gentle reverse-S, made by the long ox team swinging round at the end; later ones are straight
Headland	The strip at the end of a furlong where the plough team turned	Later ploughed as a ridge of its own
Family holding	The land of one peasant family	One family might hold about 70 strips totalling 20 acres (8 ha). The standard holding was the virgate (in the north and east of England counted as two oxgangs): nominally 30 acres (12 ha), in practice often 24–32 acres, and varying from place to place

Who held what: A survey of 104 English manors in the 13th century found that 45% of tenants had less than 3 acres (1.2 ha), 31% had half a virgate and 22% a full virgate. A family needed at least 10 acres (4 ha) to live from its land alone. Smallholders worked for wages or crafts.

Where: In England, open fields covered a broad belt from Yorkshire and Lincolnshire across the Midlands to south-

central England. Kent and Essex kept small, square, hedged fields. Open fields survive today at Laxton (Nottinghamshire, three great fields) and Braunton (Devon). Ridge and furrow is still visible under grass all over the Midlands, best seen in low sun or light snow.

Meadow, pasture, commons and waste

Land	Use	Where	Notes
Meadow	Hay, cut in early summer to feed oxen and horses in winter	Flat land along streams and rivers that floods in winter	Small but very valuable: Milton Ernest had only 81 acres (33 ha) of meadow against 1,405 acres of arable. After haymaking, many meadows were opened for common grazing (in England, “Lammas land”, from 1 August)
Pasture	Grazing	Hillsides, heath, fallow fields, stubble after harvest	Ploughs needed oxen: an 8-ox team for a heavy plough. The <i>carucate</i> , a land unit of about 120 acres (50 ha, four virgates), was the land one team could plough in a year. These units were used for tax, so real sizes varied
Commons	Shared rights over the lord's waste	Heath, moor, marsh, wood	Rights could include pasture, pannage (pigs on acorns), estovers (wood), turbary (cutting turf or peat) and piscary (fishing). Numbers of animals were often limited (“stinted”) per holding
Waste	Unploughed land: heath, moor, marsh, scrub	Poor soils and parish edges	Not useless: fuel, bracken, grazing, rabbits, reeds

Infield-outfield in the uplands

RULE OF THUMB

In hills and poor soils, one small, well-manured field near the houses was cropped every year. Bigger outfields farther out were cropped for a few years, then rested as grass.

- **Infield:** close to the farm or hamlet, given all the manure, cropped every year (oats and bere, a hardy barley).

- **Outfield:** cropped for a few years until the soil was tired, then left to grass for several years.
- **Where:** Scotland, Ireland, Wales, northern England, Scandinavia, and the sandy north of Germany and the Netherlands, where the infield (*Esch*) was built up with “plaggen”: heath turf soaked in animal dung.
- **Runrig:** in Scotland, strips of the infield were often shared out among the families of a *fermtoun* (farm hamlet) and sometimes redistributed.

The Mediterranean: terraces and irrigation

RULE OF THUMB

In dry lands, water decides everything. Gardens and orchards cluster along canals and below springs. Grain grows on dry slopes. Vines and olives climb terraced hillsides.

Feature	What it is	Real examples and numbers
Terraces	Steps built into slopes with stone walls, to hold soil and water	All round the Mediterranean, for vines, olives and grain. Dating of terrace soils shows a major phase of building c. 1100–1600
Acequia	An irrigation canal fed by a weir on a river (Spanish, from Arabic)	Valencia: 8 main canals from the river Turia water about 17,000 ha (170 km ² , 65 sq mi) of <i>huerta</i> (irrigated garden land). The Water Tribunal that settles disputes between irrigators goes back to al-Andalus and still meets today. (Modern city growth has since covered much of the huerta.)
Noria	A big wheel turned by the river current that lifts water in pots or scoops	Hama (Syria): wheels up to 20 m (66 ft) across. The Albolafia wheel in Córdoba is a medieval example (partly rebuilt today). A wheel turned by an animal walking in a circle is called a <i>saqiya</i>
Qanat	A gently sloping tunnel that brings groundwater from the foot of hills to the surface, with a line of access shafts	Iran: most are under 5 km (3 mi) long, a few up to ~70 km (43 mi); access shafts every 20–35 m (65–115 ft); the “mother well” at the head is usually 20–200 m (65–650 ft) deep. About 50,000 worked in Iran in the mid-20th century. Palermo (Sicily) has a medieval qanat network, usually credited to its Arab rulers (9th–11th centuries). The oases of the Tafilalt and the Draa valley (Morocco) have relied on qanats, there called <i>khattara</i> , since the late 14th century; the Tafilalt had about 400 of them
Coltura promiscua	Grain grown between rows of trees and vines (vines trained up the trees)	Central Italy
Chestnut woods	Chestnuts dried and ground as flour	Mountain villages of Corsica, the Apennines, the Cévennes

In Egypt, villages sat on mounds above the Nile flood, with basins of fields that the flood watered each year. In Syria,

the Ghouta of Damascus was a belt of irrigated orchards around the city, fed by channels from the Barada river.

Rice, tanks and lake gardens (Asia and the Americas)

RULE OF THUMB

Wet rice needs flat, flooded fields. On slopes people build **terraces**; on dry plains they store the monsoon rain in **tanks** (reservoirs held back by an earth dam, called a *bund*); in shallow lakes they build **raised garden beds**. Each system ties a group of villages to one shared water source, so villages, temples and fields line up along the water.

Why: A rice field (a *paddy*) must hold standing water, so it needs a level surface with a low bank round it. Water has to pass from field to field down the slope, so farmers form groups that build and run the canals together. Temples or village councils often decide who gets water and when. A paddy can be cropped every year with no fallow, so rice lands feed far more people per square kilometre than European grain fields; for the numbers, see [Beyond Europe](#).

Landscape	Where and when	Numbers and facts	How to draw it
Subak (a cooperative irrigation group)	Bali (Indonesia). UNESCO dates the system back to the 9th century; networks of water temples have managed whole river basins since the 11th century	About 1,200 subak (UNESCO; a 2019 estimate gives about 1,560). Each has 50–400 farmers who share water from one source; a subak covers from a few hectares to almost 800 ha (2,000 acres). It is built of canals, tunnels and weirs. Temples mark the water sources and the points where water is divided; the lake temple Pura Ulun Danu Batur is the chief water temple. The UNESCO site covers about 19,500 ha (195 km ² , 75 sq mi)	Terraces in contour bands down a volcano's slopes, forest above them, small temples at springs, weirs and canal splits
Mountain rice terraces	Honghe Hani terraces, Yunnan (south-west China), built up over about 1,300 years (UNESCO)	The UNESCO site covers 16,603 ha (166 km ² , 64 sq mi) with 82 villages. In places there are over 3,000 terraces between the forest edge and the valley floor	Four bands down the slope: forest on top (the water source), then villages, then terraces, then the river

Landscape	Where and when	Numbers and facts	How to draw it
Village tank cascade	Dry zone of Sri Lanka; developed over nearly 2,000 years (FAO)	A chain of small tanks down one shallow valley; each tank spills into the one below, and the paddy lies below each dam. About 14,200 village tanks are known in Sri Lanka, on average one per 2.6 km ² (1 sq mi) in the dry-zone provinces. One studied area today (Palugaswewa, 187 km ² or 72 sq mi) has 78 tanks in 12 cascades and 37 villages; each tank watered up to about 75 ha (190 acres). A village had one main tank; the local saying “tank, stupa, village, temple” names the four parts of every village	A curved earth dam across a shallow valley, the water behind it, a sluice in the dam, a fan of paddy below; the village on dry ground beside it; several tanks in a chain
Royal tank	Parakrama Samudra, at Polonnaruwa (Sri Lanka). King Parakramabahu I (reigned 1153–1186) joined several tanks into one lake; its oldest part, Topa Wewa, dates from about 386 AD	About 22 km ² (2,200 ha, 8.5 sq mi) of water. The bund is about 14 km (8.5 mi) long and 12 m (40 ft) high. It waters about 7,300 ha (18,000 acres) of paddy	A big lake behind a long dam beside the royal city, with canals fanning out below
Chinampas (raised beds built in a lake)	Lakes Xochimilco and Chalco, Valley of Mexico. The earliest securely dated beds are from 1150–1350; they spread over the lake beds after about 1400, when the Aztec Triple Alliance needed food for its capital, Tenochtitlan	Long, narrow beds built up from layers of plants and mud, with canoe canals between them. Britannica gives 6–10 m (20–35 ft) wide and 100–200 m (330–650 ft) long; other sources give smaller beds. Willows planted at the edges hold the soil	Narrow parallel green strips with thin blue canals between them, in the shallow edge of a lake, with canoe routes to the city
Angkor (Khmer Empire)	Cambodia, 9th–16th centuries	A low-density settlement spread over about 1,000 km ² (390 sq mi) of water works (Evans et al. 2007): houses on low mounds with household ponds; local temples (a mound about 20 m square, with a moat and a small pond to the east); rice fields, embankments, canals and roads. Huge rectangular reservoirs (<i>barays</i>): the West Baray, begun in the 11th century, is about 7.8 × 2.1 km (4.8 × 1.3 mi). Whether the barays mainly watered fields or were mainly religious symbols is debated	A grid of rice fields and embankments dotted with small mounds, each with a pond; temple mounds with moats; straight canals and raised roads; one or two giant rectangular reservoirs near the centre

MAP TIP

Draw rice terraces as tight bands that follow the contour lines, and keep forest on the hilltops above them as the water source. Draw a tank as a curved dam across a shallow valley, with the water behind it, a sluice in the dam and a fan of paddy below; put the village on dry ground at one side

and its temple close by. Chain several tanks down one valley. Put water temples at springs, weirs and the points where canals split, and put houses on low mounds above the flooded fields.

Orchards, vineyards and gardens

- **Gardens and orchards** sit next to the houses (ring 1). Orchards give apples, pears, cherries and plums in the north, and figs, almonds and citrus in the south. Normandy and the west of England are cider country.
- **Vineyards** need sun and warm summers. The great medieval wine regions were around Paris, Champagne, Burgundy, the Rhine and Moselle, and Gascony (Bordeaux). Gascon wine was shipped to England in large amounts after 1152, when Henry of Anjou (King Henry II from 1154) married Eleanor of Aquitaine.
- **The northern limit was farther north than today's wine map.** Domesday Book (1086) lists over 40 vineyards in England, and 139 were recorded in 1509, when Henry VIII became king. In the 14th century, vineyards worked in Poland, for example in Silesia and at Kraków, Toruń and Zielona Góra. Most of these made thin wine for

monks and lords. Put northern vineyards on south-facing slopes, near monasteries or castles. Commercial wine regions are in [Industry, Resources and Special Towns](#).

- **Olives** mark the true Mediterranean climate. They do not grow where winters are hard. If your region has olives, it has dry summers, terraces and probably biennial fallow.

MAP TIP

Draw open fields as large unhedged areas with faint hatching in different directions (each block of hatching is a furlong). Draw hedged, enclosed fields as a patchwork of small irregular shapes. Draw meadow as a pale green ribbon along streams, and heath or moor with tufts. In dry lands, put green gardens along canals and below qanat outlets, and leave the slopes above them brown. Draw a qanat as a straight line of tiny rings across dry country: each ring is a shaft mouth with its heap of spoil (dug-out earth) around it, and from the air a qanat looks like a line of holes or anthills. Put the village and its gardens where the line ends. Keep the rings much smaller than the small open circle used for a market town.

LATER ERA (1500S+)

In England, landlords enclosed many open fields for sheep pasture in the 15th and 16th centuries. How much was enclosed, and when, is debated. J. R. Wordie (1983) estimated that about 45% of England was already enclosed by 1500, 47% by 1600 and 71% by 1700. Other historians think more than half the farmland was still open in 1700. Either way, most remaining open fields went in the parliamentary enclosures of the 18th–19th centuries. “Floated” water meadows (channels that flooded meadows on purpose to get an early crop of grass) appear in the early 1600s; Rowland Vaughan described his in Herefordshire in 1610. Maize from the Americas spreads in Iberia and northern Italy in the 16th century. See [What Changes After 1500](#).

FANTASY TWIST

Magic that doubles harvests halves the land a village needs. Its fields shrink to about a 1.5 km (1 mi) circle and more land is left for woods and pasture. It also lets villages sit closer together. Keep the ring pattern anyway: gardens near, fields out, woods at the edge.

Woods, forests and hunting grounds

RULE OF THUMB

A medieval wood is not wild. Almost every wood belongs to someone and is cut on a regular cycle. A medieval “forest” is a **legal** area for royal hunting, which may contain villages, fields and heath as well as trees.

Managed woodland

Practice	What it is	Numbers	On the map
Coppice	Trees (hazel, ash, oak, hornbeam, maple, lime) cut to the ground; many new poles grow from the stump	Cycles depended on the tree and the product: birch for faggots (bundles of sticks) every 3–4 years, oak poles up to about 50 years. Medieval cycles were mostly short, often under 10 years (Rackham); they grew longer in later centuries. Products: firewood, poles, wattle, hurdles, charcoal	Compact woods with sharp edges, bounded by a bank and ditch to keep animals out after cutting
Coppice with standards	Coppice plus scattered tall trees left to grow into timber	A 1544 English statute required 12 standards per acre (about 30 per hectare) to be left, and the wood to be fenced after cutting	Same as coppice
Pollards	Trees cut well above the ground, usually about 2–4 m (6–13 ft) up, out of reach of grazing animals	Cut every 2–6 years for leaf fodder, or 8–15 years for poles	Scattered big trees in open grassland

Practice	What it is	Numbers	On the map
Wood-pasture	Grassland with pollards and old trees, grazed by cattle, sheep or deer	Most commons, parks and forests	Trees scattered over pasture
Pannage	Pigs turned into woods in autumn to eat acorns and beechmast	Domesday Book (1086) measured many woods by how many pigs they fed, e.g. "wood for 40 pigs". The New Forest still has a pannage season of at least 60 days	Pig herders in oak and beech woods

Real example: Hatfield Forest (Essex). A royal forest made between 1086 and 1225, 403 ha (996 acres) today. The ecologist Oliver Rackham called it "almost certainly unique": all the elements of a medieval forest survive there, with deer,

cattle, coppice woods, pollards, scrub, timber trees, grassland and fen.

Royal forests

- **What:** an area under special **forest law**, which protected deer and wild boar ("venison") and the vegetation that fed them ("vert"). Clearing land, cutting trees or hunting without permission were offences.
- **How big:** At the peak (late 12th to early 13th century), royal forest covered about **one third of southern England**. At one stage in the 12th century all of Essex was forest. In 1086 Sherwood covered perhaps a quarter of Nottinghamshire.
- **Dates:** The Normans brought forest law (the New Forest was made by William I). The Assize of the Forest came in

1184. The **Charter of the Forest (1217)** cut the forests back and protected commoners' rights. The forest bounds were cut back again around 1300.

- **Officials:** wardens, foresters, verderers (forest judges), regards (inspectors who looked for illegal clearing and building) and agisters (who ran the grazing and pannage).
- **Elsewhere:** German kings and princes claimed similar hunting rights (*Wildbann*). French kings kept great forests such as Fontainebleau and Compiègne.

Parks, chases, warrens and lodges

Feature	What it is	Numbers and examples
Deer park	A private enclosed hunting ground and venison farm, needing a royal licence to "empark"	Domesday Book (1086): 36 parks. Rackham's estimate for c. 1300: about 3,200 parks , covering nearly 2% of England, one per ~40 km ² (15 sq mi) on average. Usually 40–80 ha (100–200 acres); some much larger. Hampton Court (Herefordshire) got a 15th-century licence for 1,000 acres. Mostly stocked with fallow deer
Park pale	The boundary: a bank with a wooden fence on top and a ditch on the inside	Outlines usually have rounded corners. "Deer leaps" let deer in but not out
Chase	An unfenced hunting ground held by a great lord, not the king (though a chase could pass to the Crown)	Cranborne Chase (Dorset, Wiltshire and Hampshire), held by great lords such as the earls of Gloucester and later the earls of March
Rabbit warren	Rabbits farmed for meat and fur	The Normans brought rabbits in the late 11th or 12th century. In the 13th century one rabbit was worth more than a workman's daily wage. Warrens sit on dry, sandy, poor land such as the Breckland (Norfolk and Suffolk). Artificial burrows are long low mounds called pillow mounds . Thetford Warren Lodge (c. 1400) is a fortified warren's house
Hunting lodge	A house for the lord's hunting visits, inside a forest or park	Clarendon (Wiltshire): a hunting lodge stood in its park by 1130; Henry II and Henry III turned it into a royal palace

MAP TIP

Do not paint a royal forest as solid trees. Draw its boundary as a green dash-dot line with a large italic name (“Forest of ...”), and inside it show woods, heath, villages and fields. Do not use a dotted line: dotted lines mean tracks and other routes. Give a county 1–3 royal forests in the hunting country nearest the king’s palaces. Draw deer parks as rounded pale (fence) lines 0.5–1 km (0.3–0.6 mi) across, with scattered trees inside, next to castles and great manor houses, and put a lodge inside. Mark a royal hunting lodge with a small crown. Put rabbit warrens on sandy heaths.

LATER ERA (1500S+)

Charcoal for blast furnaces and glassworks raised the value of coppice, and more woods were enclosed and managed. Many deer parks became landscaped parks around country houses. Forest law faded away by the mid-17th century.

How much of the map is fields and woods

RULE OF THUMB

In a settled lowland kingdom c. 1300, colour about **a third** of the land as ploughland and about **a tenth** as woods. Most of the rest is pasture, heath, moor and marsh. Hill country has far less ploughland. After a great plague, cut the ploughland by a quarter to a third and turn it into pasture and scrub.

Why: Ploughland follows people. Each person needed about 1 ha (2.5 acres) of arable land (see [Population and Settlement Sizes](#)), so the fields were largest when the population peaked, c. 1300. Woods were cleared for new fields until the Black Death stopped it. One open-field parish can be almost all arable, like Milton Ernest ([above](#)), but a whole kingdom also has hills, heaths, marshes and wooded parish edges.

England, the best-measured case

England covers about 13 million ha (32 million acres, 130,000 km²).

Date	Arable land, fallow included	Share of England	Sown that year	What was happening
1300	11.5 million acres (4.7 million ha)	~36%	7.4 million acres (3.0 million ha)	Population near its medieval peak (about 4.5–5 million)
1380	8.7 million acres (3.5 million ha)	~27%	5.2 million acres (2.1 million ha)	After the Black Death (about 2.5 million people in 1377)
1450	7.6 million acres (3.1 million ha)	~24%	4.6 million acres (1.8 million ha)	Population at its low point (about 1.9 million)
1600	8.9 million acres (3.6 million ha)	~28%	6.7 million acres (2.7 million ha)	Population growing again; less land left fallow

These figures come from Broadberry, Campbell and others (2010, Table 1); the shares are worked out from England’s area. “Arable” here counts the fallow field too. Only the

sown part grew a crop in any one year, but on a map fallow looks like part of the open fields, so colour it as fields.

Date	Woodland and wood-pasture	Notes
1086	~15% of England	Rackham’s estimate from Domesday Book
c. 1349	perhaps ~7%	About half the 1086 figure, after two and a half centuries of clearing (Royal Forestry Society summary; a rough estimate)
After 1349	Stopped falling; some land went back to scrub and wood	Pollen records from north-west Europe show birch and hazel spreading over abandoned grazing land by the late 14th century (Yeloff and van Geel 2007)

Woods were spread very unevenly. Heavy clays, steep slopes and parish edges kept more wood; “champion” (open-field) parishes such as Milton Ernest had almost none.

A “forest” is not tree cover. At its peak, royal forest covered about a third of southern England, but this was a legal zone

that included villages, fields and heath as well as woods (see [Royal forests](#)). Do not add it to the woodland share.

A land budget for a whole realm

Land	Share of a settled lowland kingdom c. 1300	Basis
Ploughland (open fields, fallow included)	35–40%	England: ~36% in 1300
Woodland and wood-pasture	7–15%	England: ~15% in 1086, perhaps ~7% by 1349
Meadow	about 5%	Milton Ernest: 81 of 1,600 acres. More along big rivers, less in dry or hilly country
Deer parks	nearly 2%	Rackham's ~3,200 parks (see above); parks overlap with wood-pasture
Villages, gardens and roads	a few per cent	7% in Milton Ernest (84 acres of village and 28 of roads); less where people are few
Pasture, heath, moor, fen, marsh and other common waste	the rest, roughly a third to a half	Worked out as what is left

Hill country is different. An upland parish of 50–100 km² (20–40 sq mi) with 100–500 people needs only about 100–500 ha of arable, so only about 1–10% of its land is ploughed (worked out from this chapter's own numbers). The rest is rough grazing, moor and wood.

MAP TIP

On a kingdom map of settled lowland, colour about a third of the land as fields, in broad patches around each cluster of villages. Spread the woods in many separate patches (on parish edges, heavy clay, steep slopes and hunting grounds) adding up to about a tenth, not one solid block. Give the rest to pasture, heath, moor and marsh, mostly on poor soil, hills and wet ground. In hill country, shrink the fields to small patches in the valleys. Show royal forests as a named green dash-dot boundary, not as solid trees.

After a plague: England's arable shrank by about a quarter between 1300 and 1380, and by about a third by 1450. The lost fields became sheep and cattle pasture, and some went back to scrub and wood. Whole villages were deserted (see [chapter 01](#)).

LATER ERA (1500S+)

By 1600 English arable had climbed back to about 28% of the land, and much less of it lay fallow: 2.2 of 8.9 million acres in 1600, against 4.1 of 11.5 million in 1300. More of the sown land grew peas, beans and other crops. Enclosure also turned some fields into hedged pasture (see [above](#)).

Mills and water management

RULE OF THUMB

By 1300 there was roughly **one mill per village**. Water-mills sit on streams with a reliable flow; windmills stand on open, exposed high ground, especially in flat country with weak streams.

Why: Grinding grain by hand with a quern took hours every day. A mill saved that labour, and it made the lord a

steady income. Tenants had to grind at the lord's mill (in England, "suit of mill") and pay a toll in grain called **multure**, usually around a twentieth of the grain (often quoted as 1/16 to 1/24). Lords fought hand-mills. At St Albans, in a long quarrel with the townspeople in the late 13th and 14th centuries, the abbot had their querns seized and used the stones to pave the floor of his parlour.

How many mills

Date	Mills in England	Notes
1086 (Domesday Book)	5,624 watermills (Hodgen 1939); later counts give ~ 6,000 or more	Mills in about 4,000 places; about one mill for every 40–50 recorded households. Some “mills” were extra pairs of millstones in one building
c. 1300	~ 10,000–15,000 mills, most of them water-mills	Estimates vary (see Holt 1988 and Langdon 2004). Either way, England had roughly as many mills as parishes. Windmills were now common in the east
After 1350	Fewer	After the Black Death there was less grain to grind, and many mills closed

Kinds of mill

Type	Where	Dates and facts	How to draw it
Watermill	On streams and small rivers with a steady flow; a weir and leat create the fall of water the wheel needs	Vertical wheels in most of Europe: <i>undershot</i> (the water pushes the bottom of the wheel) or <i>overshot</i> (the water pours onto the top); small horizontal-wheeled “Norse” mills in Ireland, Scotland, Scandinavia and the mountains	A wheel symbol on the stream, with its leat
Windmill	Flat, open or dry country: eastern England, Flanders, northern France, the Low Countries	Earliest certain reference in northern Europe: 1185 , at Weedley in Yorkshire. The first type was the post mill (the whole body turns on a post). Masonry tower mills appeared by c. 1300. Dutch drainage mills appeared from 1408	An X of four sails on a dot or small mound, on high ground away from trees
Tide mill	Estuaries and sheltered coasts with a strong tide	Nendrum (Northern Ireland) c. 619, rebuilt c. 787; Ebbsfleet (Kent) 691–692 (dated by tree rings); Wood-bridge (Suffolk) first recorded 1170. A pond fills at high tide through a one-way gate and drives the wheel as the tide falls	A mill wheel on a dam across a tidal creek
Horse or donkey mill	Castles, besieged towns, dry places with no stream	Slow but independent of water	No special symbol; inside a castle or town
Ship mill and bridge mill	Big rivers with no space for weirs	Floating mills on the Tiber, Danube and other large rivers; Paris had a bridge lined with mills	Small mill wheels in midstream, labelled “ship mills”; mill wheels on the bridge symbol. Do not draw boats: a small boat on a river marks the head of navigation

Leats, ponds and weirs

- **Weir:** a low dam across the river that raises the water and turns part of it aside.
- **Leat (mill race, head race):** an artificial channel that takes water from the weir to the mill. It can run from a few hundred metres to over a kilometre (0.6 mi). It runs along the valley side, falling more gently than the river, so the water arrives at the mill higher than the river.

MAP TIP

Give almost every village a mill. Draw the river, a weir, the leat running parallel to the river as a thin blue line, the mill as a wheel on the water, and the tail race rejoining the river. On a good stream, mills can follow each other every 1–2 km (0.6–1.2 mi). In flat, dry country, put windmills (an X of four sails on a dot) on the highest ground near the village, often on an old mound. Label them “Mill”, “Mill Hill” or “Windmill Hill”. Put tide mills on creeks near a coastal town.

- **Millpond:** stores water so the mill can run when the stream is low.
- **Tail race:** returns water to the river below the mill.
- **Conflicts:** mill weirs blocked boats and fish. Rivers with many mills were hard to navigate (see [Trade Routes, Roads and Transport](#)). Mills also drove fulling, forges and other industry (see [Industry, Resources and Special Towns](#)).

LATER ERA (1500S+)

Wind-pumped drainage made the great Dutch land reclamations possible (see below). Wind-driven sawmills appeared in Holland from the 1590s. Mills multiplied again for paper, gunpowder and metalworking.

Other rural features

RULE OF THUMB

Status features (moats, dovecotes, fishponds, parks, warrens) cluster around manor houses and monasteries. Working features (barns, folds, kilns, crosses, paths) follow the fields, the flocks and the roads.

Quick list

Feature	What it is	Where to put it	Numbers and examples
Fishponds	Ponds for breeding and storing freshwater fish for fast days	Near manors, monasteries and castles, in a valley floor fed by a stream; often a chain of 2–12 ponds joined by channels	About 2,000 are recorded in England, thought to be only a small part of the medieval total. Building peaked in the 12th century. Groups of up to 12 ponds in a line or a cluster; big storage ponds and small, shallow breeding ponds. Example: a grange at North Kelsey (Lincolnshire) had a pond 65 × 50 m (210 × 165 ft) and another about 75 m (250 ft) long
Dovecotes	Towers for breeding pigeons for meat and manure	Next to manor houses, parsonages and monasteries	Only manorial lords could keep them in medieval England. In France this was the <i>droit de colombier</i> . Medieval ones are usually round and stone-built. Kinwarton (Warwickshire, 14th century) has over 580 nesting holes
Moated sites	A house platform (the “island”) surrounded by a wet ditch	Clay lowlands; manor houses, granges, rectories	About 6,000 known in England, most dug c. 1250–1350, with by far the most in central and eastern England; a status symbol as much as a defence. Small: the island is usually about 25–50 m (80–165 ft) across, inside a moat about 7–15 m (23–50 ft) wide, so the whole site is only about 40–80 m (130–260 ft) across. Lincolnshire examples: Heapham, a square island 24 m across in a moat 8–12 m wide; Withcall, islands of 40 × 30 m and 45 × 20 m. A few high-status sites are much bigger: South Kelsey Hall’s main island is 110 × 100 m. See Footprints
Monastic grange	An outlying farm run by a monastery, often worked by lay brothers (monks who did the manual work) or by paid labourers	Next to the abbey or wherever it held land, up to a day’s journey or more away	First appeared in the 12th century, led by the Cistercians, and used until the English monasteries were dissolved in 1536–1541. Byland Abbey (North Yorkshire) had an industrial grange at Denby Dale, some 50–60 miles (80–95 km) away
Tithe barn	A great barn for crops paid as tithe (one tenth) or rent	Next to a church, rectory or grange	About 200 medieval barns (built before 1550) survive in Britain; not all were tithe barns. Great Coxwell (Oxfordshire), built c. 1292 (roof timbers felled 1291–92) for Beaulieu Abbey’s grange: 46 m long, 13 m wide, 15 m high (152 × 43 × 48 ft)
Sheepfolds and sheepcotes	Pens and long sheep-houses for winter	On downs, wolds and moors; on monastic sheep granges	In Norfolk, lords had the right to fold their sheep on tenants’ fields at night to manure them
Wells	Village water	Village centre or each toft	Clay villages used ponds and shallow wells; chalk villages needed deep ones
Wayside crosses	Stone crosses at junctions, boundaries and moor paths	Crossroads, parish edges, routes over moors (Dartmoor)	Mostly 13th–14th century in Britain. Many hundreds survive, often only the base or a broken shaft; Dartmoor alone has about 130 known crosses. In Germany, small <i>Sühnekreuze</i> (penance crosses) mark a killing
Limekilns	Kilns that burn limestone into lime	Near limestone and fuel, by a creek or harbour, or at a big building site	Mostly for mortar in the Middle Ages. Spreading lime on fields became practical first in coal-mining areas, where coal made lime cheap from the late 13th century; Fitzherbert describes it in his <i>Boke of Husbandry</i> (1523). See Industry, Resources and Special Towns
Pound	Pen for stray animals	Edge of village	See above

Summer pastures and transhumance

Transhumance means moving herds between winter and summer pastures every year. It shapes mountain and dry landscapes.

Type	Region	Distance	Season	On the map
Shielings (summer huts on hill pasture)	Scotland (<i>àirigh</i>), northern England, Wales (<i>hafod</i> = summer house, <i>hendre</i> = winter house), Ireland (<i>booley</i>), Norway (<i>seter</i>)	Usually within a day's walk, uphill	Scotland: families went up "about mid-summer" for "a month or six weeks" to make butter and cheese	A cluster of tiny squares (the huts) by a stream on high pasture. Scottish huts were mostly rectangular, 5.7–14 m (19–46 ft) long, of dry stone or turf. A 1995–2001 survey in Scotland recorded about 3,100 shieling huts
Alpine pastures	Alps, Pyrenees, Carpathians	From the valley village up to the high meadows	Summer	Tiny squares (huts and cheese-making dairies) above the tree line
Long-distance transhumance	Castile (the Mesta , royal charter 1273)	550–750 km (340–470 mi) from León and Old Castile to Extremadura and Andalusia; rarely more than 250 km (155 mi) from New Castile and La Mancha	South in autumn (arriving in October), north in April–May	More than 2.5 million sheep moved each year in 1436–1549. The royal drove roads (<i>cañadas reales</i>) were given a fixed width only where they crossed farmland: 90 <i>varas</i> (Castilian yards), about 75 m (250 ft)
Tratturi	Abruzzo to Apulia (southern Italy)	The main route, L'Aquila to Foggia, is about 244 km (152 mi)	Winter in the Apulian lowlands, summer in the mountains	Wide grassy drove roads (<i>tratturi</i>), drawn as broad dotted lines (two rows of dots); King Alfonso V of Aragon organised them as a royal toll system (the <i>Dogana</i>) in 1447

Place names often record shielings: English and Scots "shiel" or "shield" (Galashiels, Pollokshields), Old Norse *sætr*

("-side", "-seat") and *ærgi* ("-ergh", "-argh"), and Gaelic *àirigh* ("Airie", "Arie").

Boundaries, paths and hollow ways

- **Parish and manor boundaries** followed streams, ridges, old roads and banks, and were marked with stones, crosses, ancient trees and banks. Each spring (Rogation week) parishioners "**beat the bounds**": they walked the whole boundary and struck the markers with green boughs. Boys came along and were sometimes bumped on the boundary stones, so that they would remember the line all their lives.
- **Hedges and banks**: where fields were enclosed early, the hedges themselves are boundaries. Fields cleared from woodland have hedges full of woodland plants.

- **Hollow ways** (sunken lanes) are tracks worn deep by centuries of feet, hooves and rainwater, especially on chalk and soft sandstone. One estimate puts England's total at over 1,600 km (1,000 mi); in Germany some reach 5 m (16 ft) deep.
- **Footpaths and church paths**: every hamlet had a path to its parish church and mill. In big northern parishes, "corpse roads" carried the dead from distant hamlets to the parish churchyard.

MAP TIP

On a local map, add a manor house with a moat (a small open square with a double outline), a dovecote and a chain of fishponds (small rectangles along the stream) below it. Put a grange (a small open square topped with a cross, or the label "Grange") with a great barn in a monastery's outlying land. Mark the parish boundary with a thin dash-dot-dot line that follows streams and ridges. Where paths cross

it, put a wayside cross (a tiny cross on a base) or a boundary stone (just the letters "BS" written on the line, as on Ordnance Survey maps; a tick would read as a standing stone, and a dot as a farmstead). Draw a few thin dotted tracks from the village up to summer huts on the hills, and a drove road (a broad dotted line, two rows of dots) in sheep country.

Reclaimed land: fens, polders and assarts

RULE OF THUMB

In the High Middle Ages population pressure pushed farming into marsh, coast and forest. Reclaimed land has **straight lines**: dykes, drains and long narrow fields. Old land has **curved lines**.

Type	What it is	Real examples	On the map
Silt fen and salt-marsh reclamation	Banks built against the sea or floods; marsh drained by ditches	The Fens (about 3,900 km ² , 1,500 sq mi): medieval farmers embanked the silt land around the Wash. Romney Marsh (about 260 km ² , 100 sq mi): land “inned” (enclosed by walls) in stages; much of Walland and Denge Marsh by the 14th century	Long sea banks (bold lines with short ticks on the water side), straight drains, villages in a line along the old banks, sheep on the marsh
Peat fen	Wet inland marsh	The southern Fens stayed mostly undrained: fish, eels, wildfowl, reeds, turf, summer grazing. Monasteries (Ely, Thorney, Crowland, Ramsey, Peterborough) stood on dry “islands”; the Isle of Ely is about 60 km ² (23 sq mi)	Islands of dry land with a church or abbey, surrounded by marsh and meres
Polders and dykes	Land below water level, enclosed by dykes and drained	Netherlands: first polders in the 11th century; a cooperative to build and keep up a dam on the Rhine near Wijk bij Duurstede (south-east of Utrecht) c. 1122; the Rijnland water board in 1255; the first drainage windmill near Alkmaar in 1408. Drained peat shrinks and sinks, so dykes had to be raised again and again	A grid of drains inside a dyke. Draw the dyke as a bold line with short ticks on the water side (a plain thick line means a highway). Villages strung along the dykes; drainage windmills only from 1408
Assarts	Fields cleared from woodland	Peak in the 13th century, by single farmers, village groups, lords and monasteries (especially the Cistercians). In royal forests a licence or a fine was needed. Many went back to woodland after the Black Death	Irregular hedged fields biting into woodland, with isolated farms

Place-name clues: English assart names include Stocking, Stubbings, Ridding, Royd and Breach. Older English clearing names end in *-ley* or *-leigh* (Old English *lēah*, “wood, clearing”), and Norse ones in *-thwaite*. French *essart* and *-sart* and German *-rode*, *-rath* and *-reuth* (as in Bayreuth) mean “cleared land”. For more name endings and how to

use them, see [Naming settlements realistically](#). German eastward settlement (*Ostsiedlung*, 12th–14th centuries) cleared forest in the same way, with Waldhufendörfer and standard farm holdings (the Frankish *Hufe*, about 24 ha or 60 acres).

MAP TIP

If your kingdom is in its growth phase, show new strip villages pushing up forest valleys and new dykes on the coast. If it is after a plague, show some assarts going back to scrub and a few deserted villages (see [chapter 01](#)).

LATER ERA (1500S+)

Big drainage projects begin. The Dutch drained whole lakes with windmills: the Beemster (drained by 1612) and the Schermer (by 1635). See [What Changes After 1500](#). In England, drainage of the Fens began in earnest in the 1630s, and the Dutch engineer Cornelius Vermuyden worked on its second phase in the 1650s. Straight new rivers and huge rectangular fields replaced the old marsh.

Local administration: parish, manor, hundred

RULE OF THUMB

The **parish** is the unit of church and village life. The **manor** is the unit of lordship and rent. The **hundred** groups roughly 10–15 parishes for justice and tax. They often do not match.

Unit	Area	What it does	Seat or meeting place
Township or vill	Part of a parish in the north and uplands; equal to the parish in the lowlands	Farming community, tax unit	The village
Parish	~14 km ² (5.4 sq mi) lowland; 50–100+ km ² upland	Church, tithes, burial; ~9,500 in England c. 1300	Parish church
Manor	2–10 km ² (0.8–4 sq mi): one village, part of one, or several	Lord's land, rents, labour services, manor court	Manor house
Hundred / wapentake	Very variable: small in the south-east (Sussex, Kent), large in the Midlands and north. A rough average is ~150–200 km ² (60–80 sq mi) and 10–15 parishes (about 630 named hundreds were listed in 1316; derived)	Court for local justice and order. It met 12 times a year by the 12th century; later it met every two weeks, until a 1234 ordinance set it at every three weeks. "Wapentake" (Old Norse for "weapon-taking") is the name in the Danelaw, the part of England once under Danish law (Yorkshire, Lincolnshire, the East Midlands)	Often an open-air site: a mound, a tree, a stone, a ford, a crossroads
Lathe / rape	Groups of hundreds in Kent (7 lathes) and Sussex (4 rapes) at the Norman Conquest	Between hundred and shire	A royal centre or castle
Shire / county	See Capitals, Realms and Borders		County town

Equivalents elsewhere: the Frankish *centena*, the Scandinavian *herred* or *häräd*, the French *châtellenie* (castle district), and the Italian *pieve* (a large rural district served by one baptismal church with several chapels).

Moot sites: A *moot* is an assembly. Assemblies met in the open at places everyone could find. Secklow Mound (in

modern Milton Keynes) was the meeting place of Secklow hundred. Tynwald Hill (Isle of Man) and Thingwall (Wirral) come from the Norse *thing* (assembly). Place-name clues are -low or -law (mound), "Moot Hill", "Gallows Hill", "Court Hill", and names with Thing- or -ting.

MAP TIP

On a county map, divide the land into hundreds about 12–15 km (7–9 mi) across, smaller in rich, crowded country and larger in the hills. Draw their boundaries as thin dash-dot-dot lines. Name each hundred after its meeting place, and mark that place with a small lone tree labelled "Moot",

even when the real meeting place was a mound, a stone or a ford (a small mound symbol would read as a barrow). The site is often on a road or at a ford near the middle of the hundred and away from any village. Gallows (a small "II" on a hill) often stand nearby.

Drawing a local-area map

RULE OF THUMB

At local scale, almost every square kilometre has a human feature: a field, a path, a hedge, a mill, a wood with a name. Empty white space means marsh, moor or forest, and should be labelled as such.

What to show at each scale

Map size	Example	What to show
~5 × 5 km (3 × 3 mi)	One or two parishes	Individual houses, tofts and crofts, furlongs, the mill and its leat, ponds, pound, paths, named woods and fields
~20 × 20 km (12 × 12 mi)	A hundred or a barony	Every village and hamlet, churches, mills, manor houses, moated sites, parks, granges, woods, commons, roads and footpaths, parish boundaries
~50 × 50 km (30 × 30 mi)	A small county	Villages as dots, market towns, castles, abbeys, forests, big parks, rivers and main roads. Leave out mills and hamlets

How many features: England c. 1300 averages

These are derived from national totals and the area of England (130,000 km²). Use them as a sanity check. Real features cluster: parks and moats on lowland clays, warrens on sandy heath, windmills in the flat east.

Feature	One per ...	Per 100 km ² (10 × 10 km; 39 sq mi)	Per 20 × 20 km map (400 km ²)
Parish church and village (lowland)	~14 km ²	~7	~25–30
Hamlets and farmsteads	Varies by region	0–5 (open-field lowland) to 20+ (dispersed uplands)	0–100+
Mills (c. 1300)	~9–13 km ²	~8–11	~30–45
Deer parks	~40 km ²	~2–3	~10
Moated sites	~22 km ²	~4–5	~15–20
Market places (c. 1300, many tiny)	~75 km ²	~1–2	~5
Hundred meeting places	~150–200 km ²	~0.5–0.7	~2–3
Fishponds	Recorded: ~65 km ² (a minimum)	at least 1–2	at least 6

For castles, abbeys and roads, see [Castles, Forts and Military Outposts](#), [Religious, Cultural, Legal and Ancient Sites](#) and [Trade Routes, Roads and Transport](#).

Suggested symbols

These symbols match the guide's [master legend](#).

Feature	Symbol
Farmstead	Tiny dot
Hamlet	2–5 small house marks, no church
Village	House marks around a church; on a kingdom map, a tiny dot (notable villages only)
Parish church	Small church with a tower (a small plain cross with no tower means a chapel)
Manor house	Small open square; double outline if moated (a real moated site is only about 40–80 m across, so on a 1 cm = 1 km map this is a symbol, not to scale)
Grange	Small open square topped with a cross, or the label “Grange”

Feature	Symbol
Watermill	Wheel on the stream, with the leat as a thin blue line
Windmill	X of four sails on a dot or small mound
Tide mill	Mill wheel on a dam across a tidal creek
Open fields	Unhedged areas with blocks of hatching
Enclosed fields	Small irregular polygons
Meadow	Pale green band along streams
Vineyards	Hatched strips on south-facing slopes
Wood	Tree symbols with a firm edge (managed wood)
Wood-pasture	Scattered single trees on grass
Deer park	Rounded pale (fence) line with scattered trees inside
Royal forest	Green dash-dot boundary with a large italic name; villages and fields inside
Commons, heath, moor	Tufts or stipple
Fishponds	Small rectangles in a chain along a stream
Shielings (summer huts)	Cluster of tiny squares on high pasture
Track or footpath	Thin dotted line
Drove road	Broad dotted line (two rows of dots)
Wayside cross	Tiny cross on a base, set on the road or path
Boundary stone	The letters "BS" on the boundary line
Moot (assembly) site	Small lone tree, labelled "Moot"
Gallows	Small "П" on a hill
Deserted village	Grey church alone in a field, labelled "(lost village)"
Dyke or sea bank	Bold line with short ticks on the water side
Qanat	Straight line of tiny rings (shaft mouths) ending in green gardens
Rice terraces	Tight parallel lines that follow the contours
Tank (reservoir)	Curved dam line with water behind it and a fan of small fields below
Parish, manor or hundred boundary	Thin dash-dot-dot line along streams and ridges

A method for a local map

1. Draw relief and water: streams, springs, flood plains, marsh.
2. Choose the settlement pattern for each part (nucleated, dispersed, linear).
3. Place villages 1.5–4 km (1–2.5 mi) apart on good land, near water and on the edge of the flood plain. Place hamlets and farms in the uplands, forest and marsh.
4. Draw the parish boundaries (thin dash-dot-dot lines) through the land between villages.
5. Draw the rings: fields around each village, meadow along streams, pasture and wood at the parish edges.
6. Add one mill per village, with its leat or on its hill.
7. Add manor houses, moats, dovecotes, fishponds and one or two deer parks near the richest manors.
8. Add a royal forest or chase, rabbit warrens on heath, granges in monastic land, and shielings in the hills.
9. Add paths: village to church, mill, market town and neighbours; drove roads to summer pastures.
10. Add the hundred moot site, wayside crosses, boundary stones and a few names.

FANTASY TWIST

Other peoples would farm differently. Dwarves in the mountains could farm like Mediterranean terrace villages, with irrigation tunnels like qanats. Elves might manage forests as wood-pasture and coppice on very long cycles, with tightly enforced “forest law”. In a land with flying mon-

sters, open-field villages would need shelters in the fields and lookouts on the church tower. In each case keep the basic rules: water, walking distance and rings of land use. See [Fantasy Variants](#).

Quick summary

- 85–95% of people lived in the countryside. On a local map, almost all the land is fields, pasture, wood or waste, and almost all of it has an owner and a name.
- Open-field arable gives tight nucleated or green villages; uplands, forest and marsh give hamlets and farmsteads; valleys and dykes give strip villages in a line.
- A typical lowland village has 30–60 households (150–300 people), a territory of 5–15 km² (2–6 sq mi), and lies 1.5–4 km (1–2.5 mi) from its neighbours.
- Land use forms rings: gardens and orchards, then arable within about 2 km (1.3 mi) of the houses, then pasture, then wood and waste at the parish edge. Meadow follows the streams. Beyond 3–4 km (2–2.5 mi), people build outlying farms or switch to pasture.
- Across a whole settled lowland kingdom c. 1300, about a third of the land is ploughland (England: ~36%, fallow included), 7–15% is woodland, and most of the rest is pasture, heath, moor and marsh. After a plague, ploughland shrinks by a quarter to a third.
- Open-field strips were about 200 m (220 yd) long and ¼–1 acre in size, ploughed into ridge and furrow. Systems had two or three great fields, one left fallow each year.
- Uplands used infield-outfield and summer shielings; the Mediterranean used terraces, canals, norias and qanats; wet-rice lands used terraces, tanks and shared canals with water temples (Bali, Sri Lanka, Angkor), and the Aztecs built raised beds in lakes; Castile moved over 2.5 million sheep each year, some of them 550–750 km (340–470 mi).
- Mills: 5,624–6,000+ in Domesday England (1086); about 10,000–15,000 by 1300 (estimates vary), roughly one per village. Windmills in England from 1185; tide mills from the 7th century.
- A medieval “forest” is a legal hunting area, not just trees. Royal forest covered about a third of southern England at its peak. There were about 3,200 deer parks c. 1300, usually 40–80 ha (100–200 acres) with rounded outlines.
- Many manors have moats (about 6,000 moated sites are known in England, most only 40–80 m across), dovecotes and fishponds. Monasteries have granges and great barns.
- Parish, manor and hundred are different units. Hundreds of very roughly 150–200 km² (60–80 sq mi) met at open-air moot sites.
- Reclaimed land (fens, polders, assarts) has straight lines and new names; old land has curved boundaries.

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CHAPTER 7

INDUSTRY, RESOURCES AND SPECIAL TOWNS

Most medieval towns grew because of farming and trade. Some towns grew for a different reason: there was silver in the hill, salt in a spring, herring in the sea or a fast stream to drive a mill. This chapter tells you what people dug, cut, boiled, wove and caught, where each industry sat on the land, and what kind of settlement it created. Use it to give your map “odd” towns in odd places, each with a clear reason to exist.

The baseline period is the High and Late Middle Ages in Europe (c. 1000–1500). Settlement sizes follow [Population and Settlement Sizes](#). Roads, rivers, ports and transport costs are in [Trade Routes, Roads and Transport](#). Ordinary village mills and woodland use are in [Villages, Farms and the Countryside](#). The map symbols for this chapter are listed in [Suggested symbols](#).

IN THIS CHAPTER

The basic rules: what pulls industry to a place

Mining and metals

Salt

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Fishing

Shipbuilding, naval stores and arsenals

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The basic rules: what pulls industry to a place

RULE OF THUMB

Industry goes to the heaviest thing it needs. That is usually the ore, the fuel or the water power, not the customer.

Why: Moving goods over land was very expensive. In England c. 1300, moving goods by land cost about 8 times as

much as by sea and 2 times as much as by river (see [chapter 05](#)). So a smelter (a furnace that heats ore, the metal-bearing rock, to get the metal out) is built next to the mine and the woods, not next to the city. Only the finished, lighter product (silver bars, iron blooms, salt, cloth) travels.

Three things decide almost everything:

Input	Why it matters	What it means for the map
Raw material (ore, clay, stone, brine, fish)	Heavy, bulky, often worthless until processed	Industry sits on top of the resource
Fuel (wood, charcoal, peat, coal)	Smelting, salt boiling, glass, brick and lime use huge amounts	Industry sits in or next to forest, peat bog or coal outcrop
Water power (a fast stream with enough fall)	Drives bellows, hammers, fulling mills, ore crushers, mine pumps	Industry sits in hill valleys, on streams, behind small dams and ponds

Then there are two more pulls:

- **Water transport.** Bulk goods follow rivers and coasts. A quarry or coal pit near a navigable river can sell far away; one 30 km (19 mi) inland usually sells only locally.

- **Skilled people and capital.** Fine cloth, armour and glass need trained workers and merchants with money. These stay in towns, even when the raw material comes from far away.

The master table

Resource	What it needs	Where it goes on the map	What settlement it creates
Silver, lead-silver	Ore veins; wood for pit props and smelting; water to drain mines and turn wheels	Old mountains and hill country, up side valleys	Chartered mining town, a mint, many mining hamlets around it
Gold	Veins in mountains, or gold-bearing river gravel	Mountains; river sands	Mining town with a royal mint
Copper	Ore and a lot of fuel for roasting	Hills and mountains	Mining town with bare, smoke-killed land around it
Iron	Ore (common), charcoal, water power	Wooded hills with fast streams	Scattered bloomeries and forges; iron villages; an iron-trading town where the valley meets a big river
Tin	Tin-bearing gravels and granite moors	Moorland valleys	Tinners' camps; "coinage towns" at the edge of the moor
Coal	Seams that reach the surface	Coalfields near a coast or navigable river	Pit hamlets; a coal port
Salt from brine	A salt spring plus a great deal of fuel	Lowland springs	A rich salt town; "salt roads" leading away from it
Rock salt	Salt beds in the ground	Foothills and mountains	Mining town with a royal official's castle
Sea salt	Flat shore, sun, dry summers	Lagoons and sheltered bays, mostly in the south	Salt-pan villages and a loading port
Building stone	A good outcrop close to water	Hill edges near rivers or the sea	Quarry hamlet; stone shipped by boat
Clay	Clay plus fuel	River valleys, estuaries, glacial lowlands	Brickfields outside towns; potters' villages
Lime	Limestone or chalk plus fuel	Limestone edges, building sites, ports	Kilns, not towns
Timber	Forest plus a river that can float logs	Upland forests above rivers	Rafting villages; a timber market in a river town
Charcoal, glass, potash, tar	Woodland	Deep forest	Temporary camps; glasshouses that move
Peat (turf)	A fen or bog, plus a river or canal to a fuel-hungry town	Wet lowlands with few trees	Turf commons beside fen villages; flooded pits that become lakes
Wool	Grazing land	Downs, limestone hills, uplands, dry plateaus	Sheep villages, monastic farms, wool market towns
Cloth	Wool, fulling water, dyes, skilled labour	Towns; fulling mills in hill valleys	Cloth towns; rural weaving villages
Dye plants (woad, madder)	Rich arable land	Plains near cloth regions	Woad towns and woad mills
Herring	Seasonal shoals plus salt	Shallow seas and straits	Seasonal beach markets and curing stations
Cod (stockfish)	Cold, dry, windy spring weather	Arctic coasts	Fishing stations, drying racks, one big trading port
Ships	Oak, pine, tar, hemp, iron	Sheltered estuaries near forest and rivers	Shipyards; walled state arsenals
Wine	Sunny slopes and a river for export	South-facing river slopes	Wine villages; an export port downriver
Sugar cane	Frost-free land, irrigation water, a crushing mill, fuel for boiling, a port	Irrigated coastal plains in the warm south (Cyprus, Sicily, Valencia)	A fortified estate centre with a mill and boiling house; cane villages; a loading port
Beer	Clean water and grain	Grain regions; towns	Brewing towns

MAP TIP

For every industry you draw, check three things on the map: the resource, the fuel and the way out (a river, a coast or a road to one). If one is missing, move the industry or add what is missing.

Mining and metals

What a mining town is like

RULE OF THUMB

Mines are found in old, rugged hill country, up side valleys, far from the best farmland. The mining town must buy almost all its food.

Why: Metal ores form in old, folded and broken rocks: the Harz, the Ore Mountains (Erzgebirge), the Slovak and Bohemian hills, Cornwall, the Pennines, the Alps. The valleys give water power. The hills give wood.

Special laws. Rulers wanted miners badly, so they offered freedoms:

- **Free mining.** In German lands this was called *Bergfreiheit* (mining freedom): any settler could dig for ore, even without owning land, if he paid a share (often a tenth) to the lord. The silver had to be sold to the lord's mint.
- **Own courts and law.** Mining codes (*Bergordnung*) set out claims, duties and miners' rights. Tinnners in Cornwall and Devon had their own stannary courts from King John's charter of 1201. Derbyshire lead miners had the Barmote Court.

Size:

Type	Population	Notes
Mining camp or hamlet	20–300	Huts near shafts. Many vanish when the vein runs out.
Ordinary mining town	1,000–5,000	The usual case. Most mining towns stayed small.
Boom town	10,000–20,000	Rare. Two examples from just after 1500: Annaberg (Saxony), founded in 1496, grew from about 3,000 people c. 1500 to about 12,000 by 1540. Joachimsthal (Bohemia), founded in 1516, had about 20,000 people by 1534 and was then the second town of Bohemia after Prague.

Boom towns are the main exception to the rule that towns over 10,000 need navigable water. Silver is light and valuable, so it pays for food carted in over the hills. Mining towns boom and bust. A rich vein can create a town in 20

- **Lower taxes and extra rights**, such as market, brewing and customs rights. Mining towns often had more self-government than ordinary towns.

What it looks like:

- Streets on slopes. Older mining towns are often irregular, because they grew fast around mine shafts. Later ones founded by a ruler (such as the Ore Mountain towns of c. 1500) were often planned, with a grid of streets and a square market place.
- Heaps of waste rock (spoil heaps) on the hillsides.
- Smelting huts and ore-crushing mills along the stream below.
- A **mint**, because silver and gold were turned into coin on the spot. A strong kingdom often kept a mint at each big silver mine, as well as in the capital. See [How many mints?](#)
- A large, rich church for its size. Often a royal castle or official's house to watch the money.
- Little farmland. Food came in from lowland markets.

years. Flooding, deep shafts and exhausted ore can empty it again. Drainage was the big limit. Medieval pumps could not drain deep shafts, so miners dug long drainage tunnels (adits) out to the valley side.

Silver

Silver was the money metal of medieval Europe, so silver mines made kings rich.

Mine	Dates	What happened
Rammelsberg, Goslar (Harz, Germany)	Mining recorded c. 968; much older	Ore with silver, lead, copper and zinc. Henry II built an imperial palace at Goslar from 1005 because of the silver. Over 100 miners died in a collapse in 1376. Worked until 1988.
Freiberg (Saxony)	Silver found c. 1168	Found at the village of Christiansdorf. The margrave of Meissen offered free mining; miners poured in, and within about 20 years the town of Freiberg ("free mountain") existed. Its first boom faded in the 14th century.
Kutná Hora (Bohemia)	Silver mined from c. 1260; great rush in the 1290s	Near Sedlec Abbey. King Wenceslas II issued a mining code and created the Prague groschen coin around 1300, struck at the central mint in the "Italian Court". In the late 1200s the district produced perhaps a third of Europe's silver. It became Bohemia's second town after Prague.
Iglesias (Villa di Chiesa, Sardinia)	13th century	Founded by the Pisan count Ugolino della Gherardesca. Ruled by its own law code, the <i>Breve</i> (a 1327 copy survives). Estimated at about a tenth of Europe's silver in the early 1300s.

Population estimates for Kutná Hora vary hugely. Popular books give up to 60,000. That is very likely too high. Treat it as a large town of many thousands.

Gold

Medieval Europe had little gold. The main source in the 14th century was the Kingdom of Hungary, in the mountains of today's Slovakia and Transylvania.

- Around 1330, Hungarian mines produced about 1,400 kg of gold a year. That was more than 30% of the world's output (an estimate).
- **Kremnica** (Slovakia) was one of the most important of these gold mines. It got royal town privileges in 1328. From 1331 it was the seat of the royal official (the "chamber count") who ran all mines and mints in twelve counties.

From 1335 its mint struck gold florins, later the famous Kremnica ducats. Kremnica's own yearly output is not well recorded; treat any exact figure with caution.

- Kremnica sits at 564 m (1,850 ft) in a mountain valley, about 20 km (12 mi) in a straight line from Banská Bystrica. Several mining towns stood within a day's travel of each other.
- Gold also came from washing river sand, a small, scattered trade along certain rivers.

Copper

- **Falun** (Sweden): the "Great Copper Mountain". Copper was dug here by the mid-1200s, and possibly as early as about 1000. A document of 1288 records a share in the mine, which was run by a group of share-owning master miners. The first known royal charter dates from 1347.

- Copper ore was roasted in open fires, which gave off sulphur smoke and killed the plants for a long way around. Draw a bare zone around a big copper mine.
- Copper mixed with tin makes bronze (bells, cannon); mixed with zinc ore (calamine) it makes brass.

LATER ERA (1500s+)

Falun peaked in the 1600s, when it produced a large share of Europe's copper (popular accounts say about two-thirds of world output in the mid-1600s; treat this as a rough claim). New silver towns boomed too: Joachimsthal (1516) gave its

name to the *thaler* coin and so to the word "dollar". After about 1550, cheap silver from Spanish America (Potosí, found in 1545) hurt many European mines.

Iron

RULE OF THUMB

Iron ore is common. Charcoal and water power are the real limits. Put ironworks where forest, fast streams and ore meet.

Why: Medieval iron was made with charcoal. Bloomeries (simple clay furnaces) burned roughly as much charcoal as

ore, by weight. Estimates of the total vary widely, but they often fall around 10–20 kg of charcoal for each 1 kg of finished iron. Since charcoal is made from about four or more times its weight in wood, that means very roughly 50–150 kg of wood for 1 kg of iron. Wood is too bulky to move far, so ironworks moved to the wood.

Region	Features
Erzberg (Styria, Austria)	A whole mountain of iron ore, worked in open pits. First recorded in 1171. Two mining communities, Innerberg (today Eisenerz) and Vordernberg, smelted ore in water-powered “wheel works” (<i>Radwerke</i>) along the valley. The iron trade was given to privileged towns: Steyr for Innerberg iron (1287) and Leoben for Vordernberg iron (1314).
Siegerland (Germany)	Iron mining and smelting in wooded hills. Communities managed their oak and birch woods as <i>Hauberg</i> : coppice (trees cut to the stump so they regrow) cut every 16–20 years, with a year of rye sown after each cut. Formal forest ordinances date from 1562 and 1565.
Basque country (Biscay and Gipuzkoa, Spain)	Water-powered ironworks (<i>ferrerías</i>) in steep, rainy, wooded valleys from the 1300s. Iron was exported to England and Flanders through ports such as Bilbao (founded 1300).
The Weald (south-east England)	Bloomeries from early times; the first blast furnace in England c. 1490. See Forest industries .

What to draw: iron districts are not one big town. They are many small sites (a furnace, a hammer forge, a pond, char-

coal hearths) scattered along every stream, plus one or two towns that trade and finish the iron.

Tin

- Most of Europe’s medieval tin came from **Cornwall and Devon**. The Ore Mountains (Saxony and Bohemia) became the other big source, especially in the late Middle Ages.
- Most medieval tin was “streamed”: washed out of gravels in valley floors, not mined deep. Tanners turned moorland valleys into rows of spoil heaps and water channels.
- **Stannaries** (tin-mining districts) had their own law. King John’s charter of 1201 let tanners dig for tin on

almost anyone’s land and placed them under the Lord Warden of the Stannaries and his own courts, not the ordinary courts.

- All tin had to be brought to **coinage towns** to be weighed, tested and taxed. In Cornwall these were Lostwithiel, Bodmin, Liskeard, Truro and Helston. In Devon, Tavistock, Ashburton and Chagford (1305) and Plympton (1328).

MAP TIP

Put coinage towns on the edge of the moor, where tin came down from the uplands and met a road or a tidal river. Do not put them in the middle of the moor.

Lead

- Main English fields: **Derbyshire** (the Peak, around Wirksworth) and the **Mendip Hills** (Somerset). Also the northern Pennines.
- Derbyshire had free mining: anyone who showed the *barmaster* (the mining official) good ore could open a mine and keep it while he worked it. Disputes went to

the Barmote Court. Lead at Wirksworth and Bakewell is already recorded in Domesday Book (1086).

- Lead was smelted in **boles**: open hearths on windy hill-tops and ridges, using the wind as bellows. Many “Bole Hill” place names in the Peak mark them.
- Uses: roofs of churches and castles, gutters, water pipes, the strips that hold stained glass. Lead ore often holds silver, which was separated in a furnace.

Coal

Coal was a small, local fuel in the Middle Ages, used mainly by smiths, lime burners and the poor in coal districts. It mattered in two places above all:

- **Newcastle upon Tyne** (England). Seams came to the surface along the coast and river banks, so the English called it “sea coal”. Sea coal from the north-east was reaching London by 1228, and the coal trade by sea grew through the 1200s. In 1306 a royal proclamation banned London craftsmen from burning sea coal in their furnaces because of the smoke. By the early 1500s (before

1539) the Tyne shipped roughly 15,000 tonnes a year; medieval shipments were probably of this size or smaller.

- **Liège** (today’s Belgium). The first texts on coal here date from the 12th century, and coal digging was regulated by 1229. Miners dug drainage tunnels called *areines*. A special court, the *Voir-Jurés du charbonnage*, judged mining disputes.

Medieval coal pits were small: “bell pits” (a shaft widened at the bottom, abandoned when it got dangerous) and shallow tunnels into hillsides.

LATER ERA (1500S+)

Coal boomed when wood became scarce. Tyne shipments grew from about 15,000 tonnes a year before 1539 to about 240,000 tonnes in 1600 and 425,000 in 1634. Draw many col-

liery (coal-mine) hamlets along the Tyne, each with the mine symbol (crossed hammers), and a sea lane (a dotted line at sea) to London labelled “colliers” (coal-carrying ships).

Mining districts on the map

Scale	What to show
Continent	A mining-district symbol (crossed hammers or a pick) on 3–6 mountain areas. Not every range has ore.
Kingdom	1–3 mining districts. In each, a cluster of 2–7 mining towns, each 10–40 km (6–25 mi) apart, plus a mint town (see How many mints?). The central Slovak mining towns and the towns of the Ore Mountains are real clusters like this.
Local area	Shafts, spoil heaps, adit mouths in the valley side, ore-crushing mills on the stream, smelting huts, charcoal hearths in the woods, the miners’ town on a slope, a castle or official’s house.

FANTASY TWIST

Dwarves who can drain or shore up deep mines remove the medieval depth limit, so their mines last for centuries instead of decades. Magic ores (mithril and the like) behave like gold or silver: valuable enough to pay for long, guarded roads, so a remote mine can still support a large town. A dragon or monster in the hills is a good reason why a known ore field is *not* worked. More in [chapter 10](#).

MAP TIP

Draw a mining town up a side valley, with a road down to the nearest market town or river. Label the mine with a name (in German lands, mine names were often saints or wishes: “St Anne’s Find”, “Good Hope”). Add a few abandoned shafts (the mine symbol drawn grey, labelled “(old shaft)”) and a ghost hamlet (grey house marks, labelled “(lost hamlet)”). Mining landscapes are full of failed attempts.

Salt

RULE OF THUMB

Everyone needs salt, and few places have it. Every realm on your map needs at least one salt source, or a trade route that brings salt in.

Why: People and animals need salt to live. Salt also preserved fish, meat, butter and cheese for winter and for fast days. A medieval region without salt had to import it, so salt towns became rich, and salt was taxed heavily.

Three ways to make salt

Method	How it works	Fuel	Where	Examples
Brine springs	Brine (very salty water) comes up from salt beds underground. It is boiled in large pans until salt crystals form.	Huge: wood, peat or (later) coal	Inland lowlands and valleys over buried salt beds	Lüneburg, Droitwich, Halle, Salins
Rock salt mines	Salt is dug as rock, or water is pumped into the mountain to dissolve it and the brine is boiled	Some (for boiling brine)	Foothills and mountains	Wieliczka and Bochnia (Poland), Hallstatt and other Alpine salt towns
Sea salt pans	Seawater is let into shallow clay pans and the sun evaporates it	Almost none	Flat, sheltered coasts with hot, dry summers	Bay of Bourgneuf (France), Venetian lagoon (Chioggia), Ston (Croatia)

In the cold, wet north, people also boiled seawater or burned salt-soaked peat (in Zeeland and Frisia) to get salt. This produced much less.

Salt towns

Brine-spring towns:

- **Lüneburg** (northern Germany). Salt making dominated the town from the 12th century. A brine spring was surrounded by 54 boiling huts, each with 4 pans: 216 pans in all. The workers lived in their own walled “Salt Quarter” of the town; they numbered several hundred at the works’ height. The salt went to Lübeck on the Baltic, mostly to salt herring. In the 16th century about 19,000 tonnes a year made that trip.
- **Droitwich** (England). Brine here is nearly saturated: about 7–10 times saltier than seawater (often quoted as “ten times”). Domesday Book (1086) shows the king holding the brine pits, and many manors across the West Midlands owning salt pans or rights to salt. Salt was carted along “saltways” to neighbouring counties and to the River Severn at Worcester.
- **Halle** (Saale, Germany). The name is usually linked to an old word for salt. Four brine wells stood around the market. The salt workers, the *Halloren*, were a famous guild. Halle was in the Hanseatic League by 1281.
- **Salins** (Franche-Comté, France). Two saltworks were working by 1115. Salt made Salins the second town of the region after Besançon, and the title “lord of Salins” was prized by the counts and dukes of Burgundy.
- **Bay of Bourgneuf** (south of the Loire, France). “Bay salt” was cheap, grey and coarse. In the late Middle Ages, fleets of Hanseatic and later Dutch ships (the “Bay fleet”) came every year to load it for the Baltic fisheries. The bay was a major salt producer in the 15th and 16th centuries.
- **Venetian lagoon and Chioggia**. Salt pans made Chioggia rich in the 12th–14th centuries. Venice tried to control the salt trade of the whole northern Adriatic.

The fuel problem

Brine boiling burned enormous amounts of wood. Brine towns were often blamed for clearing the forests around them. Lüneburg Heath is the classic example, but the story

Rock-salt towns:

- **Wieliczka and Bochnia** (near Kraków, Poland). Rock salt mining began in the 13th century. King Casimir the Great issued a mining ordinance in 1368. In the 14th century the Kraków saltworks gave the Polish crown about a third of its income. At its height (16th to mid-17th century) Wieliczka had about 2,000 miners and produced over 30,000 tonnes a year.
- **Hallstatt** (Austria). Salt mined since prehistoric times. A tiny town squeezed between a lake and a mountain, reached only by boat or narrow paths until the 19th century.

Sea-salt districts:

- **Ston** (Croatia). Dubrovnik bought the Pelješac peninsula in 1333 for its salt pans. From 1358 it built walls more than 7 km (4.3 mi) long, with 40 towers, across the neck of the peninsula; 5.5 km (3.4 mi) survive. The twin planned towns of Ston and Mali Ston sit at each end. The salt pans were one of Dubrovnik’s main sources of income.

is not proven: the heath probably began with much older clearances, back in the Bronze Age. When wood ran out, there were two answers: bring fuel in by water, or pipe the brine to the forest.

LATER ERA (1500s+)

Brine pipelines appear. From 1595 to 1607 a wooden pipeline was built to carry brine from Hallstatt down to Ebensee, where wood was plentiful. Sources give its length as

34–40 km (21–25 mi). From 1775, brine from Salins was piped 21 km (13 mi) to the new Royal Saltworks at Arc-et-Senans, built next to the large Forest of Chaux.

Salt roads

- Salt moved overland farther than other bulk goods because everyone had to buy it. The **Old Salt Route** from Lüneburg to Lübeck was about 90–100 km (55–60 mi). In 1398 the Stecknitz Canal opened to carry salt by water.
- Around Droitwich, the Upper and Lower Saltways led east across the Midlands and south over the Cotswolds.
- Salt roads usually lead from the salt source to the nearest navigable river or port, then by water to the big markets. Many old roads are still called “Salt Way”, *Salzstraße* or *Via Salaria*.

FANTASY TWIST

Fire magic or magical heat removes the fuel problem of brine boiling. Then brine towns can grow without eating their forests, and a salt spring in a treeless land becomes valuable. A “salt mage” guild would be as powerful as a mint.

MAP TIP

Place a brine town on lowland near a spring, with woods or peat bogs nearby (or a river to bring fuel). On a local map, mark the spring with a small blue dot labelled “Brine Pit”; on a kingdom map, add “(salt)” to the town’s label instead. Draw 2–4 roads radiating from it, each labelled “Salt Road” or “Saltway” on the first stretch. Sea-salt pans need flat, sheltered shore and a warm, dry summer: draw a checkerboard of small rectangles along a lagoon. In the north, keep sea salt small and import “bay salt” by ship. Place names with “Hall”, “Salz”, “Sal” or “Saline” often, but not always, mark salt. In England, the “-wich” brine towns of Cheshire and Worcestershire (Droitwich, Nantwich, Middlewich, Northwich) are a real cluster.

Stone, clay and lime

Quarries

RULE OF THUMB

Ordinary buildings use local stone. Only fine stone for great churches and castles travels far, and almost always by water.

Why: Stone is the heaviest common building material. Moving it often cost more than quarrying it, so builders used the nearest stone that would do.

Real examples:

- **Caen stone** (Normandy). A fine, cream-coloured limestone. The quarries lay close to the River Orne, so ships could carry it across the Channel. Canterbury Cathedral was rebuilt in it from 1070.
- **Barnack stone** (Northamptonshire). Used for Peterborough and Ely cathedrals and many fenland churches, moved by river and fen waterways.
- **Purbeck marble** (Dorset). A dark, polishable limestone used for thin columns in many English cathedrals. Shipped from the coast.
- **Millstones**. Hard lava stone from the area between Mayen and Andernach (Eifel, Germany) was carried by cart to the Rhine at Andernach and shipped as far as England and Scandinavia. Rough sandstone (gritstone) from the Peak District made English millstones.
- **Slate** for roofs came from places with natural splitting rock, such as Cornwall and north Wales. Most medieval roofs were thatch, wooden shingles or tiles instead.

What to draw: a quarry is a notch in a hillside, with a track down to water and a small hamlet of quarrymen. For a big

church project, masons set up a temporary lodge (workshop) at the quarry or at the building site.

Brick

RULE OF THUMB

Where there is no good stone, people build in brick. Look for flat lowlands with clay.

Why: The lands around the southern Baltic, the North Sea coast, the Po plain and the Garonne valley have deep clay and few hard rocks.

Brick region	Examples
Northern Germany, Poland, the Baltic states, Denmark, southern Sweden ("Brick Gothic")	Lübeck, Gdańsk, Toruń; the Teutonic Order's castle at Malbork (begun in the late 13th century), the largest brick castle
Low Countries	Most towns of Holland and Flanders
Lombardy and the Po plain	Milan, Pavia, Cremona, Bologna
Southern France	Toulouse ("the pink city"); Albi Cathedral (1282–1480)
Eastern England (late medieval)	Hull; Tattershall Castle (1430s–40s)

Brick and tile kilns stood outside the town walls, near clay pits and water, because of fire risk and smoke. Old clay pits became ponds.

Lime kilns

- Lime (made by burning limestone or chalk) was needed for mortar, plaster and limewash, and sometimes spread on fields.
- Kilns stood next to limestone or chalk outcrops, or next to big building sites, or at ports where stone and fuel could both arrive by boat. Lime burners were among the first medieval users of coal.

- A lime kiln is a small industrial site, not a settlement. On a local map, mark kilns along a limestone edge with the word "Limekiln" (or "Limekilns" for a row), not with a picture. A small tower would read as a castle, and a small mound as a barrow (an ancient burial mound) or a motte (a castle mound).

Pottery

- Everyday pots were made in **potters' villages** near good clay, with woodland or heath for fuel. Most sold within a day's travel by cart.

- A few kinds of pottery were traded across Europe: Rhineland stoneware (Siegburg, Langerwehe, Raeren) went down the Rhine to the North Sea and the Baltic; lustreware from Málaga and Manises (near Valencia) went around the Mediterranean and to England.

MAP TIP

On a kingdom map, mark only famous quarries (with the quarry symbol, a notch in the hillside, and a name like "Marble Quarries") and the brick-built towns of clay lowlands. On a local map, add clay pits and kilns outside the walls (labelled "Clay Pits", "Brick Kilns"), a quarry track to

the river, and a lime kiln at a limestone edge. If you colour your town symbols, use a pale stone colour (buff or cream) in a region of stone and brick red in a clay lowland. Do not use grey: the guide's **master legend** keeps grey for ruins and abandoned places.

Forest industries

Medieval forests were not empty wilderness. They were working landscapes full of woodcutters, charcoal burners, iron smelters, glassmakers, tar boilers and pig herders. Ordinary village woods (fuel, pannage (letting pigs feed on

acorns), timber for houses) are in [chapter 06](#); this section covers the industries.

Timber

- Big building and ship timber came from forests close to floatable rivers. Logs were tied into rafts and floated downstream to towns.

Charcoal burning

- Charcoal was made in **clamps**: a dome of stacked logs covered with turf and earth, burned slowly for 6–8 days (several weeks for very large ones).
- It takes roughly 4–8 kg of wood to make 1 kg of charcoal. A traditional clamp turns about a quarter of the wood's weight into charcoal on average; small, careless burns do worse.
- Charcoal is light but crumbles if carried far, so it was made in the woods near where it would be used. The

- Venice drew timber from Alpine forests down rivers like the Piave and the Adige.
- English and Dutch builders imported Baltic oak boards (“wainscot”) from Prussia and Livonia in the 14th–15th centuries.

burners lived in huts next to the clamp to watch it day and night.

- Woods for charcoal were usually **coppiced**: cut to the stump and allowed to regrow, giving poles every few years. Medieval cycles were mostly under 10 years; the 10–20-year cycles of charcoal woods (such as the Siegerland *Hauberg*) belong mostly to the 1500s and later (see [Managed woodland](#)).

Iron smelting in the forest

Bloomeries were small furnaces that produced a spongy lump of iron (a bloom), which was then hammered. Early blooms weighed a few kilograms. Water-powered bellows and hammers, spreading in the High Middle Ages, allowed much bigger furnaces. Late medieval blooms could reach a few hundred kilograms.

Blast furnaces melt the iron completely and run for months. In Sweden, Lapphyttan is an early example, dated

somewhere between about 1150 and 1350. In England the first one was built in the Weald (at Queenstock, Buxted) about 1490. Blast furnaces need a dam and a pond for a waterwheel, and a steady supply of charcoal.

How much forest does an ironworks need? In the Weald, historians estimate that a post-medieval blast furnace needed, for a sustainable supply, the wood from about a 5 km (3 mi) radius, in a landscape that was a quarter to a third woodland.

LATER ERA (1500S+)

Blast furnaces spread. The Weald had nearly 180 furnace and forge sites over its history and produced over 9,000 tonnes of iron a year in the 1590s. Cast-iron cannon were made there from 1543. Furnace ponds and “hammer ponds” (millponds for forge hammers) dot the map of the Weald to this day.

Forest glass

- From about 1000 to 1700, northern European glass was made in the forest from sand and wood ash (potash). This is called *forest glass* (*Waldglas*).
- It used enormous amounts of wood: estimates are 150–200 kg of wood for 1 kg of glass, counting both fuel and ash. A large glasshouse might burn about 67 tonnes of wood a week, for about 40 weeks a year.

- Glasshouses moved to new woods when the old ones were cut out. The Spessart (Germany) is a good example of a glass forest. In the early 1400s just four glassworks in its Mainz-ruled part made around 230,000 pieces of blown glass a year, plus flat (window) glass. Other glass districts were Bohemia, the Black Forest, the Jura and the Weald.

LATER ERA (1500S+)

A 1615 English proclamation banned wood fuel in glassmaking. In the same year Sir Robert Mansell got a monopoly on English glass and used sea coal instead of wood. Glassmakers moved to coalfields.

Potash, tar and pitch

- **Potash**, made by burning wood and leaching the ash, was needed for glass, soap, bleaching linen and dyeing. The great forests of Prussia, Poland and Lithuania exported it through Baltic ports such as Gdańsk (Danzig).

FANTASY TWIST

If the forest is home to elves, spirits or monsters, charcoal burners, glassmakers and tar boilers become the frontier people of your world. They need permission, guards or a truce. A realm that cannot cut its forests cannot make iron or glass, and must import them.

- **Tar and pitch**, made by slowly burning pine wood and stumps in pits or kilns dug into slopes, waterproofed ships, ropes and roofs. Norway, Sweden, Finland and the eastern Baltic were the main sources.

MAP TIP

Draw a forest with small clearings, each labelled with its work: “Charcoal Hearths”, “Glasshouse”, a “Forge” by a stream with its pond (draw the forge as a watermill wheel on the stream), a woodcutters’ hamlet. Add one track out of the forest to a river. Mark a few clearings as abandoned (“Old Glasshouse”). Forest industries leave no towns, only hamlets with names like Coalpit, Furnace, Hammer or, in German lands, Glashütte (glass hut).

Peat diggings

RULE OF THUMB

Where wood is scarce in wet lowlands, people dig peat for fuel. Big peat diggings fill with water and turn into lakes. So some “natural” lakes on a real map are really old fuel pits.

Why: Peat (also called turf) is half-rotted plant matter that builds up over thousands of years in fens and bogs. Cut into blocks and dried, it burns well. Where the woods had

been cleared and towns needed fuel (eastern England, Holland, Flanders, Frisia), peat was dug on a large scale and sent to town by boat. In low land a pit soon goes below the water table, so it fills with water. Waves then eat away the soft peat banks, and the lakes grow.

Fen villages often had **turbary** (the right to cut turf or peat on a common; see [chapter 06](#)). Each village dug inside its own boundary, so the diggings follow parish lines.

Real example: the Norfolk Broads (England)

- The Broads are 63 shallow lakes, most less than 4 m (13 ft) deep, joined by rivers. Today they have over 200 km (120 mi) of navigable waterways.
- They are flooded medieval peat pits. Peat was dug here for fuel from the 12th to the 14th centuries, by monasteries and villages, and sold to **Norwich** and **Great Yarmouth**. The records of Norwich Cathedral Priory show that large amounts of peat were dug, in what was then one of the most crowded parts of England.
- Each parish dug inside its own boundary, so the shapes of the broads still follow old parish boundaries. Banks of uncut peat (baulks) were left between neighbouring dig-

gings. Some still separate one broad from the next, and roads cross the wet ground on causeways.

- The water rose (a wetter climate and a rising sea level). By the end of the 14th century almost all the diggings had been given up and had flooded.
- People forgot. Until the 1950s most people thought the Broads were natural. The botanist Joyce Lambert and her colleagues showed that they have almost vertical sides and flat floors, like dug pits, and published the proof in 1960.
- Size: most broads are small, from a few hectares up to about 1.4 km² (0.5 sq mi) for the largest, Hickling Broad.

Real example: the Haarlemmermeer (Holland)

Between Haarlem, Leiden and Amsterdam, peat cutting for fuel (much of it by people from Haarlem) and storms turned a few small lakes into one big inland sea. The Dutch called the growing lake the “water wolf”.

Date	What the map shows
13th century	Three or four separate peat lakes (the old Haarlemmermeer, the Spieringmeer, the Leidsemeer and the Oude Meer), with farmland and villages on the strips of land between them. One Dutch timeline gives about 9,100 ha (91 km ² , 35 sq mi) for all of them around 1250; other estimates are lower.
1477–1508	Storms (such as the flood of 1477) break through the strips. By 1508 the old Haarlemmermeer and the Spieringmeer have joined.
Late 1500s	The lakes have become one lake.
c. 1650	About 15,000 ha (150 km ² , 58 sq mi).
1848	About 16,850 ha (169 km ² , 65 sq mi). It was then pumped dry by 1852.

Villages such as Nieuwerkerk and Rietwijk were swallowed, and the village of Aalsmeer lost more than half of its land.

LATER ERA (1500S+)

After about 1530, Dutch diggers began to dredge peat from under the water with a net on a long pole. They dried the wet peat on long, narrow strips of land left between the pits (*legakkers*). This made whole lake districts shaped like a comb of thin strips and long water channels, such as the Loosdrechtse and Vinkeveense Plassen in the north-west of the province of Utrecht. From the 1600s, Dutch investors pumped many lakes dry with windmills (the Beemster by 1612); see [Drainage and new land](#).

MAP TIP

Draw peat lakes with straight or blocky edges, not smooth round ones, and leave narrow banks (baulks) and causeways between them. Put a “turbary” or “Turf Fen” common beside each fen village (the commons symbol, tufts or stipple, with that label), and a river or canal carrying turf boats to the nearest town. The Broads lie within about 30 km (19 mi) of Norwich, along its rivers. For a map set a century or more after the digging, turn the pits into a chain of lakes, and mark a drowned village or two: a grey church standing in the water (the master legend’s symbol for a lost village), labelled, for example, “Old Fenby (lost village)” or “Drowned Church”.

Textiles

Cloth was the biggest manufacturing industry of medieval Europe. It needs wool (or flax, cotton or silk), water power for fulling, dyes, and many skilled workers.

Wool regions

RULE OF THUMB

Sheep go where land is too dry, too thin or too high for the plough: chalk downs, limestone hills, uplands and dry plateaus.

Region	Features
England	Europe's main source of fine wool in the 13th–14th centuries. Customs records show England exporting on average about 25,000–40,000 sacks a year between the 1280s and the 1330s (five-year averages; some estimates for peak years reach 45,000). One sack weighed 364 lb (165 kg), the fleeces of about 200–260 sheep. So the exported wool alone came from something like 5–10 million sheep. Famous wool areas: the Cotswolds, Lincolnshire, the Welsh Marches, Yorkshire.
Castile (Spain)	Large flocks moved between summer pastures in the north (Old Castile and León) and winter pastures in the south (Extremadura and Andalusia), walking 550–750 km (340–470 mi) each way. The herders' guild, the <i>Mesta</i> , received its royal charter in 1273. Wool exports began in the 14th century; fine merino wool became a big export to Flanders and England in the late 15th century.
Uplands everywhere	Scotland, Wales, the Alps, the Pyrenees, the Apennines: wool mainly for local cloth.

Many English wool estates belonged to Cistercian monasteries (see [chapter 08](#)), which ran large sheep farms called

granges. After the Black Death, many abandoned villages became sheep runs.

Cloth towns

Cloth centre	When	Notes
Flanders: Ghent, Bruges, Ypres, plus Lille and Douai	12th–14th centuries	The densest cloth region in Europe. Fine cloth from imported English wool. Ghent had perhaps 40,000–65,000 people in the 14th century (estimates vary widely; see chapter 02). Bruges perhaps 35,000–46,000. Ypres perhaps 20,000–30,000 c. 1300 (older books say up to 40,000), but only about 10,000–11,000 by 1412, after its industry declined.
Florence	13th–15th centuries	The chronicler Giovanni Villani wrote that in the 1330s Florence had over 200 wool workshops making 70,000–80,000 cloths a year, and that over 30,000 people lived from the trade. Wool came from England and Spain.
English cloth districts	14th–15th centuries	After about 1350 England made more of its own cloth. Production spread into hilly countryside with fast streams: the West Country (Stroud valleys, Wiltshire), East Anglia, the West Riding of Yorkshire.

What a cloth town looks like:

- A **cloth hall**, where cloth was inspected and sold. Ypres's huge Cloth Hall was built in the 1200s.
- **Tenter grounds:** open fields beside the town, filled with wooden frames where wet cloth was stretched on hooks to dry ("tenterhooks").

- Dyers' streets along the river, often downstream.
- Weavers' quarters with long, narrow houses and many small workshops.

Fulling mills and water power

Fulling means beating woollen cloth in water with a special clay (fuller's earth) to clean it and thicken it. At first people did it by trampling the cloth with their feet. A **fulling mill** used a waterwheel to lift wooden hammers.

- Fulling mills are recorded in Normandy in the late 11th century (about 1086). In England the first certain ones appear in a survey of 1185, on lands of the Knights Templar.

- A water-powered mill did the work of many fullers, so water fulling was much cheaper than fulling by foot.
- The historian Eleanora Carus-Wilson argued (1941) that fulling mills pulled the English cloth industry out of the old towns and into the countryside, where fast streams were. Later historians think she exaggerated, but the pattern of cloth villages along hill streams is real.

MAP TIP

Put fulling mills on fast streams in hilly country near sheep land. A cloth district on the map is a hill valley with a string of villages and fulling mills along the stream (draw each mill as a watermill wheel; English names for a fulling mill

include “Walk Mill” and “Tuck Mill”, and in Wales a *pandy*), a market town at the valley mouth where cloth is sold, and tenter grounds on the slopes.

Dyes and mordants

Dye or mordant	Colour or use	Where it came from
Woad	Blue	Thuringia: the five “woad towns” of Erfurt, Gotha, Tennstedt, Arnstadt and Langensalza; the Lauragais (the triangle between Toulouse, Albi and Carcassonne), which made Toulouse merchants rich; Picardy; parts of England
Madder	Red	Zeeland and Flanders; many other lowland areas
Weld	Yellow	Widespread
Kermes	Expensive crimson, from an insect	Mediterranean
Alum	Mordant (fixes colours)	Phocaea in Asia Minor, run by Genoese merchants from the 1270s; after 1460–62, the papal mines at Tolfa near Rome

Woad needed rich, well-manured arable land and a woad mill to crush the leaves. Woad merchants in Erfurt helped

found its university; in Toulouse they built grand town houses.

Linen, cotton blends and silk

- **Linen** came from flax, grown on damp, rich soils. Retting (rotting the stems in ponds or streams) stank and polluted water. Main linen regions: Swabia and Lake Constance (Ulm, Konstanz, St Gallen), Westphalia and Flanders.
- **Fustian** (a cloth of linen and cotton) was made in southern German towns such as Ulm and Augsburg from the 14th century, with cotton imported through Venice.

LATER ERA (1500S+)

Protestant refugees from Flanders brought the “New Draperies” (lighter cloths) to English towns such as Norwich and Colchester. English cloth replaced raw wool as the country’s great export.

- **Silk** came first from the Byzantine and Islamic worlds (Constantinople, Syria, Muslim Spain: Almería, later Granada). In Italy, Lucca led silk weaving in the 13th–14th centuries. After political trouble in Lucca in 1314, many of its silk workers moved to Venice, Florence and Bologna.

MAP TIP

On a kingdom map, draw one or two textile regions, not more. Show sheep country (downs, hills, dry plateaus) with a few wool market towns; a cloth region with several close-set towns and mill streams; and, if you like, a dye district on good plains nearby (“Woad Country”). Silk belongs in rich southern or eastern cities that trade with the Byzantine or Islamic world. On a local map, add tenter grounds (an open field beside the town labelled “Tenters”), a cloth hall, dyers by the river below town and fulling mills on the side streams.

Fishing

RULE OF THUMB

Fresh fish is eaten near the water. Preserved fish (salted, dried or smoked) travels across a continent. Big fisheries need salt, barrels and a market.

Herring

- The great medieval herring fishery was off **Scania** (southern Sweden, then Danish). Every year from about 1200, a huge market was held on the beaches between the small towns of Skanör and Falsterbo. The official season is usually given as 24 August to 9 October, but in practice it could start in late July and run into late October.
- The Danish king gave Hanseatic towns their own plots of beach (a *fytte* or *vitte*) to gut, salt and pack fish. Counts vary, but around 25–30 towns had plots at the peak; the largest (Lübeck's and Danzig's) covered 6–10 hectares

Stockfish and cod

- **Stockfish** is cod dried on wooden racks in cold, dry, windy northern springs, without salt. It keeps for years.
- The main source was **Lofoten** (northern Norway). Fishermen came every winter and spring to fishing stations. Traders held summer fairs at Vågan.

Fishing villages, weirs and curing

- **Fishing villages** sat on beaches where boats could be pulled up, in sheltered coves, and at river mouths. Many were seasonal stations with huts, not true villages.
- **Fish weirs** were fences of stakes or stone in rivers, estuaries and tidal shores that trapped fish. They often

LATER ERA (1500S+)

Dutch “herring busses” (large decked fishing boats that salted the fish at sea) made Holland rich from the 1400s on. From the 1500s, Basque, Portuguese, French and English ships sailed to Newfoundland for cod.

Why: The Church forbade meat on fast days: Fridays, often Wednesdays and Saturdays, and all of Lent. That is well over a third of the year. Inland towns needed huge amounts of preserved fish.

(15–25 acres) each. Lübeck dominated, bringing salt from Lüneburg and barrels.

- A good 14th-century year may have produced about 300,000 barrels of herring (an estimate). A contemporary claimed 300,000 people worked there during the season, surely an exaggeration. By one estimate, the market supplied about a third of the Danish king's income at its peak.
- The herring left Scania in the early 15th century, and the fishery moved to the North Sea.
- English herring centred on **Great Yarmouth**, with a large autumn herring fair.
- The fish was shipped south to **Bergen**, which controlled the trade. Stockfish exports are estimated at about 4,000 tonnes a year in the 1300s. German Hanseatic merchants ran a trading post there (the *Kontor*, from about 1350–60). Nearly 1,000 Germans lived in its houses and warehouses on the main wharf, and more came in the summer sailing season.

blocked boats. Magna Carta (1215, clause 33) ordered fish weirs removed from the Thames, the Medway and all of England, except on the sea coast. (Fishponds at manors and monasteries are in [chapter 06](#).)

- **Curing** needs salt, barrels (so cooperers and timber), drying racks or smokehouses, and space on the shore.

MAP TIP

Put herring markets on a beach or spit next to a narrow sea passage, with a small town or two and a large empty area labelled “fair ground” or “curing booths”. Put stockfish stations along Arctic coasts, with their racks labelled “Fish Racks”, and one big trading town with a German (or foreign) quarter at the end of the route. Draw fishing hamlets in every sheltered cove, roughly a few kilometres apart along good coasts, fewer along cliffs.

Shipbuilding, naval stores and arsenals

RULE OF THUMB

Shipyards need a sheltered shore, timber that can be floated in, and a town with skilled carpenters. A state arsenal is a walled factory inside a great port.

What ships need:

Material	Use	Source
Oak	Hull frames and planks	Lowland forests near rivers; Baltic oak by sea
Pine and fir	Masts, spars, decks	Mountain and northern forests: the Alps, Scandinavia, the Baltic
Tar and pitch	Waterproofing	Pine forests of the north and east (see Forest industries)
Hemp	Rope and caulking	Damp farmland
Iron	Nails, anchors, fittings	Iron districts
Sailcloth	Sails	Linen and hemp regions

Small yards: Most medieval ships were built on open beaches, creek sides and river banks near towns, not in special yards. Every port town had some.

Great arsenals:

- **Venice Arsenal.** Begun around 1104 (the exact date is uncertain). Expanded with the *Arsenale Nuovo* (c. 1320) and again in 1473. It covered about 45 hectares (110 acres), around 15% of the city, and was surrounded by a wall about 3.2 km (2 mi) long. Popular accounts say it employed “some 16,000 people” in the early 1500s and could produce nearly a ship a day using standard parts.

The regular skilled workforce was probably smaller; estimates vary.

- **Barcelona:** the royal shipyards (*Drassanes*), ordered by Peter III in 1285 and enlarged in 1328–90.
- **Seville:** the *Atarazanas* (shipyards), built under Alfonso X from 1252 outside the walls, by the river, with 17 long brick halls.
- **Rouen:** the French royal galley yard (*Clos des Galées*), founded in the 1290s (usually dated 1294) with Genoese help.

MAP TIP

Put an arsenal at the edge of a big port city, on its own basin, behind its own walls. Draw a timber road or river from the forests to it. On a kingdom map, mark 1–3 royal shipyards or arsenals on the coast facing the realm’s main

enemy, each with an anchor plus a castle tower. Ordinary beach and creek yards need no symbol; on a local map, just label them “Shipyard”. Add tar and timber ports in the northern forests if your realm builds ships.

Metalworking, crafts, leather, drink and sugar

Metal and craft towns

RULE OF THUMB

Raw metal is made near the mine and the forest. Fine metal goods are made in towns with skilled workers, merchants and good roads.

Town	Famous for	Notes
Milan	Armour	Probably the leading armour centre of the 15th century. Workshops such as that of the Missaglia family (famous in the 1400s) exported armour all over Europe. Workshops specialized in helmets, gauntlets and other parts. Brescia came second.
Nuremberg	Mail armour, metal goods, later firearms	A trading city where many metal crafts met.

Town	Famous for	Notes
Passau and Solingen	Sword blades	Passau's "running wolf" mark was already on its blades before 1300 and meant quality. Later, Solingen smiths copied it. Solingen lies in hilly, wooded country on the Wupper river, with many fast streams for grinding mills.
Cologne	Swords and mail (from the 12th century)	A great Rhine trading city.
Dinant (Meuse valley)	Brass and copper vessels, called <i>dinanderie</i>	Used zinc ore from near Aachen. Sacked by the Burgundians in 1466.
Toledo	Sword blades	Famous in the late Middle Ages and after.
Fabriano (central Italy)	Paper	Making paper by 1276, with water-powered mills. Paper making had reached Europe through Muslim Spain, where Xàtiva (near Valencia) made paper by the 12th century.

Metal craft towns sit on trade routes, close to (but not inside) iron or copper districts: Solingen near the Bergisch

hills, Steyr at the edge of the Styrian iron district, Nuremberg between several mining regions.

Tanneries and other dirty trades

- **Tanning** turned hides into leather by soaking them for many months in pits of water and crushed oak bark (tanbark), after removing hair with lime or dung. It smelled terrible and poisoned water.

- Tanneries needed running water, hides (from butchers and cattle markets) and oak bark (from coppiced woods, as in the Siegerland *Hauberg*).
- Towns often forced tanners to work outside the walls and downstream. In Bristol, tanners, dyers, fullers and weavers were grouped in the Redcliffe and Temple area across the river.

Breweries

- Beer was brewed everywhere, but some towns brewed for export. Hopped beer, which keeps longer, spread from north German towns to the Low Countries and England in the 14th–15th centuries.
- **Hamburg** was called "the brewhouse of the Hanse". Around 1360 it had over 450 breweries, producing more than 25 million litres a year, most of it for export.
- **Einbeck** (Lower Saxony) exported strong beer from Antwerp in the west to Riga in the east, and from Stockholm

in the north to Munich in the south. Its exports are recorded from 1351, and more than half of the town's houses had the right to brew. The word *bock* (a strong beer) is usually said to come from "Einbeck", as pronounced in Munich.

- Breweries need clean water, grain (barley, wheat or oats) and fuel. Put them inside towns, by wells or clean streams.

Wine regions

RULE OF THUMB

Vineyards grow on sunny slopes above rivers. The river carries the wine to a port.

- **Bordeaux** (Gascony, ruled by English kings 1152–1453). The early 14th century was the peak of the Bordeaux wine trade. In good years Bordeaux shipped tens of thousands of tuns of wine (a tun is about 950 litres, or 252 gallons). The record usually quoted is about 100,000 tuns (102,724) in 1308–09; treat these as estimates from customs records. Hundreds of ships sailed from Gas-

cony each year in the wine fleets. Wine from inland (the "high country") came down the Garonne and Dordogne rivers.

- **The Rhine and Moselle.** Rhenish wine was shipped down the Rhine through Cologne to the Low Countries, England and the Baltic.
- **Burgundy.** The Côte around Beaune. In 1395 Duke Philip the Bold banned the Gamay grape to protect the quality of Pinot Noir.
- Vines grew farther north than today. Domesday Book (1086) lists over 40 vineyards in southern England.

MAP TIP

Draw vineyards as hatched strips on south- or south-east-facing slopes along a river, with villages at the foot of the slope. Put a wine port where the river meets the sea or a big river, with warehouses and a customs house. Label the

region's wine ("Claret", "Rhenish"). For beer, there is no need to draw anything; just note "famous breweries" for a town of 2,000 or more.

Sugar estates (Mediterranean)**RULE OF THUMB**

Sugar is a southern crop with a factory attached. It needs a warm coastal plain, river water for irrigation, a mill, fuel and a port, all close together.

Why: Sugar cane needs a frost-free climate and a lot of water. Mediterranean summers are dry, so the fields must be irrigated from a river. Cane is heavy and bulky, so it is crushed near the fields, in a mill turned by water or by animals. The juice is then boiled in big copper cauldrons, which burns a lot of fuel. The finished sugar was a luxury, sold far away, so the estate needs a port.

Sugar came to the Mediterranean from the Islamic world. Crusaders met it in the Holy Land, and early in the 12th century Venice acquired villages near Tyre (in today's Lebanon) and set up sugar estates there. In the late Middle Ages the main Christian producers were Cyprus, Sicily and, from the 15th century, the coast of Valencia. Venice was Europe's chief sugar refining and trading centre in the 15th century.

Real examples:

- **Kolossi** (Cyprus). In 1210 King Hugh I gave the fief of Kolossi, about 60 villages from the fertile plain at the mouth of the Kouris River up into the hills, to the Knights Hospitaller. It became their main estate on the island. Beside the castle are the remains of a sugar factory, a 14th-century mill for crushing cane and an aqueduct that brought water to it. The tower that stands today was built in 1454. Kolossi is about 14 km (9 mi) west of the port of Limassol.
- **The great Cypriot owners.** Sugar was one of Cyprus's main exports. The three best-studied sugar works belonged to the island's biggest landowners: the Lusignan kings at Kouklia, the Hospitallers at Kolossi and the Venetian Cornaro family at Episkopi. The royal works at Kouklia-Stavros was built in the late 13th century and worked until the end of the 16th century.
- **Gandia** (Valencia, Spain). In the early 15th century, sugar-making methods were brought from Sicily with the help of Sicilian masters. Around Gandia there were 4 sugar mills by 1433 and 14 by about 1500. In 1500 the

industry there employed some 500 people and 220 animals.

LATER ERA (1500S+)

Sugar moves out into the Atlantic: to Madeira (cane brought in by the Portuguese in the 1420s; a large producer by the 1490s), the Canary Islands, and then Brazil and the Caribbean, on plantations worked by enslaved Africans. The Mediterranean sugar estates, with their higher costs, slowly decline.

MAP TIP

In a warm southern kingdom, draw one or two sugar districts on irrigated coastal plains: cane fields as small enclosed plots (the enclosed-fields symbol), thin blue irrigation channels from a river (drawn like a mill leat, thinner than a canal), and a sugar mill and boiling house fed by an aqueduct. Draw a water-driven mill as a water-mill wheel labelled "Sugar Mill". Put a fortified estate tower beside the works, drawn as a small solid square (the tower-house symbol); if a military order owns the estate, as the Hospitallers owned Kolossi, use the commandery symbol instead, a small church with a shield. (The older castle at Kolossi was wrecked by Mamluk raiders in the 1420s.) Put a loading port within a short cart trip, like Kolossi and Limassol. A sugar district is a cluster of several mills, not one big town; name the estate as well as the mills ("the Commandery").

Where industry sits in and around a town

RULE OF THUMB

Clean water upstream, dirty trades downstream. Fire and smoke outside the walls. Big land-hungry works in the suburbs and fields.

Trade	Where	Why
Drinking water, breweries	Upstream, or at wells and springs	Needs clean water
Grain mills, fulling mills	On the river at weirs, above and below town; on streams in the countryside	Need a fall of water
Butchers and slaughterhouses	At the edge of town, near the river	Blood and waste dumped in the water
Tanners, dyers, flax retting	Downstream and outside the walls	Smell and pollution
Smiths, potters, brick and tile kilns, lime kilns, glass	Outside the walls or at the edge	Fire risk and smoke
Tenter grounds, rope making, timber yards, bleaching fields	Suburbs and open fields	Need a lot of space
Smelters, charcoal, glasshouses	At the mine and in the forest, not in town	Fuel and ore are too heavy to move
Shipyards, timber yards	Waterfront, often at the edge of the port	Need shore and space

Street names often record these trades: Tanner Street, Dyers' Lane, Shambles (butchers), and in Florence the *Corso dei Tintori* (dyers' street) by the Arno.

MAP TIP

On a local map of a town and its surroundings, draw the river flowing through. Upstream: mills and a water intake. In town: markets, brewers, guild halls. Just below town: butchers and tanners. Outside the walls: kilns, tenter

grounds and rope walks. Then draw a few kilometres more: fulling mills on side streams, quarries and clay pits on the slopes, charcoal hearths in the woods.

Special town types and how to spot them

Most towns are market towns. These are the special ones. Each has a “signature” a reader can spot on the map.

Town type	How to spot it on the map	Typical size	Where it sits	Real examples
Mining town	Up a valley in the hills; spoil heaps, shafts, mint, big church; little farmland	1,000–5,000; boom towns 10,000–20,000	Ore-bearing hills, near wood and fast water	Goslar, Freiberg, Kutná Hora, Kremnica, Iglesias
Salt town	Brine spring or salt mine; boiling houses; salt roads radiating out	Small to middle-sized, but rich for its size	Over salt beds; sea-salt pans on warm, sheltered coasts	Lüneburg, Droitwich, Halle, Salins, Wieliczka, Ston
Port	Harbour, quays, warehouses, lighthouse or harbour chain	Any size; the largest cities were ports	Estuaries, bays, river mouths	See chapter 05
Fishing village	Beach or cove, drying racks or smoke-houses (label them; do not draw boats, because a small boat marks the head of navigation)	50–500 (a village or hamlet); seasonal stations swell in season	Sheltered coast, river mouths	Lofoten stations, Skanör and Falsterbo
Cloth town	Cloth hall, tenter grounds, fulling mills on nearby streams, dyers by the river	2,000–10,000 usually; the Flemish giants 20,000–65,000	Near wool, water power and trade routes	Ghent, Bruges, Ypres, Florence, English cloth towns
Fair town	Big open fairground (often a meadow outside the walls), many inns and warehouses, several roads meeting	Often only a small or middle town	Where regions and routes meet	Troyes, Provins, Lagny, Barsur-Aube; Frankfurt; Stourbridge Fair at Cambridge
Iron or craft town	Forges and grinding mills along streams; guild halls	1,000–10,000; bigger trade cities	Edge of iron or wood districts, on a trade route	Steyr, Solingen, Dinant, Nuremberg
Wine town	Vineyards on slopes behind, cellars, river quay	Small; the export port can be large	South-facing river valleys	Beaune; villages along the Rhine and Moselle; Bordeaux as port
Arsenal or naval town	Walled shipyard basin, rope walks, timber stores	Part of a big port	Sheltered coast facing the main sea enemy	Venice, Barcelona, Seville
Spa (bath) town	Hot or mineral springs, bath houses, many inns	Small	Over hot springs, often in hills	Bath, Aachen, Baden (Switzerland), Karlsbad (Karlový Vary, privileges 1370)
University or pilgrimage town	Colleges, great shrine church, hostels	Varies	Varies	See chapter 08

Fair towns

A **fair** was a big market held once or a few times a year, often for days or weeks, where long-distance merchants met. A **market** was weekly and local (see [Trade nodes](#)).

- The **Champagne fairs** (eastern France, late 12th and 13th centuries) were six fairs a year in four towns (Lagny, Barsur-Aube, Provins and Troyes) where Flemish cloth met Italian merchants. Their calendar, trading phases and decline are described in [The Champagne fairs](#).

Spa towns (briefly)

Hot springs attracted bathers from Roman times. Medieval spa towns were small but busy with visitors: Bath in Eng-

- Later great fairs: Frankfurt (autumn fair privileged in 1240, a spring fair added in 1330), Geneva and then Lyon in the 15th century, Medina del Campo in Castile.
- Fair towns are often **not** big cities. Their signature is a large open space, many inns and storage buildings, and good roads from several directions. Neighbouring Champagne fair towns were about 45–65 km (28–40 mi) apart in a straight line, two or three days by road, so merchants could move from one fair to the next.

land, Aachen (Charlemagne's favourite residence), Baden in Switzerland (described as a lively resort by the Italian scholar Poggio Bracciolini in 1416) and Karlsbad in

Bohemia. In the Islamic world, public bath houses (*hammams*) stood in every city, so baths alone do not make a spa town there.

Suggested symbols

These symbols match the guide's [master legend](#).

Many small works (kilns, racks, forest clearings) are best shown with a word, not a picture. A new little picture can

easily look like a symbol that already means something else: a small tower reads as a castle, a mound as a barrow, a boat as the head of navigation.

Feature	Symbol	Show at which scale
Mine or mining district	Crossed hammers (or a pick)	Continent (3–6 districts), kingdom, local
Mining town	Its usual town symbol, with crossed hammers beside it	Kingdom, local
Old shaft or worked-out mine	Crossed hammers drawn grey, labelled "(old shaft)"	Local
Mint or coinage town	No extra symbol: add "(mint)" or "(coinage town)" to the town's label	Kingdom, local
Salt pans	Checkerboard of small rectangles along the shore	Kingdom, local
Brine spring or hot spring	Small blue dot, labelled "Brine Pit" or "Hot Spring"; on a kingdom map, add "(salt)" or "(baths)" to the town's label instead	Kingdom (label only), local
Salt road	The ordinary road line, labelled "Salt Road" or "Saltway" on its first stretch	Kingdom, local
Quarry	Notch in a hillside, with a track to water	Kingdom (famous ones), local
Lime, brick or tile kilns; clay pits	Words only: "Limekiln", "Brick Kilns", "Clay Pits"	Local
Water-powered works (forge, furnace, ore-crushing mill, fulling mill, sugar mill)	Watermill wheel on the stream, labelled with the trade ("Forge", "Furnace", "Walk Mill", "Sugar Mill"); a hammer pond as one blue pond with a straight dam at its lower end	Local
Charcoal hearths, glasshouse, tar kilns	A clearing in the wood symbol, labelled with the trade	Local
Peat diggings (turbary)	Commons symbol (tufts or stipple), labelled "Turf Fen"; flooded pits as blue lakes with straight, blocky edges	Local
Drowned village	Grey church standing in the water, labelled "(lost village)"	Kingdom (a few), local
Tenter grounds, fair field, fish racks	Open ground labelled "Tenters", "Fair Field" or "Fish Racks"	Local
Fair town	Its usual town symbol, with the fair months in the label ("Provins: May and September fairs")	Kingdom
Vineyards	Hatched strips on south-facing slopes	Kingdom, local
Port (for wine, coal, fish or timber)	Large anchor with a quay (head port), or a small anchor (member port)	Continent, kingdom
Arsenal or royal shipyard	Anchor plus a castle tower	Continent, kingdom
Ordinary shipyard	The label "Shipyard" on the shore; no symbol	Local
Fortified estate tower (for example on a sugar estate)	Small solid square (tower house); a small church with a shield if a military order owns it	Local

FANTASY TWIST

Flying mounts or teleport circles make light, valuable goods (gold, silver, silk, potions) even less tied to roads, but bulk goods (salt, grain, stone, timber) still follow water. So magic adds new special towns (an airship yard, a mage-crafting town at a crystal mine) without removing the old ones. See [chapter 10](#).

MAP TIP

Every special town needs its “why” visible on the map: a mine symbol, salt pans, a fairground, vineyards, a harbour. Use the symbols in the table above, and give each a name that hints at its trade (Silverdale, Saltwich, Clothford, Forgeham). On a kingdom map, aim for something like 1–3 mining districts, 1–3 salt sources, one cloth region, one or two fair towns and several ports. That is enough to make the economy feel real without crowding the map.

Quick summary

- Industry goes to the heaviest input: ore, fuel (wood, charcoal, peat, coal) or water power. Only the light, finished product travels.
- Mining towns sit up valleys in old hill country, buy their food, have special privileges (free mining, own courts, a mint) and boom and bust. Most had 1,000–5,000 people; rare boom towns reached 10,000–20,000.
- Iron is limited by charcoal: very roughly 50–150 kg of wood per 1 kg of bloomery iron (estimates vary). A Wealden blast furnace needed the woods of about a 5 km (3 mi) radius.
- Every realm needs salt: brine springs (with huge fuel needs), rock-salt mines, or sea-salt pans on warm coasts. Salt roads run from the source to the nearest navigable water.
- Stone travels far only by water; clay lowlands build in brick. Kilns and clay pits sit outside the walls.
- Forests are industrial: charcoal hearths, forges with ponds, moving glasshouses (150–200 kg of wood per kg of glass), tar and potash.
- In wet, treeless lowlands people dig peat, and big diggings flood into blocky lakes: the 63 Norfolk Broads (dug 12th–14th centuries) and the ever-growing Haarlemmermeer.
- Cloth needs wool, fulling water and dyes. Fulling mills put cloth villages along fast hill streams; great cloth towns (Ghent perhaps 40,000–65,000; Florence) imported wool.
- Preserved fish travels far: Scania herring (a beach market every autumn), Lofoten stockfish through Bergen (about 4,000 tonnes a year in the 1300s).
- Shipyards are everywhere along shores; state arsenals (Venice, Barcelona, Seville) are walled factories in great ports.
- Clean water upstream, dirty trades (tanners, dyers, butchers) downstream and outside the walls; fire trades at the edge.
- Give every special town a visible reason on the map: a mine, pans, a fairground, vineyards, a harbour, or (in the warm south) a sugar mill among irrigated cane fields.

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- Paphos Life: *The ancient sugar factory at Kouklia* — https://www.paphoslife.com/blog/sugar_factory/2
- Wikipedia: *History of sugar* (Tyre estates, Venice, Madeira) — https://en.wikipedia.org/wiki/History_of_sugar
- Wikipedia: *Trapiche* (sugar mills at Gandia) — <https://en.wikipedia.org/wiki/Trapiche>
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- Wikipedia: *Champagne fairs* — https://en.wikipedia.org/wiki/Champagne_fairs
- Wikipedia: *Frankfurt Trade Fair* — https://en.wikipedia.org/wiki/Frankfurt_Trade_Fair
- Wikipedia: *Karlovy Vary* — https://en.wikipedia.org/wiki/Karlovy_Vary
- Wikipedia: *Jáchymov* — <https://en.wikipedia.org/wiki/Jáchymov>

CHAPTER 8

RELIGIOUS, CULTURAL, LEGAL AND ANCIENT SITES

Castles and markets show who rules and who trades. Churches, monasteries, shrines, gallows and ruins show what people believe, how they are judged and how old the land is. This chapter tells you where to put these sites, how many to draw, how far apart they stand, and why.

The baseline period is the High and Late Middle Ages in Europe (c. 1000–1500), with examples from the Mediterranean and the Islamic world. Most counts come from England, because English records are the best. Village and town numbers are in [Population and Settlement Sizes](#). Coronation and royal burial churches are in [Capitals, Realms and Borders](#).

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The religious map at a glance

RULE OF THUMB

Every village has a church. Every town has a hospital. Every big town has friars. Every region has a few monasteries. Every kingdom has a handful of cathedrals and one or two universities.

The Church was the biggest landowner and builder in medieval Europe. Bishops and abbots held roughly a quarter of England in 1086 (estimates vary; see [chapter 03](#)). Clergy were only about 1–2% of the people, but their buildings were everywhere.

How many of each, per area

These densities come from England, where the counts are good. England covers about 130,000 km² (50,000 sq mi).

Feature	Number in England	One per ...	Per 1,000 km ² (386 sq mi)	Notes
Parish church	~9,500 (c. 1300)	~14 km ² (5.4 sq mi)	~73	About 4 km (2.5 mi) church to church in the lowlands
Market (chartered)	1,746 recorded by 1300	~75 km ² (29 sq mi)	~13	Shown for comparison (baseline)
Religious house (monks, canons, nuns, friars)	~900 (1530s)	~145 km ² (56 sq mi)	~7	Average only ~13 people each
Hospital or almshouse	800–1,100 founded c. 1080–1540 (estimates vary)	~120–160 km ² (46–62 sq mi)	~6–8	Not all open at the same time
Castle in use	500–600 at one time (England and Wales)	~250–300 km ² (95–115 sq mi)	~3–4	See chapter 04
Leper house	~300 (estimates vary)	~430 km ² (166 sq mi)	~2	Mostly founded 1100–1350
Friary	183 (1530s)	~700 km ² (270 sq mi)	~1.4	Only in towns

Feature	Number in England	One per ...	Per 1,000 km ² (386 sq mi)	Notes
Cathedral	17 (from 1133)	~7,600 km ² (2,950 sq mi)	0.13	Plus 4 in Wales
University	2 (Oxford, Cambridge)	~65,000 km ² (25,000 sq mi)	0.015	Very rare everywhere

Take a sample county of about 2,500 km² (965 sq mi), a square 50 km (31 mi) on a side. At English densities it would have:

- about 180 parish churches;
- 15–20 religious houses;
- 3–4 friaries in its 1–3 biggest towns;
- 15–20 hospitals and almshouses over time;
- about 6 leper houses;
- 8–10 castles in use;
- about a one-in-three chance of a cathedral.

MAP TIP

Do not draw all of these at every scale. On a kingdom map, show cathedrals, great abbeys, major shrines and universities. On a local map, show every parish church, plus the abbeys, friaries, hospitals, gallows and ruins. See [Drawing it all](#).

Parish churches and chapels

RULE OF THUMB

One parish, one church, one priest. In good farmland the churches stand about 3–5 km (2–3 mi) apart. In hills they stand about 7–10 km (4–6 mi) apart, more in the emptiest uplands, with small chapels in between.

Why every village has a church

A **parish** is the area served by one church and its priest. The people of the parish paid **tithes** (one-tenth of their crops and young animals) to support the church. In return the parish church had the right to baptise their babies and bury

their dead. Most English and French parishes were fixed by about 1200 and then changed very little for 600 years.

So a parish needed enough farmland to feed a priest from one-tenth of its harvest. That is why parishes are small in rich land and huge in poor land.

Numbers

Place and date	Parishes	Average size	Notes
England c. 1300	~9,500	~14 km ² (5.4 sq mi)	Lowland parishes 5–15 km ² (2–6 sq mi); upland parishes 50–100+ km ² (20–40+ sq mi)
France (late Middle Ages to 1789)	roughly 40,000 (an estimate; counts vary, some higher)	~13 km ² (5 sq mi)	Most of France's modern communes grew out of these parishes
Northern and central Italy	Older system: the <i>pieve</i>	A <i>pieve</i> covered several villages	A <i>pieve</i> was a rural "mother church" with the only font (baptism basin); smaller chapels around it served the villages
Medieval towns in England	Many tiny parishes	Some under 1 ha (2.5 acres)	London had over 100 parish churches inside its walls by the 1100s; Norwich about 56; Lincoln over 40; York ~45 in 1300 (39 by 1428)

Old English towns like London, Norwich, York and Lincoln were full of tiny parishes, because rich townsmen founded many small churches before 1200. Norwich packed 56 parish churches into about 1.5 sq mi (4 km²) inside its walls.

New planted towns of the 1200s usually got only one church, and sometimes only a chapel that belonged to an older village parish.

Chapels of ease and other chapels

A **chapel of ease** is a smaller church inside a parish, built for people who live too far from the parish church. It usually had no right to bury the dead or baptise, so people still went to the mother church for those.

- **Where:** in hamlets and distant villages of big upland parishes, and in new settlements on cleared land. Many were founded from the late 1200s by the villagers themselves.

Where the church stands in the village

- On the **highest dry spot** in or next to the village, often slightly apart from the houses.
- **Next to the manor house.** The lord often built the first church, and the two buildings stand side by side.

Church towers as landmarks

Church towers and spires were the tallest things most people ever saw. Travellers and sailors steered by them.

Building	Height	How far you can see the top (flat land, clear day, eye at 1.7 m)
Small village tower	15 m (50 ft)	up to ~18 km (11 mi)
Large village or town tower	25–40 m (80–130 ft)	up to ~22–27 km (14–17 mi)
Boston “Stump”, Lincolnshire (tower built c. 1450–1520)	81 m (267 ft)	up to ~37 km (23 mi); a landmark for travellers in the Fens and ships in the Wash
Lincoln Cathedral spire (tower finished 1311, spire added after; blown down 1548)	reputedly about 160 m (525 ft)	up to ~50 km (31 mi); possibly the tallest building in the world in its day, though some historians doubt the height

The distances above are the geometric limit (the curve of the earth). Haze, trees and hills cut them a lot. In real life a village tower is a useful landmark for 5–10 km (3–6 mi). In flat lowland, where churches stand about 4 km (2.5 mi)

- **Private chapels:** inside castles, manor houses, hospitals and bridges (bridge chapels collected tolls and prayers for travellers).

- **Wayside crosses and shrines:** at crossroads, on parish boundaries and on moorland tracks. In Catholic Central Europe small roadside shrines are common.

- **Holy wells:** springs dedicated to a saint, common in Wales, Cornwall, Ireland and Brittany.

- On an **older sacred site.** At Rudston (Yorkshire) the Norman church stands beside a prehistoric standing stone about 7.6 m (25 ft) tall in its churchyard.

- In towns, near the **market place**, and at the gates.

- If the village later moved or died, the church often stayed behind, alone in the fields. A lonely church is a sign of a deserted or shrunken village (see [chapter 01](#)).

apart, you can usually see several neighbouring towers from any one of them.

LATER ERA (1500S+)

Parishes barely change. In Protestant lands, side altars, chantries (endowed chapels for prayers for the dead) and many shrines go; England dissolved its chantries in 1547. Printed county maps such as Christopher Saxton's atlas of England and Wales (1579) mark towns and villages with small church-tower symbols, because the church was what defined a place.

MAP TIP

On a local map, give each village one church symbol: a small church with a tower. Use a spire for richer towns, and a small plain cross with no tower for chapels. In hill country, draw fewer churches and add 2–4 chapel symbols per parish in the outlying hamlets. In an old town, scatter many small church symbols inside the walls; in a planted grid town, draw only one.

Dioceses, cathedrals and archbishops

RULE OF THUMB

Old Christian lands around the Mediterranean have many small dioceses. Newly converted lands in the north and east have few huge ones. So in Italy a cathedral may sit in a town of 2,000 people; in England it means one of the realm's main cities.

A **diocese** (also **see** or **bishopric**) is the area ruled by one bishop. His main church is the **cathedral**, which holds his *cathedra* (throne). An **archbishop** is the senior bishop of a **province**, a group of dioceses.

Why dioceses differ so much

The first bishops sat in Roman cities. Each Roman *civitas* (city with its own territory) got a bishop. Italy and southern France had many small Roman cities, so they got many

small dioceses. Lands converted later (England, Germany, Scandinavia, Poland, Hungary) had few towns, so a bishop's diocese was drawn around a whole kingdom, tribe or county.

Numbers and sizes

Region	Dioceses	Average area	Notes
England	17 from 1133 (plus 4 in Wales)	~7,600 km ² (2,950 sq mi)	2 archbishops: Canterbury and York. Lincoln diocese ran from the Thames to the Humber, over 200 km (125 mi).
France (within today's borders)	roughly 130–150 over 1200–1500 (the number changed over time)	~3,700–4,200 km ² (1,450–1,650 sq mi)	About 15 archbishops (Reims, Sens, Rouen, Tours, Lyon, Bourges, Bordeaux, Narbonne and others). Many small sees in the south; Pope John XXII created new ones in 1317.
Umbria (central Italy)	12	~850 km ² (330 sq mi); range 100–2,500 km ² (40–965 sq mi)	Terni's diocese covered only ~100 km ² (40 sq mi). Città di Castello was a see with about 5,000 people.
Italy as a whole	Hundreds	Often under 1,000 km ² (390 sq mi)	Italy still had about 325 dioceses before the mergers of 1986 (101 were suppressed then); many were of medieval origin. Southern Italy had the most.
Holy Roman Empire (c. 1500)	Roughly 50 bishops and archbishops (counts vary with date and borders)	~15,000–20,000 km ² (5,800–7,700 sq mi) (estimate)	Many bishops were also princes ruling land (see chapter 03).
Scandinavia	Few	Very large	Archbishoprics at Lund (1103/04), Nidaros/Trondheim (1150s) and Uppsala (1164).
Scotland	~13	Large	No archbishop until St Andrews (1472) and Glasgow (1492).

Italian sees stayed even when the town shrank. In Tuscany, Grosseto fell to about 1,500 people after 1350, and

Cortona from about 10,000 to about 3,000, but both kept their bishops (Paganelli 2025).

Cathedrals move to towns

Bishops wanted to sit in towns. After the Norman Conquest, the Council of London (1075) moved English sees out of villages: Selsey to Chichester, and Sherborne to Old Sarum. Dorchester-on-Thames moved to Lincoln (1072) and

Thetford to Norwich (1090s). Old Sarum later moved again, down to the river at Salisbury (from 1220; see [chapter 01](#)). A small village with a “-minster” name or a ruined great church may be a former see.

What a cathedral city looks like

- A cathedral much bigger than any parish church. Winchester Cathedral is about 170 m (560 ft) long. Cluny's great abbey church in Burgundy (not a cathedral) was the largest church in Europe until the 1500s.

- A walled **close** (precinct) around it, with houses for the canons (the cathedral clergy), the bishop's palace, a school and often a gate that was locked at night.
- In Islamic cities, the equivalent is the **Friday mosque** (the main congregational mosque), in the centre beside the main market; big cities later had several.

- In England, nine cathedrals were also Benedictine monasteries (Canterbury, Durham, Winchester, Norwich, Ely, Worcester, Rochester, Bath and Coventry), and Carlisle was served by Augustinian canons. This was rare elsewhere.

- Extra business: courts of the church, lawyers, pilgrims, craftsmen, and a big building site for centuries.

MAP TIP

On a kingdom map, mark each cathedral with a two-towered church or a mitre, and each archbishop's seat with a double cross beside the city symbol. For an English-style realm, use about one diocese per 5,000–10,000 km² (2,000–4,000 sq mi). For an Italian-style land, use one per

500–2,000 km² (190–770 sq mi), and let many cathedral towns be small. Draw diocese borders only on special church maps; they often follow old Roman or tribal borders, not the current political ones.

Monasteries, friaries and military orders

RULE OF THUMB

Monks go to the countryside. Canons go to the edges of towns. Friars go inside towns. The more friaries a town has, the bigger it is.

The orders and where they built

Order	Founded	Where they built	Typical size	England (1530s or total)
Benedictines (monks following the Rule of St Benedict, c. 530s)	Old houses from before 1000	Often in or beside towns that grew up at their gates (Bury St Edmunds, St Albans, Peterborough)	20–60 monks in big houses	Part of ~260 houses of monks
Cluniacs (reformed Benedictines)	Cluny, Burgundy, 910	Network of priories under one abbey; 314 houses paid allegiance to Cluny in the 1100s	Small priories of 5–20 monks	A few dozen small priories
Cistercians ("white monks")	Cîteaux, Burgundy, 1098	Remote valleys with a river, "far from the concourse of men"; they drained land and built water channels and mills	Rievaulx under Abbot Aelred (1160s): 140 monks and 500 lay brothers	About 75–80 abbeys in England and Wales in all (13 of them in Wales)
Carthusians (hermit monks)	Grande Chartreuse, French Alps, 1084	The most isolated sites: mountains, woods, the foot of escarpments. Each monk lived alone in a cell with a garden	12–24 monks plus lay brothers; Mount Grace (1398) had 17 monks' cells and 6 for lay brothers	Only 9 charterhouses
Augustinian canons (priests living by a rule, serving parishes and hospitals)	Spread from c. 1100	Near or in towns, at hospitals, at shrines, at castles	Often 6–20 canons	Part of ~300 houses of canons
Premonstratensian canons ("white canons", with a strict rule close to the Cistercians)	Prémontré, 1120	Rural, like Cistercians	Small	Part of the canons above
Nuns (all orders)	—	Mostly rural and small; often poorer than male houses	Average ~14 nuns	142 nunneries, ~2,000 nuns
Friars: Dominicans, Franciscans, Carmelites, Austin Friars	1210s–1250s; in England from 1221 (Dominicans) and 1224 (Franciscans)	Inside towns, usually on cheap land at the edge: near the walls, at a gate, or in a suburb	10–60 friars	183 friaries, ~3,000 friars
Military orders: Templars (suppressed 1307–12), Hospitallers	1110s–1120s	Commanderies (also called preceptories): farm estates with a chapel in good farmland, plus houses in big towns	A few brothers and many servants	By about 1300 the Templars had at least 970 houses (commanderies and castles) across Europe and the East

Across England in the 1530s there were nearly 900 religious houses with about 12,000 people: 4,000 monks, 3,000 canons, 3,000 friars and 2,000 nuns. That is only about 13

Cistercian valleys

Cistercians wanted three things: isolation, running water and flat land to build on. They founded abbeys in valleys of Yorkshire, Wales, Burgundy and Castile that had few people. At Rievaulx (1132) the monks moved the River Rye more than once in the 1100s to get enough level ground. The order grew fast: about 333 abbeys by 1152, when the

Granges: the monastery's farms

A **grange** is an outlying farm owned by a monastery. Early Cistercian rules said a grange should be within a day's walk of the abbey, so the **lay brothers** (working brothers who farmed it) could come back for Sunday and feast-day services. The rules gave no exact distance. A day's walk is about 25–35 km (15–22 mi) (see the travel speeds in [Trade Routes, Roads and Transport](#)). The rule was often broken.

- Fountains Abbey (Yorkshire) had 39 granges by the early 1300s. Its grange at Cowton lay about 20 mi (32 km) north of the abbey.

Friaries tell you how big a town is

Friars lived by begging and preaching, so they needed many people nearby. In 1968 the historian Jacques Le Goff suggested using the number of friaries as a rough measure

people per house on average. Most houses were small; a few great abbeys were very large.

order tried to halt new foundations, and nearly 750 at its height in the 1400s.

They often cleared or moved existing villages to get their “desert”. So a Cistercian abbey on your map should sit in an empty valley, with no village at its gate.

- Morker, a Fountains grange, held about 40 cows, 5 teams of oxen and 300–400 sheep in the late 1400s.
- Granges specialised: sheep runs, cattle farms, fisheries, iron or lead works, salt.

Farm layouts are in [Villages, Farms and the Countryside](#). Monastic mines and mills are in [Industry, Resources and Special Towns](#).

of how big and important a French town was. The table below is a simplified rule of thumb in that spirit. Treat the sizes as rough guides, not exact figures.

Number of friaries	Town size (rough guide, France)
0	Small towns of 2,000–3,000 people usually had none
1	Towns of roughly 5,000 (estimate)
2	Towns of about 10,000
4 (one of each main order)	Big towns of 20,000–40,000
5+	The great cities (London, Paris, Florence)

In England, friars reached further down the town ladder: many county towns of 3,000–5,000 had one or two friaries.

Outside Latin Europe

- **Orthodox monasteries** favour mountains and islands. Mount Athos in Greece has 20 great monasteries; the Great Lavra was founded in 963. Meteora in Thessaly perched monasteries on rock pillars from the 1300s.
- **Islamic lands** had no monks, but they had **Sufi lodges** (*khanqah*, *zawiya*, *tekke*) in towns and along roads, and **ribats**: fortified posts on frontiers and coasts (North

Africa, the Byzantine border) that later became hostels and Sufi centres.

LATER ERA (1500S+)

Protestant rulers close the monasteries: Sweden from 1527, Denmark in 1536, England and Wales in 1536–41, and many German states. Abbey churches become parish churches, country houses or roofless ruins; the land goes to nobles. In Catholic lands, new orders such as the Jesuits (1540) build colleges and churches inside towns, not in the countryside. See [What Changes After 1500](#).

FANTASY TWIST

A religious order of warrior-monks in a monster-haunted land behaves like the Templars or the Teutonic Order: fortified commanderies along dangerous roads and frontiers, each with farms to pay for it. Treat the fortified ones as castles on your map and use a castle symbol (see [chapter 04](#)); keep the small church with a shield for ordinary farm commanderies.

MAP TIP

Put about 1 religious house per 150 km² (60 sq mi) in a settled kingdom. Place Cistercian and Carthusian houses in empty, well-watered valleys away from villages. Put Benedictine abbeys in old towns or at the centre of a “monastery town”. Put friaries in towns of about 3,000–5,000 and up, near the walls or gates, and nunneries in quiet rural spots. Give each big abbey 5–40 granges, most within about 25 km (15 mi) and a few much farther. Name some farms “Grange”, “Temple ...” or “... Abbey Farm”. On a local map, a great abbey is a walled precinct 350–600 m (0.2–0.4 mi) across, not a single church; see [Footprints](#).

Pilgrimage: shrines, routes and hostels

RULE OF THUMB

Pilgrims use existing roads. A famous shrine pulls a chain of hospices, bridges and small towns along its route, about one day’s walk apart.

A **shrine** is a holy place, usually the tomb or relics (holy remains) of a saint. A **pilgrimage** is a journey to a shrine for prayer, healing or forgiveness. Every region had small local shrines; a few became international.

The great pilgrimages

Destination	What drew people	Key dates	Route and numbers
Jerusalem	Christ’s tomb (Holy Sepulchre) and the holy places	Crusader rule 1099–1187; pilgrim galleys from Venice afterwards	Venice to Jaffa by galley took 48 days in 1480 (baseline)
Rome	Tombs of Saints Peter and Paul	First Jubilee (holy year) 1300	Chronicler Giovanni Villani, who was there, claimed that Rome held about 200,000 pilgrims all through 1300, besides its own people. This was a guess, not a count
Santiago de Compostela (Galicia)	Tomb of St James, “found” in the 800s	Guidebook in the <i>Codex Calixtinus</i> , c. 1140	Four French roads (from Paris/Tours, Vézelay, Le Puy and Arles) join at Puente la Reina; about 800 km (500 mi) from Roncesvalles at the Pyrenees
Canterbury	Thomas Becket, killed 1170, made a saint 1173	Shrine 1220–1538	About 90 km (56 mi) from London
Cologne	Relics of the Three Kings, brought from Milan in 1164	1164 onward	On the Rhine trade route
Mecca (Hajj)	The Kaaba; one pilgrimage is a duty for every able Muslim	Every year	Great caravans from Cairo, Damascus and Baghdad. The Darb Zubayda from Kufa (Iraq) was a desert road of wells, cisterns and stations. It was improved under the Abbasid caliphs (late 700s–early 800s) and named after Zubayda (died 831), wife of Caliph Harun al-Rashid, who paid for much of the work

Nobody counted medieval pilgrims. Modern guesses for Santiago or Rome vary hugely, so treat any single number with caution.

How long and how far

Archbishop Sigeric of Canterbury wrote down his route home from Rome in about 990. It had about 80 stages over about 1,700 km (1,100 mi): **about 20 km (12 mi) per stage**. It

crossed the Alps by the Great St Bernard Pass. This is a good spacing for pilgrim stops on your map. It is shorter than a strong walker's day of 25–35 km (15–22 mi) (baseline), because pilgrims were often old, sick or slow.

What a pilgrim road looks like

- **Hospices** (free lodgings for pilgrims and travellers) at passes, river crossings and empty stretches. The Great St Bernard hospice was founded around 1050 at 2,469 m (8,100 ft), right at the top of the pass. Roncesvalles, below the Pyrenees pass, had a famous pilgrim hospital from the 1100s.
- **Bridges** built for pilgrims, often by holy men or monasteries. Santo Domingo de la Calzada in Spain grew up around the bridge, road and hospital that St Dominic built in the 1000s. Hospital de Órbigo is named after its pilgrim hospital.
- **Pilgrim towns** along the road, with many inns, hospitals and a “street of the Franks” for foreign settlers. Puente la Reina is named after its pilgrim bridge.

FANTASY TWIST

If miracles at a shrine are real and visible (healing, oracles), pilgrimage becomes a major industry. Make the shrine town 2–5× bigger than its farmland could support, with many inns and hospitals, and draw a pilgrim road with hospices every 20 km (12 mi) all the way to the nearest big cities.

- **At the shrine:** a great church, many inns, hospitals, souvenir sellers (pilgrim badges, scallop shells for Santiago), money changers, and often a leper hospital on the approach road. At Canterbury, the leper hospital of Harbledown (founded c. 1084–85) stood just west of the city on the road from London. Passing pilgrims gave alms there, for example to see a slipper said to have been Becket's.

LATER ERA (1500S+)

Protestant lands end pilgrimage. Becket's shrine at Canterbury and the shrine of Walsingham were destroyed in 1538. Catholic pilgrimage continues and new shrines appear, but long international pilgrimages decline.

MAP TIP

Pick one or two great shrines for your continent and several regional ones for each kingdom. Draw the pilgrim road along existing roads: keep the ordinary road line, add small scallop marks along it, and label it (“Pilgrims’ Way”, “Saint’s Road”). Put a hospice at every mountain pass and dangerous crossing, a small pilgrim town every 2–4 days’ walk, and a leper house on the last stretch before the shrine. Use a scallop shell or a small reliquary as the shrine symbol, never a star (a filled star means a capital).

Hospitals, leper houses and almshouses

RULE OF THUMB

Hospitals stand at town gates, bridges and main roads. Leper houses stand outside the town, on a main road in, often within 1–2 km (about 1 mi) of the gate.

A medieval **hospital** was a religious charity house. Most did not treat illness as we do. They gave food, a bed and prayer to the poor, the old, the sick and travellers. An **almshouse** was a small home for poor old people. A **leper house** (leprosarium, lazar house) housed people with leprosy, a disease that people feared and connected with sin.

Numbers

Place	Number	Notes
England, all hospitals and almshouses	800–1,100 founded c. 1080–1540 (estimates vary)	Over 500 almshouses before the Reformation
England, leper houses	about 300 (estimates vary)	Most founded 1100–1350; many became almshouses later
Kingdom of France	~2,000 leper houses	King Louis VIII's will (1227) left gifts to about 2,000 leper houses
All of Christendom	19,000 leper houses	A guess by the chronicler Matthew Paris for the early 1200s; not a count

Most almshouses and small hospitals had **4–13 residents**. Thirteen was a favourite number (Christ and the twelve apostles). The Hospital of St Cross at Winchester, founded

about 1132–36 for thirteen poor men, was one of the first. A few town hospitals were large, with 100 or more beds.

Where they stood

- **Ordinary hospitals:** at a town gate, by a bridge, by a market or on a main road into town. Many towns had a “St John’s Hospital” at the gate where travellers arrived.
- **Leper houses:** outside the walls, beside a main road where travellers would give alms, often near a stream. St Giles’ leper hospital (founded 1101–09) stood on the road west from London, between the City and the gallows at Tyburn. In Paris, a leper house stood near the great gallows of Montfaucon on the hill outside the walls.

- **Islamic hospitals (bimaristans)** stood in the centre of the city, funded by a religious endowment (*waqf*). Nur al-Din built one in Damascus in the 1150s; Sultan Qalawun’s great hospital in Cairo was finished in 1284.
- **Pilgrim hospices:** see above.

Place names show where these were: “Spital”, “Spittal”, “Spitalfields”, “Maudlin” (from Mary Magdalene, a common leper-house saint), “Lazar”, or “Saint-Lazare”.

MAP TIP

Give every market town one hospital, and every walled town 2–5. Put a leper house on a main road 1–2 km outside each town of 2,000+ and most market towns (England had about 300, roughly one per 430 km²); big cities may have one on

each main road. On a local map, draw each as a small house with a cross and label it (“St John’s Hospital”, “the Lazar House”). In an Islamic city, place the hospital near the great mosque and the main market instead.

Universities and schools

RULE OF THUMB

Universities are as rare as great cities. A kingdom has none, one or two. Most sit in big cities with a bishop, a court or both; a few grew in middling towns that already had famous schools.

A **university** was a guild (association) of masters and students, licensed to teach and grant degrees. The scholars of the day called it a *studium generale*. There were about 20 in Europe c. 1300 and 60–80 by 1500 (baseline). Over 80 were founded before 1500, but some failed quickly.

Major universities and when they began

Region	Universities (founding date)
Italy	Bologna (by tradition 1088; organised in the late 1100s), Padua 1222, Naples 1224, Siena 1240, Rome 1303, Perugia 1308, Pisa 1343, Pavia 1361, Ferrara 1391, Turin 1404
France	Paris (c. 1150–1200), Montpellier (medicine from 1220), Toulouse 1229, Orléans 1235, Avignon 1303, Angers 1356, Caen 1432, Bordeaux 1441
Iberia	Salamanca 1218, Lisbon/Coimbra 1290, Lleida 1300, Barcelona 1450, Zaragoza 1474, Valencia 1499
Britain	Oxford (teaching by 1096; grew fast after 1167), Cambridge 1209, St Andrews 1413, Glasgow 1451, Aberdeen 1495
Central Europe	Prague 1348, Kraków 1364, Vienna 1365, Pécs 1367, Heidelberg 1386, Cologne 1388, Erfurt (1379/1392), Leipzig 1409, Rostock 1419, Leuven 1425, Basel 1460, Tübingen 1477

Region	Universities (founding date)
North	Uppsala 1477, Copenhagen 1479
Islamic world (colleges, not universities)	Al-Qarawiyyin mosque, Fez (859); al-Azhar mosque, Cairo (970s); Nizamiyya <i>madrasa</i> (college of law), Baghdad (1065–67)

The first universities grew out of famous cathedral and city schools in Italy, France and England. After 1340 rulers

founded new ones as status symbols in their capitals (Prague, Kraków, Vienna; see [chapter 03](#)).

How big

Student numbers are uncertain. Oxford had perhaps 1,500–2,000 students around 1300 (modern estimates vary); Paris had several thousand; most universities had a

few hundred. The Baghdad Nizamiyya is said to have had about 3,000 students in 1096. A university town filled with colleges, halls, bookshops and inns, and with fights between students and townsfolk.

Schools

- **Cathedral and monastery schools** in every cathedral city and big abbey.
- **Grammar schools** (Latin schools) in many towns by 1400. Many were run by a chantry priest or a guild.

- **Song schools and parish clerks** taught reading in some villages.
- **Islamic lands:** a *madrasa* (college) in every important city, and Qur'an schools attached to mosques. Like hospitals, they were funded by *waqf* endowments.

LATER ERA (1500s+)

Universities multiply (Leiden 1575 in the Dutch Republic, and many more). Jesuit colleges appear in many Catholic towns after 1540. Protestant towns found grammar schools.

MAP TIP

Give a kingdom of 2–5 million people one or two universities, placed in the capital and in the oldest cathedral city. Most universities sat in cities of 10,000+, but Oxford and Cambridge grew in towns of only a few thousand people, so avoid towns under about 3,000. Mark it with a book or a small “U” by the city name. Every town on a local map can have a grammar school, but you do not need to draw it.

Legal and civic sites

RULE OF THUMB

Gallows stand outside the town, on high ground, beside a main road, at the edge of the lord's land. Pillories and stocks stand in the market place. Courts meet at old, well-known landmarks.

Gallows

Gallows were meant to be seen. They marked the lord's right to judge and kill (**high justice**) and warned travellers.

- **Where:** on a hill or rise, outside the town or village, near a main road, often at the boundary of the lordship or parish.
- **France:** a lord's gallows were called *fourches patibulaires*. By custom, the number of stone pillars showed his rank: 2 for a simple lord with high justice, 3 for a castellan (the lord of a castle and its district), 4 for a baron, 6 for a count, 8 for a duke, and as many as he liked for the king. This was the rule in principle; local customs varied.

- **Paris:** the great royal gallows of **Montfaucon** stood on a small hill outside the walls, with the leper house of Saint-Lazare in the same area. Viollet-le-Duc's 19th-century drawing shows it with 45 places for bodies, but it may make the gallows grander than it really was.
- **London: Tyburn** stood about 5 km (3 mi) west of Newgate prison, where two Roman roads met (today's Oxford Street and Edgware Road). Its first recorded hanging was in 1196. Close by stood “Oswulf's Stone”, which gave its name to the local hundred (district), Ossulstone.
- **Anglo-Saxon England (before 1066):** archaeologist Andrew Reynolds showed that execution cemeteries were usually on **hundred boundaries**, beside **major**

roads, and often on or next to **prehistoric barrows** (burial mounds).

Pillories, stocks and lock-ups

- **Pillory** (holds the head and hands): in the market place, often on a platform. Used for cheating traders.
- **Stocks** (hold the feet): the Statute of Labourers of 1351 ordered **every town and village** in England to have

Courts and meeting places

- **Open-air moots**: Early medieval assemblies met outdoors at a landmark: a mound, a prehistoric barrow, a standing stone, a big tree, a ford or a crossroads. By the 12th century the English hundred court met 12 times a year. Many hundreds are named after their meeting place (Ossulstone; names ending in “-low”, from an old word for mound).
- **Things**: Norse and Germanic assemblies (the *thing*) met in the open, often at mounds, near water and land

Town halls, guildhalls and market crosses

- **Town hall** (guildhall, moot hall, *palazzo comunale*, *Rathaus*): on or beside the main market place. Often with a covered market below and a council chamber above.
 - London’s Guildhall was rebuilt in 1411–1440; a guildhall is first recorded there in 1127 or 1128.
 - Siena’s Palazzo Pubblico (begun 1297) faces the Campo, where the city’s three hills meet. Its tower, the Torre del Mangia, was built in the 1320s–1340s (sources give different dates). It was meant to be one of the tallest towers in Italy, a rival to the towers of Florence and of Siena’s own cathedral.

FANTASY TWIST

Where the dead can rise, gallows and execution grounds are not left with bodies hanging. Expect burning places or burial under stones at crossroads, and a guard house or shrine nearby.

- **Names**: Gallows Hill, Gallows Green, Hangman’s Hill (England); *Galgenberg* (Germany); *Justice* (France).

stocks. They stood on the green, by the church or in the market.

- **Ducking stool** (a seat on a long pole, used to dip offenders into water) at a pond or river; a small lock-up or cage in some towns.

routes, and at territorial boundaries. Iceland’s Althing met at Þingvellir from 930. Tynwald Hill on the Isle of Man is still used.

- **Inauguration sites**: Scottish kings were made at the mound of Scone.
- **Indoor courts**: manor courts in the lord’s hall, shire courts in the county town or castle, church courts in the cathedral city.

- **Belfries**: Towns in Flanders and northern France built civic bell towers (Bruges, Ghent) to hold the town’s bells, charters and archives. They were a symbol of town freedom, a rival to the church tower. UNESCO lists 56 of them.
- **Guildhalls of crafts and merchants**: in rich trading towns, near the market or the river (cloth halls in Flanders and England).
- **Market crosses**: a stone cross, later often a roofed shelter, marking the market place and the town’s market rights. Proclamations were read there. Examples: Malmesbury (c. 1490), Chichester (1501).

MAP TIP

On a local map, put one gallows symbol (a small “II”) on a hill by the main road outside each town and at the edge of each great lord’s land; label it “Gallows Hill”. Put the town hall, market cross and pillory together in the market place. Mark the hundred or district meeting place at a mound, stone or ford roughly central to its area, often on an old road; draw it as a small lone tree labelled “Moot”.

Burial grounds

RULE OF THUMB

Christians are buried around their parish church. Jews, Muslims, plague victims and executed criminals are buried outside the town.

- **Churchyards:** around every parish church, usually walled or fenced. The rich were buried inside the church. Town churchyards filled up quickly.
- **Charnel houses:** buildings or crypts to store bones dug up to make room for new graves. Graves were often reused after about five years. A charnel house at St Bride's, Fleet Street, London, held an estimated 7,000 people's bones. Paris's Cemetery of the Holy Innocents had famous charnel galleries around its walls.
- **Plague cemeteries:** in 1348–49 London opened emergency cemeteries outside the walls at East Smithfield

MAP TIP

You do not need to draw churchyards on most maps; the church symbol implies one. On a local or town-edge map, draw a Jewish cemetery or a Muslim cemetery as a small walled plot outside a gate, with a name label (for example

and West Smithfield. The London Charterhouse (a Carthusian monastery) was founded in 1371 on the West Smithfield burial ground.

- **Jewish cemeteries:** almost always outside the walls. In England, Jews could only bury their dead in London until 1177; after that, other communities got their own, such as York's "Jewbury" outside the walls. The Jewish cemetery at Worms lay outside the medieval wall; its oldest readable gravestone dates to 1058/59.
- **Muslim cemeteries:** outside the city gates, sometimes very large (Cairo's great cemeteries east and south of the city).
- **Criminals and outcasts:** at gallows sites and parish boundaries.

"Jewbury", as at York) so it is not read as a friary or a garden. Add a plague pit or charnel chapel if your story has had a plague.

Minorities and other faiths

RULE OF THUMB

Minorities live in their own quarter. At first it is usually near the ruler's castle or the market (for protection and trade). Later it is often forced to the town edge and walled off.

Jewish communities

- **Where:** in towns, almost never in villages in northern Europe. Their quarter had many names: *Jewry* (England), *Judengasse* (Germany), *judería* or *call* (Iberia). It had a synagogue, a ritual bath, a bakery, and a cemetery outside the walls.
- **Near power:** Jews were under the king's protection in England and many other realms, so they often lived close to the castle or the market.
- **England:** communities in London, York, Lincoln, Norwich, Oxford, Cambridge, Winchester and other royal towns. There were no more than about 3,000 Jews by 1278. They were expelled in 1290.

- **Frankfurt:** Jews lived near the cathedral, at the centre of town life, until 1462. Then they were forced into the *Judengasse*, a single curved street about 330 m (1,080 ft) long and 3–4 m wide, just outside the old wall, with three gates. It started with about 110 people. It held about 260 in 1543 and about 2,700–3,000 by the early 1600s, packed into fewer than 200 houses.
- **Expulsions:** England 1290; France 1306 and finally 1394; many German cities in the 1400s; Spain 1492; Portugal 1496–97. After an expulsion, the quarter keeps its name ("Old Jewry") but not its people.

Muslims under Christian rule

- **Iberia:** Muslims under Christian rule were called **Mudéjars**. They lived in their own quarter (*moreña*) with a mosque, bath, cemetery, market and bakery, or in whole

Muslim villages. They were especially numerous in the kingdoms of Valencia and Aragon, where they were mostly farmers. In Valencia most Mudéjars were Arabic-speaking Muslim peasants who farmed for Christian lords; they may have been a majority of the countryside

in the late 1300s (see [Language lines are not borders](#)). In Aragon and Castile, by contrast, Mudéjars spoke the same Romance language as their Christian neighbours. Castile forced them to convert or leave in 1502; Valencia and Aragon in 1525–26.

- **Conquered mosques:** when Christians took a Muslim city, its great mosque usually became the cathedral: Toledo (taken 1085), Córdoba (1236), Seville (1248). Seville later replaced it with a new Gothic cathedral (decided

1401, begun 1402) but kept the minaret (the Giralda) as the bell tower.

- **Sicily:** under Norman rule (conquest 1061–1091, kingdom from 1130) Sicily had many Muslims, mosques and Arabic-speaking officials. Emperor Frederick II moved the remaining Muslims to **Lucera** in Apulia over about twenty years from 1224. Their number reached 15,000–20,000. The colony lasted until 1300, when it was sacked, most people were killed or enslaved, and its mosques were replaced with churches.

Christians and Jews under Muslim rule

Christians and Jews were “protected peoples” (*dhimmi*) in Islamic lands. They kept their churches and synagogues, usually in their own quarters. In Cairo, many Coptic

churches and an important synagogue stood in the old city of Fustat, beside the newer royal city.

Orthodox, Armenian and foreign communities

- **Constantinople:** Italian merchants had their own quarters by the Golden Horn. The Venetian quarter dated from 1082. The Genoese were given Galata (Pera), across the water, in the late 1260s and held it as a walled colony from 1273 to 1453.
- **Southern Italy:** many Greek-speaking Orthodox monasteries survived in Calabria and Apulia under Latin bishops.

- **Lviv (Lwów, Poland):** the city had Armenian, Ruthenian (Orthodox) and Jewish quarters next to the Catholic one; the Armenian cathedral was built in 1363–1370.
- **Foreign merchants** often had their own walled compound (*fondaco*, *funduq*, *Kontor*) with a chapel. See [Trade Routes, Roads and Transport](#).

LATER ERA (1500s+)

Segregation becomes stricter. Venice created the first “ghetto” in 1516; in 1555 Venice had 923 Jews among 160,000 people. Rome followed in 1555. Spain expelled the Moriscos (Muslims forced to convert) in 1609–14, emptying hundreds of villages in Valencia and Aragon.

MAP TIP

On a city or town-edge map, give a trading town of 5,000+ a small minority quarter: a cluster of houses with a synagogue (a lamp symbol, never a star), or a mosque in a reconquered land (a dome with a crescent), near the castle or market. Put its cemetery outside a gate. On a kingdom map, a border region where two faiths meet can have villages of both: use the church symbol for one and the dome with a crescent for the other. Show recent expulsions with names like “Old Jewry” or “Moorish Quarter” and no symbol for a working temple.

Ancient and ruined layers

RULE OF THUMB

Every medieval landscape is built on older ones. Prehistoric sites sit on high ground and ridges. Roman sites sit on straight roads and at river crossings. Medieval people used old things as quarries, fortresses, boundaries, meeting places and sources of legend.

What a medieval traveller would see

Old feature	Age (in 1300 unless noted)	How common	Where	How medieval people used it
Roman roads	1,100–1,600 years	About 3,200 km (2,000 mi) of main paved roads in Britain, plus many minor roads. In the whole empire, over 80,000 km (50,000 mi) of stone-paved roads (out of 400,000 km in all), with official way stations (<i>mansiones</i>) every 25–30 km (16–19 mi)	Straight lines between old towns and forts, often on a raised bank (<i>agger</i>); the road itself usually 5–8 m (16–26 ft) wide	Still main roads (Watling Street, Fosse Way, Ermine Street). English law books of the 1100s put four great roads under the king's special protection. Many parish and county boundaries follow them.
Roman towns and forts	1,000–1,800 years	Every Roman province	River crossings, coasts, road junctions	Most Mediterranean ones stayed towns. In Britain some died (Silchester, Wroxeter; see chapter 01). Forts became castles.
Roman marching camps (an army's overnight camps)	~1,100–1,250 years in Britain	Nearly 500 known in Britain, almost half of them in Scotland; most 10–18 ha (25–45 acres)	Beside Roman roads, about a day's march apart: 11–24 km (7–15 mi)	Low rectangular banks with a ditch, still visible on moors; in ploughed land most were levelled. At Chew Green, on Dere Street (the Roman road north from York) right beside today's English–Scottish border, a small medieval settlement (Kemylpeth, pottery of c. 1250–1450) and probably a chapel sat on the Roman earthworks, and the place was a meeting point for settling cross-border cases from 1249. See Roman marching camps as old earthworks .
Hillforts (Iron Age earthwork forts)	1,300–2,000+ years	4,147 in Britain and Ireland (Atlas of Hillforts, 2017): about one per 75 km ² (29 sq mi) on average, but very uneven	Hilltops and ridge ends, especially in the west and south	Some re-fortified as castles or towns (Old Sarum: castle and cathedral). Many used as sheep pasture or fair grounds.
Long barrows (Neolithic burial mounds)	~4,300–5,100 years	About 300 earthen long barrows in eastern Britain, plus stone-chambered ones in the west; most 20–70 m (65–230 ft) long	Ridges and hill brows	Landmarks, boundary marks, legends. Wayland's Smithy (then in Berkshire, now Oxfordshire) is a chambered long barrow already named after a legendary smith in a charter of 955.
Round barrows (Bronze Age burial mounds)	~2,800–3,800 years	Many thousands in England; often in groups (barrow cemeteries)	Ridges and sky-lines, chalk downs	New burials in the early Middle Ages; meeting places; gallows sites; boundary markers
Stone circles and standing stones	~3,800–4,600 years	About 1,300 circles in Britain and Ireland	Uplands, moors, river valleys	Feared, Christianised or destroyed. At Avebury, villagers toppled and buried stones in the early 1300s. One man, a barber-surgeon, was found under a fallen stone, with coins of the 1320s. A church was built just west of the Avebury henge (a prehistoric ring of bank and ditch) in the 900s.
Deserted villages	Recent: most emptied after 1350, so they matter most for maps set in 1350–1500	About 3,000 known in England; 1,500+ emptied c. 1350–1520	Clay lowlands and poor land; sheep country	House platforms, hollow lanes and a lone church in a sheep field (see chapter 01)
Abandoned castles	Up to ~250 years in 1300; more by 1500	England and Wales had about 1,700–1,800 castle sites (the Gatehouse gazetteer lists 1,761), but only 500–600 in use at once	Hills, river crossings, old borders	Grass-covered mottes (earth mounds) and ruined towers; stone robbed for houses (see chapter 04)

How old things were reused

- **As quarries.** Roman ruins were the easiest source of cut stone and brick. The Norman abbey church at St Albans was built largely of Roman brick and tile from the ruined Roman town of Verulamium, across the river. Many churches near Roman sites (Wroxeter and its neighbours) are full of Roman stone.
- **As fortresses.** At Portchester, a Roman fort of about 285–290 still stood complete. The Normans built a castle keep in one corner (probably late 1000s) and an Augustinian priory inside the walls (founded 1128; its church dates from the 1130s). In southern France, the

Roman amphitheatre of Arles became a fortress with four towers and more than 200 houses inside it.

- **As holy ground.** Churches were built at or beside pagan sites (Rudston, Avebury). Some saints' shrines sat at old Roman cemeteries.
- **As boundaries and meeting places.** Barrows, stones and Roman roads made ideal parish edges and moot sites, and gallows sites (see [Legal and civic sites](#)).
- **As legends.** People explained old earthworks with giants, devils or old heroes. Names like "Grim's Ditch" (Grim being a name for the god Woden) and "Devil's Dyke" are common in England.

Where ruins survive and where they disappear

RULE OF THUMB

Ruins near living towns get quarried away within a few generations. Ruins survive far from people, in hard stone, or where a taboo protects them.

- **Near towns:** Roman walls survive because the town still uses them; other Roman buildings vanish into new houses.
- **In empty country:** hillforts, barrows and stone circles survive, especially on moors, downs and heaths.
- **In the Middle East:** ancient cities often survive as **tells** (mounds of earlier settlements, built up over thousands

of years). Roman and Byzantine ruins stand in the desert edge, where few people needed the stone.

LATER ERA (1500S+)

The Dissolution (1536–41) creates hundreds of ruined abbeys in England and Wales, often with a new country house built from their stones. Antiquarians begin to record ruins: William Camden's *Britannia* (1586) describes Roman sites county by county. Old castles that cannot take cannon are abandoned or slighted (deliberately wrecked) in the wars of the 1600s.

FANTASY TWIST

An older, vanished race (elves, giants, an ancient empire) leaves the same kinds of layers: roads, forts on hills, tombs on ridges, stones in circles. If its ruins are still whole near living towns, there must be a reason: fear, a curse, a monster, or a law. Otherwise people would have carted the stone away. See [Fantasy Variants](#).

MAP TIP

Give a kingdom map a few long, straight "old roads" that link cities in near-straight lines, with later towns along them. On local maps, add 1–3 hillforts per 100 km² (39 sq mi) in hilly country (ovals of rings on hilltops), scattered barrows along ridges (small filled mounds), a stone circle or two on moorland (small rings of dots), a Roman camp beside an old road (a grey, closed dashed rectangle with rounded corners), a ruined motte by an old crossing (the motte symbol drawn grey), and one or two deserted villages shown by a grey lone church labelled "(lost village)". Label them in the local style: "Old Fort", "Giant's Grave", "Kings' Stones", "Castle Hill", "Old Street".

Drawing it all: symbols, density and scale

RULE OF THUMB

Draw what a traveller at that scale would notice. Cathedrals and great shrines on a continent map; abbeys and gallows on a local map; every church only when villages are shown.

Suggested symbols

These symbols match the guide's [master legend](#).

Feature	Symbol idea	Show at which scale
Archbishop's seat	Double cross beside the city symbol	Continent, kingdom
Cathedral	Church with two towers, or a mitre	Kingdom, local
Parish church	Small church with a tower; spire for rich towns	Local
Chapel	Small plain cross, no tower	Local
Wayside cross	Tiny cross on a base, set on the road or path	Local
Abbey or priory	Church with a square cloister beside it	Kingdom (great abbeys), local (all)
Friary	Name only, or a small church inside the town outline	Local and town maps
Grange	Small open square topped with a cross, or the label "Grange"	Local
Commandery (military order)	Small church with a shield	Kingdom, local
Great shrine	Scallop shell or small reliquary (never a star: a filled star means a capital)	Continent, kingdom
Pilgrim road	The ordinary road line with small scallop marks along it (not a dotted line)	Kingdom
Hospice at a pass	Small house with a cross	Kingdom, local
Hospital or leper house	Small house with a cross; leper house labelled "Lazar House"	Local
University	Book, or a "U" next to the city name	Continent, kingdom
Gallows	Small "Π" on a hill	Local
Moot or thing site	Small lone tree, labelled "Moot" (or "Thing")	Local
Mosque	Dome with a crescent	Kingdom (great mosques), local
Synagogue	Lamp (never a star)	Local, town edge
Ruin	The feature's own symbol, broken or grey, labelled "(ruin)"	Kingdom (big ones), local
Deserted village	Grey church alone in a field, labelled "(lost village)"	Kingdom (a few), local
Hillfort	Oval of rings or hachures on a hill	Local
Barrow	Small filled mound; a long barrow as a longer mound	Local
Standing stone	Small upright tick	Local
Stone circle	Small ring of dots	Local
Old or Roman camp	Grey closed dashed rectangle with rounded corners, labelled "Camp"	Local
Roman road	Straight thick solid line labelled "Street" or "(Roman)"; grey where abandoned	Kingdom, local

Footprints: how big to draw them on a local map

RULE OF THUMB

A great abbey is not one building but a walled estate 350–600 m (0.2–0.4 mi) across, as big as a large hillfort or Malbork, the largest castle by area. A friary is one town block. A moated manor or a small motte is only a dot.

Why so big? A great abbey had to feed, house and employ hundreds of people. Its **precinct** (the walled area of the monastery) held much more than the church and the **cloister** (the square, covered walk with the monks' main rooms around it). Around them were an **inner court** with guest houses and workshops, and an **outer court** with barns, mills, smithies, fishponds and orchards. At Fountains the River Skell ran through the precinct, and the

farming and industry lay in the outer court south of the river. Rievaulx's outer court held a double water-mill and fishponds, and a great gatehouse stood at the entrance. Fountains and Rievaulx both built a wall around the whole precinct in the early 1200s.

Most of England's ~900 religious houses were small prior-ies with much smaller precincts. The numbers below are for the great houses, plus other sites for comparison.

Site	Area	Rough size across	Drawn at 1 cm = 1 km	Real examples
Great Cistercian abbey precinct	12–37 ha (30–92 acres)	~350–600 m (0.2–0.4 mi)	3.5–6 mm	Roche ~30 acres (12 ha); Kirkstall ~40 acres (16 ha); Fountains ~70 acres (28 ha), inside a wall almost 3.5 m (11 ft) high built in the first half of the 1200s; Rievaulx 92 acres (37 ha)
Great Benedictine abbey	~15 ha (36 acres)	~400 m (0.25 mi)	~4 mm	Glastonbury: today's abbey grounds cover 36 acres (15 ha); parts of the medieval precinct wall still stand
Cathedral close	Up to ~32 ha (80 acres)	Up to ~570 m (0.35 mi)	Up to ~6 mm	Salisbury, laid out from 1220, has Britain's largest close: about 80 acres (32 ha). Most closes are smaller.
Friary (in a town)	~1.5–2 ha (4–5 acres)	~120–140 m (400–460 ft)	~1.5 mm	Northampton Greyfriars: about 4 acres (1.6 ha); Canterbury Blackfriars: about 1.8 ha (4.5 acres)
Moated manor site	Island often under 0.15 ha (0.4 acre)	Island ~25–50 m (80–165 ft) across, inside a moat 7–15 m (23–50 ft) wide: the whole site is ~40–80 m (130–260 ft)	Under 1 mm: use a symbol	Lincolnshire scheduled sites: islands of 24 m, 40 × 30 m and 50 × 30 m. A high-status site, South Kelsey Hall, has a main island of 110 × 100 m (about 1.1 ha)
Small motte-and-bailey castle	Bailey ~0.15–0.7 ha (0.4–1.7 acres)	Motte (earth mound) 30–90 m (100–300 ft) across at the base; whole castle roughly 100–150 m (330–490 ft) long (estimate)	~1–1.5 mm	The Mount, Ilketshall St John (Suffolk): motte 46 m across, bailey platform 53 × 30 m. Ewyas Harold (Herefordshire): bailey about 0.7 ha
Large baronial castle	~3 ha (7.5 acres)	~170 m (560 ft)	~2 mm	Castle Acre (Norfolk): about 3 ha to the outer edge of its defences. The small walled town beside it covered 3.8 ha (9.4 acres)
Hillfort reused as castle and cathedral town	~11 ha (28 acres)	400 × 360 m (1,300 × 1,180 ft)	4 × 3.6 mm	Old Sarum (see chapter 04)
The largest castle	21 ha (52 acres)	~450 m (0.3 mi)	~4.5 mm	Malbork, Teutonic Order headquarters (see chapter 04)
Deer park	40–80 ha (100–200 acres)	0.5–1 km (0.3–0.6 mi)	5–10 mm	About 3,200 in England c. 1300 (see chapter 06)

“Rough size across” is the side of a square with the same area, calculated by this guide. Real precincts were irregular: they followed rivers, roads and valley sides. For a sense of

scale, Fountains' precinct (28 ha) could hold the whole walled town of Castle Acre (3.8 ha) seven times.

MAP TIP

On a local map at 1 cm = 1 km, draw a great abbey as a walled outline 3.5–6 mm across, with a gatehouse on the road side and the church and cloister inside. At a larger scale, such as 4 cm = 1 km (1:25,000, where the precinct is 14–24 mm across), add the inner and outer courts, a mill on the stream, a chain of fishponds, barns and orchards. Draw a

friary as a small walled plot against the town wall or at a gate. A moated manor or a small motte is under 1 mm at 1 cm = 1 km, so use a symbol: a small open square with a double outline for a moated manor, and a small mound with a tower tick for a motte (see [chapter 06](#) and [chapter 04](#)).

Density checklist for a kingdom of 50,000 km² (19,000 sq mi)

Use this for an English-style kingdom of about 1–2 million people c. 1300. Scale up or down by area.

Feature	How many	Where
Archbishop	1	Oldest or richest city, often not the capital
Cathedrals	5–10 (English style) or 25–100 (Italian style)	Main cities; in the Italian style, many small towns
Parish churches	~3,500 in lowland; fewer in hills	One per village; many in old towns
Religious houses	300–350	Monks in valleys, canons at towns, friars in towns, nuns in quiet places
Friaries	60–75	Only in towns of 3,000–5,000+; 4–5 in the biggest city
Hospitals and almshouses	300–400 over time	Gates, bridges, roads
Leper houses	100–120	1–2 km (about 1 mi) outside towns on main roads
University	0–1	The biggest cathedral or capital city
Great shrine	1–3	One national saint, plus regional ones
Hillforts and barrows (if the land was settled in prehistory)	Hundreds	Hilltops, ridges, downs

Place names that tell the story

Everyday village endings (-ton, -by, -ley and so on), the common church words (-minster, Kirk-, Llan-, Kil-) and the Roman word -chester are listed in [Naming settlements](#)

realistically, with a method for naming a whole region. The table below adds only the names that point to the sites in this chapter: saints, church owners, hospitals, gallows and ancient remains.

Name element	Meaning	Examples
Münster (German), Moutier (French)	Monastery or mother church (from Latin <i>monasterium</i> , like English -minster)	Münster; Marmoutier near Tours (“the greater monastery”)
-church	Church	Whitchurch (“white church”, probably one built of pale stone)
San-, Santa-, Saint-, Sankt-	Named after a saint’s church or shrine	Saint-Denis, Santiago
Abbots-, Bishops-, Monks-, Nun-	Owned by the church	Bishop Auckland (home of the Bishops of Durham), Monkton, Nuneaton
Temple-	Owned by the Templars	Temple Bruer, Templecombe
Spital, Maudlin, Lazar	Hospital or leper house	Spitalfields
Gallows, Galgen-, Justice	Execution place	Gallows Hill
Street, Strat-, Stret- (on a road)	Roman road	Stratford, Stretton
-low, Barrow, Tump	Burial mound or mound-shaped hill	Ludlow, Taplow

MAP TIP

Check the balance at the end. If your kingdom has one splendid cathedral and nothing else religious, it will look empty. Most real kingdoms had a dense, even carpet of

small churches, a scatter of monasteries, a few hospitals at every town and ruins of older ages on the hills. Common errors are listed in [Common Mistakes](#) and [How to Fix Them](#).

Quick summary

- One parish church per village, about 4 km (2.5 mi) apart in the lowlands and about 7–10 km (4–6 mi) apart in the hills; England had ~9,500 parishes, one per ~14 km² (5.4 sq mi).
- Old towns had many tiny parishes (London over 100, Norwich about 56); planted towns had one.
- Dioceses are small and many in Italy and southern France (Umbria: ~850 km², 330 sq mi, each), large and few in England (17; ~7,600 km², 2,950 sq mi, each) and the north. In Italy a cathedral can sit in a small town.
- Religious houses: about 1 per 150 km² (60 sq mi) (England ~900 in the 1530s, average 13 people). Cistercians and Carthusians in empty, watered valleys; Benedictines in old towns; canons at town edges; friars inside towns. A great abbey's walled precinct covers 12–37 ha (30–92 acres), 350–600 m (0.2–0.4 mi) across; a friary about 1.5–2 ha (4–5 acres).
- Friaries roughly measure town size: none under ~3,000 people, 2 at about 10,000, 4 at 20,000–40,000.
- Granges were meant to lie within a day's walk of their abbey, but many lay farther (Fountains' grange at Cowton was about 32 km, 20 mi, away).
- Pilgrim roads follow existing roads, with stops about 20 km (12 mi) apart, hospices at passes, pilgrim towns and a leper house near the shrine.
- Hospitals stand at gates, bridges and roads; leper houses often 1–2 km (about 1 mi) outside towns on main roads.
- Universities are rare: ~20 in Europe c. 1300, 60–80 by 1500, mostly in big cities with a bishop or court.
- Gallows stand on high ground by a main road, outside town, at the lordship's edge; pillories and stocks in the market place; moots at mounds, stones, trees and fords.
- Jewish and Muslim quarters lie near the castle or market, often later pushed to the edge; their cemeteries lie outside the walls.
- Old layers matter: Roman roads, Roman marching camps, hillforts (4,147 in Britain and Ireland), barrows, stone circles, deserted villages and abandoned castles. Ruins near towns get quarried; ruins in empty land survive.

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medieval universities; Medieval university; Al-Nizamiyya of Baghdad; Tyburn; Gibbet of Montfaucon; Gallows; Pillory; Stocks; Moot hill; Thing (assembly); Hundred (county division); Guildhall, London; Palazzo Pubblico; Belfries of Belgium and France; Market cross; Charnel house; History of the Jews in England (1066–1290); Frankfurter Judengasse; Jewish Cemetery, Worms; Venetian Ghetto; Mudéjar; Lucera; Roman roads in Britain; Hillfort; Atlas of Hillforts of Britain and Ireland; Long barrow; Bowl barrow; Roman roads; Stone circle; Avebury; Rudston Monolith; Portchester Castle; Arles Amphitheatre; St Albans Cathedral; Spolia; Pope Boniface VIII (Villani's 200,000 pilgrims); Catholic Church in Italy (101 dioceses suppressed in 1986); Communes of France; Winchester Cathedral; Torre del Mangia; Seville Cathedral; Galata; Zubaidah bint Ja'far; Hospital of St Cross; Wayland's Smithy; Motte-and-bailey castle (motte sizes); Salisbury Cathedral; Chew Green; Marmoutier Abbey, Tours (*Majus monasterium*); Whitchurch, Shropshire ("white church"); Bishop Auckland.

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Densities per 1,000 km², the "sample county" and the "kingdom of 50,000 km²" checklist are this guide's own calculations from the English counts above, not historical counts. Tower visibility distances are geometric limits (curve of the earth only) calculated by this guide. In the footprints

table, the "rough size across" and "drawn at 1 cm = 1 km" columns are this guide's own conversions from the recorded areas.

CHAPTER 9

WHAT CHANGES AFTER 1500

Your map may be set a little later than the classic Middle Ages: a world with cannon, printed books and ships that cross oceans. This chapter explains what changed on the ground between about 1500 and 1650, and how to show it on a map. Most rules in the rest of this guide still hold, because villages, market towns and travel speeds barely change. What changes is the top of the system: bigger capitals, new kinds of fortresses, ocean ports, post roads, canals, drained land and a new religious map.

The medieval baseline numbers (settlement sizes, density, travel speeds) are in [Population and Settlement Sizes](#) and [Trade Routes, Roads and Transport](#). This chapter only explains what is different.

IN THIS CHAPTER

What changes and what stays the same

Population and bigger cities

Gunpowder and the new fortifications

Ocean trade, ports and trading posts

Posts, roads and canals

Money, printing and news

Capitals, states and borders

The countryside: enclosure, new crops and drainage

Industry: furnaces, mines and mills

Religion: the Reformation on the map

Comparison table: medieval vs 1500 to 1650

Drawing a world in transition

Quick summary

Sources and further reading

What changes and what stays the same

RULE OF THUMB

After 1500 the bottom of the map stays medieval and the top changes. Villages, fields and market towns look much the same. Capitals, ports, fortresses and long-distance links are new or much bigger.

Why: Most people (80–90% almost everywhere) still farm with the same tools and animals. On land, goods still move

by ox, horse and mule. So everything that depends on walking distance stays the same: how far apart villages are, how far apart market towns are, and how much land feeds a town. What changes is what rich states and merchants can now pay for: cannon and new walls, ocean ships, standing armies, big courts, printing presses and engineering projects.

Timeline of the changes

Date	What happens	What it means on the map
1490s	Taxis relay post (from 1490); French siege guns in Italy (1494); Columbus (1492) and da Gama (1498)	Post stations along main roads; old walls suddenly weak; new ocean routes
1500–1540s	Bastioned fortifications develop in Italy and spread north in the 1530s–1540s	Star-shaped walls around key towns
1500–1600	Population recovers and grows; Atlantic ports boom	Bigger capitals and ports; Seville, Lisbon and Antwerp, then Amsterdam and London
1517–1560s	Protestant Reformation; English monasteries closed 1536–41	Abbey ruins, abbeys turned into houses, religious borders
1560s–1600	Fixed capitals (Madrid 1561); printed atlases and county maps; Spanish rules for planting colonial towns (1573)	Palace complexes; more accurate maps; grid towns around a plaza in the colonies
1600–1650	Chartered trading companies (English 1600, Dutch 1602); lakes drained (Beemster 1612); Japan's one-castle order (1615); French edict against interior castles (1626); Thirty Years' War (1618–48)	Trading posts overseas; new polder landscapes; far fewer working castles in strong states; devastated regions

What does not change

Feature	Value (same as the medieval baseline)
Village spacing, lowland	1.5–4 km (1–2.5 mi)
Market towns in practice	9–16 km (6–10 mi) apart; England had ~650 working markets in the late 1500s
Walking traveller	25–35 km (15–22 mi) a day
Horse cart	30–40 km (20–25 mi) a day
Water vs land costs	Bulk goods still go by water; land haulage stays several times dearer
Share living in towns of 10,000+	Still under 10% in most countries; the big exceptions are the Netherlands (24% in 1600, 32% in 1650) and Belgium and Italy (15–19%)

MAP TIP

Do not redraw the whole countryside for a later setting. Keep your medieval village and market-town pattern. Then change the top layer: enlarge the capital and the main ports, add new forts, post stations, canals and drained land, and turn some castles and abbeys into ruins or country houses.

Population and bigger cities

RULE OF THUMB

Europe regains its 1300 peak during the 1500s and keeps growing until about 1600–1620. Almost all the extra growth in towns goes to capitals and ocean ports.

How many people

- **Europe:** Jan de Vries's estimates for Europe without Russia and the Ottoman lands are roughly 62 million (1500), 70 million (1550), 78 million (1600) and 75 million (1650). Counting Russia and the Ottoman Balkans, the guide's baseline is about 100 million by 1600. Other historians give somewhat different totals, but the shape is the same: growth in the 1500s, a halt in the 1600s.

- **England:** about 2.35 million (1522), 4.1 million (1600) and 5.3 million (1650). England only passed its 1348 peak around 1600–1630.
- **The 1600s are harder.** Growth stops or reverses after about 1620. The Thirty Years' War (1618–48) devastated parts of Germany. Italian plagues in 1629–31 and 1656–57 each killed 20% or more of the people in the areas they hit. Milan lost about 60,000 of its 130,000 people in the plague of 1629–31. Naples lost 150,000 or more people, about half the city, in the 1656 plague.

The big cities

Figures are estimates in thousands. Sources disagree, so treat each range as “somewhere in here”.

City	c. 1500	c. 1600	c. 1650	Why it grew or shrank
Istanbul	100–280 (Ottoman count of 1478: 80–100)	400–700	400–700	Ottoman capital, fed by sea from Egypt and the Black Sea
Paris	95–250 (often ~200)	220–300	400–500	Capital of the largest western kingdom
Naples	100–150	250–280	~300 before the 1656 plague, then about half	Spanish viceroy's capital; nobles and migrants from the whole kingdom
London	~50	~200	375–400	Capital and chief port; took over much of Antwerp's trade
Amsterdam	10–15	50–60 (105 in the 1622 count)	150–175	Took over Antwerp's trade after 1585; Baltic grain and Asian spices
Seville	45–70	110–150 (peak, late 1500s)	~130 before the 1649 plague, which killed nearly half	Legal monopoly port for the Americas from 1503
Lisbon	55–65	110–150	130–170	Spice route to India; Brazil sugar
Antwerp	~40 (1496)	~100 in 1566, then ~42 in 1589	~50–60	Boom as Europe's market, then siege (1585) and blockade of the river Scheldt

City	c. 1500	c. 1600	c. 1650	Why it grew or shrank
Madrid	under 20 (not yet capital)	75–95	100–130	Made the fixed capital in 1561 (about 18 then, about 80 by 1598)
Venice, Milan	100–150 each	110–150 each	Venice ~100–120; Milan smaller	Large but no longer growing; the 1629–31 plague killed about a third of Venice (46 of 140) and almost half of Milan (60 of 130)
Rome	55–60	~100	100–120	Papal capital rebuilt as a showpiece for pilgrims

Shifting rank: the Atlantic shift

The order of cities changes faster than in the Middle Ages. Three patterns are worth copying:

1. **Rise of ocean ports.** Seville, Lisbon, Antwerp, Amsterdam and London grow two to ten times. Seville grew from perhaps 45,000–70,000 people around 1500 to over 100,000 by the late 1500s, which made it the largest city in Spain.
2. **Rise and fall.** Antwerp grew from about 40,000 (1496) to 100,000 (1566). After the Spanish took it in 1585, the

Dutch closed the river Scheldt to sea trade, and by 1589 only about 42,000 people were left. Its merchants and craftsmen moved to Amsterdam and London.

3. **Capital pulls life out of an old city.** When Madrid became the capital, the old regional centre Toledo, the largest city in the region until about 1600, declined (Ringrose 1973).

The Mediterranean does not collapse. Venice, Genoa, Naples and Istanbul stay huge. But the fastest growth is now on the Atlantic and North Sea coasts.

Why capitals grow so much

A capital is now a **consumer city**. It holds the court, the law courts, the treasury, the nobles' town houses, foreign ambassadors, and thousands of servants and luxury craftsmen. By 1650 about one English person in thirteen lived in London (about 400,000 out of 5.3 million). In some realms

the capital becomes 5–10 times bigger than the second city. England is the extreme case: London was more than ten times the size of the next English town. In other lands (Spain, Italy) several big cities stay close in size. See [Population and Settlement Sizes](#) for the “primate city” pattern.

MAP TIP

For a 1600-style map, draw the capital 2–4 times bigger than its medieval size, and the fastest-growing port or capital up to 5–10 times bigger. Show suburbs spilling far outside the old walls. Add one “fallen” city: a port with a silted or blockaded river, a smaller population and a grand but half-empty centre.

Gunpowder and the new fortifications

Why the old walls failed

Medieval walls were **high and thin**, to stop ladders. Cannon smashes high stone walls quickly. When Charles VIII of France invaded Italy in 1494, he brought about 18,000 men (other accounts say up to 25,000). He also brought a siege

train of at least 40 guns on wheeled carriages (Parker). Castles that had once held out for months now fell in days. Round medieval towers also leave **dead ground**: spots at the foot of the wall that the defenders cannot shoot at.

RULE OF THUMB

After 1500, a wall that is tall is weak. Good walls are now low, very thick, backed with earth, sunk behind a wide ditch, and shaped so that every part can be shot at from another part.

The bastion and the star fort

A **bastion** is an arrow-shaped platform that sticks out from the wall. Guns on each bastion fire along the face of the next bastion, so there is no dead ground. Seen from above, a ring of bastions looks like a star. This system is called the

trace italienne (“Italian outline”) because it was first built in Italy. Extra **outworks** stand in or beyond the ditch, for example the **ravelin** (a triangular island fort in front of the wall).

Date	Place	Why it matters
1501–c. 1503	Nettuno (papal fort, usually credited to Antonio da Sangallo the Elder)	Early angled bastions; very thick, low walls without battlements
1508–1537	Civitavecchia (Forte Michelangelo)	Papal port fort; often called the first key example of the new style
1509	Padua	The engineer Fra Giocondo replaced the old wall with a broad ditch swept by low gun positions; a huge imperial siege army failed
1534–1537	Fortezza da Basso, Florence	Large pentagonal (five-sided) bastioned fortress by Antonio da Sangallo the Younger; often cited as an early model
1530s–1540s	Spain, France, the Low Countries, England	The style spreads out of Italy
1564–c. 1570	Turin citadel	Pentagon with five bastions (sources give finishing dates from 1566 to 1570)
1566	Valletta, Malta	New fortress capital, built after the Great Siege of 1565
1593	Palmanova (Venice)	Planned nine-pointed star town; first ring of walls (finished 1623) about 7 km (4.3 mi) round
1593	Bourtange (Dutch)	Small five-pointed fort guarding a road through a bog

Sizes and garrisons

Sizes and garrisons below are rough estimates, except the named examples.

Kind	Typical size	Garrison	Where to put it	Map symbol
Watchtower	10–15 m tall	a handful of men	Every few km along raided coasts, each in sight of the next	Small triangle
Coastal artillery fort	30–100 m across	10–30 in peace (Walmer 18, Camber 29 in 1540)	Harbour mouths, anchorages, river mouths	Small round outline with rounded lobes (a rosette , like the plan of Deal Castle, 1539–40), never pointed
Small fort (<i>sconce</i> , Dutch <i>schans</i>)	100–250 m across, 4–5 bastions	50–300	River crossings, dikes, roads through marshes	Small star outline (the star-fort symbol), never filled
Citadel	400–700 m across, 4–6 bastions	several hundred to 2,000+	At the edge of a big or rebellious city	Pentagon outline joined to the town (the bastioned-citadel symbol)
Fortress town	1–2.5 km (0.6–1.5 mi) across, 6–12+ bastions	1,000–5,000 in war	Frontiers, key river towns, big ports	Star-edged town outline, never filled

Bastions are usually spaced so that musket fire from one reaches the next, roughly 200–350 m apart (an approximation). This is why star forts look so regular.

Cost. Bastions were enormously expensive. Siena bankrupted itself in 1544 to pay for its new defences. Amsterdam's

22 bastions cost 11 million florins. Henry VIII spent £376,000 on his coastal forts (1539–47); one small block-house cost about £500, and the three forts of the Downs in Kent cost £27,092 together.

RULE OF THUMB

Only rich states can afford bastions everywhere. Put full star forts at frontiers, the capital, the main ports and a few river crossings. Ordinary towns keep their medieval walls, perhaps with a few earth bastions added on the side facing the enemy.

Citadels: forts that watch their own city

A **citadel** is a star fort built at the edge of a city, half inside and half outside the walls. It defends the city, but it also

controls the citizens. Its guns point both outward and into the streets.

- **Fortezza da Basso**, Florence (1534–37), built for the new Medici duke.
- **Turin** (from 1564), designed by Francesco Paciotto after Turin became the Savoy capital; meant to defend the city and also to overawe its people.

- **Antwerp** (1567–72), also by Paciotto, built on the Duke of Alba's orders during the Dutch Revolt. After the Spanish

troops sacked the city in 1576, the citizens tore down part of it. It was rebuilt after 1585.

MAP TIP

Draw a citadel as a pentagon outline (the guide's bastioned-citadel symbol) joined to one corner of the town wall, with an open space (an *esplanade*) between the fort and the houses. If a ruler has just conquered a rebellious city, give it a citadel.

Coastal defence

- **England:** Henry VIII's "Device Forts" (1539–47). These were dozens of forts, blockhouses and earth gun platforms at the anchorages and river mouths most open to invasion: the Downs in Kent, the Thames, the Solent and Falmouth. Twenty-four sites were already finished and manned by the end of 1540. Most were small, round, squat gun forts with tiny garrisons. Seen from above, the bigger ones look like a flower: Deal Castle (1539–40), about 70 m (230 ft) across, is a round central tower inside two rings of six rounded bastions; Walmer has four rounded bastions.

MAP TIP

On a raided coast, draw a chain of watchtowers (small triangles) 3–12 km (2–7 mi) apart, every one in sight of the next. Keep the villages a little inland and uphill. The tower spacing is an estimate. About 379 towers on roughly 2,000–2,400 km (1,250–1,500 mi) of Neapolitan mainland coast is one per 5–6 km (3–4 mi). That count comes from 1748, so it may include some towers built after 1600. About

- **Kingdom of Naples:** viceroys' orders of 1532 and especially 1563 pushed the building of watchtowers along the whole coast against corsairs (sea raiders, here mainly from North Africa). An inventory of 1748 listed 379 coastal towers. They passed warnings by smoke by day and fire by night.
- **Genoa** built about 85–90 towers around Corsica between 1530 and 1620 (90 were planned in 1531; a list of 1617 counts 86). See [Watchtowers, beacons and coastal warning](#).

85–90 towers on Corsica's roughly 1,000 km (620 mi) of coast is one per 11–12 km (about 7 mi). Put a real fort only at the harbour mouth, drawn as the coastal artillery fort symbol: a small rosette outline (rounded lobes, like Deal Castle's plan), never pointed. A harbour fort with angled bastions takes the star outline instead.

Bigger armies, longer sieges

- **Army size:** Charles VIII's 18,000 men looked huge in 1494. By the 1630s the leading states kept armies of over 150,000 (Parker). These are paper strengths. The number of men actually present was often much lower.
- **Sieges become the main event.** A fortress with bastions could hold for months. Ostend held out for three years (1601–04) because it was supplied by sea; more than 100,000 people were killed, wounded or fell sick on both sides.

- **Military corridors.** Spain moved troops from Milan to the Netherlands along the **Spanish Road**, about 1,000 km (620 mi) through Savoy, Franche-Comté and Lorraine. A march took 32–60 days, usually 40–50, so about 20–25 km (12–16 mi) a day, with food bought at stopping places (*étapes*) along the way. More than 123,000 soldiers marched this route between 1567 and 1620, against only about 17,600 sent by sea.
- **Devastated land.** Large armies living off the land ruin whole regions. Show burnt villages and empty farms along their routes.

Fortified lines on open frontiers

Where there are no mountains or rivers to block raiders, states built long lines:

- **Muscovy's Great Abatis Line** (nominally finished in 1566) stretched from Bryansk to Meshchera. It was made of felled trees laid as a barrier, ditches, earth banks, palisades and watchtowers, linking fortified towns such as

Tula, Ryazan, Serpukhov and Kolomna. In the late 1500s about 35,000 local militia guarded it (some sources give more).

- **Muscovy's Belgorod Line** (1635–1654) was the second line, built further south along the southern edge of the forest steppe (the zone where woods give way to open grassland). It ran from the Vorskla River (today in Ukraine) east to the Tambov area. It was about 800 km

(500 mi) long with its bends, or about 600 km (370 mi) in a straight line. It linked fortress towns such as Belgorod, Kozlov and Tambov (both founded in 1635–36 for the line) with earth banks, felled-tree barriers and palisades (walls of wooden stakes). It used rivers, marshes and woods as part of the barrier. Eighteen of its fortress towns were new. Sources differ a little on when it was finished; one gives 1646.

- **The Habsburg–Ottoman frontier** in Hungary and Croatia was a chain of fortresses and soldier-farmer settlements from the mid-1500s. For the medieval frontier layout it grew from (a forward line of small posts and a rear line of fortress towns), see [Outposts in wild and contested land](#).

What happened to the old castles

Fate	Real examples	How to show it
Updated with earth bastions or gun platforms	Common in frontier and war zones	Castle icon inside a star outline (the star-fort symbol)
Rebuilt as a palace or residence	Chambord (begun 1519) and the other Loire châteaux; Heidelberg's Renaissance wings (1550s and 1601–07)	Castle icon with gardens, labelled "palace"; a royal palace or hunting lodge takes the small crown instead
Replaced by a country house nearby	Elizabethan "prodigy houses" (1570–1620): Burghley (finished 1587), Hardwick Hall (1590–97)	House inside a ring of park trees; the old castle nearby, drawn broken or grey
Kept as a prison, arsenal or mint	Tower of London	Castle icon in the town, labelled with its new use ("prison", "mint")
Deliberately wrecked ("slighted")	In and after the English Civil War, Parliament slighted, or planned to slight, more than 100 buildings (castles, town walls and houses); Corfe (1646), Kenilworth (1649)	Castle icon drawn broken or grey, labelled "(ruin)"
Abandoned and used as a quarry	Many minor castles in peaceful interiors	Small castle icon, broken or grey, often near a village

MAP TIP

In a 1500–1650 world, inland castles in peaceful regions should mostly be ruins, palaces or manor houses. New gun forts belong at frontiers, coasts, capitals and river crossings. Do not put a fresh star fort next to every village. See [Castles, Forts and Military Outposts](#) for the medieval castle network.

When a strong ruler cuts the number of castles (France and Japan)

RULE OF THUMB

When one ruler finally controls the whole land, private castles stop being useful to him and become a danger. The number of working castles then falls sharply, often by a single order.

Why: A castle lets a local lord defy the ruler. A ruler with his own army and cannon no longer needs these castles for defence. So he keeps the forts on the frontier and the seats of his governors, and orders the rest pulled down.

- **France, 1626:** a royal edict, issued in Louis XIII's name under Cardinal Richelieu, ordered that all towns and castles "in the interior of our realm" that were not

important for frontier defence "shall be razed and demolished". It aimed at the strongholds of rebellious nobles.

- **Japan, 1615:** after the Tokugawa family took Osaka Castle in 1615, the shogunate (the military government) issued the "one castle per province" order (*ikkoku ichijō rei*). Each daimyo (great lord) kept one castle, normally his own residence, and demolished the others. A few of the greatest lords were allowed more. In the same year, the Laws for the Military Houses (*Buke shohatto*) said that every repair had to be reported to the shogunate, and that new building or enlargement needed its permission. The order stayed in force until the lords' domains were abolished in 1871.

Japan	Before 1615	After 1615
Working castles	about 3,000 (a rounded figure)	about 170 (more than nine in ten gone)
Who holds one	Any lord strong enough to build one	Only the great lords, usually one each
Main job	War: base and refuge	Government: the seat of a lord's domain
New building	Free	Only with the shogunate's permission

Real examples: Iwakuni Castle (western Japan) was built in 1601–08 and pulled down in 1615, when it was only seven years old. Himeji Castle, completely rebuilt in 1601–09, survived as a great lord's seat.

MAP TIP

For a 1600–1650 map of a strong, centralised realm, keep about one working castle per province (or per great lord), set inside the province's main town. Draw that town around the castle, as the province's capital. Show the other old castle sites as ruins (the castle icon drawn broken or grey) or just as a place name such as “Castle Hill”. Keep extra forts only on the frontier and the coast. A realm still split between strong nobles keeps many castles.

The castles that survived became the centres of **castle towns** (*jōkamachi*, “town below the castle”). The lord's officials, his warriors and the merchants who served them lived around the castle. These towns became the regional capitals of Japan for the next 250 years.

FANTASY TWIST

If dragons or war-mages are your world's “artillery”, the same logic applies: tall walls become weak, and defenders build low, thick, angled walls that cover each other. If the attackers **fly**, though, a star fort gives no protection. Expect roofed gun towers and underground shelters instead. See [Fantasy Variants](#).

Ocean trade, ports and trading posts

Bigger ships

Ship	Main period	Typical size	Crew	Main use
Cog (medieval)	1200s–1400s	30–200 tons; a few were larger (see chapter 05)	small	North Sea and Baltic cargo
Carrack (Portuguese <i>nau</i>)	late 1400s–1600	400–2,000 tons; Portuguese India ships were often over 1,000; the <i>Madre de Deus</i> (captured 1592) was rated at about 1,600 tons	600–700 aboard the <i>Madre de Deus</i>	India and spice route
Galleon	mid-1500s–1700	up to about 1,000 tons on the Atlantic (many smaller); Manila galleons 1,700–2,000	—	Treasure fleets, warships
Fluyt (Dutch)	from 1595 (first built at Hoorn)	medium cargo ship	about 12, where other ships of the same size needed about 30	Cheap bulk cargo (grain, timber)

Ship “tons” were measured in different ways (cargo space or the ship's own weight), so treat all these sizes as rough.

Why it matters on the map: big ships need deep water. Trade gathers in fewer, deeper harbours. Old river ports far

inland lose out, and **outports** (small harbours near the sea) grow downstream. See [Where Settlements Are Built](#).

Ocean routes and how long they took

- **Portugal to India (the *Carreira da Índia*):** about 6 months each way. Ships left Lisbon in February–April and came home in June–August of the following year, so a round trip, including the wait in India for the right wind, took about 14–18 months. There were 1,033 ship departures

from Lisbon between 1497 and 1650, about 7 a year. Many never came back: of 806 ships sent out between 1497 and 1612, only 425 returned to Portugal (many others stayed in Asian waters, and 66 were lost at sea).

- **Spain to the Americas:** from the mid-1560s (the convoy system was set up in 1564–66) ships sailed in armed con-

voys. Usually two fleets a year left Seville. A fleet had about 17 ships in 1550 and more than 50, much larger ones by 1600. The fleets met at Havana for the voyage home.

- **Across the Pacific:** the Manila galleon linked Acapulco (Mexico) and Manila (Philippines) from 1565.

Monopoly ports

A state could force all ocean trade through **one port**, to collect taxes and control it. That port then grows very fast.

- **Seville:** in 1503 the Crown gave it the monopoly of trade with the Americas and set up the **Casa de Contratación**

Chartered companies and trading posts (“factories”)

A **chartered company** is a merchant company given a monopoly of trade with one region by the state. It could build forts, keep soldiers and make treaties.

Ocean routes follow the wind belts, not straight lines. They touch land at a few famous stopping places: islands, capes and harbours for water, food and repairs.

(House of Trade) there. By the late 1500s Seville had over 100,000 people and was the largest city in Spain.

- **Lisbon:** the Crown's India House controlled the spice trade.

A **factory** in this period is not a workshop. It is a trading post run by a *factor* (agent): a walled compound with warehouses, offices and homes, on land the local ruler allows. Some grew into forts and then into colonial cities.

Power or company	Founded	Examples of posts (date)
Portuguese Crown (State of India from 1505)	—	Elmina, West Africa (1482); Goa (1510); Malacca (1511); Hormuz (1515)
English East India Company	1600	Surat, India (permanent factory 1613)
Dutch East India Company (VOC)	1602	Batavia, now Jakarta (1619), its Asian capital
Dutch West India Company	1621	Elmina (taken from Portugal 1637)

Scale: between 1602 and 1796 the VOC sent 4,785 ships to Asia. The English company, its nearest rival, sent 2,690.

Merchants from abroad also have fixed quarters at home, for example the German merchants' hall in Venice, the *Fondaco dei Tedeschi* (rebuilt 1505–08).

RULE OF THUMB

A trading post sits beside an existing local port city, not in empty land. It prefers a spot that is easy to defend from the sea: an island, a peninsula or a river mouth. It stays small (often tens to a few hundred Europeans; an estimate) unless it becomes a colonial capital.

MAP TIP

On a continent map, mark ocean routes as long curving dotted lines (the sea-lane symbol), with arrows where the wind sets the direction, a sailing season (“out Feb–Apr”) and two or three named ports of call. On the home coast, give the monopoly port the head-port symbol (a large anchor with a quay, name in capitals); on a local map add docks, a customs house and a “House of Trade”. Abroad, mark trading posts with a small fort symbol next to a local port city, labelled with the company name. Do not use a flag: a small flag (pennant) marks a duchy seat.

Planted colonial towns: the Spanish grid

Where a European power ruled the land itself, not just a trading post, it founded new towns. Spain founded the most in this period, and it planned them all to one pattern.

RULE OF THUMB

A planted colonial town is a grid of square blocks around one central **plaza** (open square). The main church and the government houses stand on or beside the plaza. Every such town in a colony looks alike.

Why: A grid is quick to lay out with a cord and stakes. It makes it easy to give each settler a plot, and the town can grow outward in any direction. One central square gives

the church and the royal officials a place to be seen, and space for markets and festivals.

The rules: Philip II's ordinances of 1573 on discovery and new settlement (148 articles, later part of the "Laws of the Indies") wrote down a pattern that had already been used for decades. Lima, for example, was founded in 1535 on a grid of $13 \times 9 = 117$ square blocks around its main square.

Where the pattern came from: Spain already had planned towns at home. The forerunner most often named is Santa Fe near Granada. In 1491, while besieging Granada, Ferdin-

and Isabella turned their camp into a small walled town on a grid, with a square in the middle and a gate in each of the four sides (see [Camps that became towns](#)). Historians of town planning have often called it the direct model for the American towns. Others say it was too small and too military, and point instead to older planned grid towns of the medieval Crown of Aragon (Bielza de Ory 2002). Either way, the idea of a planted grid town around a central square came from medieval Spain.

Rule (1573 ordinances)	What it says
Where the plaza goes	At the landing place in a port town; in the centre of an inland town
Plaza shape	A rectangle, its length at least 1.5 times its width ("best for fiestas in which horses are used")
Plaza size	Matched to the number of people expected. Not smaller than 200×300 feet; not larger than about 530×800 feet; 400×600 feet is "a good proportion". These are Castilian feet of about 28 cm (11 in), so roughly 56×84 m, 148×223 m and 111×167 m
Streets	Twelve streets leave the plaza: one from the middle of each side and two from each corner
Church	On a whole block of its own, near the plaza
Government	The royal council and town council house, the customs house and the arsenal, near the church

Real towns did not always follow every rule, but the plaza and the grid are almost everywhere. In Lima the blocks were about 110–125 m (360–410 ft) on a side (sources differ),

and the cathedral and the viceroy's palace stand on the main square.

MAP TIP

For a colony, draw small identical grid towns: a square of a few blocks around a plaza, with a church and a governor's or council house on the plaza. In a port town, put the plaza by the landing place. Draw the colonial capital to the same pattern, just with many more blocks. The regular grids should

stand out against the irregular older towns of the local people. The grid is a local-map drawing: at kingdom scale, use the ordinary town symbols. For medieval planned towns with grids (*bastides*), see [Planted "new towns" and bastides](#).

Posts, roads and canals

The post: relay stations

A **relay post** is a chain of stations where riders change horses (and sometimes riders) so that letters never stop.

- **Taxis post:** from about 1490 the Taxis family ran a line for Maximilian of Habsburg between Innsbruck and Mechelen in the Low Countries. Its centre moved to Brussels around 1500–01. The contract of 1505 added stations in Spain and France (Granada, Toledo, Blois, Paris, Lyon) and fixed delivery times for summer and winter.
- **Spacing:** stations every 20–40 km (12–25 mi).

- **Speed:** under the 1505 contract, Brussels to Innsbruck took $5\frac{1}{2}$ days in summer and $6\frac{1}{2}$ in winter, and Brussels to Paris 44 hours in summer (54 in winter). Relay posts covered up to about 150 km (95 mi) a day.
- **Public use:** the contract of 1516 let the Taxis carry private letters for profit, as long as court mail arrived on time. The emperor confirmed the Taxis monopoly in 1595, and from 1615 it was hereditary.
- **England:** there was a Master of the Posts from 1516, and post stages were about 32 km (20 mi) apart. Charles I opened the royal post to the public in 1635. The General Post Office was founded in 1660.

MAP TIP

Along a post road, draw a small horn symbol (a post station) every 20–40 km (12–25 mi). Each station is an inn with large stables. Small towns on post roads grow; towns off them fall behind.

Roads

Roads improve only a little. Some states make local communities repair them: England's Highways Act of 1555 made each parish give unpaid labour on its roads. Coaches for the rich become common in the later 1500s, but heavy

goods still go by water wherever possible. The new thing is **planned military routes** such as the Spanish Road. Travel speeds stay as in the baseline: see [Trade Routes, Roads and Transport](#).

Canals with pound locks

A **pound lock** is a chamber with a gate at each end. A boat enters, the gates close, and the water level rises or falls gently. This lets a canal climb hills. Most medieval canals were flat, or used dangerous “flash locks” (a single gate opened to let a rush of water carry boats through). Pound

locks were not completely new: Damme near Bruges had one by 1396, and one of Milan's canals (the Naviglio di Bereguardo) got 18 of them in 1452–58. What is new after 1500 is that they spread, and that whole canals start to climb over hills.

Canal	Dates	Size	Why it matters
Exeter Canal, England	1564–66	short: a few km (accounts differ; Wikipedia gives 5.2 km / 3¼ mi), 3 pound locks	First pound locks in Britain; brought small ships back to a city cut off by weirs
Briare Canal, France	1604–42 (work stopped after Henry IV's murder in 1610 and restarted in 1638)	57 km (35 mi), 36 locks, including a “staircase” (locks joined directly one after another) of 7 locks at Rogny	First summit-level canal in Europe with pound locks; joined the Loire and Seine river systems; 6,000–12,000 labourers
Dutch <i>trekvaarten</i>	1630s–1660s (the first, Amsterdam–Haarlem, opened in 1632)	a network linking the cities of Holland	Straight canals for horse-towed passenger barges; about 7 km/h (4.5 mph), about 30 passengers, on a timetable

A **summit-level canal** climbs over a watershed (the high ground between two river systems). It needs reservoirs on the summit to feed water to the locks.

MAP TIP

Draw canals as straight or gently bending blue lines, unlike rivers. Mark each lock with two short bars close together across the line (a single bar with a “T” means a toll), grouped where the land rises. A summit canal needs a reservoir lake near its highest point. Towns grow at the canal's junctions with rivers.

Money, printing and news

Exchanges and banking centres

An **exchange** (French *bourse*) is a building where merchants meet daily to trade goods, loans and bills of exchange (written orders to pay a sum of money in another city, used instead of carrying coin).

Date	Place	What
1531	Antwerp	First purpose-built exchange building
1565–71	London	Royal Exchange, copied from Antwerp's; founded by Thomas Gresham, opened by Elizabeth I in 1571
1609	Amsterdam	Exchange Bank (<i>Wisselbank</i>), a city-backed public bank
1611	Amsterdam	Exchange building

Great banking families still matter: the Fuggers of Augsburg lent to the Habsburgs, and Genoese bankers financed the Spanish crown.

RULE OF THUMB

A continent has only one or two money centres at a time. They sit in the leading trading port of the day (Antwerp, then Amsterdam; Genoa; London), which is not always the capital.

Printing towns

- By 1500, 236 towns in Europe had a printing press (a figure given by Braudel).
- Before 1650, printing took place in more than 1,300 places in Europe (Universal Short Title Catalogue data).
- Output grew roughly tenfold in the 1500s, to perhaps 150–200 million copies.

• **Big centres:** Venice (the largest c. 1500), Paris, Lyon, Antwerp, Basel, and the book-fair towns of Frankfurt and later Leipzig.

• **Ottoman lands:** printing in Arabic script was forbidden or discouraged until 1729. Jewish, Greek and Armenian presses did exist there.

A printing town needs paper (water-powered paper mills on nearby streams), customers (a university, a court, a bishop, lawyers) and good routes to send books out.

MAP TIP

Label one or two cities “book fair” or “printing centre” in your realm, usually a university or trade city. Put paper mills on fast streams within a day or two of it. Printed news, maps and pamphlets now travel along the post roads.

Capitals, states and borders

Fixed capitals and palace complexes**RULE OF THUMB**

By 1600 most big states have one fixed capital, with a ring of royal palaces, hunting parks and monasteries within a day's ride.

- **Madrid:** Philip II made it the fixed capital in 1561. It was small before the court arrived: about 18,000 people in 1561. By 1598 it had about 80,000, and estimates for about 1600 range from 75,000 to 95,000.

• **El Escorial** (1563–84), about 45 km (28 mi) from Madrid: a huge palace, monastery, royal burial church and library in one building.

• **Other examples:** the Polish court moved from Kraków to Warsaw in 1596; German princes settled in fixed **residence towns** (*Residenzstädte*). For more, see [Capitals, Realms and Borders](#).

States also become **more centralized**: more paid officials, permanent ambassadors, standing garrisons, and governors in provincial capitals. On the map this means a governor's palace and a citadel in each provincial capital, and the capital itself as a city of offices and noble houses.

Surveying and mapping

Date	Development
1533	Gemma Frisius explains triangulation : measure one baseline carefully, then use angles to fix far-away points
1569	Mercator's world map with his new projection for sailors
1570	Ortelius publishes the first modern atlas (53 maps) in Antwerp
1579	Christopher Saxton's atlas of the counties of England and Wales (35 maps, surveyed in 1574–78)
1615	Snellius measures a long chain of triangles across Holland

Borders becoming lines, slowly

Maps now show borders as lines, and rulers start to think of their realms as areas with edges. But on the ground, borders were still often defined as **lists of places**. The Treaty of the Pyrenees (1659) made the mountains the border

between France and Spain and gave France the “villages” of northern Cerdanya. Llívia was legally a town, not a village, so it stayed Spanish, and it is still a Spanish island inside France today. The exact line was only marked on the ground by the Treaties of Bayonne, 1856–68.

MAP TIP

For a 1600-style map you can draw firmer, unbroken borders for big kingdoms. Keep the dash-dot border line; a solid line means a road. Keep enclaves (the owner’s colour with a thin outline), disputed strips (two parallel dash-dot lines) and

patchwork areas, especially in a “Holy Roman Empire” type region. Put pairs of fortresses facing each other across important borders, and mark a few contested towns. See [Capitals, Realms and Borders](#) for medieval frontier zones.

The countryside: enclosure, new crops and drainage

Enclosure

Enclosure means turning open fields and common land (shared by the whole village) into private fields surrounded by hedges, walls or ditches. Often it also meant turning ploughland into sheep pasture.

- J. R. Wordie estimated that about 45% of England was already enclosed by 1500, 47% by 1600 and 71% by 1700. So the famous Tudor enclosures covered only a small area; the big new wave came in the 1600s (and continued in the 1700s).

- In the 1500s it was most visible, and most hated, in the Midlands, where landlords turned whole villages into sheep runs. Parliament passed laws against it (“tillage acts”) between 1489 and 1597. Enclosure was a cause of Kett’s Rebellion (1549) and the Midland Revolt (1607).
- In eastern Europe (Poland, Prussia, the Baltic lands) a different change happened: lords built large grain estates worked by serfs, and shipped grain down the rivers to Baltic ports for export.

New crops

Crop	Arrives	Becomes a common field crop
Maize	Spain within a few decades of 1492	A common peasant food in southern Europe by the 1600s
Potato	Canary Islands 1567; Spain about 1570; Britain and Ireland 1588–93	Ireland on a wide scale in the early 1600s; most of Europe only in the 1700s–1800s
Tomato	Spain soon after 1492; eaten in Italy by 1544	Garden crop in this period

RULE OF THUMB

Before 1650, American crops are mostly garden plants. The big exception is maize in Iberia and Italy. Do not cover your 1600 map with potato fields.

Drainage and new land

- **Dutch polders:** a **polder** is low land surrounded by dikes and kept dry by pumping. The Beemster lake in North Holland was drained between 1607 and 1612 with 43 windmills, creating about 7,200 ha (72 km² or 28 sq mi) of new farmland. The Purmer (1622), Wormer (1626) and Schermer (1635, about 48 km² or 18 sq mi) followed.

- **English Fens:** drainage began in earnest in the 1630s, led by the Earl of Bedford and his investors. A second phase in the 1650s, with the New Bedford River (completed 1652), finished the main work. The Great Level covered about 1,300 km² (500 sq mi). Local people, who lost their fishing and fowling, fought it. The drained peat shrank, so by the late 1600s parts were flooding again.

MAP TIP

Draw a polder as a perfectly regular grid of straight roads and ditches inside a ring dike, with windmills (an X of four sails on a dot) along the edge and farms evenly spaced along the roads. It should look completely different from the irregular old villages around it.

Estates and country houses

Rich families build **country houses** with parks, avenues, walled gardens and deer parks, often on former monastery land (see [Religion](#)). England's "prodigy houses" (1570–1620)

were built to host the monarch on a royal tour. Big parks can swallow whole villages and old roads. For village layouts see [Villages, Farms and the Countryside](#).

MAP TIP

Give each county-sized area one to three country houses. Draw each as a house inside a ring of park trees, an oval park 1–3 km (0.6–2 mi) across (an estimate), with a straight avenue to the nearest road. Put some on old abbey sites and

some beside a ruined castle. Show enclosed fields as small irregular polygons (hedged fields) in the sheep country, and keep open fields (unhedged areas with blocks of hatching) elsewhere.

Industry: furnaces, mines and mills

Blast furnaces

A **blast furnace** is a tall stone furnace with bellows driven by a waterwheel. It makes liquid cast iron, which can be poured into moulds, for example to make cannon. It needs iron ore, a lot of charcoal (so a lot of managed woodland) and a dammed stream.

- England's first blast furnace was at Queenstock, Buxted (about 1490–91), followed by Newbridge in Ashdown Forest (1496). Both are in the Weald of Sussex.

- They stayed few until about 1530. Then many were built: by the later 1500s the Weald had about 50 furnaces, and as many as 100 furnaces and forges together. The Weald industry peaked around 1590. The first British furnaces outside the Weald appeared in the 1550s, and British output as a whole peaked around 1620.
- England cast its first iron cannon in 1543, at Buxted.
- The method came from the Namur region (today in Belgium), through Normandy, to the Weald.

MAP TIP

Put iron works in wooded valleys with dammed streams: a chain of **hammer ponds** (mill ponds for furnaces and forges) down one valley, with charcoal burners' clearings in

the woods around them. Label them "Furnace Pond" or "Hammer Pond", so they are not read as fishponds (small rectangles in a chain). Iron districts are wooded, not bare.

Mining booms

Place	Start	Peak size	What happened
Joachimsthal (Jáchymov), Bohemia	Silver town founded 1516	about 20,000 by 1534, the second town of Bohemia after Prague	Its coin, the <i>Joachimsthaler</i> , gave us the word "thaler" (and later "dollar"). Declined because of war (the Schmalkaldic War, 1546–47), the Counter-Reformation driving Protestant miners away, deep mines that flooded for lack of pumps, and cheap American silver
Potosí (in today's Bolivia)	1545	about 130,000 by 1603 (at least 120,000 of them Andean); higher figures of 160,000–200,000 are claimed but disputed	At 4,067 m (13,343 ft); produced about 60% of the world's silver in the late 1500s; worked by forced Indigenous labour (the <i>mita</i> , about 13,000 men a year)

RULE OF THUMB

A rich strike can turn an empty valley into one of a kingdom's biggest towns within 10–20 years, and then fade almost as fast. Draw boom towns big, crowded and unwalled, in rough terrain, with a mint, mine symbols (crossed hammers) around them, and roads carrying food in and silver out.

Mills

- **Paper mills** spread with printing. They need fast, clean water and linen rags. England got its first paper mill in the 1490s (John Tate's short-lived mill near Hertford, about 1495).

- **Gunpowder mills** grind saltpetre, sulphur and charcoal. They explode, so they sit away from towns, often as separate buildings spread along a stream. Example: Chilworth, Surrey, set up by the East India Company in the mid-1620s on the Tillingbourne stream.

- **Wind sawmills:** Cornelis Corneliszoon of Uitgeest patented his crankshaft for a wind-powered sawmill in 1593–94. It cut timber into planks about 30 times faster than sawing by hand. The Zaan district north of Amsterdam

Shipyards and arsenals

State shipyards become huge. The **Venetian Arsenal** covered about 45 ha (110 acres), around 15% of Venice. Up to 16,000 workers is often quoted for its peak in the early 1500s, but normal numbers were much lower. Henry VIII founded royal dockyards at Woolwich and Deptford near London. Dutch yards on the Zaan and at Amsterdam built ships in series.

dam filled with industrial windmills and is often called the first industrial area in the world.

MAP TIP

Put shipyards at a sheltered deep-water port with timber arriving by river or sea. Draw them as a walled yard with slipways (sloping ramps where ships are built), next to a rope-walk (a very long, narrow shed for making rope). At kingdom scale, mark a state arsenal or royal dockyard with an anchor plus a castle tower. Put gunpowder mills a few kilometres outside a town on a stream, and paper mills on clean streams upriver of a printing town; draw both as watermill wheels on the stream, labelled “Powder Mills” or “Paper Mill”. A wind-mill industrial district looks like a forest of windmill symbols (an X of four sails on a dot) along a waterway.

For medieval mines, mills and industrial towns, see [Industry, Resources and Special Towns](#).

Religion: the Reformation on the map

England and Wales: the dissolution of the monasteries (1536–1541)

- About 900 religious houses: about 260 for monks, 300 for canons, 142 nunneries and 183 friaries.
- About 12,000 religious people: 4,000 monks, 3,000 canons, 3,000 friars and 2,000 nuns.

- Monasteries held about a quarter of England’s landed wealth.
- The small houses went first (243 were closed in 1536). The large ones “surrendered” by 1540.

What happened to the buildings:

Fate	Example	Map symbol
Stripped of lead roofs and left as a ruin	Fountains and Rievaulx (Yorkshire); Tintern (Wales)	Abbey symbol (church with a square cloister) drawn broken or grey, labelled “(ruin)”
Turned into a country house	Lacock Abbey	House inside a ring of park trees, name ending in “Abbey” or “Priory”
Abbey church bought by the town as its parish church	Tewkesbury	Large parish-church symbol in a small town
Made a cathedral of a new diocese	Gloucester, Peterborough, Oxford, Bristol, Chester (1540–42)	Cathedral (church with two towers, or a mitre)
Used as a quarry for local building stone	Very common	A small grey fragment of the abbey symbol, or just the name

Great shrines were destroyed (Walsingham and Becket’s shrine at Canterbury, both 1538), and pilgrim routes emptied.

Elsewhere in Europe

- **Protestant lands:** Scotland's Reformation came in 1560. Protestant German princes took over church lands, and in 1525 the Teutonic Knights' state became the secular Duchy of Prussia. Former abbeys became schools, hospitals, offices or princely residences.
- **Religious borders:** the Peace of Augsburg (1555) let each German ruler choose Lutheranism or Catholicism for his land. This created a patchwork, with churches of one faith just across the border from the other. In France the

Edict of Nantes (1598) gave the Protestant Huguenots fortified "security towns" such as La Rochelle, which fell to the king after a siege in 1627–28. In the Dutch Republic, Catholics could worship only in hidden "house churches".

- **Catholic lands:** monasteries stay and even multiply. The **Counter-Reformation** (the Catholic reform movement) adds new building: Jesuit colleges (the order was founded in 1540), new churches and new pilgrimage shrines.

MAP TIP

Make your religious map as clear as the political one. Where each ruler chose his land's faith (as in Germany after 1555), the faith border follows a political border, so the border line is already there. Give each faith a soft colour tint, not a new line, so it is never mistaken for a realm border. On the Protestant side, use ruined abbeys, country houses called

"Abbey", and big town churches. On the Catholic side, keep the working monasteries and add a Jesuit college in the main towns. Put a few pilgrim shrines (scallop shell or reliquary symbols) just inside the Catholic side of the border. For medieval religious sites, see [Religious, Cultural, Legal and Ancient Sites](#).

Comparison table: medieval vs 1500 to 1650

Map feature	Medieval baseline (c. 1000–1500)	1500–1650
Biggest city in the West	Paris ~200,000	Paris, Naples, London 200,000–400,000+; Istanbul 400,000–700,000
Capital	Often a travelling court; in strong kingdoms the capital is 3–7× the second city, while fragmented realms have rival cities of similar size	Fixed capital, 5–10× the second city in some realms, with palaces nearby
Town walls	High and thin, round or square towers	Low, thick, earth-backed, bastioned (in rich or threatened towns only)
Castles	Thousands; main military and lordly centre	Ruins, palaces, prisons or manor houses; replaced by forts and country houses
City control	Castle at the town's edge	Bastioned citadel at the town's edge
Coastal defence	Walled ports, a few towers	Lines of watchtowers; artillery forts at anchorages
Field armies	Usually 5,000–20,000	Total forces of 50,000–150,000+ for great states (paper strength)
Ships	Cogs of 30–200 tons; merchant galleys carrying about 140–250 tons	Carracks and galleons of several hundred to 2,000 tons; Dutch fluyts
Long-distance trade	Mediterranean, Baltic, overland caravans	Plus ocean routes to Asia, Africa and the Americas
Merchants abroad	Foreign quarters and halls	Plus chartered companies with fortified trading posts
Messages	Royal messengers 50–90 km (30–56 mi) a day	Relay post up to ~150 km (95 mi) a day; public letters
Canals	Short, flat, flash locks	Pound locks; summit canals; passenger barge canals
Finance	Fairs, Italian banking houses	Exchanges and public banks in the leading port
Books	Hand-copied; monastery and university centres	Printed in 1,300+ places; book-fair towns
Borders	Zones and marches	Lines on maps; still lists of villages on the ground
Maps	Rare, schematic	Surveyed county maps, atlases, triangulation
Countryside	Open fields, commons, deserted villages	More enclosure (47% of England by 1600, 71% by 1700); serf grain estates in the east

Map feature	Medieval baseline (c. 1000–1500)	1500–1650
Wetlands	Mostly left wet	Polders and drained fens
Country seats	Castles and manor houses	Country houses with parks
Monasteries	Everywhere	Ruined or converted in Protestant lands; flourishing in Catholic lands
Iron	Bloomeries (small hearths)	Blast furnaces, cast-iron cannon
Mining	Steady mining towns	Sudden boom towns (Joachimsthal, Potosí)

MAP TIP

Use this table as a checklist. Go down the left column, find each feature on your medieval map, and decide whether it stays, changes or is joined by the later version.

Drawing a world in transition

RULE OF THUMB

A real 1600 landscape has three layers of age. Old things (castles, abbeys, medieval walls) are ruined, converted or patched. Middle-aged things (villages, fields, market towns) carry on unchanged. New things (star forts, canals, polders, country houses, post roads) are geometric and stand out.

A checklist

1. **Start with your medieval map.** Keep villages, market towns and roads.
2. **Grow the top.** Enlarge the capital and one or two ocean ports. Shrink one old port whose river has silted up or been blockaded.
3. **Change the castles.** Interior castles become ruins or palaces. Draw a few country houses with parks nearby. In a strongly centralised realm, keep only about one working castle per province.
4. **Add new fortifications only where they pay:** a citadel in the capital or a rebellious city, star-fort towns on the dangerous frontier, gun forts at harbour mouths, towers along raided coasts.
5. **Mix old and new walls.** A frontier town can have a medieval wall on the safe side and new bastions on the side facing the enemy. This half-finished look is very realistic.
6. **Add the new links:** post stations every 20–40 km (12–25 mi) on the main roads, a straight canal or two, and ocean routes from your main port.
7. **Change the landscape where money went:** a geometric polder in a former lake or marsh, enclosed fields in the sheep country, a mining boom town in the hills, iron furnaces in wooded valleys.
8. **Draw the religious map:** a confessional border (a border between two faiths, shown with a colour tint for each faith), abbey ruins on one side, working monasteries on the other.
9. **Add an overseas layer if you have a world map:** trading posts beside foreign port cities, labelled with company names.

Symbols and labels

These symbols match the guide's [master legend](#).

Feature	Suggested symbol	Label example
Star fort or fortress town	Five- or six-pointed star outline , or a star-edged town outline; never filled (a filled star means a capital)	"Fort St John", "Neufort"
Bastioned citadel	Pentagon outline joined to the town	"Citadel"
Ruined castle	Castle icon drawn broken or grey	"Old Castle (ruin)"
Royal palace or hunting lodge	Small crown on its own	"Hartwood (royal lodge)"
Country house	House inside a ring of park trees	"Elmswood House", "Lacey Abbey"
Ruined abbey	Abbey symbol (church with a square cloister) drawn broken or grey	"Abbey (ruin)"
Post station	Small horn on the road	"Post"
Canal with locks	Straight blue line; each lock as two short bars close together across it	"Royal Canal"
Polder	Grid inside a dike, with mill symbols	"New Polder (drained 1612)"
Monopoly port	Large anchor with a quay; name in capitals	"SALTHAVEN"
Ocean route	Dotted line at sea, with arrows where the wind sets the direction	"To the Spice Isles (out Feb–Apr)"
Trading post	Small fort beside a foreign port city	"Company factory"
Coastal tower	Small triangles along the coast	none needed
Arsenal or royal dockyard	Anchor plus a castle tower	"Royal Dockyard"

MAP TIP

Date new things on your map or in its notes: "New Polder (drained 1612)", "Citadel (1567)". Dates make the history visible and help you keep the layers consistent.

FANTASY TWIST

In many fantasy worlds, magic plays the role of gunpowder: a new power that makes old defences useless. Use the real pattern: the rich centre adopts it first, old walls get patched, special forts appear on the frontier, and the remote countryside barely changes for generations. See [Fantasy Variants](#) and [Common Mistakes and How to Fix Them](#).

Soon after 1650: Several famous changes come just after this chapter's period. Vauban built France's great fortress belt (the *pré carré*) from the 1670s. The Canal du Midi (1666–81) linked the Atlantic and the Mediterranean. Versailles became the model palace capital from the 1680s. The potato and turnpike (toll) roads changed the countryside mostly in the 1700s.

Quick summary

- After 1500 the countryside stays medieval; the change is at the top: capitals, ports, forts and long-distance links.
- Europe grows from about 62 to 78 million (without Russia and the Ottoman lands) between 1500 and 1600, then stalls. Capitals and ocean ports grow fastest: London went from about 50,000 to about 400,000 between 1500 and 1650.
- City ranks change quickly: Antwerp rose to about 100,000 and then collapsed; Amsterdam, London, Seville and Lisbon boomed.
- Cannon made high walls weak. Bastioned “star” fortifications appeared in Italy from the late 1400s, matured there c. 1500–1540, and spread north in the 1530s–40s. They were very expensive, so only frontiers, capitals, ports and key crossings get them.
- Citadels control cities; chains of towers and gun forts guard coasts; armies grow to over 150,000 for great states, and sieges can last years.
- Old castles become ruins, palaces, prisons or are replaced by country houses. Strong rulers order most castles destroyed: France in 1626; Japan in 1615, from about 3,000 castles to about 170.
- Ocean ships of several hundred to 2,000 tons link monopoly ports (Seville, Lisbon) with trading posts abroad; chartered companies (1600, 1602) build fortified “factories”. Spain plants colonial towns on one standard grid around a central plaza (rules of 1573).
- Relay posts with stations every 20–40 km (12–25 mi) carry letters up to about 150 km (95 mi) a day; pound-lock canals start to climb hills (Briare, 1604–42).
- Exchanges (Antwerp 1531, London 1571, Amsterdam 1609–11) and printing (236 towns by 1500, 1,300+ places by 1650) create new kinds of special towns.
- Borders become lines on maps but are still lists of villages on the ground; surveying and atlases arrive.
- The countryside sees more enclosure, a few new crops (mainly maize), drained polders and fens, blast furnaces, and mining boom towns.
- The Reformation leaves ruined and converted abbeys and clear religious borders.

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spacing, trading-post sizes) are rough guides derived for this chapter, not measured averages.

CHAPTER 10

FANTASY VARIANTS: MAGIC, MONSTERS AND OTHER RACES

The other chapters describe a world without magic. This chapter shows how to add magic, monsters and non-human peoples without breaking that realism. The method is simple: apply the real rules first, then ask which **input** a fantasy element changes (water, food, defence, travel cost, threats or resources) and follow the consequences. Every adaptation here is tied to a real historical case that worked the same way.

IN THIS CHAPTER

The method: change the inputs, not the rules
 Dangerous wilderness and monsters
 Magic, effect by effect
 Non-human peoples: archetypes with real analogues
 Contact zones between peoples
 Gods, undead and dangerous seas
 What changes if: summary table
 Suggested symbols
 Consistency checklist
 Quick summary
 Sources and further reading

The method: change the inputs, not the rules

RULE OF THUMB

Fantasy does not cancel the rules of geography. It changes the numbers you feed into them. Change one input, then follow the chain of effects across the whole map.

Every settlement rule in this guide comes from six inputs. Each fantasy element changes one or more of them.

Input	The realistic rule (see chapter)	Fantasy elements that change it
Water	Every settlement needs reliable water. Desert towns exist only at oases, rivers or <i>qanats</i> (underground water channels). See Where Settlements Are Built .	Water magic, gods, cursed or poisoned rivers
Food	About 1 ha (2.5 acres) of arable land per person. Good lowland holds 30–50 people per km ² (80–130 per sq mi). A town of 10,000 needs farmland within 21–27 km (13–17 mi). See Population and Settlement Sizes .	Fertility magic, weather control, healing (more mouths)
Defence	Towns of 2,000+ are usually walled. A castle controls about 15–30 km (10–19 mi) around it. See Castles, Forts and Military Outposts .	Monsters, flying enemies, magical wards, undead
Travel cost	Land : river : sea cost $\approx$ 8 : 4 : 1. Towns over 10,000 sit on navigable water. See Trade Routes, Roads and Transport .	Portals, flying mounts, dangerous seas, guarded roads
Threats	War, raids, famine, plague	Monsters, dragons, raiding peoples, blights, angry gods
Resources	Ore, salt, stone and timber create special towns. See Industry, Resources and Special Towns .	Magic crystals, ley lines, monster parts, dragon hoards

Why this works: a reader cannot check your magic, but they can feel when a map is wrong. A city in a desert with no water looks wrong. A desert city with a great water-

temple at its centre and a ring of gardens around it looks right, because the cause is visible.

MAP TIP

For each fantasy element, write one line: “X exists, so Y changes.” For example: “Griffon riders exist, so castles have roofed towers and every lord keeps an aerie.” If you cannot write the second half, the element is only decoration. That is allowed, but it will not make the map feel real.

Dangerous wilderness and monsters

RULE OF THUMB

Danger concentrates people. The more dangerous the countryside, the fewer, bigger and better-defended the settlements, and the wider the empty land between them.

Why: a lone farm cannot defend itself against a troll or a raiding party. A wall is cheaper per person when many people share it. Fields must be close enough to a refuge that workers can run back in time. All three push people together.

How danger reshapes settlement

The real world already gives us every level of danger. Pick the row that fits each region of your map.

Threat level	Real analogue	Settlement pattern	Defences	How it looks on the map
Low (wolves, bandits)	England or France c. 1250, away from borders	Normal mix: nucleated (clustered) villages on good land, lone farms in uplands	Working castles about 20–30 km (12–19 mi) apart (see chapter 04); villages unwall	The baseline map in the other chapters
Seasonal raids (a few times a decade)	Anglo-Scottish border, 14th–16th centuries	Lone farms survive but become tower houses; villages huddle	Tower houses and fortified farmhouses. Late medieval England had over 250 tower houses, Scotland ~800, Ireland over 3,000	Small solid squares (tower houses) scattered over farmland
Frequent raids or large monsters	Latium (central Italy) c. 920–1030; Wessex in the 870s–890s; the Provence coast in the 900s	Farms abandoned; people move into walled or hilltop villages of 300–1,000	Walled villages; refuge forts within a day’s walk; watchtowers; beacons	Fewer, bigger dots on hilltops; empty valley floors; coasts deserted
Constant danger	The “Great Wilderness” between the Teutonic Order and Lithuania (13th–14th centuries); Crusader frontier castles	Only fortified towns and garrison castles; fields only near walls	Great garrisons. The Templar castle of Safed (1260s) held 1,700 people in peacetime and 2,200 in war	Islands of settlement in empty land

How far can fields be from a refuge? There is no measured figure. A useful heuristic is **about one hour’s walk: 4–5 km**

(2.5–3 mi). That gives workers time to reach the walls when the alarm sounds.

Nucleated and walled villages

The best real case is Italian *incastellamento* (the “castle-ing” of the countryside). In Latium (the region around Rome), scattered farms were regrouped into fortified hilltop villages. Pierre Toubert (1973) put the main wave between about 920 and 1030; other historians see it continuing into the 12th century (see [Where Settlements Are Built](#)). Historians still argue how much of this was caused by raids and how much by lords wanting control. For your map, the result is what matters: hilltop villages everywhere, and few houses in the open.

Where raids are seasonal, lone farms do not disappear. They turn into towers. On the Anglo-Scottish border, families built small stone tower houses and fortified farmhouses so they could shut themselves and their animals in for a few days.

The most extreme refuge is underground. In Cappadocia (central Anatolia), the underground city of Derinkuyu reaches about 85 m (280 ft) deep. It was greatly enlarged during the Arab–Byzantine wars (roughly 780–1180) as a refuge from Arab raids. It may have sheltered up to about 20,000 people with their animals and food, although that is an estimate, not a count.

Refuge networks and frontier forts

When King Alfred of Wessex faced Viking armies in the 870s–890s, he built a network of *burhs* (fortified towns). The Burghal Hidage is a list of these burhs and the land assigned to man them. It is usually dated to the early 900s, although some historians argue that it describes Alfred's own plan of about 880. Its two versions together list 33 burhs:

- Each hide (a unit of taxed land) supplied one man.
- Four men defended each pole (about 5 m or 16½ ft) of wall.
- Winchester was assessed at 2,400 hides, so 600 poles, or about 3 km (1.9 mi) of wall. This closely matches the length of its old Roman walls.
- The burhs were spread so that few country people lived more than about 24–32 km (15–20 mi), roughly one day's walk, from one.

Watchtowers and beacons

A warning network buys time. Two real systems show the range of options:

System	Date	Length and spacing	What it did
Byzantine beacon chain	9th century	About 720 km (450 mi), about nine mountain-top stations 55–100 km (35–60 mi) apart (details in Watchtowers, beacons and coastal warning)	Sent one of 12 pre-arranged messages, each tied to an hour of the day. Modern experiments suggest a message crossed the whole line in about an hour
Hungarian <i>gyepű</i>	11th–13th centuries	A belt around the whole kingdom	An outer border zone deliberately kept empty, blocked with felled trees, hedges, ditches and banks of earth and stone. Gates were the only ways through. Guard settlements of border troops (often archers) kept watch. It failed against the Mongols in 1241, after which Hungary built stone castles

How close should local watchtowers be? They must see each other's smoke or fire. In broken or wooded country that

This is the best real model for a monster-haunted kingdom: **a planned net of refuges about a day's walk apart, each manned by the farms around it.**

Denmark shows a different version. In about 970–980, King Harald Bluetooth built five large circular “ring fortresses” (Trelleborg, Fyrkat, Aggersborg, Nonnebakken and Borgring) spread across his kingdom. They were royal strongholds that showed the king's power, more than refuges for farmers. A fantasy king can do the same: a few identical, planned forts placed across the realm at once.

Rivers can carry the threat inland. Against Viking ships, the Frankish king Charles the Bald ordered fortified bridges at towns on rivers in the Edict of Pîtres (864). The best known was at Pont-de-l'Arche on the Seine. If your monsters swim or sail, expect fortified bridges and river barriers below important towns.

means a few kilometres apart. Mountain-top beacons can be 55–100 km (35–60 mi) apart. These are rough figures.

LATER ERA (1500s+)

Muscovy's Great Abatis Line (complete by 1566) and the later Belgorod Line (1635–1654; sources differ, and one gives 1646 as the end date) are real versions of this: belts of felled trees, ditches, palisades (walls of wooden stakes), watchtowers

and small forts that slowed raiders so villagers could flee and troops could gather. See [Fortified lines on open frontiers](#).

Guarded roads

Roads through dangerous country need clearing, patrols and safe stops. All three existed:

- **Clearing:** England's Statute of Winchester (1285) ordered highways between market towns to be cleared for 200 feet (about 60 m) on each side. Ditches, undergrowth and bushes had to go, so that no robber could hide close to the road. Big trees such as oaks could stay if the ground under them was kept clear. The gates of

walled towns were to be shut from sunset to sunrise, and a night watch kept.

- **Patrols:** in the 12th century, towns and peasants in northern Spain formed brotherhoods (*hermandades*) to police the pilgrim road to Santiago. In 1476 the Catholic Monarchs (Isabella and Ferdinand) set up the Santa Hermandad, a kingdom-wide police force with power over crimes on roads and in open country. The Knights Templar were founded c. 1119–1120 to protect pilgrims on the roads of the Holy Land.

- **Safe stops:** the hospice on the Great St Bernard Pass in the Alps was founded in 1049 (the traditional date) to shelter travellers. Put such stops a day's travel apart: 15–30 km (10–19 mi) on foot roads, 30–40 km (19–25 mi) on caravan routes.

The opposite also happened. Muslim raiders from al-Andalus set up a base at Fraxinetum on the Provence coast

Hunters, rangers and their lodges

Kings paid people to fight dangerous animals:

- Charlemagne's estate rules, the *Capitulare de villis* (c. 800), told stewards to hunt wolves and take wolf cubs in May. The French royal wolf-hunters (the *louveterie*) trace their origin to wolf-hunters (*luparii*) appointed by Charlemagne. French tradition dates this to 813.
- William of Malmesbury (writing c. 1125) says King Edgar (r. 959–975) took a tribute of 300 wolf skins a year from

in about 889. From about 906 until they were driven out in 972–973, they held some or all of the western Alpine passes. The chronicler Flodoard wrote in 951 that they let travellers to Rome pass only after paying tribute. **A monster that holds a pass becomes a toll-collector.** Traffic either pays, fights, or finds another pass.

the Welsh kings. He wrote over a century after the events, and the story may be exaggerated.

- Edward I of England employed Peter Corbet (1281) to destroy wolves in five western counties.
- James I of Scotland (1427) ordered three wolf hunts a year.

Wolves were rare or gone in England by about 1500. They survived in Scotland until the 1600s or later. Organised hunting with royal support worked, but it took centuries.

MAP TIP

Put hunters' or rangers' lodges (a small house mark labelled "Lodge") at forest edges, at fords and passes, and where a road enters the wild. Space them about a day's patrol apart, 15–30 km (10–19 mi), which matches a castle's control

radius. A monster-hunting order is a military order (like the Templars): give it castles on the frontier and a chapter house (its headquarters building) in a city.

"Points of light": use it for frontiers, not whole kingdoms

"Points of light" is a game-design term from *Dungeons & Dragons* 4th edition (2008): a mostly wild world with a few small, isolated islands of civilization. It is a good picture of a **frontier**, but not of a whole realm. In real history, even dangerous kingdoms had a dense core where peasants walked to market in safety.

You can estimate the size of each "point". Suppose people farm only within 4–5 km (2.5–3 mi) of their walls. That circle covers about 50–80 km². At 30–50 people per km², each point holds roughly **1,500–4,000 people**. Larger towns are possible only where a river or a protected sea route brings food from farther away.

MAP TIP

Divide your map into three zones: safe core (no shading), frontier (light hatching, the guide's symbol for a march or frontier zone) and wild (dense cross-hatching). Do not use plain colour tints for these zones, because a soft tint means a language or culture area. In the core, use the normal rules.

On the frontier, use fewer, bigger, walled settlements on hills, with watchtowers, refuge forts about a day's walk apart, and cleared roads with fortified stops. In the wild, draw only ruins, lairs and the occasional ranger lodge. A frontier band of 30–100 km (20–60 mi) is a good default.

Magic, effect by effect

Take each kind of magic separately. Ask which input it changes, how strongly, and **who controls it**. The answer to the last question tells you where the new buildings go.

Food and fertility magic

RULE OF THUMB

Fertility magic raises the ceiling on population. It does not remove the transport limit. Cities above about 50,000 still need water transport, unless magic also moves the grain.

Why: with a typical medieval yield of 4 : 1, a farmer keeps back 1 of every 4 units of grain as seed and eats or sells 3. If

magic doubles the yield to 8 : 1, the farmer keeps 7. Net food more than doubles. The land fills up with people, and the surplus feeds bigger towns.

The best real comparison is the most densely farmed land in medieval Europe: Flanders and Holland in the 15th century. The figures below come from the model in [Population and Settlement Sizes](#). The “moderate” row is interpolated.

Strength of the magic	Treat the region like	Rural density	Share living in towns	Farmland radius for a town of 10,000
None	Good lowland c. 1300	30–50 per km ² (80–130 per sq mi)	10–15%	21–27 km (13–17 mi)
Moderate (+50% yield)	Between the rows	40–60 per km ² (100–155 per sq mi)	15–20%	~15–19 km (9–12 mi)
Strong (about 2× yield)	Flanders and Holland (15th century)	50–75 per km ² (130–190 per sq mi)	20–40%	~12–15 km (7–9 mi)
Very strong (3× or more)	Nothing in medieval Europe	Higher still	Higher still	Small. The limit becomes water, fuel and transport, not food

Who controls it? If priests bless the fields, the temple takes a share, as the real Church took a tithe (a tenth). Expect rich temples in every village and a great temple-estate near the capital. If the magic is in the soil (a blessed valley), that

valley becomes the core of a realm, like the Île-de-France around Paris.

MAP TIP

Draw a blessed region exactly like Flanders: villages 1.5–2 km (about 1 mi) apart, market towns every 10 km (6 mi), and many walled towns of 5,000–20,000. Keep the edges of the blessing visible: the density should drop where the magic ends.

Water magic and desert cities

Real desert cities show what water magic would do, because a *qanat* (an underground channel that brings groundwater from the hills) is a kind of engineered spring.

- **Marrakesh** was founded c. 1070 by the Almoravids on a dry plain. From the 11th century, *khattaras* (the Moroccan name for qanats) brought groundwater to it from the direction of the Atlas Mountains. That water turned the site into an imperial capital with large gardens.
- **Yazd and Kerman** in Iran grew at the foot of mountains where qanats could be dug (see [Where Settlements Are Built](#)).

- **Valencia's** irrigation farmers still settle water disputes at a weekly Water Tribunal, held every Thursday at a door of the cathedral. Water rights become law and politics.

What changes: each magical spring acts as an oasis. Settlement forms **dots** around each water source, not a carpet. The size of each dot depends on how much water the magic gives. Food is the next limit: irrigated gardens are very productive, but there are only so many of them.

Who controls it? If one mage or one temple makes the water, the whole city depends on them. That gives them enormous power. It also makes the city fragile: real desert towns died when their water failed. A water-mage's tower in a desert city is the citadel, so draw it at the centre, walled.

MAP TIP

In a magically watered desert, draw: the water source (temple, tower, sacred well) in the centre; a ring of green gardens around it; then the town; then bare desert. Caravan stops still sit 30–40 km (19–25 mi) apart. Each one needs its own water: either a real well or a smaller magical one.

Healing magic and population growth

RULE OF THUMB

Healing magic makes a population grow faster. It does not raise the ceiling. A healed realm fills its land in a few generations, then hits the food limit.

Why: in the Middle Ages births and deaths were nearly equal. Europe's population grew only 2–3 times between 1000 and 1300, which is about 0.2–0.4% a year. If healing removes most deaths from disease and childbirth, growth can jump.

Yearly growth	Population doubles in	Real example
0.2–0.4%	175–350 years	Europe 1000–1300
1%	~70 years	—
2–3%	23–35 years	North America (the British colonies, then the United States). In 1798 Malthus noted that its population had doubled about every 25 years, because land was plentiful

What changes on the map:

- Faster frontier expansion: new villages cut into forests, and lords found planned new towns (see [Where Settlements Are Built](#)).
- More pressure on borders and more wars over land.
- No Black-Death-style collapse, so fewer deserted villages and ruins.

- Towns grow on their own. In real medieval towns, deaths outnumbered births, and towns grew only because people moved in from the countryside. Healers in a town would change that.

If healers are rare and expensive, only towns and nobles benefit. Then the towns grow and the countryside does not.

Teleport circles and portals

RULE OF THUMB

A portal is a harbour without a sea. Treat it like a port, a bridge or a strait: a fixed point where traffic gathers and where the ruler can tax it.

- **Staple right:** from 1259, Cologne forced passing merchants to stop, unload and offer their goods for sale for three days before moving on. Any city that owns a portal will try the same.

What grows around a portal: a customs house, warehouses, inns, money-changers, a market, a guard fort, and very often a temple. The portal town becomes a hub, like a crossroads town. If one end is in another realm, expect a **twin town** at each end.

Capacity decides everything. Compare the portal with a ship. A typical cog (the main northern cargo ship) carried 30–200 tons (see [chapter 05](#)).

Real analogues:

- **Øresund (the Sound):** from 1429 every foreign ship passing between the North Sea and the Baltic had to stop at Helsingør and pay the Danish king. A castle (later rebuilt as Kronborg) guarded the strait.

If a portal moves...	It acts like...	Effect on the map
A few people and letters a day	A royal messenger service	Capitals govern farther (see Capitals, Realms and Borders). Trade barely changes
Cartloads of valuable goods	A mountain pass or a caravan route	Luxury trade (spices, silk, gems) moves through portals. Portal towns get rich. Bulk goods still go by water
Shiploads of grain every day	A great river or sea lane	Inland cities can grow beyond 50,000 without rivers. The capital can sit wherever the portal network meets

Effect on capitals: a realm with portals will put its capital at the portal hub or build the network around the capital, the way Roman roads led to Rome. A portal into the capital is also a weak point. Guard it like a harbour mouth: Con-

stantinople closed its harbour, the Golden Horn, with a chain.

Flying mounts and airships

RULE OF THUMB

Flying moves people, messages and luxuries. It does not move bulk. A griffon carries about what a pack horse carries.

Why: a pack horse carries about 100–120 kg. A flying mount strong enough to carry a rider can carry a similar load. One cog carried 30–200 tons, which equals roughly **250 to 2,000** flying loads. Grain, timber, stone and wine stay on rivers and seas.

What changes:

Change	Real analogue	Map result
Fast messages	Nur ad-Din ran a regular pigeon post between Baghdad and Syria by 1167. The Mamluk sultan Baybars (r. 1260–1277) rebuilt the state relay post, the <i>barid</i>	A chain of relay aeries (nests or stables for flying mounts) a day's flight apart. If a griffon flies about 300 km (190 mi) a day, as in Capitals, Realms and Borders , the capital can govern much farther
Trade over mountains	Mule trains over Alpine passes carried mostly high-value goods	Gems, spices and letters fly straight over the range. Pass towns lose travellers and post but keep bulk freight
Aeries	Royal mews (falcon houses) and pigeon lofts sat in castles and towns	Aeries on crags and tower tops, with pasture or herds nearby to feed the animals
Defence against the sky	No direct real case	Roofed towers and wall-walks, bolt-throwers on towers, small courtyards, nets, lookouts on every peak, and keeps that can be held against an attack from above

How fast do fortresses change? The best real comparison is gunpowder. After the French invasion of Italy in 1494, new artillery forts with low, thick walls and angled bastions spread across Italy within a few decades. If flying enemies

are new in your world, show a mix: old open-topped castles beside new roofed ones.

LATER ERA (1500s+)

The bastion fort (*trace italienne*) is the real example of fortresses redesigned for a new threat. See [What Changes After 1500](#).

Weather control

Real analogue: around 815, Archbishop Agobard of Lyon wrote *On Hail and Thunder* against a popular belief in *tempestarii*, “storm-makers” who could raise or turn away hail. He complained that peasants paid them a share of their crops for protection. So even an imaginary weather-mage got paid like a tax collector.

What changes:

- **Fewer famines.** The Great Famine of 1315–17 was caused by years of heavy rain. Weather control prevents that kind of disaster, so there are fewer deserted villages.
- **Farming moves into marginal climates.** The Norse settled Greenland c. 985 and abandoned it in the 15th century, partly because the climate cooled. Weather

magic would keep such colonies alive and push farming higher up mountains and farther north.

- **Safer seas.** Reliable winds make sea trade cheaper and more ports viable.
- **Weather as a weapon.** The Mongols tried to invade Japan by sea in 1274 and 1281. A storm hit the 1274 fleet as it withdrew (how much damage it did is debated). In 1281 a typhoon destroyed much of the second fleet at anchor. The Japanese called it *kamikaze*, the “divine wind”. Coastal realms with storm-mages would be hard to invade by sea.

Who controls it? Like the lord's mill, which peasants were often forced to use and pay for, a weather-mage would be a lordly monopoly. Put weather towers on hilltops overlooking the most valuable farmland.

Magical resources: crystals, ore and ley lines

RULE OF THUMB

A magic crystal is ore. A ley line is a river. Place them, then apply the real rules for mines and rivers.

Crystals as ore. Real silver strikes show how fast a boom town grows:

Real boom town	Timeline	Size and features
Kutná Hora (Bohemia)	Mining from c. 1260; royal mining code in 1300; central mint soon after 1300; main mining zone closed in 1543	Grew fast and became the second town of Bohemia after Prague. Around 1300 its mines may have produced about a third of Europe's silver. (Some popular sources say 60,000 people. That is unlikely for a town that was smaller than Prague, so use a much lower figure.)
Novo Brdo (Serbia)	Saxon miners brought in under King Milutin (1282–1321); named in 1326; taken by the Ottomans in 1455 after a 40-day siege	About 6,000 kg of silver a year by 1450. In the 15th century 4,000–5,000 people lived inside the walls and 8,000–10,000 more in satellite settlements around them
Freiberg (Saxony)	Silver found in 1168	Grew into a major mining town

What a crystal boom brings: miners from far away with their own laws; a fortress and garrison; a mint; roads for food coming in; charcoal burners stripping the forests; and, when the lode runs out, decline. See [Industry, Resources and Special Towns](#).

Ley lines as rivers. The modern idea of ley lines comes from Alfred Watkins in the 1920s. He saw them as old straight tracks. Only in the 1960s were they reimagined as lines of

earth energy. Archaeologists reject them as real features, but in fantasy they are useful. Treat them like rivers:

- Towers and towns form **chains** along the line.
- A **nexus** (where lines cross) acts like a confluence (the place where two rivers meet): it attracts the biggest settlement and the strongest fortress.
- A place where a line is easy to reach acts like a ford: expect a tower, a toll and a fight over who owns it.

Mage towers and academies

RULE OF THUMB

Place academies like universities (in big cities) and lone mage towers like monasteries (remote, but supplied).

Type	Real analogue	Numbers	Where
Academy of magic	Universities: about 20 in Europe c. 1300, 60–80 by 1500. England had only two (Oxford and Cambridge)	1–3 per kingdom	In cities of 10,000+: they need food, lodging, books, patrons and students
Order of mages with many houses	Cistercian monks: about 500 houses by 1300 and nearly 750 at their peak in the 1400s; roughly 75–80 abbeys of monks in England and Wales (about one per 2,000 km ² or 770 sq mi)	Dozens per kingdom, if magic is as common as monasteries	Remote valleys, but with granges (outlying farms run by the monastery). Some granges were next door; others were tens of kilometres away. Their size varied greatly
Dangerous experiments	Leper houses and plague hospitals, placed outside towns or on islands	A few per city	Outside the walls, downwind (so smoke and smells blow away from the town), across water

Magical defences

Wards (magical barriers) behave like walls. A wall's cost depends on its **length**, but the space inside depends on its **area**. So big wards are cheaper per person than small ones. This favours fewer, bigger warded towns, the same logic as the burhs, where each hide paid for a share of the wall.

Ask two questions:

1. **What does the ward stop?** If it stops monsters but not armies, towns still need walls, and castles stay. If it stops armies too, castles shrink and frontier forts become ward-stone posts that are watched and maintained.
2. **Who pays for its upkeep?** English towns paid for their walls with *murage*, a toll the king allowed them to

charge. A warded town needs a ward-tax and a mage-guild, and it falls when the mages leave.

MAP TIP

Add magic in this order: (1) mark magical resources (crystals, ley lines, springs) like ore and rivers; (2) place the towers, academies and boom towns that use them; (3) adjust

density for fertility or healing magic; (4) add the new hubs (portals, aeries); (5) adjust the fortresses. Do this after the realistic steps in [Step-by-step](#), not instead of them.

Non-human peoples: archetypes with real analogues

Each common fantasy people matches a real way of life.

Use the real way of life to decide where they live, how many there are and how their land looks.

People	Real analogue	Settlement type	Density	Map signature
Dwarves	Medieval mining communities (Saxon miners in Bohemia and Serbia); Capadocian underground refuges	Mountain holds plus trade towns at the mountain gates	Holds like mining towns: a few thousand to perhaps 15,000	Gate towns on valley floors, mine symbols (crossed hammers), smelters, cleared slopes
Elves	Forest peoples of the Baltic: pagan Lithuanians and Prussians	Small clearings, hill-forts, sacred groves	1–5 per km ² (3–13 per sq mi) or less	Big forest, few dots, no roads, wide border wilderness
Halflings	Free peasant communities of fertile lowlands: Dithmarschen, Flanders	Dense villages, mills, small market towns	30–75 per km ² (78–190 per sq mi)	Many small villages, few castles
Orcs and other “horde” peoples	Steppe nomads: Mongols, Cumans, Crimean Tatars	Seasonal camps, a few royal cities	Under 1–2 per km ² (under 3–5 per sq mi)	Pastures, camps, raid trails, frontier markets
Merfolk	Sea nomads such as the Orang Laut of the Strait of Malacca	Reefs and sheltered waters, shore markets	—	Shore market towns, safe or unsafe straits
Giants and dragons	Raiders who are paid tribute (Vikings in England); predators that need huge ranges	Lairs	—	Empty rings around lairs, tribute villages beyond

Dwarves: mountain holds and gate towns

Real model: medieval mining towns. German (“Saxon”) miners settled in Bohemia, Hungary and Serbia under special privileges. They had their own mining law, their own courts and their own churches. Novo Brdo and Kutná Hora (see the table above) show the size: a few thousand people inside the walls and more in the settlements around them.

The food problem: a hold of 10,000 people needs farmland within 21–27 km (13–17 mi) on good lowland, or about 55 km (35 mi) on marginal land. Mountains are marginal

land. So a dwarf hold must buy food. That creates a **gate town**: a market town on the nearest valley floor, ideally on a river, where grain comes in and metal goes out. Treat it like a pass town or a mining town’s market.

Underground roads: real tunnels were short. The tunnel said to link Derinkuyu with the next underground town, Kaymaklı, is about 8–9 km (5–5.6 mi) long. A long underground road between holds is a wonder of the world. Draw a few, not a network.

MAP TIP

For each dwarf hold, draw one gate town on the valley floor, a road climbing to the gate (a small, clearly rounded arched doorway set into the mountainside, so it is not read as a flat-topped gallows), a mine symbol (crossed hammers),

smelter symbols on the stream below, and bare slopes where trees were cut for charcoal. Group holds into a league (like the Hanse) with one leading hold.

Elves: forest realms

Real model: the forest and marsh lands of pagan Lithuania (Christian only from 1387) and the Baltic Prussians. Density was low: 1–5 people per km² (3–13 per sq mi). Their realm had no sharp line border. Between the Teutonic Order and Lithuania lay the “Great Wilderness”, a forest belt emptied by raids from both sides and crossed by war parties in summer and winter.

What it looks like: a large forest with few, small clearings; hill-forts and sacred groves; paths but no roads; and a wide

empty border. Human villages pushing into the forest edge with new clearings are a source of conflict.

Contact: reclusive peoples may trade without meeting. Herodotus (5th century BC) described “silent trade” on the West African coast: Carthaginian traders left goods on the beach, lit a smoky fire and withdrew, and local people left gold beside the goods. Silent trade continued in the West African gold trade in the Middle Ages. Mark a few **trading stones** at the forest edge.

Halflings: fertile farmland

Real model: self-governing peasant societies on rich land. Dithmarschen, a marshland on the North Sea coast of Germany, was a peasant republic from the 13th century until 1559. Its parishes ruled themselves. In 1500, at Hemmingstedt, its farmers used dykes and flooded ditches to defeat a Danish royal army.

What it looks like: density like Flanders (50–75 per km²) or good lowland (30–50 per km²). Villages 1.5–4 km (1–2.5 mi) apart, many mills, market towns 9–16 km (6–10 mi) apart. Very few castles. Their neighbours, however, build castles along the halfling border.

Orcs and other “horde” peoples

Do not draw them as mindless. Model them on real steppe societies, which had laws, crafts, trade and some of the best logistics of the Middle Ages.

How steppe societies worked:

- **Herd:** life rested on sheep (meat, milk, wool, fuel) and horses (milk, riding, war). Each warrior rode with several horses, so he could change to a fresh one. Estimates vary: 3–4 horses per Mongol soldier, or 5–8 per steppe warrior (Devereaux).
- **Few people, much land:** density was under 1–2 per km² (under 3–5 per sq mi). In 1206 the whole Mongol army probably numbered under 100,000 men (Timothy May, cited by Devereaux).
- **Seasonal movement:** camps moved in a regular yearly cycle, usually between summer pastures (uplands or open steppe) and sheltered winter camps (river valleys). For real camp sizes, how often camps moved and how far, see [Steppe and grassland](#). Use those figures for orc camps too.
- **Dependence on farmers:** nomads needed grain, metal and cloth that they could not produce. They got them by trade or by raiding (Khazanov 1984). Ming China refused the Mongol leader Altan Khan’s requests for trade for years, and he raided instead; in 1550 he even burned the suburbs of Beijing. In 1571 the Ming gave him a title and trading rights, and border horse markets opened at Datong, Xuanfu and other garrisons. After that, raiding fell sharply.

- **Cities too:** nomad rulers built capitals and trade towns. Karakorum was built in the 1220s–1230s. Batu founded Sarai on the lower Volga in the 1240s–1250s. The Franciscan friar William of Rubruck, who passed by in 1253–54, described it as a new town that Batu was still building. The word “horde” comes from *ordu*, the ruler’s camp and court.

What it looks like: open grassland with seasonal camps (small open half-circles for round tents, never the dashed rectangle that means an army camp); winter camps along rivers; a royal camp or capital; raid trails into farmland (like the Muravsky Trail used by Crimean Tatar raiders into Muscovy); frontier markets and defence lines on the farmers’ side.

RULE OF THUMB

If a horde raids constantly, ask why the frontier markets are closed. Raids and trade are two ways of getting the same goods.

Merfolk: coastal trade

Real model: the Orang Laut (“sea people”) of the Strait of Malacca. For centuries they served the trading states of Srivijaya and later Malacca as pilots, fishermen and a naval force. They kept the straits safe for friendly ships and dangerous for others.

What it looks like: markets at sheltered bays and river mouths where land and sea peoples meet. The sea people sell fish, pearls, coral and salvage. The land people sell metal, grain, timber and cloth. A land realm allied with merfolk controls the nearby strait. An enemy has to fight for every harbour.

Giants and dragons: no-go zones and tribute

Tribute: a power you cannot defeat, you pay. England paid the Vikings huge sums in silver: 10,000 pounds in 991, 16,000 in 994, 36,000 in 1007 and 48,000 in 1012 (*Anglo-Saxon Chronicle*). A dragon that takes tribute creates a belt of paying villages, an official who collects the tribute, and perhaps a treaty stone or shrine where it is delivered.

Food: a big predator needs a big range. Here is a rough example. Suppose a dragon eats one cow a week, or 52 a year. A herd can lose perhaps 10–20% of its animals a year

without shrinking (an assumption). The dragon then needs a herd of about 250–500 cattle. That is the cattle of many villages. A Domesday plough-team, for comparison, had eight oxen.

No-go zones: around a lair (a small skull with the creature’s name), draw an empty ring, then a thin belt of tribute villages, then normal land. Giants in the mountains work the same way. Their lands look like a dangerous frontier, with watchtowers on the human side and ruined farms in between.

MAP TIP

Give each non-human people its own density and pattern. Then draw the **edges**: gate towns for dwarves, trading stones for elves, frontier markets for nomads, shore markets for merfolk, tribute villages for dragons. The edges are where most of the story happens.

Contact zones between peoples

RULE OF THUMB

Peoples who distrust each other trade at fixed places, under fixed rules, near the edge of one realm. Draw those places.

Trading posts and merchant compounds

Real example	Dates	How it worked	Fantasy use
Peterhof, the Hanseatic <i>kontor</i> (trading post) in Novgorod	c. 1200–1494	A palisaded compound of wooden houses around a stone church of St Peter, which was also the warehouse. About 150–200 German merchants per season, arriving as summer and winter groups. They lived under their own written rules (the <i>Schra</i>)	A dwarf or human merchant compound inside an elven or orc town
Fondaco dei Tedeschi, Venice	From 1228	A building beside the Rialto where German merchants had to lodge and store goods, so Venice could watch and tax them. The word comes from the Arabic <i>funduq</i> (an inn and warehouse)	A “strangers’ inn” where foreign races must stay
Galata (Pera), the Genoese colony across the harbour from Constantinople	1267/ 1273–1453	A self-governing colony with its own governor. The Genoese walled it despite a ban, and built the Galata Tower in 1348	A foreign quarter that becomes a fortress
Caffa (Kaffa), the Genoese port in Crimea	1266–1475	Founded with the consent of the Golden Horde (the Mongol khanate that ruled the steppe). It traded grain, furs, slaves, silk and spices between the steppe and the Mediterranean. Some sources claim over 70,000 people, but figures for medieval Caffa are very uncertain and probably exaggerated; treat it as a large port town	A human port on the edge of orc or nomad lands

Other kontors stood in London (the Steelyard), Bruges and Bergen (Bryggen, which closed in 1754).

Frontier markets

- **China's frontier markets:** from the Song dynasty (960–1279), China traded tea for Tibetan horses at fixed markets in Sichuan and Gansu. The Ming briefly opened horse markets for the Mongols in 1551 but soon closed them again, and raids went on. In 1571 lasting border

markets opened at Ming garrisons such as Datong and Xuanfu.

- **Silent trade:** for peoples who will not meet (see [Elves](#)).

Frontier markets sit **at a garrison or gate**, often open only on fixed days, with officials, interpreters and a strong guard.

Mixed cities

Real mixed cities kept quarters. Toledo after its conquest by Castile in 1085 had Christians, Muslims and Jews. Norman Palermo (from 1072) had Latin, Greek and Arabic speakers.

Many cities had a Jewish quarter and foreign merchant quarters, often walled or gated. In a fantasy city, give each people its own quarter with its own temple, market and gate, and put the shared market in the middle.

Borders between peoples

The Anglo-Scottish border is a good model for a tense border between two peoples. In 1249, six English and six Scottish knights wrote down the “laws and customs of the march” (the border zone). Later agreements (1398) called for

monthly “days of truce” where wardens met to settle crimes across the border. The meeting places were fixed: fords, streams and border villages such as Kershopefoot and Redden Burn. For border types in general, see [Capitals, Realms and Borders](#).

FANTASY TWIST

A shared magical language, or telepathy, removes the need for interpreters and silent trade. It does not remove the need for rules, guards and fixed meeting places. Distrust, not language, is what makes contact zones.

MAP TIP

Along each border between peoples, mark: two or three frontier markets at gates or fords; one treaty or truce meeting place; a merchant compound in the largest town on each side; and a mixed town where the main trade route crosses the border.

Gods, undead and dangerous seas

Gods with real power

Real baseline: the medieval Church was already a huge landowner. In England in 1086, bishops and abbeys held about a quarter (25–26%) of the land, measured by value. The Peace of God movement (first at Charroux in 989) declared churches, clergy and peasants off-limits to fighting. Later councils added pilgrims and merchants. The Truce of God (first at Toulouges in 1027) at first banned fighting on Sundays. Later truces stretched the ban from Wednesday evening to Monday morning, and to holy seasons. These rules were often broken. A god who **enforced** them would change the map:

- **Sanctuaries become real safe zones.** Expect villages and markets clustered around temples, and roads under divine protection busier than other roads.

- **Temple towns grow at sacred sites**, even where trade does not favour them (see the “why is it here?” test in [Where Settlements Are Built](#)). They still need food and water.
- **Temples own more land.** If temples hold a third or half of the realm, draw temple estates and granges across the best farmland.
- **Theocracies (states ruled by priests or religious orders) appear.** Real church states existed: the Papal States, and the Teutonic Order's state in Prussia.

Pilgrimage roads, hospices and shrines are covered in [Religious, Cultural, Legal and Ancient Sites](#).

Undead, plagues and blights

A spreading undead curse or a magical blight works like the Black Death: 30–60% of people dead, towns too small for their walls, and, over the next 150 years, about 1,500 or more deserted villages in England alone. Real societies responded with isolation:

- **Quarantine:** in 1377 Ragusa (Dubrovnik) made travellers from plague areas spend 30 days on the islet of Mrkan or in the town of Cavtat before entering. The word “quarantine” comes from the later 40-day period (Italian *quaranta*, forty).

Dangerous seas

If sea monsters or storms make the sea too risky, trade goes inland, even though water transport is otherwise cheaper.

- **Ming China:** grain for Beijing and the northern armies once went partly by sea, a route that suffered from storms and pirates. In 1411–1415 the Ming state rebuilt the Grand Canal with about 165,000 workers. When it reopened in 1415, sea transport of grain to the north ended. Thousands of canal barges (about 15,000 ships, by one count) carried the grain instead. The canal was slower but safer and easier to control. A dangerous sea creates canals and river ports.
- **Convoys:** Venice sent its long-distance trade in state-organised galley convoys (*mude*) from 1315 to 1533. The

- **Plague hospitals on islands:** Venice opened its first *lazaretto* (plague hospital) on a lagoon island in 1423.
- **Outsiders outside:** leper houses stood outside town gates.

What changes on the map: if the dead can rise, graveyards move outside the walls and get their own walls and watch. Expect quarantine stations on roads and islands, guard posts at the edge of blighted land, burned or salted villages, and a “dead land” that slowly returns to forest.

ships left in spring and came back in autumn. The Flanders convoy carried spices and cotton to England and Flanders and brought wool cloth home. A dangerous sea creates convoy ports, escort fleets and fixed sailing seasons.

- **Coast-hugging:** ships stay near shore and stop each night. A galley averaged about 55 km (34 mi) a day. Expect harbours of refuge about a day’s sail apart.

What changes on the map: more canals and portages (places where boats or goods are carried overland between two waterways); fortified harbours with chains across the mouth; coastal towns set back from the shore (as on the raided Mediterranean coasts; see [Coasts](#)); lighthouses and watchtowers on headlands; and a few great convoy ports instead of many small ones.

MAP TIP

For each of these three, mark the **new places** they create: shrines and sanctuary towns for gods; walled graveyards, quarantine islands and blighted zones for undead; canals, convoy ports and fortified harbours for dangerous seas. Then mark the **places they destroy**: ruins, empty coasts, abandoned villages.

What changes if: summary table

If your world has...	What changes on the map	Real analogue
Monsters in the countryside	Fewer lone farms; walled hilltop villages of 300–1,000; empty valley floors	Incastellamento, Latium, mainly c. 920–1030
Seasonal raiders or beasts	Tower houses and fortified farms	Anglo-Scottish border, 14th–16th centuries
A monster-haunted frontier	Refuge forts a day’s walk apart; beacons; emptied border belt	Alfred’s burhs; Hungarian <i>gyepű</i>
Monsters along roads	Cleared verges; patrols; hospices a day apart	Statute of Winchester 1285; <i>hermandades</i> ; Great St Bernard hospice
A monster holding a pass	Tolls, or traffic shifts to another pass	Fraxinetum raiders in the Alps, 906–973
Fertility magic	Density like Flanders (50–75 per km ²); more and bigger towns	Flanders and Holland
Water magic	Desert cities as dots around magical springs	Qanat cities: Marrakesh, Yazd, Kerman

If your world has...	What changes on the map	Real analogue
Healing magic	Faster growth; fast frontier settlement; towns grow by themselves	Colonial North America (doubling in ~25 years)
Portals	Hub towns with customs, warehouses and forts; twin towns	Øresund toll (1429); Cologne staple right (1259)
Flying mounts	Aeries; relay posts; roofed towers; luxury trade over mountains	Pigeon post (1167); Mamluk <i>barid</i>
Weather control	Fewer famines; farming in marginal climates; storm-defended coasts	Agobard's <i>tempestarii</i> (c. 815); the storms that hit the Mongol fleets off Japan (1274, 1281)
Magic crystals	Boom towns with forts, mints and imported food	Kutná Hora; Novo Brdo
Ley lines	Chains of towers; a big city at the nexus	Settlements along rivers and at confluences
Mage academies and towers	Academies in big cities; towers in remote valleys with granges	Universities; Cistercian abbeys
Wards	Fewer, bigger warded towns; ward-taxes	Burghal Hidage; murage
Dwarves	Mountain holds plus gate towns on valley floors	Saxon mining towns
Elves	Low-density forest realm; wide border wilderness; trading stones	Lithuania; silent trade
Halflings	Dense farm villages, few castles	Dithmarschen; Flanders
Steppe "hordes"	Seasonal camps; raid trails; frontier markets; defence lines	Mongols; Ming horse markets (1571)
Merfolk	Shore markets; controlled straits	Orang Laut
Dragons or giants	Empty rings around lairs; tribute villages	Danegeld (tribute paid to the Vikings, 991–1012)
Other peoples living side by side	Merchant compounds, strangers' inns, quarters, truce days	Hanseatic kontors; <i>fondaci</i> ; Anglo-Scottish march law
Gods with real power	Sanctuary towns; rich temples; theocracies	Peace of God (989); Papal States
Undead or blights	Walled graveyards outside towns; quarantine islands; deserted villages	Black Death; Ragusa (1377); Venice (1423)
Dangerous seas	Canals; convoy ports; fortified harbours	Ming Grand Canal (1415); Venetian <i>mude</i>

MAP TIP

Pick no more than three or four rows from this table for any one region. Each row adds visible features. A region with every fantasy element at once looks crowded and loses its character. Give each region one main fantasy "problem" and show how its people solved it.

Suggested symbols

RULE OF THUMB

One symbol, one meaning. Draw ordinary things (castles, churches, ports, tolls, hospices) with the normal symbols. Give magic its own shapes (diamonds for towers, ley lines and wards; a spiral for portals), so it never looks like a town, a border or a road.

These symbols match the guide's [master legend](#). Symbols marked † are fantasy-only. They are listed in the fantasy group of the master legend.

Danger and defence:

Feature	Symbol	Notes
Tower house or fortified farm	Small solid square	Many in raided borderlands
Walled hilltop village †	Village symbol on the hilltop (a tiny dot at kingdom scale); at local scale, a battlemented wall line around the house marks	Use the town symbol only if it really is a town
Refuge fort or burh	Town symbol with a battlemented ring; a fort with no town inside takes the small castle icon	About a day's walk apart
Watchtower; beacon	Small triangle; flame dot	Draw chains, each in sight of the next
Hunters' or rangers' lodge †	Small house mark labelled "Lodge"	A royal hunting lodge takes the small crown instead
Frontier and wild zones	Frontier: light hatching (the march symbol). Wild †: dense cross-hatching	Never a plain colour tint, which means a language area
Monster lair †	Small skull with the creature's name	Leave an empty ring around it

Magic:

Feature	Symbol	Notes
Mage tower or weather tower †	Slim tower topped with a small open diamond	No battlements (castle) and no spire (church)
Academy of magic †	Book beside the city name (the university symbol), labelled "Academy"	Only in cities of 10,000+
Magic crystal mine	Crossed hammers, labelled with the crystal	The normal mine symbol
Ley line and nexus †	Straight chain of small open diamonds in one colour (for example violet); a larger open diamond at the nexus	Not dotted, dashed or dash-dot, so it cannot be read as a road or a border
Ward stones †	Small solid diamonds spaced around the warded area	Not the upright tick, which means a standing stone
Magical spring or sacred well †	Small blue drop, with a green ring of gardens around the town	
Portal †	Small spiral; at a portal town, put it beside the town symbol	Never a star (capital) or a circle (town)
Aerie †	Small pair of wings on a crag or on a castle icon	Relay aeries in a chain, a day's flight apart

Peoples and contact:

Feature	Symbol	Notes
Dwarf hold gate †	Small arched doorway (n) in the mountainside	Draw the arch clearly rounded and set into the mountain, unlike the flat-topped gallows "Π" on a hill. The gate town below uses the normal town symbol
Smelter †	Small chimney with a curl of smoke	On the stream below the mine
Underground road †	Thin brown dotted line between hold gates, labelled "(underground)"	Draw a few, not a network
Nomad camp †	2–3 tiny open half-circles (round tents); the ruler's camp as a large labelled ring of them	Never filled (a filled mound is a barrow) and never the dashed rectangle (an army camp)
Raid trail †	Red dotted line with arrowheads pointing into the farmland, named ("Muravsky Trail")	
Frontier market or trading stone †	Small pair of scales; label the market days or "Trading Stone"	At a gate, ford or garrison
Truce meeting place	Small lone tree (the moot symbol, since it is also a meeting place), labelled "Truce Day"	At a ford, stream or border village

Gods, undead and dangerous seas:

Feature	Symbol	Notes
Temple or sanctuary	The church symbols; a great sanctuary takes the great-shrine symbol (scallop or reliquary)	Never a star
Walled graveyard †	Small rectangle with a solid wall line and tiny crosses inside, labelled "Graveyard"	Outside the town walls
Quarantine station or plague island	Small house with a cross, labelled "Lazaretto"	The master legend's hospital symbol
Blighted or dead land †	Pale grey wash with small black dead-tree ticks, labelled ("the Blight"); watchtower triangles along its edge	Old villages inside drawn grey, labelled "(ruin)"
Canal	Straight blue line	China's Grand Canal shows canals are not only a 1500s+ feature

MAP TIP

Put a small legend on the map itself, listing every fantasy symbol you used. Readers know what a castle icon means; they cannot guess what a spiral or a diamond means.

Consistency checklist

If something exists in your world, other things must exist or change too. Check each line that applies.

- **If monsters roam the countryside**, then there are fewer lone farms, more walls, watchtowers, ranger lodges and cleared roads. Total population is lower than on a safe map.
- **If the whole world is "points of light"**, then most towns are small (about 1,500–4,000 people). Larger towns exist only where water brings food safely.
- **If fertility magic exists**, then density is higher and towns are bigger. Big cities still sit on water, unless magic also moves grain.
- **If healing magic exists**, then the frontier fills fast, there are more new towns and land wars, and fewer ruins from plagues.
- **If water magic makes desert cities**, then whoever makes the water holds power, and the city dies if the magic stops.
- **If portals exist**, then each portal has a town, a customs house and a guard. The capital is connected to them, or is in the wrong place.
- **If flying mounts exist**, then castles have roofed towers, lords keep aeries with food animals nearby, and messengers are fast. Bulk freight still uses rivers.
- **If weather magic exists**, then someone controls it and takes payment, and famines are rare unless the weather-mages are at war.
- **If magic crystals exist**, then there are boom towns, mining law, mints and fortresses at the deposits, and ghost towns where deposits ran out.
- **If dwarves live under mountains**, then grain flows up to them through gate towns, and metal flows down.
- **If elves rule a forest**, then the forest has few roads, a wide border and a few fixed trading places.
- **If nomads raid farmland**, then there are frontier markets (or the reason they are closed), defence lines and burned villages near the steppe.
- **If dragons exist**, then there are lairs with empty land around them, tribute arrangements, and lost herds.
- **If gods enforce sanctuary**, then temples are rich and towns cluster around them.
- **If the dead can rise**, then graveyards are outside the walls and guarded, and there are quarantine places.
- **If the sea is deadly**, then canals and convoy ports replace many small harbours.
- **If two peoples meet**, then there are markets, strangers' inns or compounds, and agreed meeting places on the border.

MAP TIP

After finishing the map, read down this list and check every “if” that applies to your world. Most fantasy maps fail one or two lines, usually the food supply of a magical city or the missing edge where two peoples meet. More common errors are in [Common Mistakes](#) and [How to Fix Them](#).

Quick summary

- Apply the real rules first. Then ask which input each fantasy element changes: water, food, defence, travel cost, threats or resources.
- Danger concentrates people: fewer lone farms, walled hill-top villages, refuge forts about a day’s walk (24–32 km, 15–20 mi) apart, watchtowers, cleared roads.
- “Points of light” describes a frontier. Each point, if fields stay within 4–5 km (2.5–3 mi) of the walls, holds about 1,500–4,000 people.
- Fertility magic raises the population ceiling (treat a strongly blessed land like Flanders: 50–75 people per km²). Healing raises the growth rate, not the ceiling.
- Portals act like harbours and straits: hubs, tolls, warehouses, forts. Their capacity compared with a ship (a cog carried 30–200 tons) decides whether they change bulk trade.
- Flying mounts carry about a pack horse’s load (100–120 kg). They change messages, travel and luxury trade, not bulk freight, and they force castles to defend against attack from above.
- Magic crystals are ore (boom towns, forts, mints). Ley lines are rivers (chains of towers, a big city at the nexus).
- Place academies like universities (big cities, a few per kingdom) and mage towers like monasteries (remote, but with farms).
- Model each non-human people on a real way of life: dwarves on mining towns, elves on Baltic forest peoples, halflings on free peasant lowlands, orcs on steppe nomads with real logistics, merfolk on sea nomads, dragons on tribute-taking raiders.
- Peoples meet at fixed places under fixed rules: kontors, strangers’ inns, frontier markets, truce days.
- Gods, undead and deadly seas each create new places (sanctuaries, quarantine islands, canals) and destroy others (empty coasts, deserted villages).
- Draw ordinary features with the guide’s master legend. Give fantasy features their own non-clashing symbols (diamonds for magic, a spiral for portals, a skull for lairs) and explain them in a legend on the map.

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CHAPTER 11

COMMON MISTAKES AND HOW TO FIX THEM

Most fantasy maps go wrong in the same few ways. Experienced mapmakers on forums, blogs and in mapping guides point out the same problems again and again: cities with no water, towns sprinkled at random, roads that go nowhere and a castle on every hill. This chapter lists 30 common mistakes with human-made features. For each one it explains why it looks wrong, how to fix it, and when it can be right, because almost every rule was broken somewhere by real people for a good reason.

Each mistake has three parts: **Why it looks wrong**, **Fix** and **Exception** (a real case where the “mistake” was true). The baseline is Europe c. 1000–1500, with other regions where they help. The full explanations live in the other chapters, which are linked in each section.

IN THIS CHAPTER

Water and food: the mistakes that break a city

How many settlements, and where

Capitals and borders

Roads, routes and ports

Castles and town walls

Economy and purpose

Time: ruins, layers and eras

Drawing and labels

Sanity check: 20 questions for a finished map

Quick summary

Sources and further reading

Water and food: the mistakes that break a city

RULE OF THUMB

Before you draw a town, point to its water and to its fields. If you cannot point to both, move the town.

1. A city with no water

Why it looks wrong: People need water every day for drinking, animals, brewing, washing and crafts such as dyeing and tanning. Water is heavy, and deep wells were expensive. A town symbol far from any river, lake, spring or coast makes a careful reader ask how anyone lives there.

Fix: Put every village on a stream, a lake shore or a spring line (a row of springs at the foot of chalk or limestone hills). Put towns on rivers. If the site must be dry, such as a hilltop fortress, draw the spring, cistern (a tank for rainwater) or aqueduct that serves it. See [Fresh water](#).

Exception: Siena in Tuscany had no usable river nearby. It still grew to about 50,000 people (estimates vary: 50,000–70,000) before the Black Death of 1348. It did this

with springs and the *bottini*: underground tunnels that collect groundwater. They are first recorded in 1226, and in 1343 their water reached the main square, the Piazza del Campo. The whole system is about 25 km (16 mi) long. Venice collected rainwater in public cisterns and brought extra water by boat. Constantinople used aqueducts and huge cisterns. If you skip the river, you must draw the engineering.

A note on rivers: Check the rivers your towns sit on. Rivers join as they flow downhill. They almost never split, except in deltas and marshes. The fantasy author K. M. Alexander lists the splitting river among the “river sins” of fantasy maps: rivers only split for good in deltas. Put towns at confluences (where two rivers join), not at imaginary forks.

2. A desert city with no oasis or river

Why it looks wrong: A desert gets too little rain for farming without irrigation. A city in open sand, with no blue line or

green patch near it, has nothing to drink and nothing to eat.

Fix: Give every desert town one of four water sources, and draw the green belt of gardens and palms that it feeds:

Water source	How it works	Real example
A river crossing the desert	River floods or canals water a narrow strip	Cairo on the Nile; Baghdad on the Tigris
An oasis	Springs or shallow groundwater	Sijilmasa in the Tafilalt oasis (Morocco), the northern end of the Sahara caravan trade
A mountain-foot river	Snow-fed stream spread into irrigated orchards	Damascus and the Ghouta, an oasis of gardens and orchards watered by canals from the Barada river
Qanats	Gently sloping tunnels, with a line of shafts above them, bring groundwater from the hills	Yazd and Kerman (Iran)

Exception: Taghaza, a salt-mining village in the Sahara. When Ibn Battuta passed it in 1352, he found no trees, brackish (slightly salty) water and houses built of salt slabs roofed with camel skins. The miners lived on dates brought from Sijilmasa and the Draa valley, millet brought from the

Sahel (the dry lands south of the Sahara) and camel meat. Sijilmasa was 25 days away by caravan. A desert place with no food of its own works only if it produces something valuable and lies on a caravan route.

FANTASY TWIST

A magic spring, a water-making temple or a bound water spirit can justify a desert city. But then the water source becomes the most important and most fought-over place in the city. Draw it, wall it, and put the palace or main temple next to it.

3. No farmland around cities

Why it looks wrong: In the Middle Ages, 85–95% of people lived in the countryside, and most were farmers. Each farmer could feed only a few town-dwellers. A city surrounded by forest, swamp or “wilderness” right up to its walls has nothing to eat.

Fix: Draw a ring of fields, villages and mills around every town. Put gardens and orchards next to the walls, grain farther out, and pasture and woodland at the edge. As a

rule, a town of 1,000 needs farmland within about 7–8 km (4–5 mi) on good lowland, a town of 10,000 within 21–27 km (13–17 mi), and anything over 50,000 needs river or sea supply. The full table, including poor land, is in [Feeding towns: the hinterland](#).

Exception: Cities with good water transport could eat grain grown hundreds of kilometres away (see mistake 4). Mining camps, salt works and frontier fortresses in barren land imported their food, like Taghaza.

4. A huge city with no water transport

Why it looks wrong: Moving grain overland was slow and costly. In England c. 1300, carrying goods by land, river and sea cost roughly 8 : 4 : 1 per ton-mile (Masschaele 1993). In other words, a cart cost about twice as much as a river boat and eight times as much as a ship. For the Roman world, the historian A. H. M. Jones wrote that it was “cheaper to ship grain from one end of the Mediterranean to the other than to cart it 75 miles” (120 km). So towns above 10,000 people almost always stood on navigable water or the coast. Cities above 50,000 always needed river or sea supply.

Fix: Put every city over 10,000 on a navigable river, a lake or a sheltered coast. If you want a big inland capital, give it a canal or a river port, and draw it.

Exception: Some big cities managed without a natural waterway, but each had a special answer:

- **Milan** (100,000–150,000 c. 1300) dug the Naviglio Grande, a 50 km (31 mi) canal from the river Ticino. It was begun in 1177, reached Milan in 1258 and was fully navigable from 1272.
- **Siena** (about 50,000) lived on the farmland of its own hilly countryside and on grain carted in overland.
- **Granada** (estimates vary: 50,000–150,000 c. 1300) fed itself from the irrigated plain (*vega*) around it.
- **Damascus and Aleppo** were fed by irrigated gardens and by pack camels. In the Middle East and North Africa, pack camels had largely replaced wheeled carts by late antiquity, because they were cheaper to run (Bulliet 1975). This made long overland hauls more practical there than in Europe.

LATER ERA (1500S+)

Canals with pound locks (a chamber with a gate at each end, which raises or lowers boats) appeared in the Low Countries in the late 14th century. Eighteen were built on a Milanese canal, the Naviglio di Bereguardo, in 1452–58. After 1500, canals and canalised rivers (rivers deepened and fitted with locks) spread, so an inland city can grow if you draw its canal. Madrid, made the permanent seat of the Spanish court in 1561, grew large without a navigable river, but feeding it was a constant problem. See [What Changes After 1500](#).

MAP TIP

Draw water first, then farmland, then towns. Shade a farmland ring around each town using the radii above. Then check: does every town over 10,000 touch navigable water? If not, add a canal, an irrigated plain or a caravan route, or make the town smaller.

How many settlements, and where

RULE OF THUMB

Most dots on a realistic map are small. For every city there are several towns, dozens of market towns and hundreds of villages.

Use this table to check the numbers on your map. It is for settled lowland in the High Middle Ages, c. 1300. The full method is in [Population and Settlement Sizes](#).

Per 10,000 km ² (a square 100 × 100 km, 62 × 62 mi)	Count	Typical spacing	Basis
People	200,000–400,000	20–40 per km ² (50–105 per sq mi)	Baseline densities
Villages with a church	500–700	1.5–4 km (1–2.5 mi)	England: about 14 km ² per parish
Market places	45–130	9–16 km (6–10 mi)	Many were only villages with a weekly market; only 39% still had a market c. 1600
Towns of 2,000–10,000	3–8 (estimate)	Towns of 5,000+: 80–95 km (50–59 mi) in France and Germany	Derived from urbanization rates
Cities of 10,000+	0–1 (France, Germany); 1–3 (Flanders, northern Italy)	130–170 km (80–105 mi) in France and Germany; 60–90 km (37–56 mi) in Flanders and Italy	Derived from Buringh's city data
Great city of 50,000+	Usually none; at most one per kingdom	—	Only ~22 in all of Europe, the Middle East and North Africa c. 1300

5. Too few settlements

Why it looks wrong: Many fantasy maps show five towns in an area the size of England. Real England c. 1300 had about 9,500 parishes, more than 600 boroughs (towns with special legal rights) and about 1,746 places with a market grant, all on 130,000 km² (50,000 sq mi). A traveller in settled lowland passed a village every 1.5–4 km (1–2.5 mi).

A related mistake is to space villages a day's journey apart, about 25–35 km (15–22 mi). That is a traveller's day on foot, not the distance between villages.

Fix: At kingdom scale you do not need to draw every village. Draw the towns, and use a farmland texture or tiny dots to

stand for the villages. At local scale, draw villages 1.5–4 km (1–2.5 mi) apart in good land. See [Villages, Farms and the Countryside](#).

Exception: Empty land was real in uplands, steppe, forest and war zones. In 1086, northern and western England had under 5 people per km² (13 per sq mi). One reason was William I's "Harrying of the North" (winter 1069–70), a campaign of deliberate destruction; Domesday Book (1086) still recorded much of Yorkshire as "waste". Poland-Lithuania in 1500 averaged 6.6 people per km² (17 per sq mi), and steppe nomads lived at under 1–2 per km² (3–5 per sq mi).

6. Too many settlements, or too many big ones

Why it looks wrong: This is the opposite error: a city every 30 km (19 mi), each with 20,000 people or more. Around 1300 there were only about 230 cities of 10,000+ in Europe, the Middle East and North Africa together, and only 8 of 100,000+.

Fix: Keep the ratio. For every city of 10,000+, draw about 10 towns of 2,000–10,000 (see chapter 02), dozens of market

towns and hundreds of villages. Big symbols should be rare.

Exception: Flanders and northern Italy really were crowded. Flanders had 33–40% of its people in towns, towns of 5,000+ only 35–45 km (22–28 mi) apart, and cities of 10,000+ 60–90 km (37–56 mi) apart. A rich trading region can look busy, but it is the exception on a continent.

7. Evenly spaced settlements

Why it looks wrong: Towns set at equal distances look like a board game. Real settlements cluster along rivers, coasts, spring lines, good soils and roads, and leave gaps on heath, forest, marsh and upland.

Fix: Space by the land, not by the ruler. Draw chains of villages along valleys and coasts and leave gaps on hills. Central place theory (Christaller, 1933) is the idea that each market town serves the countryside around it, so market towns end up at fairly even distances. But it predicts a regular net

only on a flat, uniform plain. Real rivers and hills break the pattern. See [Patterns and spacing](#).

Exception: Planned colonisation made regular patterns. Nearly 700 *bastides* (planned new towns) were founded in south-western France between 1222 and 1372, most on grid plans. During the German eastward settlement (12th–14th centuries), lords laid out planned villages. Their houses stood in rows along a road or a forest clearing, each with a long strip of land behind it. Even these follow rivers and ridges.

8. Everything on the coast

Why it looks wrong: Mapmakers often draw the coastline first and then line it with cities. But most people were farmers living inland on the best soils. A ring of ports around an empty middle looks like a colony, not a kingdom.

Fix: Put ports only where there is a good harbour: estuaries, bays, river mouths. Put most villages and many towns inland along rivers. The coast may have the biggest cities; the interior has most of the people.

Exception: Where mountains come down to the sea, life really was coastal: Norway, Dalmatia, and Norse Greenland (settled c. 985; the last written record is from 1408, and the colony died out in the 15th century). Both Greenland settlements lay at the heads of fjords. A sea-trading power such as Venice could rule a chain of islands and coastal forts (the *Stato da Màr*, “State of the Sea”) with almost no inland territory.

9. Nothing on the coast

Why it looks wrong: A long coast with no ports or fishing villages ignores the cheapest transport there was. In England c. 1300, sea transport cost about one-eighth as much as land transport.

Fix: Every good harbour near farmland or a river mouth gets a port. Add fishing villages along sheltered shores.

Exception: Some coasts were avoided because they were:

- **Unhealthy:** marshy, malarial coasts such as the Tuscan Maremma were thinly settled.
- **Harbourless:** long, straight dune coasts, such as the Landes of Gascony, had few ports.
- **Raided:** on Mediterranean coasts exposed to raiders, people lived in hilltop villages a few kilometres inland and used a small landing place on the shore.

LATER ERA (1500s+)

Barbary corsair raids in the 16th century led the Spanish rulers of southern Italy and Spain to build chains of coastal watchtowers. Draw them as small triangles (the guide’s watchtower symbol) on headlands within sight of each other.

10. Empty interiors

Why it looks wrong: The middle of the continent is blank, or filled with one “Great Forest”, while the edges are

crowded. In reality, interiors held some of the richest farmland: the Paris Basin, the Po valley, the Hungarian plain.

Fix: Follow rivers inland and put a chain of towns on every navigable river, with village networks across the plains. Leave an interior empty only for a reason you can name: desert, high mountains, cold forest or marsh.

Exception: Real empty interiors existed: the Sahara and Arabian deserts, the highlands of Iceland, the Scandinavian mountains and the northern forests of Russia (1–5 people per km², or 3–13 per sq mi). Even there, draw the routes, oases and stations that cross them.

11. City sizes wrong for the era

Why it looks wrong: Fantasy capitals of 500,000 or a million people, or ordinary towns of 20,000+, break the scale of the period.

Place, c. 1300	People (estimates vary)
Cairo, the largest city in the region	220,000–600,000
Paris	usually given as ~200,000
Venice, Milan, Florence, Genoa	85,000–150,000 each
London	60,000–100,000
York, Bristol (1377, after the plague)	~11,000 and ~9,500
The next ten or so English towns (1377)	about 3,500–5,500 each

Fix: In England and France c. 1300, the largest city held only about 1–2% of the realm's people (this guide's estimate from baseline numbers). Size the capital from the realm's population, then make the next cities much smaller. See [The rank-size rule and primate cities](#).

LATER ERA (1500s+)

Capitals grew fast. By c. 1600, Istanbul had 400,000–700,000 people, London about 200,000 and Paris 220,000–300,000.

Exception: Imperial capitals were in a class of their own. Constantinople c. 1000 had 150,000–400,000 people, and Cairo c. 1300 even more. They were fed by sea and by the Nile, and by taxes from a whole empire.

MAP TIP

Count your symbols. If you have more city symbols than town symbols, or no village marks at all, rebalance using the table at the top of this section. Leave gaps where the land is poor, and say why in a short label (“moor”, “fen”, “royal forest”).

Capitals and borders

RULE OF THUMB

Capitals sit where the wealth is, not where the middle is. Borders follow old claims and the land, not a ruler.

The full treatment is in [Capitals, Realms and Borders](#). Here are the three most common errors.

12. The capital at the exact centre

Why it looks wrong: A capital placed at the geometric centre of the realm looks planned and artificial. Real capitals grew where the wealth, water, trade and the ruler's home lands were, and that was often near an edge. Paris lay in the north of the French kingdom, London in the south-east of England, Kraków near Poland's southern edge, and Vienna near Austria's eastern border with Hungary. All four stood on navigable rivers.

Fix: Put the capital in the richest core region, on a navigable river or coast, near the ruling family's home lands, or facing the main frontier. See [Centre, fortress or frontier?](#)

Exception: Madrid became the permanent seat of the Spanish court in 1561, partly because it lay near the centre of the peninsula (historians still debate the reasons). Remember, too, that many medieval kings had no fixed capital at all. Their courts travelled.

13. Straight borders without a reason

Why it looks wrong: Medieval borders grew out of estates, parishes, river stretches, watersheds (the high ground that divides two river systems), forests and old church districts. Few medieval rulers could survey a long straight line, and land was claimed piece by piece. A ruler-straight border across hills and valleys looks modern.

Fix: Make borders wiggly in settled land. Use crests, river stretches, marsh and forest edges, and old district lines. In empty land, use a wide frontier zone instead of a line. See [Borders and frontiers](#).

14. Borders that ignore the land, or follow every river

Why it looks wrong: A border that crosses a mountain range at a random point looks arbitrary. But the opposite error is also common: making every river a border. Big navigable rivers more often joined their valleys than divided them. For most of its course the Rhine had German lands on both banks, and the Danube ran through Austria and Hungary. Smaller units, like English counties, did often use rivers as boundaries.

Fix: Use crests and watersheds for long stretches. Use a river where the border is old or the river is a real obstacle.

MAP TIP

For every border segment, ask “what is this line following?” Mark the answer in your notes: crest, river, forest, old county line or treaty. If the answer is “nothing”, bend the

Exception: A strong state with surveyors could draw a straight line. The Roman frontier in Germany was the Upper Germanic Limes: a palisade and ditch with watchtowers, built in the mid-2nd century AD. One stretch ran almost dead straight for 81 km (50 mi), across valleys and hills, from near Walldürn to the Haghof south of Welzheim. In 1494, the Treaty of Tordesillas split newly found lands between Spain and Portugal along a meridian (a north–south line). The line lay 370 leagues, about 2,100 km (1,300 mi), west of the Cape Verde Islands. It was a straight line drawn over land that nobody had seen.

Example: the Anglo-Scottish border follows the River Tweed for part of its length. Its line was settled by the Treaty of York (1237) and has changed little since, apart from Berwick and the Debatable Lands.

Exception: Realms that controlled a pass usually held both sides of it. Savoy held lands on both sides of the western Alps, Tyrol on both sides of the Brenner Pass, and Navarre on both sides of the western Pyrenees. If a realm lives from a pass, its border runs beyond the pass, not along the crest.

line to the nearest real feature. Draw borders as dash-dot lines, and frontier zones as soft, wide bands of hatching or fading colour rather than hard lines.

Roads, routes and ports

RULE OF THUMB

Roads follow the easy ground between places that matter. Ports need shelter. Every journey takes days, so count them.

15. No roads at all

Why it looks wrong: A map with towns but no roads suggests that nobody travels or trades. Medieval England had a real road network. The Gough Map of Britain (conventionally dated c. 1360–70; a handwriting study (Smallwood 2010) suggests one or two decades after 1400, with parts redone or added later in the 15th century) shows about 190 thin red route lines between towns. Nine out of ten have the distance between the towns written on them. A modern study traced the real roads behind these lines: about 4,540 km (2,820 mi) of route across England and Wales, split into 455 segments. About a third of that length runs on

Full details are in [Trade Routes, Roads and Transport](#).

Roman roads, still in use 1,000 years after they were built. See [How much main road to draw](#).

Fix: Connect every town to its neighbours. Draw main roads between cities and along valleys. At kingdom scale you can leave out minor tracks. As a guide from the Gough Map, draw about 30 km of main road for every 1,000 km² of settled land (about 48 mi per 1,000 sq mi).

Exception: Where water is the road, roads matter less: river routes in Russia, the Venetian lagoon, the Fens of eastern England and archipelagos all relied on boats. Steppe and desert had tracks and caravan routes rather than built roads.

16. Roads that go nowhere, or run ruler-straight

Why it looks wrong: A road that stops in empty land has no reason to exist. A road that runs straight across hills, swamps and rivers suggests engineering that medieval roads did not have.

Fix: Every road should end at something: a settlement, port, ferry, ford, bridge, pass, mine, quarry, monastery or

the map edge. Medieval roads bend. They follow valleys, ridges and dry ground and go to river crossings. Long straight roads are Roman, or later.

Exception: Dead-end roads were real when they led to one valuable place, such as a mine, a quarry, a monastery, a pilgrimage shrine or a castle on a rock spur. Draw that destination at the end of the road.

17. Roads over mountains with no pass

Why it looks wrong: A road straight over the highest part of a range cannot be used by people on foot or with pack animals. Mountain crossings were few, slow and often closed in winter.

Fix: Run the road up a valley to a low saddle (a pass). Put a hospice (a shelter for travellers, often run by monks or canons), toll station or small castle at or near the top, and towns at the foot of the pass on both sides.

Pass (Alps)	Height	Medieval note
Brenner	1,370 m (4,495 ft)	The lowest of the major passes in the Eastern Alps
Great St Bernard	2,469 m (8,100 ft)	Hospice for travellers founded c. 1049–1050
Gotthard	2,106 m (6,909 ft)	Opened as a trade route c. 1220–1230, when the first wooden bridge was built across the Schöllenen gorge

A new pass can make a region rich. The Gotthard route ran through Uri, and control of it helped Uri, Schwyz and Unterwalden, the valleys at its northern end, to form the core of the Swiss Confederacy.

Exception: Some travellers used high, hard passes because they were shorter or avoided enemies and tolls. Mark such passes “summer only”.

18. Bridges everywhere

Why it looks wrong: On many maps every town on a big river has a bridge, and roads cross wide rivers wherever they like. In reality stone bridges were costly and valuable, and on the widest rivers they were few. London Bridge (stone, finished 1209) was the only road crossing of the Thames below Kingston upon Thames until Putney Bridge opened in 1729. That is more than 500 years with one bridge for the whole of London. Upstream, where the river is narrower, bridges were more common. Above Kingston the Thames had a bridge town about every 20 km (12 mi) by c. 1300, and more were added later, such as Chertsey

(licensed 1410). On smaller rivers, too, bridges were more common than many people think: in England, most important crossings had a bridge by the 13th century (see [The Thames example](#)).

Fix: Cross wide rivers at fords (shallow places), ferries and a few bridges at important towns. A bridge attracts roads, trade and tolls, so it usually has a town or castle beside it. Small streams can have many small bridges.

Exception: On a local map, a busy town on a small river may have several bridges. Some rich cities also built several bridges across their own river.

19. Desert or mountain trade routes with no water or stops

Why it looks wrong: A caravan route drawn as one line across 500 km (310 mi) of sand, with nothing on it, cannot be used. Pack animals need water, rest and fodder every day.

Fix: Put a stop at every day's stage: 30–40 km (19–25 mi) apart on plains and desert, and 10 km (6 mi) or less in mountains. Each stop needs a well, cistern or oasis. Along trade roads these stops were often *caravanserais*: walled

inns built around a courtyard for travellers and their animals.

Real example: The Darb Zubayda pilgrim road ran about 1,300 km (800 mi) from Kufa in Iraq to Mecca. It had 27 main stations, on average about 50 km (30 mi) apart, with smaller rest stops between them, plus wells, cisterns, reservoirs and dams. It was built and kept up mainly under the Abbasid caliphs, between 750 and 1258. Trans-Saharan caravans moved from oasis to well. Ibn Battuta took 25 days from Sijilmasa to Taghaza in 1352.

Exception: A route can cross a waterless stretch of a few days if travellers carry water in skins and know the way.

20. Ports on cliffs or exposed coasts

Why it looks wrong: Ships need deep, sheltered water out of the wind and waves, and a beach, quay or river mouth where they can unload. A port on top of a cliff, or on a straight coast battered by waves, cannot work.

Fix: Put ports in estuaries (wide, tidal river mouths), river mouths, bays behind headlands, behind islands or spits (long, narrow ridges of sand or shingle), or in drowned valleys (fjords and rias). Where a coast is mostly cliffs, put the port in the gap where a valley cuts through. Dover sits in the valley of the little River Dour, between white cliffs.

Real warnings: Exposed and silting harbours really did fail:

- **Dunwich** (Suffolk) was a leading English port in the 13th century. Storm surges in 1286 and 1287, and a great

Draw it as a dangerous “dry stretch” and write the number of days on it.

storm in 1328, ate into the town. Sand and shingle carried along the coast pushed the river mouth, and so the harbour, about 4 km (2.5 mi) to the north. Another storm in 1347 swept about 400 houses into the sea. With no harbour, the town was largely abandoned.

- **Bruges** slowly lost its access to the sea as the Zwin inlet silted up from the late 13th century onwards. By about 1500 the channel was nearly unusable, and the trade of the Low Countries moved to Antwerp.

Exception: A cliff-top town can have a port below it. Bonifacio in Corsica (founded as a fortress in 828) stands on limestone cliffs about 70 m (230 ft) high. Its harbour lies below, in a narrow, fjord-like inlet sheltered behind a long promontory. Draw the town on top and the quay below.

21. Ignoring travel time and scale

Why it looks wrong: Many maps have no scale, or a scale that breaks the story: an army crosses a 2,000 km (1,250 mi) empire in a week, or a “nearby” town is 300 km (190 mi) away. Readers notice.

Use the speeds in [Speed per day](#): for example, a walker 25–35 km (15–22 mi) a day, an army with baggage 13–20 km (8–12 mi), a royal messenger 50–90 km (30–56 mi), and a sailing ship 40–100 km (25–60 mi) as a voyage average.

Fix: Draw a scale bar first. Measure the important journeys and write them down. A kingdom 500 km (310 mi) wide is

15–20 days’ walk across, and 25–40 days for an army. Popular gaming rules are often too fast. *Dungeons & Dragons* (5th edition) has travellers on foot cover 24 miles (39 km) a day at a normal pace and 30 miles (48 km) at a fast pace. That is a good day for a rider on a road, but too fast for walkers over many days, and much too fast for armies and carts.

Exception: Messages could move much faster with relays. The Mongol relay system was claimed to cover 200–300 km (125–190 mi) a day. Travel downstream and by sea was also much faster than by land.

FANTASY TWIST

Teleport circles, flying mounts or fast magic messengers shrink travel time only for the few who can use them. Grain, stone and armies still move at cart and boat speed, so towns still need farmland and waterways. See [Fantasy Variants](#).

MAP TIP

Draw main roads as solid lines (thicker for more important roads) and tracks as thin dotted lines. Put an inn (a small house on the road) or a village every 15–30 km (10–19 mi) on main roads. In desert, put a caravanserai (the same small house, labelled “Caravanserai” or “Han”) every 30–40 km (19–25 mi), and draw the caravan route as a dotted line with a larger dot at each stage. Put a hospice (a small house with a cross) at each high pass. Write travel days on long routes: for example, 1,300 km (800 mi) by camel caravan at 25–40 km (15–25 mi) a day takes about 5–7 weeks. Check every port: is it in a bay, an estuary, a river mouth or a valley gap?

Castles and town walls

RULE OF THUMB

A castle controls something. Ask what it controls, who pays for it and who lives in it.

22. Castles on random hills

Why it looks wrong: Castles were built to control something: a town, a river crossing, a road, a pass, a harbour, a frontier, or the farms and tenants of a lord's estate. A castle on a random hill in the wilderness controls nothing and cannot feed its garrison. Also, not every castle stood on a hill. England and Wales had about 741 motte-and-bailey castles. A motte is a man-made earth mound with a wooden tower on top; the bailey is a fenced yard beside it. Mottes were typically built on low ground, many of them in towns or by rivers, and streams were dammed or diverted to fill their moats.

23. Too many castles

Why it looks wrong: A castle on every hill and by every village, and every one a great stone fortress. Stone castles were very expensive, and it was usual for one to take the best part of ten years to finish. Every castle also needed a garrison and land to pay for it.

Fix: Keep big royal fortresses rare: one or two per region, at key towns, passes and frontiers. Peacetime garrisons were small, often 5–20 men. Fill the rest with smaller lordly castles, tower houses (tall, small stone towers lived in by

Full details are in [Castles, Forts and Military Outposts](#).

Fix: For each castle, write down what it controls. Put castles in or beside towns, at bridges and fords, at the mouths of passes, at harbours, along frontiers and at the centre of a lord's lands.

Exception: Hilltop castles were common where the hills overlooked a route. The Rhine Gorge, 65 km (40 mi) between Bingen/Rüdesheim and Koblenz, has about 40 hilltop castles. Many of their lords lived on tolls from river traffic. The toll castle of Pfalzgrafenstein (its tower built 1326–27 for King Louis IV) even stands on an island in the river at Kaub. The hills were chosen because of the river, not at random.

minor lords) and fortified manor houses. Use the density table under mistake 24.

Exception: Where lordship was split among many small nobles, castles really were everywhere. The German-speaking lands had about 14,000 castles over the whole period, and Switzerland, where no ruler controlled who could build one, about 4,000. Most were small towers, not great fortresses, and not all were in use at the same time. Late medieval border zones filled with tower houses: about 800 in Scotland and over 3,000 in Ireland.

24. Too few castles

Why it looks wrong: A realm of millions has three castles, one in each corner. Some popular gaming rules make this worse. *Medieval Demographics Made Easy* (S. John Ross, first written 1993 and revised several times since) suggests one

working castle per 50,000 people. For Norman England, with about 1.7–2 million people, that gives only 35–40 castles. In fact, England and Wales had 500–600 castles in use at any one time after 1066: more than ten times as many.

Situation	One castle per	Average spacing	Real basis
Whole realm after a conquest: castles in use at one time (11th–12th centuries)	250–300 km ² (95–115 sq mi), or about 3,000–4,000 people	about 16–17 km (10–11 mi)	England and Wales after 1066: 500–600 in use on about 150,000 km ² (58,000 sq mi)
March or contested border (all castle sites)	17–28 km ² (7–11 sq mi)	4–6 km (2.5–4 mi)	Welsh March (see chapter 04)
Peaceful core of a kingdom (castles in use)	—	20–30 km (12–19 mi) (estimate)	Eastern England: a castle in each county town plus a few baronial seats (see chapter 04)
Royal castle chain along a frontier	—	20–40 km (12–25 mi) along the line	Edward I's North Wales castles, all supplied by sea

Situation	One castle per	Average spacing	Real basis
Fragmented lordship (many small towers, all periods together)	~10 km ² (4 sq mi)	3–4 km (2–2.5 mi)	Switzerland: ~4,000 sites on 41,000 km ² (16,000 sq mi)
Popular gaming rule	50,000 people	—	Less than one-tenth of what Norman England actually had

Fix: Do not spread the national average evenly. It mixes very different zones: the Welsh March had roughly 5–10 times as many castle sites per km² as peaceful eastern England. Put most castles along borders, at river crossings and around rich towns, and keep the safe middle of the realm thinner.

25. Every town walled

Why it looks wrong: Many fantasy maps put a wall around every dot. Walls were expensive to build and to maintain. From 1220 onwards, English towns usually paid for them with a royal grant of *murage* (the right to charge tolls on goods to pay for wall-building). Many of the early grants went to towns near the Welsh border, which faced a real threat of raids.

Numbers:

- **England and Wales:** Mike Salter's survey of walled towns (2013) describes 99 defended towns in England and 33 in Wales. It also lists 54 more English places with slighter or less certain defences. So England had roughly 100–150 defended towns.
- **Compare:** England had more than 600 boroughs and about 1,700 places with market grants. So only about one English borough in four to six was defended, and most small market towns had no walls at all.
- **Many “defended” towns were not stone-walled.** They had only a bank and ditch, with gates or bars (barriers across the street) where tolls were collected.

FANTASY TWIST

Flying enemies or monsters change the rules. If dragons or flying mounts are common, walls matter less than towers, roofs and shelters, and castles move to places that are hard to see from the air. If monsters roam the countryside, more villages get palisades (fences of sharpened stakes) and fewer people live on isolated farms. See [Fantasy Variants](#).

Exception: Peaceful, centralised realms needed fewer working castles. Many early earth-and-timber castles were abandoned within a century or two. So a late medieval map can show fewer working castles and more castle ruins.

- **South-western France:** most of the nearly 700 bastides started without walls. In the lands of the Count of Toulouse, the Treaty of Paris (1229) at first forbade new fortifications. Many bastides were walled only during the Hundred Years' War (1337–1453).

Fix: Wall the big cities, the regional capitals, the frontier towns and the ports facing raids. Leave most market towns open, or give them a ditch and gates.

Exception: In rich or contested regions most real towns were walled: northern Italy, Flanders and much of the Holy Roman Empire. In conquered frontier zones such as North Wales, Edward I built walled towns next to his castles, for example at Conwy and Caernarfon.

LATER ERA (1500S+)

Gunpowder changed walls. Low, thick walls with angled bastions (pointed platforms for cannon) appeared in Italy in the later 15th century and spread beyond Italy in the 1530s–40s. These “star forts” and garrison towns replace castles as the main strongholds. See [What Changes After 1500](#).

MAP TIP

Next to each castle symbol, write one word for what it controls: “ford”, “pass”, “port”, “town”, “march” (border province). Use a large castle icon (three towers) for great royal castles, a small castle icon (one tower) for lords' castles, and the same icon broken or grey, labelled “(ruin)”, for ruins. Put battlements on a town's ring only if it was walled: walled cities and frontier towns get them, unwalled towns get a plain ringed circle, and market towns a small open circle. Do not use a plain dot for a market town; at kingdom scale a tiny dot means a village.

Economy and purpose

RULE OF THUMB

Every region feeds itself first. Every town has a reason to be where it is, and you should be able to say it in one sentence.

26. Single-product economies

Why it looks wrong: “The mining kingdom”, “the wine province” or “the fishing town” where nothing else happens. In the Middle Ages almost every region grew its own grain first, and its speciality sat on top of farming. A whole region doing only one thing would starve.

Fix: Give every region farmland plus one or two specialities. Give every town several trades: a market, smiths, millers, tanners, brewers and the local speciality. If a place does specialise, draw how its food arrives, by road, river or sea.

27. Isolated towns with no reason to exist

Why it looks wrong: A town far from all others, off every road and river, with no special resource. Readers ask how it got there and who buys from it.

Fix: Give every isolated settlement one strong reason, and draw its sign: crossed hammers for a mine, a small plain

Exception: Real remote places, each with its reason:

Place	Reason	Date
Taghaza (Sahara)	Salt mine	Visited by Ibn Battuta in 1352
Kutná Hora (Bohemia)	Silver	Mining from c. 1260; boom in the late 13th and 14th centuries
Santiago de Compostela (Galicia)	Pilgrimage to the tomb of St James	Tomb reported in the 9th century; pilgrims from beyond the Pyrenees from the mid-11th century; a highly organised mass pilgrimage by the 12th century
Great St Bernard hospice (Alps)	Shelter at a high pass	Founded c. 1049–1050
Mount Athos (Greece)	Monasteries	Great Lavra, the first large monastery, founded 963

FANTASY TWIST

A rare magic resource, such as a mana spring or a dragon-bone quarry, works just like a silver mine. It creates a boom town in a bad place, a road to reach it, a castle to guard it, and food caravans to feed it.

Full details are in [Industry, Resources and Special Towns](#).

Exception: Specialised places existed when trade fed them:

- **Bergen** (Norway) exported stockfish (dried cod) from northern Norway and imported grain. From about 1350 the trade was run by Hanseatic (north German) merchants from their own quarter on the harbour, the Bryggen.
- **The Flemish cloth towns** imported English wool and grain from abroad.
- **Kutná Hora** (Bohemia) grew from a silver rush that began c. 1260. From 1300 it held the royal mint, and it became the second most important town in the kingdom after Prague.
- **Taghaza** (Sahara) lived only on salt.

cross for a chapel, a scallop shell for a great shrine, a castle icon for a fortress, an anchor for a harbour, or a blue “W” for a well. The usual reasons are a mine or salt, a pilgrimage shrine, a monastery, a fortress guarding a pass, a river crossing, an oasis or a harbour.

MAP TIP

Write a one-line “why it’s here” note for every town on the map. If you cannot, move the town to a river, crossing or resource, or delete it.

Time: ruins, layers and eras

RULE OF THUMB

A real landscape is old. Every map should show some things built long ago, and nothing built too early.

28. No ruins or older layers

Why it looks wrong: If everything was built at once, the map looks like a stage set. Real medieval landscapes were full of older things:

Older layer	Numbers (Britain and Ireland)
Iron Age hillforts (earth-walled hilltop enclosures, c. 700 BC–AD 43)	4,147 in Britain and Ireland (Atlas of Hillforts, 2017; see Ancient and ruined layers), most of them in Britain (Scotland about 1,700, England about 1,220)
Roman roads, forts, walls and towns	Many roads still in use in the Middle Ages
Early castles abandoned or replaced	Of about 1,700–1,800 castle sites in England and Wales (the Gatehouse gazetteer lists 1,761; see chapter 04), only 500–600 were in use at any one time
Deserted villages	More than 3,000 known in England from all periods; at least 1,500 abandoned c. 1350–1520, most in the 15th century when fields were turned into sheep pasture

Medieval Demographics Made Easy suggests this many ruined castles: (population ÷ 5 million) × the square root of the realm's age in years. Take a realm of about 5 million people where castle-builders have lived for 400–900 years. The formula gives only about 20–30 ruins. That is far fewer than the real landscape had. More than 1,000 castles were built in England and Wales in the 150 years after 1066. Only 500–600 were in use at any one time, so hundreds of early castles already stood abandoned by c. 1200.

Fix: Add a layer of ruins and older features: old forts on hill-tops, an old straight road that new towns ignore, a deserted village, a ruined abbey or castle, an old border dyke (a long earth bank and ditch). Let some towns stand on top of older ones. See [Religious, Cultural, Legal and Ancient Sites](#).

Exception: Truly new land, such as a frontier colony or fresh forest clearings, has few ruins. Iceland, settled from c. 870, had almost no earlier human layer.

FANTASY TWIST

Long-lived races and fallen magical empires multiply ruins. If elves or an ancient empire ruled for thousands of years, add more ruins, older roads and abandoned cities, and let new towns be built from their stone.

29. Mixing eras without reason

Why it looks wrong: Star forts next to new earth-and-timber castles, windmills in an early medieval world, or canal locks everywhere in 1100. Each feature has a date, and well-read readers know many of them.

Feature	First appears in Europe	Common by
Watermill	Roman times	About one per village in England by c. 1300 (over 10,000 mills)
Windmill (post mill: the whole mill body turns on a post to face the wind)	1185, first certain record (Weedley, Yorkshire)	13th–14th centuries
Friaries in towns	1210s in Italy; 1220s in England and northern Europe	Late 13th century
Universities	Bologna, late 11th century	About 20 c. 1300; 60–80 by 1500
Gunpowder cannon	First European picture, 1326	Decisive in sieges by the mid-15th century
Canal pound locks	Low Countries, late 14th century	15th century in northern Italy
Bastioned “star” forts	Italy, late 15th century	Spread beyond Italy in the 1530s–40s

Fix: Choose a date for your map. Use only features invented by that date. Old features can survive (layers are good), but new ones cannot appear early.

Exception: Real landscapes did mix eras. Roman roads were still used 1,000 years later. At Berwick-upon-Tweed, new walls with bastions were begun in 1558, while the old

medieval castle was left outside them. Along one stretch the new works were never finished, so the medieval walls and towers were repaired and kept in use instead. In a fantasy world, uneven technology can be a deliberate choice, such as one dwarven city with cannon in a world of knights. If you do it, make it a story point and explain it.

LATER ERA (1500s+)

If your map is set after 1500, see [What Changes After 1500](#) for star forts, postal relays, ocean ports and much bigger capitals.

MAP TIP

Put a date in the map's title box ("The Kingdom of X, Year 1312"). Then use three styles: working features in normal ink; old but still used features, such as a Roman road labelled "Street" or "(Roman)"; and ruins, drawn as the feature's own symbol broken or grey and labelled "(ruin)". Do not use dashed lines for ruins: in this guide, dashes mean fords and camp outlines.

Drawing and labels

30. Label clutter

Why it looks wrong: Every village is labelled in the same large font. Labels cross rivers and coasts, names are all in upper case, and the map uses five decorative fonts. The

map becomes hard to read, and the important places disappear.

Fix: Follow the classic rules of the Swiss cartographer Eduard Imhof (1975) and standard lettering practice:

Rule	What to do
Show importance	Make labels bigger and bolder for bigger places. Use UPPER-CASE letters only for realms and the largest features, because upper case is harder to read (one set of lecture notes puts it at about 13% less readable).
Point labels (towns)	Classic best position: above and to the right of the symbol. If that space is taken, try the other corners or directly above or below. A 2024 study with nearly 800 readers found they actually preferred labels placed directly above the point, so both work. Keep the label close to its symbol.
Line labels (rivers, roads)	Follow the line on a fairly straight stretch, just above it. Water names in italics, traditionally blue.
Area labels (realms, forests, seas)	Keep the name inside the area, spread across it and curved to fit its shape. Gaps between letters should be no more than about 4 times the letter height.
Fonts	Use 2–3 typefaces at most. Change size, weight or italics instead of adding fonts.
Size	Nothing smaller than about 6 point in print.
Spacing	Names should be neither densely clustered nor evenly spread (Imhof).
What to label	At kingdom scale, label cities, towns and major features. Leave most villages unlabelled.

Exception: Dense labelling is fine on a local map, or in a gazetteer (a list of places) beside the map. Old maps such as

the Gough Map crowd names together, so a crowded "antique" look can be a deliberate style.

MAP TIP

Place labels last, from the biggest to the smallest: realms, seas and mountain ranges first, then cities, rivers and towns, and villages only if there is room. If a label must cross something, let it cross land texture, not a river, coast or border line. Print the map at its final size and read it at arm's length.

Symbols for the fixes in this chapter

RULE OF THUMB

One symbol, one meaning. If a reader can read a symbol two ways (a star for a capital and also for a shrine), change one of them.

The fixes above name many symbols. Here they are in one place. These symbols match the guide's [master legend](#).

Scale: K = kingdom or region, L = local area.

Feature	Symbol	Scale	Mistakes
Village	Tiny dot (notable villages only); a farmland texture stands for the rest	K	5, 11
Small market town	Small open circle, no battlements	K	25
Town	Circle with a ring; battlements on the ring only if walled	K	6, 25
City	Circle with a thick battlemented ring	K	6, 25
Realm capital	Filled star in place of the tier symbol; name in bold capitals	K	12
Realm border	Thick dash-dot line	K	13, 14
March or frontier zone	Hatching, or a fading colour band 10–50 km (6–30 mi) wide	K	13, 14
Royal forest	Green dash-dot boundary with a large italic name	K, L	11
Main road	Solid line, thicker for more important roads	K	15, 16, 21
Track or mule path	Thin dotted line	K, L	15, 21
Roman road	Straight thick solid line labelled "Street" or "(Roman)"; grey where abandoned	K, L	15, 28, 29
Caravan route	Dotted line with a larger dot at each stage	K	19
Navigable river	Thicker river line up to the head of navigation, with a small boat there	K	4
Canal	Straight blue line	K, L	4
Bridge / ford / ferry	"(" with the road through it / short dashed line across the river / short dotted line with "F"	K, L	18
Toll	One bar across the road or river, with a "T"	K, L	17, 22
Inn or caravanserai	Small house on the road; label "Caravanserai" or "Han" in desert	K, L	19, 21
Hospice at a pass	Small house with a cross	K	17, 21
Port	Anchor (a large anchor with a quay for a main port)	K	8, 9, 20
Great royal castle	Large castle icon (three towers)	K	23, 25
Lord's castle	Small castle icon (one tower with battlements)	K, L	22–24
Tower house	Small solid square	K (border zones), L	23
Watchtower or coastal tower	Small triangle	K (chains), L	9
Star fort (1500s+ only)	Star outline, never filled	K, L	25, 29
Mine	Crossed hammers (or a pick)	K, L	26, 27
Chapel	Small plain cross, no tower	L	27
Abbey or priory	Church with a square cloister beside it	K (great abbeys), L	27
Great shrine	Scallop shell or reliquary, never a star	K	27
Any ruin	The feature's own symbol, broken or grey, labelled "(ruin)"	K (big ones), L	24, 28, 29
Deserted village	Grey church alone in a field, labelled "(lost village)"	K (a few), L	28
Hillfort	Oval of rings or hachures on a hill	L	28

These three features use the master legend's symbols:

Feature	Suggested symbol	Mistakes
Well or spring	Blue "W" for a well, "Spr" for a spring (the abbreviations used on Ordnance Survey maps)	1, 2, 27
Qanat	Straight line of tiny rings, much smaller than a market-town circle, ending at the village gardens (see chapter 06)	2
Old border dyke (a long earth bank and ditch)	Line with short ticks on the ditch side, grey if no longer used, labelled with its name	28

MAP TIP

Copy only the symbols you actually use into a small legend box in a corner of the map. Before you add a new symbol, check that its shape is not already used for something else.

Sanity check: 20 questions for a finished map

Answer yes or no. Every "no" is a place to look again.

- Does every settlement touch water (a river, stream, spring, lake, coast or a drawn well or cistern)?
- Does every desert town have a river, oasis or qanat, with green around it?
- Does every town have farmland and villages around it?
- Does every city over 10,000 sit on navigable water, or have a canal, irrigated plain or caravan route explained?
- Are there far more villages than towns, and far more towns than cities?
- Are settlements clustered along rivers, coasts and good soil, with gaps on hills, forest and marsh?
- Does the interior have towns and farms, unless it is desert, mountain or cold forest?
- Are your largest cities realistic for the era (only a few over 50,000, and at most one over 100,000 per kingdom)?
- Can you say why the capital is where it is, and is the reason not just "it's in the middle"?
- Does every border segment follow something (crest, river, forest, old district line, treaty)?
- Do roads connect all towns, and does every road end somewhere meaningful?
- Do mountain roads cross at passes, with a hospice, toll point or fort?
- Are big rivers crossed only at a few bridges, plus fords and ferries?
- Do desert and steppe routes have a stop with water about every 30–40 km (19–25 mi)?
- Are all ports in sheltered water: a bay, estuary, river mouth, fjord or valley gap?
- Have you drawn a scale bar and checked travel times for the main journeys?
- Can you say what each castle controls?
- Are walls drawn only around cities, frontier towns and exposed ports, not around every market town?
- Does each region have food plus a speciality, and does each remote town have a clear reason to exist?
- Are there ruins and older layers, and is every feature possible for the date on your map?

Quick summary

- Point to the water and the fields before you draw a town. Desert towns need a river, an oasis or qanats.
- A town of 10,000 needs farmland 21–27 km (13–17 mi) around it. Cities above 10,000 need navigable water, or a canal, a rich irrigated plain or caravans.
- Most dots are small. Per 10,000 km² of good land c. 1300: 500–700 villages, dozens of market places, 3–8 towns and 0–1 cities (1–3 in Flanders or northern Italy).
- Cluster settlements along rivers, coasts and good soil, with gaps on poor land. Do not use a ruler and do not line only the coast.
- Capitals sit in the richest region, not in the geometric centre. Borders follow land and old claims and are rarely straight.
- Roads end at something and cross mountains at passes. Bridges over big rivers are few, and desert routes need water every 30–40 km (19–25 mi).
- Ports need shelter. Dunwich and Bruges show what happens when a harbour fails.
- Draw a scale bar. Walkers make 25–35 km (15–22 mi) a day and armies only 13–20 km (8–12 mi).
- Every castle controls something. After 1066 England and Wales had about one castle in use per 250–300 km² (95–115 sq mi), and contested borders far more (a castle site every 4–6 km, or 2.5–4 mi, in the Welsh March). Only about one English borough in four to six was walled.
- Give every region food plus a speciality, and every remote town a reason.
- Add ruins and older layers, pick a date, and keep later inventions off the map.
- Label from big to small, above-right of (or directly above) points, with 2–3 fonts at most. Give each symbol one meaning only, as in the master legend.

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the North; Treaty of York; Eastern Settlement (Norse Greenland); Bryggen; Kutná Hora; Camino de Santiago; Great Lavra.

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Population, density, spacing, travel, transport-cost and castle figures follow the guide’s shared baseline (Bosker, Buringh & van Zanden 2013; Broadberry, Campbell et al. 2015; Masschaele 1993; Dyer; the *Gazetteer of Markets and Fairs*). Figures marked “estimate” are derived by this guide and are not measured values.



Daravel c. 1300, drawn from the layout file with the symbols of the *master legend*. It shows the capital, the 3 cities, all 38 towns and 26 of the ~180 market towns; the ~7,700 villages appear only as texture. Also shown: the 13 cathedrals, the White Keep, the 31 county castles, the 13 other royal and frontier castles, 25 baronial castles, the 9 ports (4 head ports), the 4 bridges on the Ambre and the beacon chain. The two passes take the master legend's saddle mark ("") ("" means a bridge).

CHAPTER 12

WORKED EXAMPLE: POPULATING A KINGDOM

This chapter builds one invented kingdom from an empty terrain map to a full kingdom map and one local map, and then redraws it at four dates. It uses the guide's own numbers, shows every calculation in a table, and links each number to the chapter that owns it. Treat it as a template: put in your own areas and land types and redo the arithmetic. The kingdom, its places and its history are invented; the real examples quoted come from the other chapters and from the sources at the end.

The general method is in [Step by Step](#). The coordinates of every place, road, river and border are in [worked-example-layout.json](#) (canvas 1000 × 800 units, 2 units = 1 km).

IN THIS CHAPTER

The kingdom at a glance

Step 1: Land and population

Step 2: Cities, towns and villages

Step 3: The capital

Step 4: Duchies, counties and the North March

Step 5: Castles and fortifications

Step 6: Roads, rivers and the sea

Step 7: The Church

Step 8: Industry and special towns

Step 9: Ruins and older layers

Step 10: Zoom in on the Leyford area

Step 11: A fantasy variant

Step 12: The same kingdom at four dates

Quick summary

Sources and further reading

The kingdom at a glance

RULE OF THUMB

Draw the terrain first and decide the date. Every number below depends on the land and on the year.

Daravel is a centralised feudal kingdom in the year **1300**, the crowded peak before the Great Famine and the Black Death (see [Change over time](#)).

Feature	Daravel
Area	About 116,500 km ² (45,000 sq mi): about 390 km (240 mi) east to west and 380 km (235 mi) north to south. England is 130,000 km ² .
Coast	East and south, on the Grey Sea
Estuary	The Ambre estuary, 83 km (52 mi) from the open sea to its head
Main river	The Ambre, navigable for river boats 158 km (98 mi) above the estuary; the Lisk joins it at Liskmeet
Mountains	The Fells along the whole west border. The crest is the border. One cart pass (the Wyndgap) and one mule path (the Carrow Gap).
Forest	The Harnwood, about 11,000 km ² in the north
Marsh	The Holm Fens north of the estuary, the Saltings salt marsh south of it, the Wend Levels
Other land	The Whitridge (a limestone ridge cut by the Ambre), Brackenheath (sandy heath), the Wend Downs and Brim Wolds (chalk and limestone hills)
Neighbours	Thelland to the north (hostile), Morvane to the west across the Fells (at peace)

MAP TIP

Before any dot goes down, mark the navigable stretch of each river, the tidal limit, the fords, the pass and the land types. This is step 1 of the method in [Where Settlements Are Built](#).

Step 1: Land and population

Method: measure each land type, multiply by a density from [Population density](#), and add up ([calculator](#), [steps 1–3](#)).

Land type	Area	Density used (range)	People
Fertile lowland (Ambre vale, coastal plains)	44,500 km ² (17,200 sq mi)	35 per km ² (30–40)	1,557,500
Mixed farmland and rolling hills (downs, wolds, foothills)	33,500 km ² (12,900 sq mi)	20 (15–25)	670,000
Heath and poor sandy land	3,000 km ² (1,200 sq mi)	10 (7–15)	30,000
High uplands and mountains (the Fells)	20,500 km ² (7,900 sq mi)	5 (2–5)	102,500
Forest (Harnwood, Kingswood)	11,500 km ² (4,400 sq mi)	4 (1–5)	46,000
Marsh and fen	3,500 km ² (1,400 sq mi)	5 (2–5)	17,500
Total	116,500 km² (45,000 sq mi)	~21 per km² (54 per sq mi)	~2.4 million (range 1.9–2.8 mil- lion)

Checks:

- A realm the size of England should have 2–5 million people ([Realm populations](#)). Daravel, at 2.4 million, is less crowded than England in 1290 (~36 per km²) and close to the European average of about 20 per km² c. 1300. ✓
- **Food:** at about 1 ha (2.5 acres) of ploughland per person, Daravel needs about 24,000 km² of arable, under a third of its 78,000 km² of lowland and mixed land. At 2–3 ha (5–7 acres) of all land per person it needs 48,500–73,000 km² ([Carrying capacity](#)). It has room to grow. ✓

MAP TIP

Shade the map lightly in these six land types before you place anything. The shading is your population map: crowded vale, ordinary hills, thin Fells, nearly empty forest and fen.

Step 2: Cities, towns and villages

Method: choose an urban profile and split the people into tiers, then divide by a typical size ([calculator](#), [steps 4–5](#)).

Daravel uses the **average** profile (England, France and Germany c. 1300).

Tier	Share	People	Typical size	Number of places
Cities of 10,000+	~3%	~73,000	sized by rank (below)	4
Towns of 2,000–10,000	~5%	~121,000	÷ 3,000–4,000	30–40 → 38
Small market towns	~6%	~145,000	÷ ~800	~180
Villages, hamlets, farms	~86%	~2,084,000	÷ 250 (lowland) or 100–150 (hills)	~7,700 villages

Quick-ratio check for an average realm, per million people: 1–2 cities, 12–18 towns, 60–90 market towns, 2,500–3,500

villages. For 2.4 million that is 2–5, 29–44, 145–220 and 6,000–8,500. Daravel fits every row. ✓

The top of the ladder

The capital of a strong kingdom holds about 1–2% of its people and is 3–7 times bigger than the second city ([Rank-size rule](#)). The rank-size rule predicts the 2nd place at $\frac{1}{2}$ of the largest, the 3rd at $\frac{1}{3}$, and so on.

Rank	Place	People	Rank-size prediction	Why it is here
1	Hallowbridge (capital)	40,000 (1.65% of the realm)	—	Lowest bridge on the Ambre, at the tidal limit
2	Liskmeet	12,000	20,000	Confluence; old imperial city
3	Wendmouth	11,000	13,300	River-mouth harbour
4	Brimhaven	10,000	10,000	Wool port
5	Tenterford	8,000	8,000	Cloth town on a fast hill stream
6	Ridgegate	7,000	6,700	Gap town; second bridge
7	Norburgh	6,500	5,700	March fortress and muster town
8	Wyndfoot	5,000	5,000	Pass town below the Wyndgap
9	Holmstow	4,500	4,400	Shrine on a fen island
10–13	Saltwich 4,000 (brine), Hawkridge 4,000 (march fortress), Cheapford 3,800 (ford, head of navigation), Silverhope 3,500 (silver)			

Hallowbridge is 3.3 times Liskmeet: a primate capital, like London c. 1377 at 3–4 times York. Below it the rule fits closely. ✓

The other 29 towns (2,000–3,600 each, 74,700 people in all):

Region	Towns
Crownlands	Kingsmoat 3,000; Ashenden 2,800; Ambremouth 2,400
Duchy of Lisk	Forgeham 2,800; Brockhurst 2,700; Wythmoor 2,600; Harnbury 2,500; Sedgeworth 2,400; Kingsferry 2,200
Duchy of the Fells	Carrowdale 2,400; Leadgill 2,200; Fellstead 2,000
North March	Skelbridge 2,800; Coldhaven 2,200; Scarwick 2,100; Wolfhope 2,000
Earldom of Brim	Woldby 3,300; Gullhaven 3,000; Eastwold 2,800; Stowford 2,500; Brimstead 2,300
Duchy of Wend	Wendbury 3,600; Merrow 3,100; Meremouth 2,800; Abbotsmere 2,800; Downholme 2,700; Bramling 2,400; Seacombe 2,300; Saltcove 2,000

Villages

Rural people per zone are the zone's total minus the townspeople living there. Village sizes follow [step 5 of the calculator](#); spacing uses the hexagon rule from [How far apart](#) (each place “owns” $0.87 \times \text{spacing}^2$).

Zone	Rural people	Share in villages	Typical village	Villages	Land per village	Spacing
Fertile lowland	~1,300,000	90%	250	~4,700	~9.5 km ²	~3.3 km (2 mi)
Mixed farmland	~613,000	75%	200	~2,300	~14.6 km ²	~4.1 km (2.5 mi)
Heath	~29,000	70%	150	~140	~21 km ²	~5 km (3 mi)
Fells	~85,000	50%	125	~340	~60 km ²	~8.3 km (5 mi)
Forest	~44,000	50%	125	~180	~64 km ²	~8.6 km (5 mi)

Zone	Rural people	Share in villages	Typical village	Villages	Land per village	Spacing
Marsh	~13,000	60%	125	~60	~58 km ²	~8 km (5 mi)
Total	~2,084,000	~83%		~7,700		

The other ~355,000 country people live in hamlets (10–75 people) and farmsteads (5–15), mostly in the Fells, the forest and the fen edge.

Spacing checks (Patterns and spacing):

- Lowland villages ~3.3 km apart (rule: 1.5–4 km, 1–2.5 mi); Fell parishes ~60 km² (rule: 50–100+ km²). ✓
- 222 market centres (market towns, towns and cities) give one per ~525 km², about 25 km (15 mi) apart. Add 100–200 villages with a weekly market (chapter 02's

mini-example adds 50–100 for half as many people), and markets in the vale are 14–16 km (9–10 mi) apart: a 6–10 km walk at most. ✓

- The 8 places of 5,000+ are about 130 km (80 mi) apart, like Britain; the 4 cities about 180 km (115 mi), between France and Britain. ✓
- All four cities are on navigable water or the coast. Inland Tenterford stays under 10,000 (the “why is it here?” test). ✓

LATER ERA (1500s+)

By 1600 draw the capital 2–4 times bigger (80,000–160,000) with suburbs far outside its walls, and grow Brimhaven or Wendmouth fastest if ocean trade arrives (Population and bigger cities).

MAP TIP

On the kingdom map, draw the capital, the 3 cities, all 38 towns and only 30–40 of the ~180 market towns (the schematic map at the top shows 26, to keep its labels readable). Show villages as texture in the vale, not as 7,700 dots.

Step 3: The capital

Hallowbridge stands at the lowest bridging point of the Ambre, at the tidal limit, 83 km (52 mi) up the estuary. On tidal coasts big ports sat 50–120 km (30–75 mi) inland (Notes on site types). It meets all five needs from What makes a capital:

Need	Hallowbridge
Rich core region	The lower Ambre vale: about 10,000 km ² of lowland in the Crownlands (the royal domain)
River or sea access	Sea ships to its quays; river boats 158 km (98 mi) up the Ambre. A city of 40,000 needs an overland hinterland of roughly 40–55 km (25–35 mi), so water supply is essential (Feeding towns).
Defensible core	The White Keep on a low bluff on the north bank, with the river in front
Loyalty	The Crownlands are the dynasty's home lands
Legitimacy	Royal burial abbey at Westhallow; coronation in the old city of Liskmeet

It is **not central** (Common mistakes, no. 12). The far north-west corner is 322 km (200 mi) away in a straight line,

about 350–400 km by road: 4–8 days for a royal messenger, a normal kingdom (How big can a realm be).

The royal landscape (Around the capital):

Feature	Place	Distance	Rule
Palace quarter and royal burial abbey	Westhallow	4 km (2.5 mi) upstream	Like Westminster, ~3 km from London; burial church 0–20 km
Royal residences	Elmhurst manor, Redwater Castle, Kingsmoat castle, Kingswood lodge	14, 31, 37 and 43 km (9, 19, 23, 27 mi)	3–6 residences within ~60 km
Royal forest	Kingswood Chase (a great lord's chase that passed to the Crown; the old name stayed)	about 20–70 km (12–43 mi)	10–100 km
Coronation city	Liskmeet cathedral	119 km (74 mi)	50–150 km; Reims is ~130 km from Paris

MAP TIP

Draw Hallowbridge as two linked parts: the walled merchant city with the White Keep at its upstream corner, and the palace and abbey of Westhallow a little upstream. Put the port quays below the bridge, never above it ([Bridges](#)).

Step 4: Duchies, counties and the North March

Method: divide the realm into 4–8 great fiefs, then into counties of 50–70 km (30–45 mi) across, each with a seat near its middle, and add 1–3 special zones ([Political units](#)).

People below come from each region's land types × the densities of step 1.

Region	Area	People	Counties	Seat	Character
Crownlands (royal domain)	13,900 km ² (5,400 sq mi)	~400,000	5	Hallowbridge	Lower Ambre vale
Duchy of Lisk	17,600 km ² (6,800 sq mi)	~420,000	6	Liskmeet	Middle Ambre, Lisk valley, south Harnwood
Duchy of Wend	24,100 km ² (9,300 sq mi)	~650,000	7	Wendmouth	South coast, downs, heath
Earldom of Brim	9,400 km ² (3,600 sq mi)	~240,000	3	Brimhaven	Wolds and north-east coast; small but rich
Duchy of the Fells	25,000 km ² (9,700 sq mi)	~330,000	5	Wyndfoot	Mountains, pass, mines, cloth valley
North March	23,900 km ² (9,200 sq mi)	~365,000	6	Norburgh	Frontier facing Thelland
Liberty of Holmstow	2,600 km ² (1,000 sq mi)	~13,000	1	Holmstow	The abbot's church territory in the fens
Kingdom	116,500 km² (45,000 sq mi)	~2.4 million	33		

Checks: duchies of 9,400–25,000 km² (rule: 10,000–50,000; Flanders was ~10,000 km²). 33 counties of ~3,500 km² (1,370 sq mi), about 59 km (37 mi) across, with ~73,000 people each (England: ~3,300 km² and ~120,000; the guide suggests 30–40 counties for an England-sized realm). ✓ The North March has ~15 people per km² against 27–29 in the Crownlands and Wend: emptier, as a march should be ([Marches](#)).

Smaller units: about 750 hundreds (district courts) of ~155 km², each with 10–11 parishes and an open-air moot site ([Local administration](#)); 150–200 baronies, each with a castle at its head manor (*caput*); one to three manors per village.

MAP TIP

Draw the realm border as a thick dash-dot line, duchy borders as medium dash-dot lines and county borders as thin ones (the schematic map leaves the county borders off to stay readable). Borders are always dash-dot lines, so no one mistakes them for roads ([master legend](#)). Mark each seat

Borders ([Borders and frontiers](#)):

- **West:** the crest of the Fells, with paired castles 17 km (11 mi) apart on either side of the Wyndgap pass.
- **North:** hill crests and the Harnwood in the west, the River Skel in the east, with a hatched **frontier band 30 km (19 mi) deep** (the guide's band is 10–50 km).
- **Oddities:** the **Debatable Land**, a disputed strip 16 × 6 km (10 × 4 mi), and the Liberty of Holmstow.
- **Inside:** duchy borders follow the Whitridge, the foot-hills, the fen and forest edges, never straight lines ([Common mistakes](#), no. 13).

with a small pennant and each county town with a dot in the middle of its town symbol. Label the North March and the Debatable Land by name, fill the Debatable Land with stripes in the two claimants' colours, and give the Liberty its own colour.

Step 5: Castles and fortifications

Method: apply the densities per 1,000 km² from [How many to draw](#) zone by zone. About one-third of all castle sites are in use at once.

Zone	Area	Rate used	Sites ever built	Active c. 1300	Gap between active castles
Peaceful core (Crownlands, Lisk, Wend, Brim, Liberty)	67,600 km ²	6 sites, 2 active per 1,000 km ² (rule: 4–8 and 1–3)	~410	~135	~24 km (15 mi)
Duchy of the Fells (mostly mountain)	25,000 km ²	4 sites, 1 active (low end of the core rule)	~100	~25	~34 km (21 mi)
North March	23,900 km ²	One site per 110–130 km ² , like the Anglo-Scottish border	~200	~65	~20 km (12 mi)
Total	116,500 km ²		~710	~225	

Core castles 20–30 km (12–19 mi) apart match the guide's peaceful core ([Borderlands](#)). Daravel has ~1.9 active castles per 1,000 km² against ~4 in England and Wales: fewer people, and no Welsh-style march of tiny lordships. A gam-

ing rule of one castle per 50,000 people would give only ~48 ([Gaming rules](#)).

What the ~225 active castles are ([Types of fortification](#)):

Type	Number	Where and why (its one-word job)
Royal citadel: great stone tower and curtain wall	1	White Keep, Hallowbridge: <i>town, bridge</i> , royal armoury
County-town castles: stone keeps, shell keeps on old mottes, curtain walls	31	One at the edge of every county town except the capital and Holmstow. Norburgh, Hawkridge and Skelbridge are rebuilt as concentric castles.
Pass fortress (concentric)	1	Wyndgap Castle: <i>pass</i>
Forward line on the north border	10	Small castles and towers at fords and passes, 29–43 km (18–27 mi) apart, about a day's march
Estuary castle	1	Ness Castle at the estuary mouth: <i>harbour</i>
Royal residence castle	1	Redwater Castle
Baronial castles and fortified manor houses	~180	At the head manors of baronies: <i>estate</i>

- **Garrisons:** 5–20 men per ordinary castle, 30–60 at the 14 frontier castles: 1,500–5,000 men in peace, about 0.06–0.2% of the people ([Garrisons and armies](#)).
- **Field army:** 0.2–0.5% of 2.4 million = **5,000–12,000** soldiers; an all-out 1% (~24,000) lasts only weeks.
- **Behind the front:** muster towns at Norburgh and Wyndfoot, the armoury in the White Keep, the fleet at Brimhaven ([Military geography](#)).

LATER ERA (1500s+)

Put bastioned star forts at Skelbridge and Norburgh, a citadel at Hallowbridge and low gun forts at the estuary mouth. Turn most inland castles into ruins, palaces or manor houses ([Gunpowder and the new fortifications](#)).

- **Walls:** stone walls only at the capital, the cities, Ridgigate, Norburgh, Hawkridge, Skelbridge and Wyndfoot; a ditch and gates at Tenterford and Saltwich; the rest open ([Common mistakes, no. 25](#)).
- **Beacons:** 10 beacons about 16 km (10 mi) apart from Skelbridge to the capital (rule: 5–20 km): warning within the hour, against 2–3½ days for a rider ([Beacons](#)).
- **Ruins:** the other ~485 sites are abandoned mottes and ringworks.

MAP TIP

On the kingdom map draw the White Keep, the 31 county castles, the 13 other royal and frontier castles, about one major baronial castle per 2,000–3,000 km² (about 25) and a few grey ruined mottes. Give the five great fortresses (the White Keep, Wyndgap and the three concentric castles) the large three-tower icon and every other castle the small one-tower icon; draw each beacon as a flame dot and each abandoned motte as a grey mound with a tower tick. Leave the other ~155 baronial castles for local maps.

Step 6: Roads, rivers and the sea

Water first (Rivers as roads):

Waterway	Stretch	Length	Traffic
Ambre estuary	Mouth to Hallowbridge	83 km (52 mi)	Sea ships (cogs)
Ambre	Hallowbridge – Ridlegate – Liskmeet – Cheapford	158 km (98 mi)	River boats; towpaths; Cheapford is the head of navigation
Lisk	Liskmeet to Forgeham	~55 km (34 mi)	River boats carrying iron and timber
Brim	Brimhaven to Norburgh	~100 km (60 mi)	Small boats: they supply the march fortress
Wend	Wendmouth to Wendbury	~60 km (37 mi)	Small boats

Main roads (the three royal highways, the old imperial road and the Brim Road; Roads):

Road	Route	Length
The King's Way	Hallowbridge – Leyford – Ridlegate – Liskmeet – Cheapford – Wyndfoot – Wyndgap pass	285 km (177 mi)
The North Road	Hallowbridge – Eastwold – Norburgh – Skelbridge	170 km (106 mi)
The South Road	Hallowbridge – Kingsmoat – Bramling – Wendmouth	173 km (107 mi)
Old Street (an old imperial road)	Liskmeet – Wythmoor – Abbotsmere – Wendmouth, dead straight	167 km (104 mi)
The Brim Road	Eastwold – Woldby – Brimhaven	109 km (68 mi)

About 900 km of main road needs an inn every 15–30 km (10–19 mi): **30–60 stopping places**, most in existing towns and villages (**Inns**). Regional roads (Downs Road, Fells Road, Harn Road, Salt Roads, Pilgrims' Way) link every market town to its neighbours. With them and Fell Street, the roads on the kingdom map add up to about 3,200 km (2,000 mi) inside the realm, close to the guide's 30 km of main road per 1,000 km² (about 3,500 km for Daravel; **How much main road to draw**).

- **Crossings:** the Ambre has only 4 fixed bridges: Hallowbridge, Ridlegate (71 km upstream), Liskmeet (51 km further) and Wyndfoot (96 km further). Between them: fords at Cheapford and Ambrefold, ferries at Kingsferry and Thornwick (**Crossing rivers**). This is chapter 05's pattern for a great river: a few fixed bridges, tens of km apart. A settled lowland river like the Thames had a bridge town about every 20 km (12 mi), which would put about 8 bridges between Hallowbridge and Cheapford instead of 2 (**How often is a river bridged?**).
- **Passes:** the Wyndgap is a low cart pass with a hospice near the summit, a toll, a castle and a chain of villages down to Wyndfoot. The Carrow Gap is a summer mule path with no towns (**Mountain passes**).
- **Tolls:** pontage and port customs at Hallowbridge; bridge tolls at Ridlegate and Liskmeet; the Wyndgap; the Skel bridge into Thelland (**Tolls**).
- **Ports:** about 600 km (370 mi) of open coast needs 4–6 head ports at one per 100–150 km. Daravel has 4 head

ports (the customs ports): Hallowbridge (below the bridge), Brimhaven (royal shipyard), Wendmouth and Gullhaven (herring). Its 5 member ports are Ambremouth (outport and pilots; the only lighthouse stands on Ness Castle at the estuary mouth, about 15 km (10 mi) to the north), Meremouth, Coldhaven, Seacombe and Saltcove. Sea lanes hug the coast; one open-sea crossing runs east, sailed from March to November only, as in the North Sea (**The sea; Sailing seasons**).

Travel times at the guide's speeds per day: walker 25–35 km, ox cart 15–25 km, royal messenger 50–90 km, river boat 40–100 km downstream and 10–20 km upstream, sailing ship 40–100 km.

From Hallowbridge to	Distance	On foot	Ox cart	Messenger	By water
Ridgegate	70 km (44 mi)	2–3 days	3–5 days	1–1½ days	Upstream 3½–7 days
Liskmeet	120 km (75 mi)	3½–5 days	5–8 days	1½–2½ days	Upstream 6–12 days; back downstream 1–3 days
Norburgh	127 km (79 mi)	3½–5 days	5–9 days	1½–2½ days	—
Brimhaven	167 km (104 mi) road; 182 km (113 mi) sea	5–7 days	7–11 days	2–3½ days	Sailing ship 2–5 days
Skelbridge (border)	170 km (106 mi)	5–7 days	7–11 days	2–3½ days	—
Wendmouth	173 km (107 mi) road; 373 km (232 mi) sea	5–7 days	7–12 days	2–3½ days	Sailing ship 4–9 days
Wyndfoot	218 km (135 mi)	6–9 days	9–15 days	2½–4½ days	—
Wyndgap pass	285 km (177 mi)	8–11 days	11–19 days	3–6 days	—

An army with baggage (13–20 km a day) needs 9–13 days to reach the Skel bridge, so the North March must hold out alone for about two weeks.

MAP TIP

Draw the navigable Ambre thicker up to Cheapford, with a small boat at the head of navigation. Draw the 4 bridges as “) (“ with the road running through, each ford as a short dashed line across the river and each ferry as a short dotted line with an “F” ([Symbols and labels](#)). Draw head ports as a large anchor with a quay and write their names in capitals;

member ports get a small anchor. Sea lanes and the mule path are dotted lines; the season goes in the label (“2–5 days by sea, Mar.–Nov.”). Write the travel days on the long roads (“Norburgh 4 days”). Every road must end somewhere: a town, a port, a pass, a mine or a shrine ([Drawing the route network](#)).

Step 7: The Church

Method: use the English densities and rules in [Religious](#), [Cultural](#), [Legal](#) and [Ancient Sites](#).

Feature	Daravel	Rule from chapter 08
Parish churches	~8,000 (7,700 villages plus ~300 in towns and cities; one per ~14.5 km ² , 5.6 sq mi)	~73 per 1,000 km ² would give ~8,500
Dioceses (cathedrals)	13, about 9,000 km ² (3,500 sq mi) each	English style: one per 5,000–10,000 km ²
Archbishop	1, at Liskmeet: the oldest city, not the capital	Oldest or richest city
University	1, at Liskmeet	1–2 for a kingdom of 2–5 million, in the capital or oldest cathedral city
Religious houses	~730 (one per ~160 km ²)	about 1 per 150 km ²
Hospitals open c. 1300	~250: one in each market town and open town, 2–5 in each of the 11 towns and cities with walls or a ditch	Every market town one, every walled town 2–5; at gates, bridges, main roads
Leper houses	~270 (chapter 08's English density, one per ~430 km ²): one outside each town of 2,000+ and most market towns, one on each main road out of the capital and cities	1–2 km outside the gate

Cathedral cities: Liskmeet (archbishop), Hallowbridge, Wendmouth, Brimhaven, Ridgegate, Norburgh, Wyndfoot,

Holmstow (a monastic cathedral, like Ely), Hawkridge, Wendbury, Merrow, Carrowdale and Woldby. In an Eng-

lish-style realm a cathedral marks an important place: 9 of the 13 are cities or towns of 4,000+, and the other 4 are county towns of 2,400–3,600 ([Dioceses](#)).

Religious houses by order (England’s monks, canons and nuns scaled to Daravel’s area; friaries counted town by town and commanderies county by county; [Monasteries](#)):

Order	Number	Where
Benedictine and Cluniac monks	~165	Old towns: Westhallow (royal), Holmstow (shrine), Abbotsmere (monastery town)
Cistercian monks	~60	Empty, watered valleys: Harnvale, Greywater, Coldwater, Stillwater, Mirefield
Carthusians	1	Sallowhope, deep in the Fells. England had only two charterhouses by 1300 (Witham 1178, Hinton 1227).
Augustinian and other canons	~270	Town edges, hospitals, castles
Nunneries	~130	Quiet rural sites
Friaries	~75	In towns: 6 in the capital, 3 per city, 2 per large town, 1–2 in most county towns
Military-order commanderies	~25	One farm estate per inland county

Each big abbey has 5–40 granges (outlying farms), most within about 25 km (15 mi).

The pilgrimage: the shrine of the invented national saint, St Aldwen, is at **Holmstow Abbey** on its fen island. The Pilgrims’ Way runs 87 km (54 mi) from the capital to the

shrine: 67 km (42 mi) to the pilgrim town of Stowford, then about 20 km (12 mi) by causeway or boat. That is 4–5 stages of about 20 km (12 mi), with a hospice at Stowford and a leper house on the last stretch ([Pilgrimage](#)).

MAP TIP

At kingdom scale, show only the 13 cathedrals (a church front with two towers; a double cross beside Liskmeet for the archbishop), the great abbeys (a church with a square cloister beside it), Temple Ambre (a small church with a shield), the shrine (a scallop shell, never a star) and the uni-

versity. A small crown beside Liskmeet labelled “(corona-tions)” and beside Westhallow labelled “(royal tombs)” marks the two royal churches. Save the 8,000 parish churches and the friaries for local and town maps.

Step 8: Industry and special towns

Method: for every industry, find the resource, the fuel and the way out ([The basic rules](#)). The guide suggests 1–3 mining districts, 1–3 salt sources, one cloth region, one or two

fair towns and several ports per kingdom ([Special town types](#)).

Industry	Resource, fuel and way out	Special place (people)
Silver and lead (Mining)	Veins, wood and fast streams in the Fells; road down to Wyndfoot	Silverhope with the royal mint (3,500), Leadgill (2,200), Orsdale (~900), 25–28 km (16–17 mi) apart; mining towns usually have 1,000–5,000
Brine salt (Salt)	Brine springs; fuel by water (timber, fen peat); three Salt Roads, one to an estuary landing 26 km (16 mi) away	Saltwich (4,000)
Sea salt	Salterns boiling sea water on the Saltings; small output, as in the north	Marsh-edge hamlets
Stone and lime (Stone)	Whitridge limestone right beside the Ambre	Chalkhythe quarries and kilns
Iron, charcoal, glass (Forest industries)	Ore, coppice and streams of the Harnwood; iron goes down the Lisk	Forgeham (2,800), forge ponds, a glasshouse

Industry	Resource, fuel and way out	Special place (people)
Wool and cloth (Textiles)	Sheep on the Fells, downs and wolds; fulling mills on the Tenter Water	Tenterford (8,000), Cheapford wool fair in September (3,800), Brimhaven wool exports
Fish (Fishing)	Autumn herring; Saltwich salt	Gullhaven herring fair, late September to November (3,000); fishing hamlets a few km apart
Ships, wine, brick	Harnwood oak via the Brim; south slopes; estuary clay	Royal shipyard at Brimhaven; monastic vineyards at Abbotsmere; brickfields at Hallowbridge

Every region still grows its own grain first; the mining towns buy food from the vale through Wyndfoot ([Common mistakes](#), no. 26).

The fair dates copy real ones. Stourbridge Fair near Cambridge, one of the biggest English fairs, was held around

Holy Cross Day (14 September), after the harvest. The free herring fair at Great Yarmouth began at Michaelmas (29 September) and lasted about 40 days, the main herring season on that coast ([Fishing](#)).

MAP TIP

Give each special place a visible reason: crossed hammers at Silverhope, salt pans (a small checkerboard) along the shore of the Saltings, “(salt)” in Saltwich’s label (a brine town takes the label, not the pan symbol, at kingdom scale), a quarry notch at Chalkhythe, a mill wheel on the Tenter Water, an

anchor with a castle tower for the royal shipyard at Brimhaven. A fair gets no symbol of its own: write the months in the town’s label (“Cheapford, wool fair, Sept.”). Label the three roads out of Saltwich “Salt Road”.

Step 9: Ruins and older layers

RULE OF THUMB

Every medieval landscape stands on older ones. Old things survive far from towns; near towns their stone is carted away ([Ancient and ruined layers](#)).

Layer	In Daravel	Real model
Older empire’s roads	Old Street and Fell Street: straight, still used	Watling Street
Old towns and forts	Liskmeet and Hawkridge stand on old imperial sites; Wendchester is a walled ruin in the fields about 10 km (6 mi) from Wendmouth	Wroxeter, 8 km from Shrewsbury (Reuse)
Hillforts	Hundreds; show 6. Old Harrow above Wyndfoot has a ruined church: the bishop moved down to the river town.	Old Sarum and Salisbury
Barrows and stones	Barrows on the Wend Downs ridge; the Grey Wives stone circle in the Fells	Avebury
Old border	The Thelling Dyke, 25 km (16 mi) inside today’s border	Offa’s Dyke
Abandoned castles	~485 mottes and ringworks; show about 10	
Deserted villages	Few in 1300: Old Harnvale (cleared by Cistercians), Lostwick (royal forest), Skelby (burnt in raids)	Most came after 1350 (Deserted villages)

MAP TIP

Draw ruins in grey with their own symbol and label them “(ruin)”: Wendchester is a grey walled-town symbol, Old Harrow a grey church inside hillfort rings. A deserted village is a grey church alone in a field, labelled “(lost village)”. Keep dotted lines for paths and draw the Thelling Dyke as a grey line with short ticks on the ditch side (here the Thelland

side), labelled with its name. If you redate the map to 1450, add a few more “lost village” symbols in the clay vale (hundreds of villages are gone by then, but a kingdom map shows only a few), shrink the towns inside their walls and turn hill fields back into pasture ([Step 12](#)).

Step 10: Zoom in on the Leyford area

The local map covers 10×10 km (6×6 mi), 100 km^2 (39 sq mi), of rich vale on the south bank of the Ambre, 38 km (24 mi) west of the capital on the King's Way. At 35 people per km^2 it should hold about 3,500 people. It is

centred on the market town of **Leyford**, where the King's Way crosses the Ley Brook.

Features compared with the English averages per 100 km^2 in [How many features](#):

Feature	Leyford area	English average per 100 km^2
Market town	1 (Leyford)	1–2 market places
Villages with a parish church	9, plus Leyford's church	~7 (this is the richest part of the vale)
Hamlets	5	0–5 in open-field lowland
Mills	11 (10 watermills, 1 windmill)	~8–11
Manors	9 lay manors and 1 monastic grange; 5 of them moated	~4–5 moated sites
Deer parks	2 (about 1.1 km and 0.8 km across)	~2–3
Fishponds	2 groups (6 ponds)	at least 1–2
Hundred meeting place	1 (Ley Low, a mound by the King's Way)	~0.5–0.7
Other	Hospital, gallows, 2 wayside crosses, a rabbit warren	—

Households (4.5–5 people each; [Village territory](#)):

Place	Households	People
Leyford (market town: market cross, 2 inns, hospital, bridge)	160	720–800
Stanmere (moated hall, dovecote, fishponds, park)	75	340–375
Thornwick (two manors, Ambre ferry)	70	315–350
Kettlebury, Hollowell, Ashlade (each with a moated manor)	60, 55, 50	740–825
Crossby (windmill), Oxlade (grange nearby)	50, 45	425–475
Upper Ley, Brackenhurst (heath-edge commons)	40, 35	340–375
5 hamlets (Mill End, Fennick End, Heathcot, Woodside, Coldhams)	40	180–200
Millers, warrener, park keepers, grange lay brothers	12	55–60
Total	692	~3,100–3,450 (31–35 per km^2)

How it is laid out:

- **Spacing:** villages 1.5–3.4 km (1–2 mi) apart (rule: 1.5–4 km), all on terraces above the flood plain, each church beside its manor house ([Village forms](#)). The farthest, Brackenhurst, is 5.5 km (3.4 mi) from Leyford market.
- **Rings** ([Rings of land use](#)): tofts and crofts by the houses; two or three open fields within 1.1–1.7 km (0.7–1 mi); meadow on the Ambre flood plain and along the brooks;

four coppice woods at the parish edges; Ley Heath (common pasture and warren) on the sandy south.

- **Mills:** three brooks run north into the Ambre; the Ley's four mills are 1.6–2.8 km apart; the navigable Ambre has no mill weir; Crossby, with no stream, has the windmill ([Mills](#)).
- **Roads:** the King's Way, cleared 200 ft (61 m) each side under the Statute of Winchester; lanes from every village to Leyford like spokes; drove tracks to the heath; a lane to the Thornwick ferry.

MAP TIP

At this scale every square kilometre has an owner and a name. Draw parish boundaries as thin dash-dot-dot lines along brooks and wood edges, put a mill on every village's stream, and keep empty white space only for the heath, which you label.

Step 11: A fantasy variant

Apply the real rules first, then change one input and follow the chain ([The method](#)). Here are two changes.

A. Monsters in the Harnwood ([Dangerous wilderness](#)):

Feature	Realistic Daravel	With monsters
Forest people	~46,000 in ~180 open villages and many hamlets	Perhaps half stay (an estimate), packed into 25–75 walled villages of 300–1,000 at the forest edge; the interior is empty
Refuges	None needed	Refuge forts a day's walk (24–32 km, 15–20 mi) apart round the forest's ~390 km (240 mi) edge: about 12–16 forts
Forest roads	The Lisk Road (154 km) and Fell Street	Cleared about 60 m (200 ft) each side, fortified stops every 15–30 km, patrols
Rangers	—	Lodges at the forest edge, fords and road entries, 15–30 km (10–19 mi) apart
Iron and glass	Forgeham smelts with Harnwood charcoal	Charcoal burners need guards; iron costs more or is imported

B. Dwarves under the Fells ([Dwarves](#)):

- A hold of about 10,000 under Silverhope would need farmland within about 55 km (35 mi) on marginal land. The Fells cannot supply it, so the hold buys food.
- At 200–250 kg of grain per person a year that is 2,000–2,500 tonnes: 17,000–25,000 pack-horse loads of 100–120 kg, or 45–70 loads a day.
- Grain comes by boat to Cheapford, the head of navigation, then 90 km (56 mi) by road to a **gate town** at Fell-

bridge on the valley floor, which grows from a market town into a main Fells town. Metal and coin flow back down.

- A dwarven tunnel under the Wyndgap is a pass with no snow season ([Mountain passes](#)): it takes the traffic and the toll. Real tunnels were short (8–9 km between two Cappadocian underground towns), so draw one, not a network.

FANTASY TWIST

Change only these two inputs. The vale, rivers, capital, counties and 7,700 villages stay where the real rules put them, and that contrast makes the fantasy parts feel real.

MAP TIP

Draw the edge of the haunted Harnwood as a second hatched frontier band, its empty interior in dense cross-hatching (wild land), and the refuge forts along the edge as small castle icons. Draw Fellbridge with a road climbing to a hold gate (a small rounded arch, $\cap$) in the mountainside, and mark the tunnel mouths on both sides of the Wyndgap with the same arch, joined by a thin brown dotted line labelled "(underground)".

Step 12: The same kingdom at four dates

RULE OF THUMB

A new date changes the numbers, not the geography. Rivers, passes, bridges, the capital's site and most village sites stay where they are. The number of people, towns, markets, fields, woods and working castles changes, and some places become ruins.

Why: Each kind of place follows the population, but at its own speed ([Change over time](#)):

- **Villages come first and last longest.** Most lowland villages already existed by 1100 (Domesday Book, 1086, names 13,418 places), so growth mostly made them big-

ger. New villages appeared mainly in forest, fen, heath and upland.

- **Markets, towns, castles and abbeys multiply** in the 1100s and 1200s. In Kent, market places grew from under 20 in 1200 to more than 80 by 1350, if every charter became a working market (McLain 1997). England's religious houses grew from 61 in 1066 to about 400 by 1154 (Webber 2006).
- **After the Black Death** towns shrink inside their walls, the worst fields go back to grass or scrub, and villages on poor land are given up, mostly in the 1400s.
- **After 1500** most of the new growth goes to the capital and the ports ([What changes after 1500](#)).

The scaling factors. England is the best-measured realm, so use its curve unless you have a reason not to:

England	1086	1290–1300	1400	1450	1600
People	1.71 million	4.75 million (1290)	2.08 million	1.90 million	~4.1 million
Share of the 1290 peak	36%	100%	44%	40%	~86%
Arable land, fallow included	—	11.5 million acres (1300)	8.7 million acres (1380)	7.6 million acres	8.9 million acres
Woodland and wood-pasture	~15%	~7% (c. 1349)	stopped falling; scrub on abandoned land	—	—

Sources: people from Broadberry, Campbell and van Leeuwen (2011, Table 6), with 1600 from [chapter 09](#); arable land from Broadberry and others (2010, Table 1); woodland from Rackham's Domesday estimate and a Royal Forestry Society summary ([How much of the map is fields and woods](#)).

England fell further than most of Europe. Europe as a whole went from about 70–80 million in 1300 to about 50–60 million by 1400–1450, a loss of a quarter to a third ([Realm populations](#)). So for 1400, an England-like Daravel keeps about 1.05 million people and a Daravel with the European average keeps 1.6–1.8 million. The table uses about 1.2 million.

Daravel at four dates

The 1300 column is the kingdom of steps 1–11. The other columns are estimates made with the factors above; treat every number as a range.

Tier or feature	1100	1300	1400	1600
People	~1.0 million (0.9–1.4 million)	~2.4 million	~1.2 million (1.0–1.8 million)	~2.4 million (2.0–2.8 million)
Density	~9 per km ² (22 per sq mi)	~21 (54)	~10 (27)	~21 (54)
Capital (Hallowbridge)	10,000–15,000 (1–1.5% of the realm)	40,000	20,000–25,000 (about half, like London between c. 1300 and 1377)	80,000–160,000 (2–4 times its medieval size)
Other cities of 10,000+	0–1 (Liskmeet, the old imperial city)	3	0–1	2–4, the ports first
Towns of 2,000–10,000	10–15	38	10–15 (most small towns drop below 2,000)	30–40
Market centres (market towns, towns and cities)	50–80	~222, plus 100–200 village markets	~200; most village markets lapse	130–160 market places in all

Tier or feature	1100	1300	1400	1600
Villages	6,000–6,500, most of them smaller (100–150 people)	~7,700	7,000–7,500, many half empty; 10–25% gone by 1500	6,000–7,000
Ploughland, fallow included	~10,000 km ² (~9%)	~24,000 km ² (~21%)	16,000–18,000 km ² (a quarter to a third less)	18,000–21,000 km ² (less fallow, better yields)
Woods	15–20%	~10%	~11–12%: scrub and birch on abandoned edges	~10%
Castles in use	300–400, almost all earth and timber	~225; the important ones stone	150–200, plus tower houses in the North March	the royal fortresses, 2–3 star forts, gun forts at the estuary; most others are ruins or houses
Religious houses	~100, almost all Benedictine	~730 (75 friaries)	~700	~730 if the realm stays Catholic; none working if it turns Protestant

How the counts were made:

- **People:** 1300 × England's share of the peak (1100: about 40%), or the European loss for 1400. Density is people ÷ 116,500 km².
- **Capital:** the guide's rule that a strong kingdom's capital holds 1–2% of its people ([Rank-size rule](#)); in 1100 the court still travels, so take the low end. London fell from 60,000–100,000 c. 1300 to about 35,000 in 1377.
- **Villages:** in 1100, about 90% of the vale villages, 80% of the hill villages and a third of the villages in the Fells, forest, fen and heath (~720 in 1300). For 1500, Germany lost about 23.5% of its settlements between 1300 and 1500 (about 170,000 down to 130,000), and England has more than 3,000 known deserted villages ([Deserted villages](#)).
- **Towns:** in England the towns of 1377 were only half to two-thirds of their pre-plague size, and about 30 towns

of 2,000–10,000 remained for 2.5 million people ([The settlement ladder](#)). At that rate Daravel's 3 cities fall into the town tier, and all but one of its 29 small towns (2,000–3,600 people) fall below 2,000. The 1100 count is an estimate in proportion to the people and the markets.

- **Markets in 1100:** about a fifth of the 1300 number, following Kent's growth from under 20 market places in 1200 to more than 80 by 1350.
- **Markets in 1600:** only about 39% of the market places of c. 1300 still worked c. 1600 in England. 39% of Daravel's 320–420 market places is about 130–160.
- **Ploughland:** about 1 ha (2.5 acres) per person at the peak; after the plague, England's arable fell by a quarter (1380) to a third (1450).
- **Castles and abbeys:** see [Types of fortification](#), [Timeline of the changes](#) and [the dissolution of the monasteries](#). The 1100, 1400 and 1600 castle counts are estimates.

What to redraw at each date

Date	Add	Remove, shrink or turn grey
1100	An earth-and-timber motte at most county towns and at many manors; old Benedictine abbeys; a bigger Harnwood and more woods in the clay vale; a travelling court with several royal residences (What exists by which date)	Two-thirds or more of the market towns and towns; all friaries (they arrive in the 1220s) and the Cistercian valley abbeys; stone county castles (only the White Keep stands in stone); the Tenter Water fulling mills (first recorded in England in 1185); most villages in the Fells, fen and heath
1300	Everything in steps 1–11: farms high up the valley sides, suburbs outside the gates	Only the older layers of step 9
1400	Grey "(lost village)" churches on poor land (200–700 of the 7,700 villages are gone or shrunk to hamlets by now, about 800–1,900 by 1500; at kingdom scale draw only a few); sheep pasture on old ploughland; scrub at the forest edges; gardens and empty plots inside the walls; tower houses and fortified manors in the North March	Most village markets; some small towns drop to market-town size; castles no longer kept up turn grey
1600	A capital 2–4 times bigger with suburbs far outside the walls; star forts at Skelbridge and Norburgh, a citadel at Hallowbridge, gun forts at the estuary mouth; a post station every 20–40 km (12–25 mi) on the main roads (about 23–45 on 900 km); country houses in parks	Most inland castles (ruins, palaces or manor houses); if Protestant, the abbeys become grey ruins labelled "(ruin)" or houses called "Abbey"; St Aldwen's shrine is destroyed and the Pilgrims' Way empties; more than half the market places of 1300

LATER ERA (1500S+)

The 1600 column is the medieval map with a new top layer, not a new map. Villages, market spacing and walking distances stay the same; the capital, the ports, the forts and the post roads change ([What changes and what stays the same](#)).

MAP TIP

Keep one base map with the terrain, rivers, roads and village dots. Draw each date as a separate overlay, and change the symbols, not the coastline: smaller or fewer town symbols in 1100 and 1400, grey ruins and lost villages from 1400, star forts and post horns in 1600. Put the date in the title cartouche, because a reader cannot judge the map without it.

Quick summary

- **Land × density:** 116,500 km² (45,000 sq mi) gives about 2.4 million people (1.9–2.8 million), ~21 per km² (54 per sq mi).
- **Average urban profile:** 4 cities, 38 towns, ~180 market towns and ~7,700 villages, 3.3 km (2 mi) apart in the vale.
- **Primate capital:** Hallowbridge, 40,000 (1.65% of the realm, 3.3 times the second city), at the lowest bridge, 83 km (52 mi) up the estuary.
- **Royal landscape:** palace and burial abbey 4 km (2.5 mi) away, 4 more residences within 43 km (27 mi), a royal forest, coronation 119 km (74 mi) away in the archbishop's city.
- **Politics:** 5 great fiefs plus the Crownlands, 33 counties of ~3,500 km² (1,370 sq mi), ~750 hundreds, a march with a 30 km (19 mi) frontier band, a Debatable Land, a church liberty.
- **Castles:** ~710 sites, ~225 in use: 20–30 km (12–19 mi) apart in the core, a forward line every 29–43 km (18–27 mi), a beacon chain, a field army of 5,000–12,000.
- **Routes:** water first; ~900 km of main road with an inn every 15–30 km; 4 bridges on the Ambre; one cart pass; Liskmeet is 3½–5 days' walk from the capital.
- **Church:** ~8,000 parishes, 13 dioceses, ~730 religious houses (~75 friaries), 1 university, 1 great shrine.
- **Industry:** one mining district, two salt sources, one cloth region, two fairs (months in the label), a quarry, an iron forest and 9 ports (4 head ports), each with a visible reason.
- **Older layers and fantasy:** add old roads, grey ruins, lost villages and abandoned mottes; then change one or two fantasy inputs and follow the consequences.
- **Four dates:** about 1.0 million people in 1100, 2.4 million in 1300, 1.2 million in 1400 and 2.4 million again in 1600. Keep the geography; change the counts, the ploughland, the woods, the castles and the ruins.
- **Symbols:** use the [master legend](#): a filled star for the capital, dash-dot lines for borders, solid or dotted lines for routes, “)” for bridges, grey for ruins.

Sources and further reading

All numbers come from the chapters of this guide and the shared baseline ([research/baseline-numbers.md](#)). Their full source lists are in chapters 01 to 11. The works below are the ones this example leans on most.

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CHAPTER 13

QUICK REFERENCE CHEAT SHEET

This sheet puts the guide's working numbers on one page, to keep open while you draw. Every figure is copied from the chapter that owns it, and the link under each table takes you to the full explanation. Numbers describe Europe c. 1300 unless a table says otherwise. The ranges are real: scholars disagree, so pick a value inside the range that fits your region.

IN THIS CHAPTER

The numbers you will use most
 Settlement tiers
 Population density by land type
 Urbanization and how many of each tier
 Spacing
 Travel per day and transport costs
 Political units and sizes
 Capitals checklist
 Fortifications by date and where they go
 Trade goods and route types
 Rural features
 Industry siting
 Religious sites siting
 City sizes by era
 What exists by which date
 Later-era changes (1500–1650)
 Fantasy adjustments
 Master legend and label hierarchy
 Numbers reconciled between chapters
 Sanity checklist: 15 questions

The numbers you will use most

RULE OF THUMB

Almost everyone lives in villages. A “big city” in medieval Europe is anything above 10,000 people. A place of 50,000 is one of the great cities of the known world.

What	Number
People per household (hearth)	4.5–5
Share of people living in the countryside	85–95%
Good lowland at its peak	30–50 people per km ² (80–130 per sq mi)
Whole realm, uplands included	about 7–40 per km ² (18–100 per sq mi); Europe c. 1300 about 20 per km ² (52 per sq mi)
Land per person	about 1 ha (2.5 acres) of ploughland, or 2–3 ha (5–7 acres) of all land
Village	75–500 people (usually 150–300); 1.5–4 km (1–2.5 mi) apart in lowland
Small market town	500–2,000 people; 9–16 km (6–10 mi) apart
Town of 10,000	needs farmland within 21–27 km (13–17 mi) on good land, or water transport
For every city of 10,000+	about 10 towns, 50–100 small market towns and a few thousand villages
Capital of a strong kingdom	1–2% of the realm's people; 3–7 times the second city
Walker / ox cart / army with baggage / royal messenger	25–35 / 15–25 / 13–20 / 50–90 km a day (15–22 / 10–15 / 8–12 / 30–56 mi)

What	Number
Transport cost, land : river : sea	about 8 : 4 : 1 (England c. 1300)
Inns on main roads	every 15–30 km (10–19 mi)
Working castles	20–30 km (12–19 mi) apart in a peaceful core; 4–6 km (2.5–4 mi) on a contested border
Peacetime castle garrison	5–20 men
Field army	0.2–0.5% of the realm's people
Religious houses	about 1 per 150 km ² (60 sq mi)
Cities of 100,000+ in Europe, the Middle East and North Africa	4 (c. 1000), 8 (c. 1300), 8 (c. 1500)

Source: 02 The settlement ladder, 02 How many of each kind, 02 Counting the cities, 02 The rank-size rule and primate cities, 02 Population density, 02 Carrying capacity, 02 Feeding towns, 01 How far apart: the numbers, 05 Speed

per day, 05 Cost: why water beats land, 05 Inns, hospices and caravanserais, 04 Borderlands and conquered land, 04 Garrisons and armies, 08 How many of each, per area and 08 Monasteries, friaries and military orders.

MAP TIP

Write the date in the title box and draw a scale bar before anything else. Every number on this sheet changes with the date (see 02 Change over time) and only means something once you know your scale.

Settlement tiers

RULE OF THUMB

Make the label size follow the tier, not the story. To make a small place look important, give it a special icon (castle, cathedral, shrine), not a bigger label.

Tier	People	Households	What it usually has	Symbol: continent / kingdom / local	Label
Isolated farmstead	5–15	1–2	Own fields and pasture; no church	— / — / tiny dot	none
Hamlet	10–75	2–15	A few houses; a chapel at most; belongs to a village's parish	— / — / 2–5 small house marks	6–7 pt italic, or none
Village	75–500 (English low-land usually 150–300)	15–100	Parish church, manor house, mill, smithy, open fields, common pasture	— / tiny dot (notable villages only) / house marks around a church	7–8 pt
Small market town	500–2,000	100–400	Weekly market, yearly fair, market place, burgage plots, inns, craftsmen, one parish church, often a hospital	— / small open circle / houses around a market square	8–9 pt
Town	2,000–10,000	400–2,000	Usually walled; several parishes, guilds, friaries, hospitals, often a castle, a borough charter, its own council	dot (important ones only) / circle with a ring (battlements on the ring if walled) / outline with wall and gates	9–10 pt
City	10,000–50,000	2,000–10,000	Cathedral, many parishes and friaries, suburbs, guilds, law courts, mint; sometimes a university	dot or circle / circle with a thick battlemented ring (a larger walled symbol) / full outline	11–12 pt, bold or small capitals
Great city	50,000+	10,000+	Capital or major port; several markets, university, foreign merchants' quarters, organised water supply; grain by river or sea	large circle with a ring / large walled symbol / full outline with suburbs	13–14 pt CAPITALS

A capital takes a filled star in place of its tier symbol (see the master legend).

Source: 02 The settlement ladder and 02 Map symbols and labels.

How big to draw it (towns held about 100–200 people per hectare, or 40–80 per acre):

Place	People	Built-up area	Width if roughly round
Village	150–300	5–15 ha (12–37 acres)	250–450 m (0.15–0.3 mi)
Small market town	500–2,000	5–20 ha (12–50 acres)	250–500 m (0.15–0.3 mi)
Town	2,000–10,000	20–80 ha (50–200 acres)	0.5–1 km (0.3–0.6 mi)
City	10,000–50,000	80–300 ha (200–740 acres)	1–2 km (0.6–1.2 mi)
Great city	50,000–200,000	300–1,000+ ha (740–2,500+ acres)	2–4 km (1.2–2.5 mi)

Source: 02 [How big to draw it](#).

Regional exceptions:

Region	What is different
Italy	<i>Castelli</i> (fortified hilltop villages) of 200–1,000 people. Hundreds of small bishoprics, so a cathedral does not prove a big city.
Southern Spain, Sicily, Apulia	<i>Agro-towns</i> of 2,000–10,000 people who are mostly farmers: big in people, not real towns in function.
Germany and the Low Countries	Many chartered “legal towns” had fewer than 2,000 people.
England	Only 17 bishoprics, so a cathedral does mean an important city.
Byzantine lands	Most villages well under 100 households (estimate).

Source: 02 [The settlement ladder](#).

LATER ERA (1500s+)

The ladder stays the same, but the top grows taller: by 1600 Istanbul had perhaps 400,000–700,000 people, and Paris, Naples and London 200,000–300,000 each. See [Later-era changes](#).

FANTASY TWIST

Where monsters roam the countryside, shift hamlet people into fortified villages of 300–1,000, as Italy did in its *incastellamento*. See [Fantasy adjustments](#).

MAP TIP

On a kingdom map at 1 cm = 10 km (about 1 inch = 16 mi), even a great city is only 2–4 mm across. Use symbols at that scale and draw real outlines only on local maps (1 cm = 1 km, about 1 inch = 1.6 mi, or larger).

Population density by land type

RULE OF THUMB

Above about 50 people per km² (130 per sq mi), a region needs towns, specialised commercial farming or imported grain.

Land type	per km ²	per sq mi	Real examples
Rich lowland, peak times	30–50	80–130	Northern Italy 34–40 c. 1300; England’s whole-country average already ~36 in 1290
Very urban, rich lowland	50–75	130–190	Flanders and Holland in the 15th century (estimate)
Average mixed farmland	15–30	40–80	Suffolk ~23–25 in 1086; England ~19 in 1377
Poorer land, young kingdoms	7–15	18–40	Hungary ~7 (c. 1200), ~9–11 (c. 1495); Crown of Poland 8.6 (1370), 13 (1500)

Land type	per km ²	per sq mi	Real examples
Hill country (farmed valleys with upland grazing), peak times	8–15	21–39	Cumberland and Westmorland about 15–17 in 1290 as whole counties (valleys and coastal plain included); about 6–7 in 1377
High uplands, moors, mountains	2–5	5–13	Scottish Highlands, high Alps, Pyrenees, northern Penins (estimate)
Forest or steppe frontier	1–5	3–13	Lithuania, the Rus' forest zone (estimate)
Shifting (swidden) farming in tropical forest	about 6 on average; 12–35 in busier districts; at most about 55–56	about 14 on average; 30–90; at most about 145	1950s world average about 6; modern Southeast Asia from 12 in northern Laos to 35 in northern Thailand
Steppe nomads	under 1–2	under 3–5	Herders need huge pastures
Desert	close to 0	close to 0	Except oases and irrigated valleys
Irrigated river valley	100+ on the farmed strip	260+	Egypt c. 1340
Wet-rice or lake-garden basin (outside Europe)	80–125 over a region; 230–300 in a city-and-rice-field zone	210–325; 600–780	Yangzi delta ~80 (1102); Valley of Mexico ~100–125 (1519); Greater Angkor ~230–300 (13th century)

These are typical values, not limits. Exceptional districts went far above them: Norfolk, a mostly rural county, had about 487,000 people in 1290 (a model estimate) on about 5,400 km² (2,100 sq mi), roughly 90 per km² (230 per sq mi). It still fits the rule of thumb: it had very intensive farming, one of England's largest cities (Norwich) and two busy ports (Lynn and Yarmouth). Keep such a figure for one exceptional district, not a whole realm.

Source: [02 By type of land](#) (including Norfolk, from [Broadberry, Campbell & van Leeuwen 2011](#), Table 8B: 486,920 people in 1290) and [02 Density beyond Europe](#); on Norfolk's farming, Campbell, "Agricultural progress in medieval England: some evidence from eastern Norfolk", *Economic History Review* 36 (1983).

Quick values for the population calculator (people = area × density):

Land	per km ²	per sq mi
Rich lowland	30–40	about 78–104
Average land	15–25	about 39–65
Hill country	8–15	about 21–39
High uplands, mountains, forest and marsh	2–5	about 5–13
Desert	0	0

A hex's area is about $0.866 \times (\text{width across the flat sides})^2$: a 30 km hex is about 780 km², a 30 mi hex about 780 sq mi. Multiply per km² by 2.59 to get per sq mi.

Source: [02 Step-by-step population calculator](#).

Whole realms:

Realm and date	per km ²	per sq mi
England 1086	~13	~34
England 1290 (peak)	~36	~94
England 1377 / 1450 (after the plague)	~19 / ~15	~50 / ~38
France c. 1328	~35–40	~90–105
Byzantium 1025	~7	~19
Poland-Lithuania 1500	6.6	17
Europe c. 1300	~20	~52

Source: 02 By realm and date.

Food: how far a town's farmland reaches (assumes 30 people per km² on good land, 10 on marginal land; real radii are bigger where towns compete):

Town size	Overland radius, good lowland	Overland radius, marginal land	With water transport
1,000	7–8 km (4–5 mi)	~18 km (11 mi)	Not needed
10,000	21–27 km (13–17 mi)	~55 km (35 mi)	Food can come 2–10 times farther along a river or coast
50,000	46–59 km (29–37 mi)	~125 km (78 mi)	Must have river or sea supply
100,000	65–84 km (40–52 mi)	~180 km (110 mi)	Only possible by water

Source: 02 Feeding towns: the hinterland.

MAP TIP

Before placing villages, lightly shade the map into crowded lowland, ordinary farmland, thin hill country and empty land. Then draw a faint circle at the overland radius around each town. If two circles overlap heavily, shrink one town or

put it on water. Do both in pencil or on a layer you hide later, and remove them from the finished map: in the [master legend](#) a soft tint means a language area and a ring around a place belongs to the town symbol.

Urbanization and how many of each tier

RULE OF THUMB

For every city of 10,000+, expect about 10 towns, about 50–100 small market towns and a few thousand villages.

Region	c. 1300	c. 1500
Europe	~5% in cities of 10k+; ~9.5% in all towns	5.6% in 10k+
England	~2% in 10k+ (only London); ~10% in towns of 2k+; 15–20% counting small boroughs	3.1% in 10k+
France	~3–4% in 10k+; ~7–9% in 2k+	~4.2% in 10k+
Low Countries	Flanders ~33–40% in towns of all sizes	Belgium 21%, Netherlands 16% in 10k+; Holland ~45% urban
Northern Italy	~18% in 10k+ (Centre-North); ~20% in 5k+	~15–16% in 10k+
Iberia	~8–10% in 10k+ (inflated by agro-towns)	Spain ~6%, Portugal ~3% in 10k+
Germany	~2–3% in 10k+	3.2% in 10k+
Denmark	~10% in all towns	~14% in all towns
Poland, Hungary	under 2% in 10k+	~1–2% in 10k+
Arab Middle East and North Africa	~7% in 10k+	~8% in 10k+

Source: 02 Urbanization by region.

Urban profiles (choose one per realm):

Share of people	Frontier (Poland, Hungary, Scotland, Scandinavia)	Average (England, France, Germany c. 1300)	Highly urban (Flanders, Lombardy, Tuscany)
Cities 10,000+	~1%	~3%	~15%
Towns 2,000–10,000	~3%	~5%	~12%
Small market towns 500–2,000	~4%	~6%	~8%
Villages, hamlets, farms	~92%	~86%	~65%

Per 1 million people	Frontier	Average	Highly urban
Cities 10,000+	0–1 (largest often 8–15k)	1–2 (largest 15–30k)	5–10
Towns 2,000–10,000	8–12	12–18	30–40
Small market towns	40–60	60–90	80–100
Villages	3,000–4,500, plus many hamlets and farms	2,500–3,500	2,000–2,500

To turn shares into places, divide by a typical size: towns about 3,000–4,000, market towns about 800, villages about 250 in lowland and 100–150 in hill country. The capital of a strong kingdom holds about 1–2% of all its people;

in a fragmented realm the largest city holds about 0.5% or less.

Source: [02 Step-by-step population calculator](#).

A model kingdom: England c. 1300 (4.75 million people on 130,000 km², 50,000 sq mi):

Tier	Approximate number	One per ... people
Parishes (each with a village church)	~9,500	~500
Places with a market grant	~1,750 (many only village markets)	~2,700
Boroughs (places with town status)	about 600, many tiny	~8,000
Towns of 2,000–10,000	roughly 50–60 (estimate)	~80,000–100,000
Cities of 10,000+	~6: London (60–100k) plus about five of 10–15k	~800,000

Source: [02 How many of each kind](#) and [02 The rank-size rule and primate cities](#).

MAP TIP

On a kingdom map you do not draw every village. Draw the capital, the cities, all the towns and the 30–40 most important market towns, then add villages as texture so the viewer feels the crowding. Count your symbols at the end: far more villages than towns, far more towns than cities.

Spacing**RULE OF THUMB**

Many small places close together, a few big places far apart. Stretch the pattern along rivers and roads, squeeze it on rich land and spread it on poor land.

Feature	Typical spacing	Notes
Villages, lowland	1.5–4 km (1–2.5 mi)	One parish per ~14 km ² (5.4 sq mi); about 4 km (2.5 mi) church to church
Parish churches in hills	about 7–10 km (4–6 mi), more in the emptiest uplands	Upland parishes cover 50–100+ km ² (20–40+ sq mi)

Feature	Typical spacing	Notes
Peasant walk to market	6–10 km (4–6 mi) one way	There and back in one day
Market towns, legal rule	6½ mi (10.7 km)	Bracton (13th century); rarely enforced
Market towns, in practice	9–16 km (6–10 mi); up to about 25 km (15 mi) in a thinly settled realm	1,746 English markets by 1300; ~650 working markets in the late 1500s
Towns of 5,000+	Flanders 35–45 km (22–28 mi); northern Italy 50–60 km (31–37 mi); France, Germany, Iberia 80–95 km (50–59 mi); Britain 130+ km (80+ mi); Poland 180–250 km (110–155 mi)	An upper bound from city data
Cities of 10,000+	Flanders and Italy 60–90 km (37–56 mi); France, Germany, Iberia 130–170 km (80–105 mi); Britain and Poland 260–400 km (160–250 mi)	Same basis
Rival city-states	50–90 km (30–55 mi)	Each territory 1–3 days across
Inns and stops on main roads	15–30 km (10–19 mi)	Roman <i>mansiones</i> 25–30 km (16–19 mi); pilgrim stages about 20 km (12 mi)
Caravanserais	30–40 km (19–25 mi) on plains and desert; 10 km (6 mi) or less in mountains	One day's stage
Bridge towns on a settled lowland river	about 20 km (12 mi) of river; gaps 10–35 km (6–22 mi)	Thames c. 1300: one bridge every 20–22 km (12–14 mi)
Head ports	about 1 per 100–150 km (60–95 mi) of coast	15 for all of late medieval England
Galley watering ports	100–300 km (60–190 mi)	Venice–Jaffa stops 100–450 km (60–280 mi) apart
Main roads on a kingdom map	about 30 km per 1,000 km ² (about 48 mi per 1,000 sq mi)	From the Gough Map of Britain
Rhine toll stations, Mainz–Cologne, 1250	about 15 km (9 mi)	At least 5 km (3 mi) apart
Active castles, England and Wales after 1066	one per 250–300 km ² (about 95–115 sq mi); about 16–17 km (10–11 mi) apart	500–600 in use at once
Working castles, peaceful core	20–30 km (12–19 mi) (estimate)	A castle per county town plus a few baronial seats
Castle sites, march or contested border	4–6 km (2.5–4 mi)	Welsh March, 1070s–1280s
Castles in a conquest chain	20–40 km (12–25 mi), one day's march	Each reachable by water
Beacons	about 5–20 km (3–12 mi) in hilly country; 55–100 km (35–60 mi) between high peaks	Estimates
Religious houses	about 1 per 150 km ² (60 sq mi)	Monks in valleys, friars in towns
Leper houses	1–2 km (about 1 mi) outside each town of 2,000+	On a main road
Mills on a good stream	every 1–2 km (0.6–1.2 mi)	About one mill per village
Mining towns in one district	10–40 km (6–25 mi)	2–7 towns plus a mint town
Champagne fair towns	45–65 km (28–40 mi)	Two or three days by road

Source: 01 How far apart: the numbers, 08 Parish churches and chapels, 05 Inns, hospices and caravanserais, 05 How often is a river bridged?, 05 Port ranks, 05 Coastal hopping and open sea, 05 How much main road to draw, 05 Tolls and customs, 04 How many castles and how far apart, 04

Watchtowers, beacons and coastal warning, 08 Monasteries, friaries and military orders, 06 Mills and water management, 07 Mining districts on the map, 07 Fair towns, 03 Drawing the unusual kinds.

How many in a square of 100 × 100 km (62 × 62 mi, 10,000 km² or 3,860 sq mi):

Feature	Region type	Land per place	Number
Parish villages	Lowland farmland	~14 km ²	~700
Parish villages	Upland	50–100+ km ²	100–200
Market places	England at the c. 1300 peak (many tiny)	~75 km ²	~130
Market places	Busy markets only (late 1500s)	~230 km ²	~45
Towns of 5,000+	Flanders / northern Italy / France, Germany, Iberia	1,100–1,750 / 2,200–3,100 / 5,500–7,800 km ²	6–9 / 3–5 / 1–2
Towns of 5,000+	Britain / Poland	15,000+ / 28,000–54,000 km ²	under 1 / 0–1
Cities of 10,000+	Flanders, Italy / France, Germany, Iberia / Britain, Poland	3,100–7,000 / 15,000–25,000 / 59,000–139,000 km ²	1–3 / usually 0–1 / usually none

Source: 01 [How far apart: the numbers](#).

LATER ERA (1500s+)

Add post stations every 20–40 km (12–25 mi) on main roads and, on raided coasts, towers 3–12 km (2–7 mi) apart, each in sight of the next. See 09 [Posts, roads and canals](#) and 09 [Coastal defence](#).

MAP TIP

If two market towns are only 3 km (2 mi) apart, make one of them a village. If two towns on a road are more than 40 km (25 mi) apart, add a village or an inn between them.

Travel per day and transport costs

RULE OF THUMB

Over land a traveller covers about 30 km (19 mi) a day and a loaded ox cart about 20 km (12 mi). By water, goods move at least as fast for a small part of the cost.

Mode	km per day	mi per day	Notes
Walking traveller	25–35	15–22	Pilgrim stages were shorter: about 20 km (12 mi)
Peasant to market	6–10 one way	4–6	Out, trade and home in one day
Ox cart	15–25	10–15	Slow, strong, cheap to feed
Horse cart or wagon	30–40	20–25	Horses can double the speed of oxen
Pack horse or mule train	20–40	12–25	Loads: horse 100–120 kg, mule 150–180 kg, donkey 80–100 kg
Camel caravan	25–40	15–25	About 230 kg per camel
Cattle or sheep drove	16–30	10–20	About 3 km/h (2 mph), grazing on the way; mostly 1700s–1800s evidence
Army with baggage	13–20	8–12	—
Army living off the land	10–26	6–16	—
Forced march, a few days only	30–55	20–35	—
Mounted force with spare horses	55–65	35–40	—
Rider, many days	30–50	20–30	—
Royal messenger with fresh horses	50–90	30–56	30–40 mi normal, 52–56 mi when urgent
Inca relay runners	240–300	150–190	No horses or wheels

Mode	km per day	mi per day	Notes
River boat, downstream	40–100	25–60	Estimate; daylight travel
River boat, upstream, towed	10–20	6–12	Vienna to Passau: about 300 km (190 mi) in 20–31 days
Galley along a coast	about 55 (voyage average with stops)	about 30–35	Venice to Jaffa: 48 days (1480)
Sailing ship (cog or round ship)	40–100 voyage average; 90–220 while under way	25–60; 55–135	Can wait two weeks in port for wind

Source: [05 Speed per day](#); for army speeds, [04 Speed and season](#); for relay runners, [05 Roads without wheels](#).

How long to cross your map (add rest days, Sundays, bad weather and waiting for wind):

Distance	On foot	Ox cart	Courier with fresh horses	Galley	Sailing ship
100 km (62 mi)	3–4 days	4–7 days	1–2 days	about 2 days	1–3 days
300 km (190 mi)	9–12 days	2–3 weeks	3–6 days	5–6 days	3–8 days
1,000 km (620 mi)	4–6 weeks	6–10 weeks	1.5–3 weeks	2.5–3 weeks	1.5–3.5 weeks

Source: [05 How long does it take to cross your map?](#)

Transport costs:

Comparison	Ratio or cost	Where and when
Land : river : sea, per ton-mile	about 8 : 4 : 1	England, 1290s–1340s
Water against land	water about one-tenth of land	England
Sea : river : land	about 1 : 5 : 28–56	Roman Empire, Price Edict of 301 AD
Carting wheat	+3.3% of its value per 10 mi (16 km); price doubled after about 480 km (300 mi)	England c. 1300 (best case: horses, carts, flat land)
Wagon haulage of wheat	+55% per 100 Roman miles; price doubled after about 270 km (170 mi)	Roman Empire
Grain by sea, Exeter to London	about 560 km (340 mi) by sea cost the same as 65–70 km (40–45 mi) by cart	England
Bulk goods overland	more than 30–50 km (19–31 mi) only if valuable, or in war or famine	General rule

Source: [05 Cost: why water beats land](#).

LATER ERA (1500s+)

Relay posts with fresh horses at every stage carried letters up to about 150 km (95 mi) a day. Goods still moved at the old speeds. See [09 The post: relay stations](#).

FANTASY TWIST

A griffon or a teleport circle that carries 100 kg is a pack horse with wings. It changes the speed of news and luxuries, not the price of bread. See [10 Flying mounts and airships](#).

MAP TIP

Write travel days on long routes (“12 days to the capital”). That is how travellers thought, and it shows at once if your map is too big or too small for the story.

Political units and sizes

RULE OF THUMB

Kingdom > duchy > county > barony > manor, each several to a few dozen times smaller than the one above. But medieval units overlapped and were scattered: do not expect a neat pyramid.

Unit	Typical area	Typical population (c. 1300)	Seat
Empire	0.8–1.7 million km ² (300,000–650,000 sq mi)	5–15 million	One great capital, or none (Holy Roman Empire)
Kingdom	10,000–450,000 km ² (4,000–175,000 sq mi)	0.1–17 million	Capital or travelling court
Duchy or principality	10,000–50,000 km ² (4,000–19,000 sq mi)	0.5–2 million (estimates)	Ducal city with castle, cathedral and court
County (shire)	1,000–10,000 km ² (400–4,000 sq mi); England averaged ~3,300 km ²	20,000–300,000; England averaged ~120,000	County town with castle, sheriff and court
Hundred or wapentake	~130–150 km ² (50–60 sq mi) in 1086; ~150–200 km ² (60–80 sq mi) by 1316	2,000–10,000	Open-air meeting place: mound, ford, crossroads, stone
Barony, lordship	a few manors to 100+, often scattered over several counties	—	<i>Caput</i> (head manor), usually with a castle
Knight's fee	1–2 manors; no fixed size	—	Manor house
Manor	2–10 km ²	50–500	Manor house, often moated
Parish	~14 km ² in lowland; 50–100+ km ² in uplands	75–500	Parish church

Source: [03 The units, top to bottom](#).

How many units: England c. 1300 had 1 kingdom, 39 shires, about 630 hundreds (812 traceable in 1086), ~9,500 parishes and 5,000–6,000 knights' fees. France c. 1200 had the royal domain plus 6 great lay fiefs. Hungary had about 70 counties in the 15th century. The Holy Roman Empire in 1521 had about 400 imperial estates. Song China had about

1,230 counties, each about 2,000–2,500 km² (770–970 sq mi), with seats about 45–55 km (28–34 mi) apart. Europe c. 1500 had roughly 500 political units.

Source: [03 How many units in one kingdom?](#), [03 A different model: the bureaucratic empire](#), [03 Realistic realm sizes](#).

Real realms for comparison:

Realm	Area	People	Farthest point from the capital
Andorra	468 km ²	tiny	under 1 day
Navarre	~10,000–12,000 km ²	~100,000 (c. 1300)	2–3 days
Scotland	~79,000 km ²	0.5–1 million (c. 1300)	5–8 days
Portugal	~89,000 km ²	~1.0 million (1500)	4–7 days
England	130,000 km ²	4.75 million (1290)	6–11 days
Poland	~240,000 km ² (1370)	~2 million	6–10 days
Hungary	280,000–325,000 km ²	2–3 million	8–12 days
France	smaller than modern France's 551,000 km ²	16–17 million (1328)	10–14 days
Holy Roman Empire	0.8–1.1 million km ²	11–14 million (c. 1300)	weeks; no single capital
Byzantium (1025)	1.675 million km ²	~12 million	weeks, helped by sea routes

Europe as a whole had 32–40 million people c. 1000, 70–80 million c. 1300 and 70–90 million c. 1500.

Source: [03 Realistic realm sizes](#) and [02 Realm populations and areas](#).

Distance from the capital:

Distance	Royal messenger	Army	What it means
100 km (60 mi)	1–2 days	5–8 days	Core: royal domain, residences, hunting forests
300 km (190 mi)	3–6 days	15–23 days	Counties ruled by royal officers
600 km (370 mi)	7–12 days	30–45 days	Edge of a large kingdom: great lords, marcher lords, governors
1,000+ km (600+ mi)	11–20 days	50–75+ days	Only empires with sea routes, relay posts or semi-independent governors

Direct rule reaches roughly 500–700 km (300–450 mi), about 8–12 days at a messenger's normal pace (a heuristic, not a measured law).

Source: 03 [How big can a realm be.](#)

Kinds of state and how they look:

Kind	How it looks on the map
Feudal kingdom	One outline with a patchwork of great fiefs inside; royal domain around the capital
City-state and its <i>contado</i>	A city with a territory of roughly 20–50 km (12–30 mi) radius; many side by side
League of towns	A network: member towns keep their normal symbols, with names underlined in the league's colour, linked by sea routes; no coloured area
Confederation	A cluster of small units, some jointly ruled subject lands
Composite empire	A big outer border with hundreds of small internal units
Church state	Its own colour, often in scattered pieces
Military-order state	A frontier territory with many castles and commanderies
Maritime empire	A home city plus a chain of coastal dots and islands; only these take its colour
Nomadic empire	Huge vague area: pastures, a few capitals, a ring of tributary states in a lighter tint of the empire's colour

Source: 03 [Kinds of states and how they look on a map.](#)

MAP TIP

For an England-sized kingdom (130,000 km², 50,000 sq mi), draw 30–40 counties, each 50–70 km (30–45 mi) across with its town near the middle, grouped into 4–8 duchies or great fiefs. On a continent map use at least three kinds of state: one or two feudal kingdoms, one fragmented zone and one unusual kind.

Capitals checklist**RULE OF THUMB**

A capital sits in the richest, most loyal region of the realm, on navigable water, at a place that is easy to defend. Centrality matters less than you think.

#	Check	Rule and numbers
1	Rich core region	The court, officials and garrison must be fed from the ruler's own estates
2	River or sea access	A capital grows into a big city, and big cities need grain by water
3	Defensible core site	Hill, ridge, island or river loop for the palace, treasure and archives
4	Loyal land	In the dynasty's home region, which may be off-centre (Paris, London, Kraków)
5	Legitimacy	An old sacred or royal site: coronation church, saint's tomb

#	Check	Rule and numbers
6	Type	Central, core-region, or forward (moved toward a frontier the ruler is pushing)
7	Size	Strong kingdom: 1–2% of the realm's people and 3–7 times the second city. Loose realm: the largest city holds about 0.5% or less, and may not be the capital
8	Fixed or travelling	Before about 1200 most western courts travelled: draw 5–15 royal residences about a day apart (about 30 km, 19 mi)
9	Inside the capital	Palace or castle, cathedral or great abbey, treasury, writing office and archives, law courts, mint, later a university
10	Palace and merchant city	Often two linked parts, such as Westminster about 3 km (2 mi) from London
11	Royal landscape	3–6 royal residences within about 60 km (37 mi); hunting forests 10–100 km (6–60 mi) away; burial church 0–20 km (0–12 mi); a coronation town 50–150 km (30–90 mi) away if you want one
12	Mints	Strong kingdom: the capital, 1–3 church or regional centres and one at each big silver mine. Loose realm: a mint in every princely seat
13	University	One or two per kingdom of 2–5 million, in the capital and the oldest cathedral city
14	Reach	Anything beyond about 12 days' messenger travel (roughly 700 km or 430 mi by road) needs a viceroy or becomes semi-independent

Source: 03 What makes a capital, 03 Travelling courts and fixed capitals, 03 What is in and around a capital, 03 How many mints?, 02 The rank-size rule and primate cities, 08 Universities and schools.

LATER ERA (1500S+)

Fixed capitals win everywhere (Madrid from 1561), and some capitals grow to 5–10 times the second city. See 09 Why capitals grow so much.

FANTASY TWIST

A capital chosen for magic (a ley-line crossing, a dragon-proof mountain) still needs food, water transport and loyal land, unless you show how magic replaces them.

MAP TIP

In a strong kingdom, make the capital the biggest city on the map. In a loose realm, put the court in a middling town and let a trading port or cloth city be bigger: the contrast tells the viewer how the realm works.

Fortifications by date and where they go

RULE OF THUMB

Every fortification controls something: a road, a crossing, a town, a border, an estate, a refuge or a lookout. If you cannot name it, move or delete the fort.

Type	Dates	What it is	Where it goes
Ringwork	10th–12th centuries	Earth bank and ditch with a timber fence; no mound	Early castles; common in south-west England and south Wales after 1066
Motte-and-bailey	From c. 1000 (Loire, Normandy); England after 1066	A motte (man-made earth mound) with a timber tower, and a bailey (fenced yard)	Beside towns, villages and crossings; about 80% of Norman castles in England
Stone keep	From c. 950 in France; England from the 1070s	Massive stone tower; about ten years to build	Royal and great baronial castles
Shell keep	Mostly 11th–12th centuries	Stone ring wall on top of a motte	Rebuilt mottes
Curtain-wall castle	Late 12th–13th centuries	Strong wall with flanking towers	Important royal and baronial sites
Concentric castle	Byzantine 11th century; crusader from the 1160s; Wales 1268–1295	Two or more rings of walls	Great royal and crusader fortresses on conquest frontiers
Tower house	14th–17th centuries	Tall stone house of 3–6 storeys	Raided borderlands: Ireland over 3,000, Scotland about 800, England over 250

Type	Dates	What it is	Where it goes
Peel tower and bastle house	Peels 14th–17th; bastles mostly 16th–17th centuries	Small border tower; fortified farmhouse	Anglo-Scottish border
Fortified manor house	13th–15th centuries	Moat, gatehouse, battlements; stops a raid, not an army	Lowland estates; about 460 English licences to crenellate
Town walls and gates	All periods; big wave 12th–14th centuries	Wall or bank and ditch with gates	Cities, regional capitals, frontier towns, raided ports; English murage tolls from 1220
Citadel	All periods	Fortress at the edge of a city, watching it too	Capitals and big or rebellious cities
Fortified bridge	12th–14th centuries	Bridge with a gate tower	Main-road crossings; collects tolls
Fortified church	13th–17th centuries	Thick walls, strong tower, often a ring wall	Raided villages (Transylvania, Slovenia)
Reused hillfort or Roman fort	Whenever handy	New castle inside old walls	Old Sarum, Pevensey, Portchester
Crusader castle	1100–1291	Large stone castles on spurs and hills	Hostile frontier, few defenders
Islamic <i>ribat</i>	8th–11th centuries	Fortified lodge with a signal tower	Chains along coasts and borders
Military-order castle	12th–15th centuries	Castle and monastery in one	Crusading frontiers (Malbork, Calatrava)
Siege castle	All periods	Fort built to block an enemy castle or town	180–270 m (200–300 yd) from the target, just beyond bow range
Watchtower and beacon	All periods	A handful of men who see, signal and run	Passes, fords, wells, hilltops, headlands

Source: [04 Types of fortification](#).

Where to put a castle:

Site	Why it works
River crossing (ford or bridge)	Every road and every boat must pass
Mountain pass or gap	The only easy way between two regions
Spur or promontory	Cliffs on two or three sides; one ditch across the neck
River bend (meander)	Water on three sides
Confluence	Controls traffic on both rivers
Island or lake shore	Water as a moat; controls the river or lakeside road
Coastal headland or harbour mouth	Supply by ship; watches the harbour
Hill above a town	Protects the town and watches it
Flat wet lowland	Dig a moat or flood the ground (a “water castle”)

In the German lands about two-thirds of castles stood on high ground; in England most stood on low ground beside villages and towns.

Source: [04 Where to put a castle](#).

How many to draw, per 1,000 km² (about 390 sq mi), c. 1300:

Zone	All sites ever built	Active castles	Kingdom map	Local map
Peaceful lowland core	4–8	1–3	County-town castle plus 1 major baronial castle per 2,000–3,000 km ²	All active castles plus ruins
Average (England and Wales)	~12	~4	Royal and major castles, about 1 per 1,000 km ²	All

Zone	All sites ever built	Active castles	Kingdom map	Local map
Hilly land with many small lords (German type)	~25	~8–12 (estimate)	Major and toll castles	All, on spurs and river cliffs
March or contested border	35–60	12–20	A castle-studded band	All, including mottes and towers
Raided border, late Middle Ages	8–10 castles plus many tower houses	Most of them	Castles and the main tower houses	Every tower house and bastle
Thin conquered frontier (Prussia type)	1–3	1–3	All, each paired with a town	All

On a continent map, show only about 1 fortress per 20,000–50,000 km² (8,000–19,000 sq mi). England and Wales had 1,761 castle sites in all, but only 500–600 in use at once; draw one or two ruins or earthworks for every active castle in long-settled land.

Source: [04 How many to draw](#), [04 Counts and densities](#), [04 Ruined and active castles](#).

Garrisons and armies:

Situation	People
Peacetime caretaker garrison	5–20
New frontier castle in peacetime	30–60 (Harlech 1284: 36)
Small castle under siege	20–150
Great castle in a major war	150 to about 1,200 (Kenilworth 1266)
Great crusader fortress	1,500–2,200 (Safed: 1,700 in peace, 2,200 in war)
Frontier fortress city	4,000–5,000 (Tarsus)
Field army	0.2–0.5% of the realm's people; about 1% is an all-out effort that cannot last

Source: [04 Garrisons and armies](#).

LATER ERA (1500s+)

Low, thick bastioned “star” forts appeared in Italy from the late 1400s and spread north in the 1530s–1540s. Only frontiers, capitals, main ports and key crossings got them. See [09 Gunpowder and the new fortifications](#).

MAP TIP

Next to each castle, write its job in one word: *road*, *bridge*, *town*, *border*, *estate*, *refuge* or *watch*. Put the castle at the edge of its town, on the high point or by the river, with the market at its gate.

Trade goods and route types

RULE OF THUMB

The higher the value per kilogram, the farther a good can travel over land. Cheap, heavy goods stay local unless they can float.

Good	Value per weight	How far over land	How far by water
Grain, flour	Very low	Usually under 30–50 km (19–31 mi); routine sales within about 17 km (10 mi)	Hundreds of km by river and sea
Firewood, timber, stone, brick	Very low	A few km	Far by water
Salt	Low	About 100 km (62 mi) or more on “salt roads”	Across Europe

Good	Value per weight	How far over land	How far by water
Wine	Medium	Short distances	Very far
Salted herring, stockfish	Medium	Short to medium	Continent-wide
Wool	Medium to high	By pack horse and cart to ports	England to Flanders and Italy
Metals (copper, tin, silver, iron)	Medium to high	Long distances	Long distances
Cloth, furs, wax, amber	High	Any or long distances	Any or long distances
Spices, silk, dyes, gems, sugar	Very high	Across continents	Across oceans
Live animals	Walk themselves	Hundreds of km along drove roads	—

Source: 05 Bulk goods and luxury goods.

Route types:

Route	What it is	Where
Old Roman road	Straight, engineered, often decayed; still used	Former Roman lands
Royal highway	Main route between cities under the ruler's protection	Between major towns
Local road or lane	Village to market town, village to village	Everywhere
Hollow way	Lane worn below the land, up to 5 m (16 ft) deep	Soft rock on slopes near towns and fords
Causeway	Raised road across marsh or flood plain	Fens, flood plains, bridge approaches
Pack-horse track or bridle way	Narrow track for animals, not carts	Hills and mountains
Drove road	Wide grassy route, 12–27 m (40–90 ft) wide, halts about 16 km (10 mi) apart	From uplands to towns and fairs
Caravan route	Track from well to well and caravanserai to caravanserai	Desert and steppe
Navigable river	Highway with a one-way discount: fast downstream, towed upstream	Up to the head of navigation
Portage	Boats or cargo carried overland between two rivers or round rapids	Watersheds, rapids
Sea lane	Coast-hugging route with short open-sea crossings	Coasts and island chains

As a rough guide, a village has 2–3 roads, a market town 3–5 and a city 5–8 or more. Each mountain range gets 1–3 main passes and 2–5 minor ones.

Source: 05 Kinds of roads, 05 Drove roads, 05 Mountain passes, 05 Which places become hubs?.

Trade nodes, from small to great:

Node	How often	How many
Weekly market	Weekly	Market towns 9–16 km (6–10 mi) apart
Yearly fair	Once a year, days to weeks	Many market towns had one
International fair cycle	A cycle through the year	One or two fair regions per continent at a time
Staple town	Permanent	One per river stretch or export trade
Entrepot (great storage-and-reshipping port)	Permanent	Only a handful in all of Europe and the Mediterranean c. 1300
Merchant colony (<i>Kontor</i> , <i>fondaco</i>)	Permanent	One per foreign nation in each big port

Source: 05 Trade nodes: markets, fairs, staples and entrepots.

Ships, ports and seasons:

Item	Number
Cog (North Sea and Baltic, 12th–14th centuries)	30–200 t; the Bremen cog (1380) about 90–130 t; about 1 sailor per 10 t
Merchant great galley (Mediterranean, c. 1290s–1530s)	about 140–250 t of cargo; up to about 250 crew
Caravel (15th century)	50–60 t
Head ports	15 for all of England; about 1 per 100–150 km (60–95 mi) of coast; several member ports under each; landing places in most creeks; one or two named anchorages
Mediterranean sailing season	roughly April to October
North Sea and Baltic	most winter sailing banned, roughly mid-November to late February
Indian Ocean	one voyage each way per year, timed to the monsoon

Source: 05 Ship types, 05 Port ranks, 05 Sailing seasons.

LATER ERA (1500s+)

Ocean routes open (India by sea 1498; Seville's monopoly of the Americas trade 1503), and Atlantic ports grow fast. See 09 Ocean trade, ports and trading posts.

MAP TIP

For each region, write one line: “exports ..., imports ...”. Then draw the route that carries each export to the nearest water or market. The biggest cities sit where networks meet: where land meets water, at confluences, at both ends of a pass or portage.

Rural features**RULE OF THUMB**

At local scale almost every square kilometre has a human feature. Empty white space means marsh, moor or forest, and should be labelled as such.

What is in a village:

Feature	How common	Where it sits
Parish church	One per parish	Slight rise, often at one end, next to the manor house
Manor house	One per manor (a village could have two or three)	Next to the church; often moated
Mill	About one per village (England c. 1300)	On the stream with its leat; windmill on rising ground
Smithy	Most villages	Main street, green or road
Green, pond	Many villages	In the middle; pond at the lowest point
Pound	Most villages by the 16th century	Edge of the village or green
Tofts and crofts	All villages	House plots along the streets, gardens behind

Source: 06 What is in a village.

Village territory:

Settlement	People	Territory, lowland	Farthest regular fields
Hamlet	10–75	1–5 km ² (0.4–2 sq mi)	under 1 km (0.6 mi)
Small village	75–150	2–7 km ² (0.8–2.7 sq mi)	1–1.5 km (0.6–1 mi)
Typical lowland village	150–300	5–15 km ² (2–6 sq mi)	1.5–2.5 km (1–1.5 mi)

Settlement	People	Territory, lowland	Farthest regular fields
Large village	300–500	10–20 km ² (4–8 sq mi)	2–3 km (1.2–1.9 mi)
Upland parish	100–500, in many hamlets	50–100+ km ² (20–40+ sq mi)	Infields under 1 km away; summer pastures much farther
Agro-town	2,000–10,000	~50–250 km ² (20–100 sq mi)	5–10 km (3–6 mi)

Beyond 3–4 km (2–2.5 mi), people build an outlying farm or turn the land to pasture.

Source: [06 Village territory](#).

Rings of land use around a village:

Ring	Distance from the houses	What is there
1. House plots	0–100 m (0–330 ft)	Tofts and crofts, gardens, orchards, dovecote, fishponds
2. Arable	0.2–2 km (0.1–1.2 mi)	Open fields, or the infield in the uplands
Meadow	Wherever the flood plain is	Hay; follows water, not distance
3. Pasture and commons	1–3 km (0.6–2 mi)	Common pasture, heath, rough grazing
4. Wood and waste	Parish edge, 2–4 km (1.2–2.5 mi)	Coppice, wood-pasture, moor, marsh
5. Summer pasture	5–50+ km (3–30+ mi)	Shielings, alps

Source: [06 Rings of land use around a village](#).

Land budget of a settled lowland kingdom c. 1300:

Land	Share
Ploughland (open fields, fallow included)	35–40% (England ~36% in 1300: 11.5 million acres, 7.4 million sown)
Woodland and wood-pasture	7–15%
Meadow	about 5%
Deer parks	nearly 2%
Villages, gardens and roads	a few per cent
Pasture, heath, moor, fen and marsh	the rest, roughly a third to a half

Hill country ploughs only about 1–10% of its land. After a great plague, cut the ploughland by a quarter to a third.

Source: [06 A land budget for a whole realm](#) and [06 England, the best-measured case](#).

How many features, England c. 1300 averages:

Feature	One per ...	Per 100 km ² (39 sq mi)	On a 20 × 20 km (12 × 12 mi) map
Parish church and village (lowland)	~14 km ²	~7	~25–30
Hamlets and farmsteads	varies	0–5 (open fields) to 20+ (dispersed uplands)	0–100+
Mills	~9–13 km ²	~8–11	~30–45
Deer parks	~40 km ²	~2–3	~10
Moated sites	~22 km ²	~4–5	~15–20
Market places (many tiny)	~75 km ²	~1–2	~5
Hundred meeting places	~150–200 km ²	~0.5–0.7	~2–3
Fishponds (recorded minimum)	~65 km ²	at least 1–2	at least 6

Source: [06 How many features: England c. 1300 averages](#).

Footprints on a local map:

Site	Size across	At 1 cm = 1 km
Deer park	0.5–1 km (0.3–0.6 mi); 40–80 ha (100–200 acres)	5–10 mm
Great Cistercian abbey precinct	~350–600 m (0.2–0.4 mi)	3.5–6 mm
The largest castle (Malbork)	~450 m (0.3 mi)	~4.5 mm
Large baronial castle	~170 m (560 ft)	~2 mm
Friary in a town	~120–140 m (400–460 ft)	~1.5 mm
Small motte-and-bailey castle	roughly 100–150 m (330–490 ft) long (estimate)	~1–1.5 mm
Moated manor site	~40–80 m (130–260 ft)	under 1 mm: use a symbol

Source: 08 Footprints: how big to draw them on a local map.

LATER ERA (1500s+)

About 45% of England was enclosed by 1500, 47% by 1600 and 71% by 1700 (Wordie's estimate). Show hedged fields in sheep country. Polders behind dykes are medieval (Low Countries, from the 11th century), but whole lakes drained by windmills are new (Beemster, by 1612), and so are the drained English Fens. See 09 The countryside: enclosure, new crops and drainage.

MAP TIP

For each village, imagine a circle of about 2 km (1.3 mi) radius: open fields inside, a ribbon of meadow along the stream, woods and heath in the gaps between neighbouring circles. Draw parish boundaries through that outer land, along streams, ridges and wood edges.

Industry siting**RULE OF THUMB**

Industry goes to the heaviest thing it needs: the ore, the fuel or the water power, not the customer. Only the light, finished product travels.

Resource	Where it goes on the map	What settlement it creates
Silver, lead-silver	Old mountains and hill country, up side valleys	Chartered mining town, a mint, mining hamlets around it
Gold	Mountains; river sands	Mining town with a royal mint
Copper	Hills and mountains	Mining town with bare, smoke-killed land around it
Iron	Wooded hills with fast streams	Scattered bloomeries and forges; an iron-trading town where the valley meets a big river
Tin	Moorland valleys	Tinners' camps; "coinage towns" at the edge of the moor
Coal	Coalfields near a coast or navigable river	Pit hamlets; a coal port
Salt from brine	Lowland springs, with wood or peat nearby	A rich salt town; salt roads leading away
Rock salt	Foothills and mountains	Mining town with a royal official's castle
Sea salt	Lagoons and sheltered bays with dry summers	Salt-pan villages and a loading port
Building stone	Hill edges near rivers or the sea	Quarry hamlet; stone shipped by boat
Clay and brick	River valleys, estuaries, stoneless lowlands	Brickfields outside towns; potters' villages
Lime	Limestone edges, building sites, ports	Kilns, not towns
Timber	Upland forests above floatable rivers	Rafting villages; a timber market in a river town
Charcoal, glass, potash, tar	Deep forest	Temporary camps; glasshouses that move

Resource	Where it goes on the map	What settlement it creates
Peat	Wet lowlands with few trees	Turf commons; flooded pits that become lakes
Wool	Downs, limestone hills, uplands, dry plateaus	Sheep villages, monastic farms, wool market towns
Cloth	Towns; fulling mills in hill valleys	Cloth towns; weaving villages along mill streams
Dye plants (woad, madder)	Rich plains near cloth regions	Woad towns and woad mills
Herring	Shallow seas and straits	Seasonal beach markets and curing stations
Cod (stockfish)	Arctic coasts	Fishing stations, drying racks, one big trading port
Ships	Sheltered estuaries near forest and rivers	Shipyards; walled state arsenals in great ports
Wine	South-facing river slopes	Wine villages; an export port downriver
Sugar cane	Irrigated, frost-free coastal plains (Cyprus, Sicily, Valencia)	Fortified estate with mill and boiling house; a loading port

Fuel numbers: very roughly 50–150 kg of wood per 1 kg of bloomery iron; 150–200 kg of wood per 1 kg of forest glass;

a Wealden blast furnace needed the woods of about a 5 km (3 mi) radius.

Source: 07 The master table, 07 Iron, 07 Forest industries.

In and around a town:

Trade	Where	Why
Drinking water, breweries	Upstream, or at wells and springs	Clean water
Grain and fulling mills	At weirs above and below town; on country streams	A fall of water
Butchers	Edge of town, near the river	Waste into the water
Tanners, dyers, flax retting	Downstream and outside the walls	Smell and pollution
Smiths, potters, brick, tile and lime kilns, glass	Outside the walls or at the edge	Fire and smoke
Tenter grounds, rope walks, timber yards	Suburbs and open fields	Space
Smelters, charcoal, glasshouses	At the mine and in the forest	Fuel and ore too heavy to move
Shipyards	Waterfront, at the edge of the port	Shore and space

Source: 07 Where industry sits in and around a town.

Special towns:

Town type	Typical size	Signature on the map
Mining town	1,000–5,000; boom towns 10,000–20,000; camps 20–300	Up a valley; spoil heaps, shafts, mint, big church, little farmland
Salt town	Small to middle-sized, rich for its size	Brine spring or salt mine; salt roads radiating out
Fishing village	50–500; seasonal stations swell in season	Beach or cove; drying racks or smokehouses as labels ("Fish Racks"). Do not draw boats: a small boat marks the head of navigation
Cloth town	2,000–10,000; the Flemish giants 20,000–65,000	Cloth hall, tenter grounds, fulling mills, dyers by the river
Fair town	Often only a small or middle town	Big open fairground, many inns, several roads meeting
Iron or craft town	1,000–10,000	Forges and grinding mills along streams
Wine town	Small; the export port can be large	Vineyards on slopes behind, river quay
Spa town	Small	Hot springs, bath houses, many inns

Per kingdom, aim for 1–3 mining districts (each 2–7 mining towns plus a mint town), 1–3 salt sources, one cloth region, one or two fair towns and several ports. On a continent map, put mining symbols on 3–6 mountain areas.

LATER ERA (1500s+)

Blast furnaces, coal and boom towns: the Weald had about 50 furnaces by the later 1500s, and Joachimsthal reached about 20,000 people by 1534. See [07 What a mining town is like](#) and [09 Industry: furnaces, mines and mills](#).

Source: [07 Special town types and how to spot them](#) and [07 Mining districts on the map](#).

MAP TIP

For every industry, point to three things on the map: the resource, the fuel and the way out (a river, a coast or a road to one). If one is missing, move the industry or add what is missing. Every realm needs salt, or a route that brings it in.

Religious sites siting

RULE OF THUMB

Every village has a church. Every town has a hospital. Every big town has friars. Every region has a few monasteries. Every kingdom has a handful of cathedrals and one or two universities.

How many, per area (England):

Feature	Number in England	One per ...	Per 1,000 km ² (386 sq mi)
Parish church	~9,500 (c. 1300)	~14 km ² (5.4 sq mi)	~73
Religious house	~900 (1530s)	~145 km ² (56 sq mi)	~7
Hospital or almshouse	800–1,100 founded c. 1080–1540	~120–160 km ² (46–62 sq mi)	~6–8
Leper house	~300	~430 km ² (166 sq mi)	~2
Friary	183 (1530s)	~700 km ² (270 sq mi)	~1.4
Cathedral	17 (plus 4 in Wales)	~7,600 km ² (2,950 sq mi)	0.13
University	2	~65,000 km ² (25,000 sq mi)	0.015

Source: [08 How many of each, per area](#).

Where each one goes:

Feature	Where	How many
Parish church	Highest dry spot in the village, next to the manor house; in towns near the market and gates	One per parish; 7–10 km (4–6 mi) apart in hills
Chapel of ease	Hamlets and distant villages of big upland parishes; new clearings	2–4 per upland parish
Cathedral	Main cities (English style) or many small towns (Italian style), with a walled close	English style 1 per 5,000–10,000 km ² (2,000–4,000 sq mi); Italian style 1 per 500–2,000 km ² (190–770 sq mi)
Archbishop	Oldest or richest city, often not the capital	1 per kingdom of about 50,000 km ²
Benedictine abbey	In or beside old towns, often a “monastery town” at its gate	Part of the 1 per 150 km ² (60 sq mi)
Cistercian or Carthusian house	Empty, well-watered valleys, with no village at the gate	Same
Augustinian canons	Near or in towns, at hospitals, shrines and castles	Same
Nunnery	Quiet rural spots	Same
Friary	Inside towns of about 3,000–5,000 and up, near the walls or a gate	4–5 in the biggest city

Feature	Where	How many
Grange (monastery farm)	Most within about 25 km (15 mi) of the abbey, a few much farther	5–40 per big abbey
Military-order commandery	A manor farm with a chapel in good farmland	One per inland county
Hospital or almshouse	Town gates, bridges, main roads	One per market town; 2–5 per walled town
Leper house	1–2 km (about 1 mi) outside the town on a main road, often near a stream	One per town of 2,000+; big cities one per main road
Pilgrim hospice	Every mountain pass and dangerous crossing	Pilgrim stops about 20 km (12 mi) apart
Great shrine	Its own town on existing roads, with inns and hospitals	1–2 per continent; 1–3 per kingdom
University	Capital or oldest cathedral city; avoid towns under about 3,000	0–2 per kingdom of 2–5 million
Gallows	On a hill by the main road outside each town; at the edge of a great lord's land	One per town and great lordship
Moot (assembly) site	A mound, stone or ford roughly central to its district, often on an old road	One per hundred
Jewish or Muslim cemetery	Outside a town gate	One per community
Minority quarter	Near the castle or market of a trading town of 5,000+	One per such town, if you want one

Source: 08 Parish churches and chapels, 08 Dioceses, cathedrals and archbishops, 08 Monasteries, friaries and military orders, 08 Pilgrimage, 08 Hospitals, leper houses and almshouses, 08 Universities and schools, 08 Legal and

civic sites, 08 Burial grounds, 08 Minorities and other faiths, 04 Military-order houses.

Friaries tell you how big a town is (rough guide, France):

Number of friaries	Town size
0	Small towns of 2,000–3,000 usually had none
1	Roughly 5,000 (estimate)
2	About 10,000
4 (one of each main order)	20,000–40,000
5+	The great cities

In England, many county towns of 3,000–5,000 had one or two.

Source: 08 Friaries tell you how big a town is.

Checklist for an English-style kingdom of 50,000 km² (19,000 sq mi), about 1–2 million people:

Feature	How many
Archbishop	1
Cathedrals	5–10 (English style) or 25–100 (Italian style)
Parish churches	~3,500 in lowland; fewer in hills
Religious houses	300–350
Friaries	60–75
Hospitals and almshouses	300–400 over time
Leper houses	100–120
University	0–1

Feature	How many
Great shrine	1–3
Hillforts and barrows (if settled in prehistory)	Hundreds

Source: 08 Drawing it all: symbols, density and scale.

LATER ERA (1500s+)

Protestant lands close their monasteries (England and Wales 1536–41) and end pilgrimage: draw ruined abbeys and country houses called “Abbey” on one side of the faith divide, working monasteries and Jesuit colleges on the other. Show each faith as a soft colour tint, never as a new line. See 09 Religion: the Reformation on the map.

MAP TIP

On a kingdom map show only cathedrals, great abbeys, great shrines and universities. Save parish churches, friaries, hospitals and gallows for local maps.

City sizes by era

RULE OF THUMB

In any period only about 4–8 cities in Europe, North Africa and the Middle East had 100,000+ people. Nearly all were capitals or great ports on the sea or a big river.

Year	Cities of 10k+	Cities of 50k+	Cities of 100k+
c. 1000	~130 (Europe 81, Middle East and North Africa 48)	~11	4
c. 1300	~230 (Europe 172, Middle East and North Africa 60)	~22	8
c. 1500	~250 (Europe 195, Middle East and North Africa 52)	~31	8

Source: 02 Counting the cities.

The biggest cities (thousands of people; central estimate in brackets where ranges are wide):

Rank	c. 1000	c. 1300	c. 1500
1	Constantinople 150–400 (300)	Cairo 220–600 (220)	Cairo 150–400 (180)
2	Baghdad 125–500 (300), declining	Paris 75–270 (250); ~200 is the most common figure	Istanbul 100–280 (280); a 1478 count suggests ~80–100
3	Cairo-Fustat 135–200	Granada 50–150 (150)	Paris 95–250 (200)
4	Córdoba 100–450 (100)	Venice 90–120	Naples 100–150 (125)
5	Seville 70–90	Milan 100–150	Edirne ~100–127
6	Tinnis and Damietta ~80 each	Genoa 85–100	Venice 100–150
7	Palermo 60–75	Constantinople ~100	Milan ~100
8	Alexandria ~60	Florence 95–120	Tunis ~65–100
9	Kairouan, Salerno ~50	Baghdad ~95 (after the 1258 sack)	Aleppo 65–80
10	Venice 17–45	Seville ~90	Granada ~70; Valencia ~75
11	Antioch ~45; Thessaloniki, Regensburg, Damascus ~40	London 60–100	Lisbon, Palermo, Fez, Damascus 55–65
12	Rome 30–35; Kiev 20–45	Fez, Marrakesh, Tunis, Naples, Aleppo, Córdoba 55–70; Ghent 40–65	Genoa, Rome, Florence, Bursa ~55–60; London ~50

Source: 02 The biggest cities.

A middle-sized kingdom: England in 1377 (after the plague; before it, all were about 1.5–2 times larger):

Place	People
London	~35,000
York	~11,000
Bristol	~9,500
Coventry	~7,000
Norwich	~6,000
About 10 more towns	roughly 3,500–5,500 each

Source: 02 [The settlement ladder](#).

Beyond Europe:

City	Date	People
Kaifeng (Northern Song capital)	c. 1100	600,000 to over 1 million (estimates vary)
Hangzhou (Southern Song capital)	1270s	about 186,000 families in a 1270 count; perhaps over 1 million by 1276
Greater Angkor	13th century	700,000–900,000 over about 3,000 km ² (1,160 sq mi)
Tenochtitlan	1519	200,000–250,000 (estimates run to 400,000) on 8–13.5 km ²
Kilwa Kisiwani	14th century	at least 10,000, perhaps 20,000
Great Zimbabwe	13th–15th centuries	under 10,000 (older estimates 18,000–20,000)

Source: 02 [Cities beyond Europe](#).

MAP TIP

On a continent map of a “medieval Europe” world, mark only 3–8 great cities of 50,000+, almost all on coasts, at river mouths or on big navigable rivers, each with a clear reason to be great: a court, a strait, a delta or an export trade.

What exists by which date

RULE OF THUMB

Pick a date for the map. Old things may survive (layers are good), but nothing may appear before its date.

Feature	First appears	Common by
Watermill	Roman times; 5,624–6,000+ in England in 1086	About one per village in England c. 1300 (~10,000–15,000 mills)
Tide mill	Nendrum c. 619; Ebbsfleet 691–692	Estuaries and sheltered coasts
Stone keep	c. 950 in France; England from the 1070s	Important castles by 1300
Motte-and-bailey castle	c. 1000 (Loire, Normandy); England after 1066	11th–12th centuries; many abandoned by 1300
Three-field system	—	Spread in the 12th–13th centuries
Stone bridges	Boom from the 12th century: Regensburg 1135–1146; London 1176–1209	Most important English lowland crossings bridged by the late Middle Ages
Windmill (post mill)	1185, Weedley (Yorkshire); tower mills by c. 1300	13th–14th centuries
Fulling mill	Normandy c. 1086; England 1185	Cloth valleys, 14th–15th centuries
Gotthard pass as a trade route	First Schöllenen bridge, c. 1220	—

Feature	First appears	Common by
Murage (tolls for town walls)	1220 (Shrewsbury)	About 1,470 grants recorded
Friaries	1210s in Italy; England 1221 (Dominicans) and 1224 (Franciscans)	Late 13th century
Bastides (planned towns of south-west France)	1222 (Cordes)	Nearly 700 by 1372
Pontage (bridge tolls)	1228	About 370 grants to the 1440s
Concentric castle	Byzantine 11th century; crusader 1160s	Wales 1268–1295
Moated manor sites	—	Most dug c. 1250–1350
Universities	Bologna (traditionally 1088)	About 20 c. 1300; 60–80 by 1500
Paper mill	Xàtiva (Spain) by the 12th century; Fabriano (Italy) by 1276	—
Gunpowder cannon	First European picture, 1326	Decisive in sieges by the mid-15th century
Tower houses	14th century	14th–17th centuries
Polder (drained land behind a dyke)	Low Countries, 11th century; Rijnland water board 1255	Sea banks and reclaimed marsh spread in the High Middle Ages (Low Countries, the Fens, Romney Marsh)
Navigable canal	Naviglio Grande (Milan): begun 1177, fully navigable 1272	Short canals in Lombardy and the Low Countries; long canals mostly after 1500
Pound lock	China by 984; Vreeswijk 1373 or 1385; Damme 1396	18 on a Milanese canal, 1452–58
Canal over a watershed	Stecknitz Canal, 1391–98 (mostly flash locks)	—
Drainage windmill	1408 (Holland)	—
Hopped beer	North German towns	Low Countries and England, 14th–15th centuries
Sound Dues (Danish strait toll)	1429	—
Blast furnace	Lapphyttan (Sweden), c. 1150–1350; England c. 1490	Later era (see below)
Relay post (Taxis)	1490	Later era (see below)
Printing press	—	236 towns by 1500
Bastioned “star” fort (later era)	Italy, late 1400s	Beyond Italy in the 1530s–1540s
Exchange building (later era)	Antwerp 1531	London 1571; Amsterdam 1611
Pound locks in Britain (later era)	Exeter Canal, 1564–66	—
Dutch fluyt cargo ship (later era)	1595	—
Lake drained as a polder (later era)	Beemster, by 1612	Purmer 1622, Wormer 1626, Schermer 1635
Summit canal with pound locks (later era)	Briare Canal, 1604–42	—
Potato as a field crop (later era)	Spain about 1570	Ireland early 1600s; most of Europe only 1700s–1800s

Source: 06 How many mills, 06 Kinds of mill, 04 Types of fortification, 05 Bridges, 05 Mountain passes, 05 Mills, weirs and locks, 05 Tolls and customs, 07 Fulling mills and water power, 07 Metal and craft towns, 07 Breweries, 07 Iron smelting in the forest, 01 Planted “new towns” and bastides, 08 Monasteries, friaries and military orders, 08

Major universities, 29. Mixing eras without reason, 09 Timeline of the changes, 09 Bigger ships, 09 Exchanges and banking centres, 09 Printing towns, 09 Canals with pound locks, 09 Drainage and new land, 09 New crops, 06 Fields and farming systems, 06 What is in a village (moated

sites), 06 Reclaimed land, 4. A huge city with no water transport (Naviglio Grande).

MAP TIP

Put the date in the title box ("The Kingdom of X, Year 1312"), then run down this table. Use three styles: working features, old features still in use (like an ancient road), and ruins in grey.

Later-era changes (1500–1650)

RULE OF THUMB

After 1500 the bottom of the map stays medieval and the top changes. Villages, fields and market towns look much the same; capitals, ports, fortresses and long-distance links are new or much bigger.

Map feature	Medieval (c. 1000–1500)	1500–1650
Village spacing; market towns; travel speeds	1.5–4 km (1–2.5 mi); 9–16 km (6–10 mi); walker 25–35 km (15–22 mi) a day	Unchanged
Capital	Often a travelling court; 3–7 times the second city in strong kingdoms	Fixed capital, 5–10 times the second city in some realms; draw it 2–4 times its medieval size
Biggest city in the West	Paris ~200,000	Paris, Naples, London 200,000–400,000+; Istanbul 400,000–700,000
Town walls	High and thin	Low, thick, earth-backed, bastioned (rich or threatened towns only)
Castles	Thousands; main military and lordly centre	Ruins, palaces, prisons or manor houses; replaced by forts and country houses
Coastal defence	Walled ports, a few towers	Watchtower chains 3–12 km (2–7 mi) apart; artillery forts at anchorages
Field armies	Usually 5,000–20,000	Total forces of 50,000–150,000+ for great states (paper strength)
Ships	Cogs of 30–200 tons; galleys carrying about 140–250 tons	Carracks and galleons of several hundred to 2,000 tons; Dutch fluyts
Messages	Royal messengers 50–90 km (30–56 mi) a day	Relay posts every 20–40 km (12–25 mi), up to ~150 km (95 mi) a day
Canals	Short and flat, mostly flash locks; the first pound locks (Damme 1396, Milan 1452–58)	Pound locks spread; summit canals; passenger barge canals
Books	Hand-copied	Printed in 1,300+ places by 1650
Countryside	Open fields, commons, deserted villages	More enclosure (47% of England by 1600); serf grain estates in the east
Wetlands	Mostly left wet; sea banks and small polders on the coasts	Whole lakes drained as polders (Beemster by 1612); inland fens drained
Monasteries	Everywhere	Ruined or converted in Protestant lands; flourishing in Catholic lands
Iron and mining	Bloomeries; steady mining towns	Blast furnaces; sudden boom towns (Joachimsthal, Potosi)

Source: 09 What does not change and 09 Comparison table: medieval vs 1500 to 1650.

Big cities after 1500 (thousands):

City	c. 1500	c. 1600	c. 1650
Istanbul	100–280	400–700	400–700
Paris	95–250 (often ~200)	220–300	400–500
Naples	100–150	250–280	~300 before the 1656 plague, then about half
London	~50	~200	375–400
Amsterdam	10–15	50–60 (105 in the 1622 count)	150–175
Seville	45–70	110–150	~130 before the 1649 plague
Lisbon	55–65	110–150	130–170
Madrid	under 20 (not yet capital)	75–95	100–130

Source: 09 [The big cities](#).

New fortifications:

Kind	Typical size	Garrison	Where
Watchtower	10–15 m tall	a handful	Every few km along raided coasts
Coastal artillery fort	30–100 m across	10–30 in peace	Harbour mouths, anchorages, river mouths
Small fort (<i>sconce</i>)	100–250 m across, 4–5 bastions	50–300	River crossings, dykes, marsh roads
Citadel	400–700 m across, 4–6 bastions	several hundred to 2,000+	Edge of a big or rebellious city
Fortress town	1–2.5 km (0.6–1.5 mi) across, 6–12+ bastions	1,000–5,000 in war	Frontiers, key river towns, big ports

Bastions sit roughly 200–350 m apart. When one ruler controls the whole land, castles are cut sharply: France's edict of 1626 ordered interior castles razed; Japan went from about 3,000 working castles to about 170 after 1615.

Source: 09 [Sizes and garrisons](#) and 09 [When a strong ruler cuts the number of castles](#).

MAP TIP

Do not redraw the countryside for a later setting. Keep your medieval villages and market towns, then change the top layer: enlarge the capital and one or two ocean ports, shrink

one silted or blockaded port, add star forts only where they pay, post roads, a canal, a drained lake and faith areas (soft colour tints, never a new border line).

Fantasy adjustments**RULE OF THUMB**

Fantasy does not cancel the rules of geography. It changes the numbers you feed into them. Change one input (water, food, defence, travel cost, threats or resources) and follow the chain of effects.

How danger reshapes settlement:

Threat level	Real analogue	Settlement pattern and defences
Low (wolves, bandits)	England or France c. 1250	The normal map; working castles about 20–30 km (12–19 mi) apart; villages unwallled
Seasonal raids	Anglo-Scottish border, 14th–16th centuries	Lone farms become tower houses; villages huddle
Frequent raids or large monsters	Latium c. 920–1030; Wessex 870s–890s	Walled or hilltop villages of 300–1,000; refuge forts within a day's walk; beacons
Constant danger	The "Great Wilderness" of the Baltic; crusader frontier	Only fortified towns and garrison castles; fields only near walls

Source: [10 How danger reshapes settlement](#).

Key fantasy numbers:

If your world has...	What changes	Numbers
Monsters in the countryside	Fewer lone farms; fields close to a refuge	Fields within about one hour's walk, 4–5 km (2.5–3 mi), of the walls (heuristic)
A "points of light" frontier	Isolated walled settlements	Each point about 50–80 km ² of fields and 1,500–4,000 people
A monster-haunted frontier	Refuge forts manned by the farms around them	Refuges about a day's walk, 24–32 km (15–20 mi), apart; a frontier band of 30–100 km (20–60 mi)
Fertility magic	Higher density, bigger towns	+50% yield: 40–60 per km ² (100–155 per sq mi); about 2× yield: 50–75 per km ² (130–190 per sq mi), like Flanders and Holland
Healing magic	Faster growth, same ceiling	2–3% a year doubles the people in 23–35 years
Flying mounts	Fast messages and luxuries, not bulk	One load about 100–120 kg; a cog's 30–200 tons is roughly 250–2,000 flying loads; a griffon post at 300 km (190 mi) a day is 3–6 times a royal messenger
Portals	Hub towns with customs, warehouses and forts	Only shiploads of grain every day free inland cities from the water rule
Mage academies	Placed like universities	1–3 per kingdom, in cities of 10,000+
Dwarves	Mountain holds that buy food through gate towns	Holds of a few thousand to perhaps 15,000; a hold of 10,000 needs farmland within 21–27 km (13–17 mi) on lowland or about 55 km (35 mi) on marginal land
Elves	Low-density forest realm, wide border wilderness	1–5 per km ² (3–13 per sq mi) or less
Halflings	Dense farm villages, few castles	30–75 per km ² (78–190 per sq mi); villages 1.5–4 km (1–2.5 mi) apart
Steppe "hordes"	Seasonal camps, raid trails, frontier markets	Under 1–2 per km ² (under 3–5 per sq mi)
Dragons	Empty ring around the lair, tribute villages beyond	One cow a week needs a herd of about 250–500 cattle (rough example)
Undead or blights	Like the Black Death	30–60% dead; walled graveyards outside towns; quarantine islands

Use no more than three or four of these in any one region. A fantasy danger frontier (30–100 km, 20–60 mi) is wider than a real march band (10–50 km, 6–30 mi), but both are drawn with the same hatching; truly wild land beyond it takes dense cross-hatching. For the symbols, see the fantasy group of the [master legend](#).

Source: [10 Dangerous wilderness and monsters](#), [10 Suggested symbols](#), [10 Magic, effect by effect](#), [10 Non-human peoples](#), [10 Gods, undead and dangerous seas](#), [10 What changes if: summary table](#), and for the griffon post [03 How big can a realm be](#).

MAP TIP

For each fantasy element, write one line: "X exists, so Y changes" ("griffon riders exist, so castles have roofed towers and every lord keeps an aerie"). Then draw the edges where peoples meet: gate towns, trading stones, frontier markets, tribute villages.

Master legend and label hierarchy

RULE OF THUMB

One symbol, one meaning. A filled star means a capital and nothing else. Borders are always dash-dot lines; routes are always solid or dotted lines; colour fills and tints are for areas only.

The chapters suggest symbols for their own topics. This legend gives one symbol per feature for the whole guide, and chapters 02–11 now use it. If a map tip anywhere seems to use a symbol differently, this legend wins. Scale: C = continent, K = kingdom or region, L = local area. Many small works (kilns, racks, shipyards) take a word, not a pic-

ture: a new little picture easily looks like a symbol that already means something else.

Settlements and power:

Feature	Symbol	Scale
Farmstead	Tiny dot	L
Hamlet	2–5 small house marks, no church	L
Village	Tiny dot (K, notable villages only); house marks around a church (L)	K, L
Village without a parish church (pagan, Sámi, steppe or other non-Christian peoples)	Tiny dot (K); a cluster of hut marks with no church (L)	K, L
Walled hilltop village (Italian <i>castello</i> , or a refuge village in a dangerous land)	Village symbol on the hilltop (a tiny dot at K); a battlemented wall line around the house marks (L)	K, L
Small market town	Small open circle	K
Town	Circle with a ring; battlements on the ring if walled	K
Town with only a bank, ditch and gates	Circle with a plain ring, no battlements (K); at L, the town edge drawn as an earthwork (a line with short ticks on the ditch side) with gaps for the gates	K, L
City	Circle with a thick battlemented ring (a larger walled symbol)	C (dot), K
Great city	Large circle with a ring (C); large walled symbol (K)	C, K
Realm capital	Filled star in place of the tier symbol, sized by tier; name in bold capitals	C, K
Dispersed capital (Angkor type)	Filled star on a walled square core with large rectangular reservoirs (<i>barays</i>); around it a wide zone of tiny house-mound dots and small ponds among rice fields, with no outer wall	K, L
Duchy or regional seat	Tier symbol with a small pennant (flag) on top	K
County town	Tier symbol with a central dot	K
League member town (Hanse type)	Normal tier symbol; name underlined in the league's colour	C, K
Mint or coinage town	No extra symbol: add "(mint)" or "(coinage town)" to the label	K, L
Mining town	Its tier symbol with crossed hammers beside it	K, L
Coronation or royal burial town	Small crown beside the tier symbol, labelled "(coronations)" or "(royal tombs)"	K
Royal residence or royal hunting lodge	Small crown on its own (royal places only)	K, L
Lord's or ranger's lodge (not royal)	Small house mark labelled "Lodge"	L
Planted colonial grid town (1500s+)	Ordinary tier symbol (K); a small square grid of blocks around a plaza, with church and council house on the plaza (L)	K, L
Town after a great plague	Open green space inside the walls	L
Nomad camp	2–3 tiny open half-circles (round tents), never filled; the ruler's camp (<i>ordu</i>) as a large labelled ring of them	K, L
Manor house	Small open square; double outline if moated	L
Moot (assembly) site	Small lone tree, labelled "Moot" (or "Thing" for a Norse or Germanic assembly)	L
Border truce meeting place	Small lone tree, labelled "Truce Day"	L
Gallows	Small "Π" on a hill	L

Military:

Feature	Symbol	Scale
Great royal castle or fortress	Large castle icon (three towers)	C (greatest only), K
Ordinary castle	Small castle icon (one tower with battlements)	K, L

Feature	Symbol	Scale
Toll castle	Small castle icon, with a toll bar and “T” across the river or road	K, L
Refuge fort or burh (no town inside)	Small castle icon; a burh with a town inside uses the town symbol with a battlemented ring	K, L
Motte or small earthwork castle	Small mound with a tower tick on top (grey if abandoned)	L (K for notable or abandoned ones, in grey)
Tower house, peel or bastle	Small solid square	K (border zones), L
Citadel	Small castle icon at the edge of the town ring	K, L
Fortified bridge	Bridge symbol with a small tower	L
Fortified church	Church with battlements	L
Watchtower or coastal tower	Small triangle	K (chains), L
Beacon	Flame dot	K, L
Siege castle	Small dashed castle near its target	L (dated maps)
Army or siege camp	Closed dashed rectangle with rounded corners, labelled with its name or “camp”; grey for an old or Roman camp. A closed shape, never an open line	K, L (dated maps)
Arsenal, naval base or royal shipyard	Anchor plus a castle tower (C, K); at L, a walled yard with slipways next to a rope-walk	C, K, L
Battlefield	Crossed swords and the year	K, L
Star fort or fortress town (1500s+ only)	Five- or six-pointed star outline or a star-edged town outline, never filled	K, L
Bastioned citadel (1500s+ only)	Pentagon outline joined to the town	K, L
Castle refitted with gun bastions (1500s+)	Castle icon inside a star outline	K, L
Castle rebuilt as a palace (1500s+)	Castle icon with gardens, labelled “palace”; a royal palace takes the small crown instead	K, L
Castle kept as a prison or mint	Castle icon in the town, labelled with its new use (“prison”, “mint”)	K, L
Coastal artillery fort (1500s+)	Small round outline with rounded lobes (a rosette, like the plan of Deal Castle, 1539–40), never pointed; a fort with angled bastions takes the star outline	K, L

A walled town needs no military symbol of its own: use the town or city symbol with battlements. A fortified or moated manor house uses the manor-house square (double outline if moated). The coastal artillery fort is a rosette outline

(rounded bastions, like Deal Castle), never pointed; see [09 Coastal defence](#).

Religious and charitable:

Feature	Symbol	Scale
Archbishop's seat	Double cross beside the city symbol	C, K
Cathedral	Church with two towers, or a mitre	K, L
Parish church	Small church with a tower; spire for rich towns	L
Chapel	Small plain cross, no tower	L
Wayside cross	Tiny cross on a base, set on the road or path	L
Abbey or priory	Church with a square cloister beside it	K (great abbeys), L
Friary	Name only, or a small church inside the town outline	L

Feature	Symbol	Scale
Grange	Small open square topped with a cross, or the label "Grange"	L
Military-order commandery	Small church with a shield (a farm commandery); a fortified commandery or an order's frontier castle takes a castle icon	K, L
Great shrine	Scallop shell or reliquary, never a star	C, K
Hospital, almshouse, hospice	Small house with a cross; leper house labelled "Lazar House"	K (pass hospices), L
Cemetery outside the town (Jewish, Muslim, plague)	Small walled plot outside a gate, with a name label (e.g. "Jewbury")	L
University	Book, or "U" beside the city name	C, K
Mosque	Dome with a crescent	K (great mosques), L
Synagogue	Lamp, never a star	L

Economy and countryside:

Feature	Symbol	Scale
Head port	Large anchor with a quay; name in capitals	C, K
Member port	Small anchor	K
Anchorage	Anchor in open water, labelled "the Roads"	K
Ordinary shipyard	The label "Shipyard" on the shore; no symbol	L
Inn or caravanserai	Small house on the road; label "Han" or "Caravanseraï" in desert	K, L
Fair town	Tier symbol; fair months in the label ("Provins: May and September fairs")	K
Mine or mining district	Crossed hammers (or a pick)	C, K, L
Salt pans	Checkerboard of small rectangles along the shore	K, L
Quarry	Notch in a hillside, with a track to water	K (famous ones), L
Lime, brick or tile kilns; clay pits	Words only: "Limekiln", "Brick Kilns", "Clay Pits"	L
Watermill	Wheel on the stream; leat as a thin blue line	L
Water-powered works (forge, furnace, ore-crushing mill, fulling mill, sugar, paper or powder mill)	Watermill wheel on the stream, labelled with the trade ("Forge", "Furnace", "Walk Mill", "Paper Mill", "Powder Mills")	L (K for notable ones)
Hammer or furnace pond	One blue pond with a straight dam at its lower end, labelled "Hammer Pond" or "Furnace Pond"	L
Tide mill	Mill wheel on a dam across a tidal creek	L
Ship mill or bridge mill	Small mill wheels in midstream, labelled "ship mills"; or mill wheels on the bridge symbol. Never boats	L
Windmill	X of four sails on a dot or small mound	L
Forest works (charcoal hearths, glasshouse, tar kilns)	A clearing in the wood symbol, labelled with the trade	L
Peat diggings (turbary)	Commons symbol labelled "Turf Fen"; flooded pits as blue lakes with straight, blocky edges	L
Tenter grounds, fair field, fish racks	Open ground labelled "Tenters", "Fair Field" or "Fish Racks"	L
Royal forest	Green dash-dot boundary with a large italic name; villages and fields inside	K, L
Deer park	Rounded pale (fence) line with scattered trees inside	L
Wood	Tree symbols with a firm edge	K, L

Feature	Symbol	Scale
Wood-pasture	Scattered single trees on grass	L
Open fields	Unhedged areas with blocks of hatching	L
Enclosed fields	Small irregular polygons	L
Tofts and crofts	A row of narrow plot outlines along the street, each with a small house mark at the street end; crofts as long narrow plots behind	L
Meadow	Pale green band along streams	L
Commons, heath, moor	Tufts or stipple	L
Upper limit of hill farming	No line of its own: the field symbols stop and the moor tufts begin. Push this edge downhill after a plague	L
Fishponds	Small rectangles in a chain along a stream	L
Shielings (summer huts)	Cluster of tiny squares on high pasture	L
Vineyards	Hatched strips on south-facing slopes	K, L
Dyke or sea bank	Bold line with short ticks on the water side	K, L
Polder (Low Countries from the 11th century)	Grid of drains inside a dyke; villages along the dykes; drainage mills only from 1408	K, L
Rice paddies	Fine grid of small fields along rivers and deltas, with villages on the dykes and on mounds among them	K, L
Rice terraces	Tight parallel lines that follow the contours	K, L
Chinampas (lake gardens)	Narrow parallel green strips with thin blue canals between them, in the shallow edge of a lake	K (as a band), L
Shifting cultivation	Small villages far apart, with scattered clearings of different ages in the forest around them	K, L
Country house (1500s+)	House inside a ring of park trees	K, L
Overseas trading post (1500s+)	Small fort beside a foreign port city, labelled with the company	C

Water supply and irrigation:

Feature	Symbol	Scale
Well	Blue “W” (the Ordnance Survey abbreviation)	L
Spring	Blue “Spr” (the Ordnance Survey abbreviation)	L
Brine spring or hot spring	Small blue dot labelled “Brine Pit” or “Hot Spring”; on a kingdom map, add “(salt)” or “(baths)” to the town’s label instead	K (label only), L
Cistern	The word “Cistern” in blue; no picture	L
Aqueduct	Thin blue line labelled “Aqueduct”, drawn on a row of small arches where it crosses a valley	K (great ones), L
Qanat	Straight line of tiny rings (shaft mouths with spoil heaps), much smaller than a market-town circle, ending in green gardens	K, L
Oasis	Green belt of gardens and small palm marks around the town or well	K, L
Irrigation channel	Thin blue line drawn like a mill leat, thinner than a canal	L
Tank (reservoir)	Curved dam line with water behind it and a fan of small fields below; Angkor-type <i>barays</i> as large straight-sided rectangles of water	K, L

Routes and crossings (always solid or dotted):

Feature	Symbol	Scale
Royal highway	Thick solid line	K
Regional road	Medium solid line	K
Local road	Thin solid line	L
Hollow way (sunken lane)	Two thin parallel solid lines	L
Causeway	Solid road line on a bank across marsh, flood plain or shallow lake; where it joins an island city to the shore, leave gaps with bridges for boats	K, L
Roman or older-empire road	Straight thick solid line labelled “Street” or “(Roman)”; grey where abandoned	K, L
Steep or stepped section (roads without carts, Andes type)	Steps or small chevrons “>” on the road line	K, L
Track, bridle way, mule path	Thin dotted line	K, L
Drove road	Broad dotted line (two rows of dots)	K, L
Seasonal migration route (nomads, herders, Sámi)	Dotted line with arrows showing the direction of each season’s move; label the season	K, L
Caravan route	Dotted line with a larger dot at each stage	C, K
Pilgrim road	The ordinary road line with small scallop marks along it	K
Raid trail	Red dotted line with arrowheads pointing into the farmland, named (for example the Muravsky Trail, used by Crimean Tatar raids in the 1500s)	C, K
Sea lane	Dotted line at sea, with arrows where the wind sets the direction, and a season label	C, K
Navigable river	Thicker river line up to the head of navigation, with a small boat there	C, K
Portage	Short dotted line between two rivers, labelled with its length	K
Bridge	Two short curved lines “)”(“ with the road running through; label “rope bridge” where it is one	K, L
Ford	Short dashed line across the river	K, L
Mountain pass	Small saddle mark (a shallow “U” between two hill ticks) on the route, with the pass name and height; never “)(”, which is a bridge	K, L
Ferry	Short dotted line across the river with “F”	K, L
Toll	One bar across the road or river with a “T”	K, L
Canal	Straight blue line. Medieval canals were short (Naviglio Grande, Milan, 1177–1272; Stecknitz Canal, 1391–98); long canals with many pound locks belong mostly after 1500	K, L
Lock	Two short bars close together across the river or canal	L
Lighthouse	Small tower with short rays; never a star	K, L
Post station (1500s+)	Small horn on the road	K

Boundaries (always the dash-dot family):

Feature	Symbol	Scale
Realm border	Thick dash-dot line, or a coloured band along the inside edge	C, K
Duchy or great fief	Medium dash-dot line	C, K
County	Thin dash-dot line	K
Hundred, parish, manor	Thin dash-dot-dot line	K (hundreds, on a county map), L
Boundary stone	The letters “BS” on the boundary line, as on Ordnance Survey maps (which also use “BP” for a boundary post)	L
Disputed border	Two parallel dash-dot lines, or alternating colours	K

Feature	Symbol	Scale
March or frontier zone	Hatching, or a fading colour band 10–50 km (6–30 mi) wide	C, K
Enclave	Owner's colour with a thin outline	K
Language or culture area	Soft colour tint only, never a line	C, K
Faith area (mainly 1500s+)	Soft colour tint only, never a new line	C, K

The march band is the frontier itself. A whole military zone, from the forward forts back to the fortress towns a day or two behind them, can be 30–80 km (20–50 mi) deep, so fortress towns may lie behind the hatched band. A fantasy

danger frontier is a wider band than a real march: 30–100 km (20–60 mi).

Area colours and tints:

Feature	How to colour it	Scale
Realm	One pale flat colour per realm, so roads and towns stay readable	C, K
Vassal duchy	A lighter shade of its king's colour	C, K
Church land, palatinate, free city	Its own colour; do not merge it into the kingdom around it	K
League of towns	No area colour: underline the member towns' names in the league's colour	C, K
Nomadic empire	Its colour on the core pastures; the tributary ring in a lighter tint of the same colour	C
Maritime empire	Its colour only on its islands and harbour towns; the land behind keeps the local ruler's colour	C, K
Language, culture or faith areas	A separate layer of soft tints in a hue family that no realm uses (for example, realms in warm colours and languages in cool ones), or a small inset map	C, K
Frontier and wild land	Hatching (march) or dense cross-hatching (fantasy wild land); never a plain tint	C, K
Drafting aids: density zones, hinterland circles, messenger-day lines	Pencil or a hidden layer only; remove before finishing	—

Antiquities and older layers:

Feature	Symbol	Scale
Any ruin	The feature's own symbol, broken or grey, labelled "(ruin)"	K (big ones), L
Deserted village	Grey church alone in a field, labelled "(lost village)"	K (a few), L
Drowned village	Grey church standing in the water, labelled "(lost village)"	K (a few), L
Old border dyke or linear earthwork (Offa's Dyke type)	Line with short ticks on the ditch side, grey if no longer used, always labelled with its name	K, L
Hillfort	Oval of rings or hachures on a hill	L (K for big or famous ones)
Barrow	Small filled mound; long barrow as a longer mound	L
Standing stone	Small upright tick	L
Stone circle	Small ring of dots	L
Old or Roman camp	Grey closed dashed rectangle with rounded corners, labelled "Camp"	L

Fantasy (only if your world has them):

Feature	Symbol	Scale
Wild, monster-held land	Dense cross-hatching (the frontier in front of it takes the march hatching)	C, K
Monster lair	Small skull with the creature's name; leave an empty ring around it	K, L

Feature	Symbol	Scale
Blighted or dead land	Pale grey wash with small black dead-tree ticks, labelled (“the Blight”), with watchtower triangles along its edge	C, K
Mage tower or weather tower	Slim tower topped with a small open diamond (no battlements, no spire)	K, L
Academy of magic	Book beside the city name (the university symbol), labelled “Academy”	C, K
Ley line and nexus	Straight chain of small open diamonds in one colour (for example violet); a larger open diamond at the nexus	C, K
Ward stones	Small solid diamonds spaced around the warded area	L
Magical spring or sacred well	Small blue drop, with a green ring of gardens around the town	K, L
Portal	Small spiral beside the town symbol	C, K
Aerie	Small pair of wings on a crag or on a castle icon	K, L
Dwarf hold gate	Small rounded arch (n) in the mountainside	K, L
Smelter	Small chimney with a curl of smoke	L
Underground road	Thin brown dotted line between hold gates, labelled “(underground)”	K
Frontier market or trading stone	Small pair of scales, labelled with the market days or “Trading Stone”	K, L
Walled graveyard	Small rectangle with a solid wall line and tiny crosses inside, labelled “Graveyard”	L

Source: 02 Map symbols and labels, 02 Beyond Europe, 02 Change over time, 03 Symbols and line styles, 03 Drawing the unusual kinds, 03 Colour and labels, 04 Suggested symbols, 04 Frontier posts, 05 Symbols and labels, 05 Kinds of roads, 06 Suggested symbols, 07 Suggested symbols, 08 Suggested symbols, 08 Burial grounds, 09 What happened

to the old castles, 09 Planted colonial towns, 09 Symbols and labels, 10 Suggested symbols, 11 Symbols for the fixes in this chapter. Ordnance Survey abbreviations (“BS”, “BP”, “W”, “Spr”) are cited in chapters 03 and 11.

Look-alike symbols: keep them apart.

Shape	Use it only for	Not for
Star	Filled = realm capital; outline = star fort (1500s+)	Great city (large ringed circle), citadel (castle icon at the town edge), shrine (scallop), synagogue (lamp), lighthouse (tower with rays), portal (spiral)
Ring or dot in a circle	Ring = town; central dot = county town	Motte (mound with a tower tick), moot (lone tree), hinterland circles (pencil only)
Small square	Solid = tower house; open = manor house	Farmstead (tiny dot); tofts (narrow plots with a house mark)
Small rectangles in a chain	Fishponds	Hammer ponds (one pond, straight dam, labelled)
Triangle	Watchtower	Beacon (flame dot), barrow (filled mound), standing stone (tick)
Cross	Church or chapel	Windmill (X of sails on a dot)
Bars across a line	One bar with “T” = toll; two bars = lock	Bridge (“/”)
Small boat	Head of navigation	Ferry (dotted line with “F”), ship mills (wheels in midstream), fishing villages (labelled racks)
Dotted line	Routes and tracks	Borders and royal forests (dash-dot), pilgrim roads (shell marks), ley lines (diamonds)
Dashed line	Fords; closed camp outlines	Duchy borders (dash-dot); sea lanes, drove roads, portages (dotted)
Line with ticks	Dyke or sea bank (ticks on the water side); old earthwork (ticks on the ditch side, grey if old, labelled)	Highway (plain thick line)
Soft colour tint	Language, culture or faith areas	Realms (pale flat fills), frontiers (hatching), density zones (pencil)

Shape	Use it only for	Not for
Crown	Royal places only: alone = residence or lodge; beside a town = coronation or burial town	Duchy seat (pennant), lord's lodge ("Lodge")
Arch shape	∩ in a mountainside = dwarf hold gate	Gallows ("Π", flat-topped, on a hill by a road)

Label hierarchy:

Feature	Lettering	Size
Realm	Widely spaced CAPITALS across the whole territory	Largest on the map
Duchy or great fief	Smaller italic capitals	Below realm
County	Lower case	Below duchy
Great city	CAPITALS	13–14 pt
City	Bold or small capitals	11–12 pt
Town	Regular	9–10 pt
Small market town	Regular	8–9 pt
Village	Regular; label only notable ones at kingdom scale	7–8 pt
Hamlet	Italic, or none	6–7 pt
Rivers, lakes, seas	Italic, traditionally blue, along a straight stretch of the line	By importance
Marches and empty zones	Named in the area ("the West March", "moor", "fen")	Area label

Rules: nothing under about 6 pt in print; 2–3 typefaces at most; upper case only for realms and the largest features; put point labels above and to the right of the symbol, or directly above; keep area labels inside the area, curved to its shape, with letter gaps no more than about 4 times the letter height; place labels last, biggest first.

Older layers: Ordnance Survey maps have long used separate typefaces for antiquities of different periods (pre-

Roman, Roman and medieval), so ruins read as ruins without extra symbols. Copy the idea: pick one extra typeface for your older layers, and use it only for them. Saxton's printed county atlas of 1579 shows the period look: towns and villages as small church-tower symbols.

Source: [30. Label clutter](#), [03 Colour and labels](#), [02 Map symbols and labels](#), [08 Church towers as landmarks](#).

MAP TIP

Print the legend beside your map and test it: cover the labels and ask a friend what each symbol means. Any symbol they read two ways needs changing.

Numbers reconciled between chapters

Where chapters gave different values, this sheet uses the owner chapter's value. Most of the old disagreements are now fixed in the chapters; the last column says what the

other chapters say now (and what earlier drafts said), and the last two rows are small differences that remain.

Topic	Value used on this sheet	Owner chapter	What other chapters say
Cog capacity	30–200 t; Bremen cog (1380) about 90–130 t	05 Ship types	09 ("30–200 tons; a few were larger") and 10 now agree; earlier drafts said 100–300 t and 40–200 t
Merchant galley cargo	about 140–250 t	05 Ship types	09 agrees
Hundreds (English districts)	~130–150 km ² (50–60 sq mi) in 1086, with 812 traceable; ~150–200 km ² (60–80 sq mi) by 1316, with about 630 named	03 The units, top to bottom	06 gives the 1316 figure only
English arable c. 1300	11.5 million acres including fallow (about 36% of England), 7.4 million sown	06 England, the best-measured case	02 now agrees: about 2.4 acres (1.0 ha) of arable per person (an earlier draft said 10 million acres)

Topic	Value used on this sheet	Owner chapter	What other chapters say
Peasant walk to market	6–10 km (4–6 mi) one way	05 Speed per day	01 now agrees (earlier 4–7 mi)
Active castles, England and Wales after 1066	one per 250–300 km ² (about 95–115 sq mi); about 16–17 km (10–11 mi) apart	04 Counts and densities	08 and 11 now agree (earlier drafts printed “100–115 sq mi” and a spacing of 15–18 km)
Castle sites, England and Wales	1,761 (Gatehouse gazetteer)	04 Counts and densities	08 and 11 now say about 1,700–1,800 (1,761)
Castle spacing on a contested border	4–6 km (2.5–4 mi)	04 Borderlands and conquered land	04’s county table and 11 now agree (an earlier draft gave 4–5 km)
Swidden (shifting) farming density	about 6 per km ² (14 per sq mi) on average; 12–35 (30–90) in busier districts; at most about 55–56 (about 145)	02 By type of land	01 now agrees (earlier drafts gave only 6 or only 10–35)
Parish churches in hills	about 7–10 km (4–6 mi) apart	08 Parish churches and chapels	02 and 00 Step 14 now agree (00 earlier said 8–15 km)
Good lowland density	30–50 per km ² (80–130 per sq mi)	02 Population density	10 now agrees (it earlier converted it as 78–130 per sq mi)
Hill and upland density	Hill country 8–15 per km ² (21–39 per sq mi); high uplands, moors and mountains 2–5 (5–13)	02 By type of land	00 Step 2 and 01’s terrain table now agree (they earlier gave one “uplands and mountains” value of 2–10, or 5–25 per sq mi)
Inns on main roads	15–30 km (10–19 mi)	05 Inns, hospices and caravanserais	01 and 11 now agree (they earlier converted it as 9–19 mi)
Towns over 10,000 on water	Most, with inland exceptions: Coventry (~12,000), Siena (perhaps 50,000), Granada, silver boom towns	02 Feeding towns	05 now points to 02’s exceptions; 07 adds the boom towns
Towns per city of 10,000+	about 10	02 How many of each kind	11 now agrees (earlier roughly 5–15)
Ghent in the 14th century	40,000–65,000 (estimates vary widely)	02 The biggest cities	07 now agrees (earlier 55,000–70,000)
Market towns, in practice	9–16 km (6–10 mi)	01 How far apart	02 and 10 now agree (earlier 10–16 km)
Main road on a kingdom map	about 30 km per 1,000 km ² (48 mi per 1,000 sq mi)	05 How much main road to draw	11 now agrees (about 190 red lines on the map; about 4,540 km of real road traced behind them; 48 mi per 1,000 sq mi)
Corsican watchtowers	about 85–90 (a 1617 list counts 86)	04 Watchtowers	01 and 09 now agree
Religious houses, England	~900 in the 1530s: one per ~145 km ² (56 sq mi); for drawing, about 1 per 150 km ² (60 sq mi)	08 How many of each, per area	02 and 08 agree
Joachimsthal in 1534	about 20,000	07 What a mining town is like	07 and 09 agree

MAP TIP

When two numbers disagree, use the range from the chapter linked in the “Owner” column, and pick a value inside it that suits your region.

Sanity checklist: 15 questions

MAP TIP

Run this check when the map is finished. Every “no” is a place to look again before you add a single label.

#	Question	If the answer is “no”	Read
1	Does every settlement touch fresh water (river, stream, spring, lake, coast, or a drawn well or cistern)?	Move it to water, or draw the engineering that brings water	01 Fresh water
2	Does every desert town have a river, oasis or qanat, with green gardens around it?	Add the water source and its green belt	2. A desert city with no oasis or river
3	Does every town have villages and farmland within its radius (21–27 km, 13–17 mi, for a town of 10,000 on good land)?	Draw the farmland ring, or shrink the town	02 Feeding towns
4	Is every city over 10,000 on navigable water or a coast, or explained by a canal, irrigated plain, rich farmland or silver?	Move it to water or make it smaller	4. A huge city with no water transport
5	Are there far more villages than towns and far more towns than cities (about 10 towns and 50–100 market towns per city of 10,000+)?	Rebalance the symbols	02 How many of each kind
6	Is each realm's population within the range of a real realm of the same area?	Change the densities or the area	02 Realm populations and areas
7	Are settlements clustered on good land, rivers and coasts, and does every empty area have a reason you can name?	Add villages to unexplained holes; thin out crowds on bad land	01 Where people do not build
8	Is each capital in a rich, defensible core on navigable water, and sized for its kind of realm?	Move or resize it	Capitals checklist
9	Does every border segment follow something (crest, river, forest, old district line, treaty), with wide marches across empty land?	Bend it to a real feature, or widen it into a zone	03 Borders and frontiers
10	Does every road end somewhere, bend to crossings, cross mountains only at passes, and do big rivers have only a few bridges?	Redraw the road; replace extra bridges with fords and ferries	05 Drawing the route network
11	Are there stops every 15–30 km (10–19 mi) on main roads, and water every 30–40 km (19–25 mi) on desert routes?	Add inns, villages, wells or caravanserais	05 Inns, hospices and caravanserais
12	Can you name what each castle controls, and are castles sparse in the core and dense on the borders?	Move it to a choke point or delete it	04 What fortifications are for
13	Does every region have food plus a speciality, every special town a visible reason, and every realm a source of salt?	Add the resource, fuel and way out, or a trade route	07 The basic rules
14	Are there ruins and older layers, and is every feature possible by the date in your title box?	Add old roads, hillforts and ruins; remove features that are too new	What exists by which date
15	Does each fantasy element have an “X exists, so Y changes” line, and does every magical city still show where its water and food come from?	Add the consequence, or cut the element	10 Consistency checklist

For the order in which to use everything on this sheet, follow **00 Step by Step: From Blank Map to Living World**.